

4th Applied Active Inference Symposium (Part 1, Nov 13, 2024) ~ Full Upload

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hello and welcome everyone it is
November 13th
2024 we're live and kicking off the
fourth applied active inference
Symposium it will be a great program and
we will begin with Carl friston and
Lance dcosta
discussing scale-free active inference
we discussed in live stream 58.1 with
Lance last week and in 58.0 with Arun
two weeks ago a bit about this work and
today Carl thank you so much for joining
and presenting on it so looking forward
to it and throughout in the live chat
please feel free to write any comments
thanks again everyone for joining and
looking forward to this
well thank you Daniel it's a great
pleasure to uh kick off the Symposium
this year um I'm going
to rehearse um a
presentation I made at the um active
inference Workshop um a few months
ago but this time I hope in a slightly
more relaxed uh and unpack way so I'm
fondly hoping for lots of questions and
uh embellishments from the uh from the
audience and uh and from Lance um this
is a bit of a colloquial presentation
it's just going to focus on something
specifically that we've been working on
for the past few months which is
effectively addressing the question how
can one scale active inference for
high-dimensional real world problems and

the solution that we've been pursuing is to adopt a scale-free approach um this work has been written up um and put on archives um however it's also being submitted for uh publication peer reviewed publication to a special collection um in uh memory of Herman hen um and we've equipped the peer reviewed version with a little forwards I just wanted to frame or preempt the scientific part of this presentation just by motivating why why the work of Herman haken is foundational and underwrites many of the um ideas and procedures that underly this scale-free approach to active inference so Herman hen was famous for synergetics um a physics Le approach to self-organization um and in particular the importance of of understanding the coupling between different scales of Dynamics in self organization and if you go to the Wikipedia entry you will read about macroscopic and microscopic variables and how they are coupled to each other in a in the fashion of circular causality um so there's both bottom up and top down causation where the top down causation is sometimes read in terms of um the slaving principle I'm going to present this approach um to harnessing this um notion of a scale-free universe uh in terms of installing that scale-free aspect in the generative models that we bring to the table to solve decision- making to uh simulate or um emulate active inference

under discrete State space models
so let me start at the beginning um the
nature of things and the free energy
principle um and this is just to
rehearse the fundamentals of the free
energy principle it all rests upon the
notion of a Markov blanket where we can
separate internal and external states by
boundary States or states that
constitute the Markov blanket namely
the sensory States and the active States
and for those people not familiar with
um this particular partition of states
of any random dynamical system it's just
specifying systemically um a system say
a brain in terms of its inputs and
outputs and um just
formalizing the conditional dependencies
that Define what is an input to a brain
or an agent and what is its output in
terms of the sensory and active states
that thereby render the system of
interest in this instance a brain open
to the environment the econ Nish um the
heatbath if you're a physicist or a
chemist um in virtue of the fact that
the inside can influence the outside via
the um active States the outputs and the
outside can influence inside via the
sensory States or the um or the inputs
and with this
setup we can elaborate um a kind of
physics namely uh um basing mechanics
that is almost well forly um related um
and inherit exactly the same
mathematical and functional form as all
the other mechanics that we find in
physics including quantum mechanics sta

statistical mechanics and classical mechanics the the only thing that separates the um the basic mechanics is the fact that we have in mind explicitly this partition between internal States and external States and the uh blanket states that intervene and couple the internal to the external and that basic mechanics is effectively summarized here in terms of um variational and expected free energy um where we can read the expected free energy as effectively PRI beliefs about the kind of outputs or actions that um in this instance a brain could exert on the external States um and the variational free energy keeping implicitly the um the internal States a good model so it's tracking in a probabilistic sense the um the Dynamics on the outside and we can combine this to uh create a generalized free energy um that now theoretically or perhaps even anthropomorphically we can now read the gradient flows that subtend this phasing mechanics in terms of action perception just by identifying the internal and active states of any agent as autonomous States and noting that they effectively um um flow on the same free energy functionals so that's the basic setup um but what I want to do before um applying the free energy principle in the service of things like active inference is just back up and reflect upon the fundamental role of the Markoff blanket and how that leads to a scal

invariant picture or view of Dynamics

and

self-organization so what I'm going to assume is that ultimately we will assume that basic mechanics um means that the Dynamics of anything that is defined in terms of its boundary or Mark of blank it can be described in terms of Act of inference but before that we have to think about the nature of things um and in particular

um if one goes back to the particular physics paper um the nature of the states that constitute things or particles um and there's a bit of a subtlety here because before you can start talking about States you really have to Define what is a state is it the state of something is it the state of a particle um and that presents an issue because you're defining a particle in terms of this particular partition into internal external uh and Boundary or blanket States so there's a um um there's a slight toor not topology but there's a slight um problem in resolving how you define states of things or particles and particles that possess um States and the resolution of that is to adopt um a recursive definition so that the particles constitute a set of microscopic blanket and internal States while the states are a set of macroscopic igen functions igen modes or just simply mixtures of the blanket States so I'm here deliberately um using the notion of microscopic and microscopic States just in uh as a um a

nod to synergetics and in particular
kind of synergetics that um Herman hen
developed and subsequently uh championed
so this uh is cartooned or illustrated
um on the right hand side here so we
sort of take some states um and um uh
and then um apply a a particular
partition on the basis of which states
are coupled to which other states and
the conditional independences that
ensue um and then having done that we
can then take mixtures zon mixtures of
the boundary states that then generate
new States um and then we can um
effectively by taking those mixtures we
effectively reduce the dimensionality so
that the higher scale um coar grains the
lower scale and then we reduce it and
then we just rinse wash and repeat and
we do that and every time we sort of
pass through this cycle of RG operators
um reduction and course graining
operators we move to one scale above
successively and this is the scale
invariant aspect of um that I want to
leverage in in the remainder um of this
presentation um I use the word RG uh
deliberately because um one way of
looking at this or articulating this is
to use the apparatus of the
renormalization group so this is where
renormalization gets into the game that
you're recursively um applying these
operators course graining things um in a
way that conserves their Dynamics and
that's an easy um um uh that's an easy
criteria to meet for us because we know
that anything that has a particular

petition can be described in terms of a gradient flow on a variational generalized free energy so we have everything that we require in order to apply um the apparatus of the renormalization group so that what that basically does for us once we recognize that

particles are composed of states where the states are ion mixtures of particles in this scale invariant fashion gives us a view of any universe that comprises things of things of things um as um inherently uh resolving a chicken and egg problem through scal invariance by application of the renormalization group so here's just another illustration of how that actually works in practice when you write down the equations I'm not going to bother going through the equations this is just used iconically um just to make the observation that as you move

upscales um so for example in an evolutionary context this would be the sort of the Dynamics of phenotypes and then we move up to the uh an evolutionary scale um by um effectively Co graining in a way that Herman hen um would exactly prescribe in terms of taking those slow macroscopic variables that decay very very slowly um which are technically the igen mixtures equipped with very small um igen values whose real part approaches zero From Below effectively saying that these patterns Decay very very very slowly and they have a kind of persistence that lends

higher and higher scales a temporal dilation so things change more and more slowly as you increasingly uh C grain things and that this falls out of the maths um when you simply apply a particular partition and put that partition in um as the um the course grading operator in this uh in these in this scale free uh formalism so in short basic mechanics entail a generative model of external Dynamics but we've just seen that just by defining things in terms of particles that have States and defining States in terms of the I functions of particles that external Dynamics are inevitably scale invariant it can be no other way and therefore the generative model must be scale invariant or when expressed as a graphical model scale free and I just slip that in as a um a nod to the fact that people often talk about scale freeness um generally speaking when people apply the notion of scale invariance to anything if that thing is a graph then you can call it scale free so a scale free thing is a um is a graph that has has this scale invariance crucially in both time and in space because we are dealing with dynamical systems and we're doing this um Dimension reduction or course graining um using these um eigen functions that necessarily pick out these slow macroscopic sometimes called stable modes they're stable because they they don't Decay or dissipate um almost

infinitely quickly so they're stable in the sense that they Decay very very slowly so how can we harness that observation that truism at least truism under the world as described by hen um in um the construction of plausible generative models and in particular um generalized Markoff decision or partially observed Markoff decision processes um well what we can do is we can just look at the different kinds of depth um and ask at what point would we um be able to leverage um this renormalizing aspect of the cause effect architecture of Any Given world that we're trying to navigate and predict so here's uh the normal um depiction of a um a generalized um markof decision process and by generalized what I mean is that there's um an explicit representation of not just the states of things that are generating outputs or things that can be measured or observed on the um on the sensory sector of a marov blanket but also their Dynamics in terms of the paths that um specify the transitions between states as we know normally encoded in terms of a transition tensor B here um so just by putting in the paths as explicit random variables that accompany for each given Factor um the hidden States we have a generalized kind of uh Markoff decision process uh encoded in the usual way in this in um in terms of tensors um that themselves can be parameterized as dis distributions so um that

parameterization equips these kinds of models with a parametric depth in the sense that if something is deep parametrically that means that the parameters of various probability distributions are themselves random variables that have their own probability distributions so for example the likelihood mapping that Maps from sensory states to outcomes here uh usually encoded by an a tensor here mapping from latent states to observations is itself encoded in terms of um derish parameters um blending this uh kind of model A parametric depth and um one can ask what the implications in that are well the implications are another kind of depth in terms of um you now have to op optimize or the B apply basic mechanics to the latent States the hidden States and that would be inference and then you also have to um consider at a slightly slow time scale the optimization of the derish parameters and that would correspond to learning and so on um if I wanted to go to the structural learning of the model in of it in and of itself we also have um implicitly a temporal depth in the sense that we're going to be raring out into the future and that there are dynamics here um and this Temple depth is generalized in in virtue of having um these explicit uh path variables that we're going to uh exploit later on we can also have hierarchical depth in the sense that we can put

um compose many markof decision processes um in the in the sense that the outputs of one process now constitute the inputs or the empirical prior the top- down constraints the inductive biases on the Dynamics and states of the level below and the Dynamics are um um the kind of inputs or sorry outputs of the uh upper level that provide empirical PES on the paths and the initial conditions are supplied or mediated by a mapping between higher levels slower States um and the uh initial conditions or initial states of the lower uh of the lower level by this DET tensor here that you can lump together um in terms of sort of high you can lump together the D and the E tensors they play the role effectively of the a tensors in sort of um mapping from one level to the uh the next level there's also factorial depth which many people will be familiar with in terms of um for any one of these um lump States here there may be many many different factors that have an a certain Independence structure such that you have interactions between factors in determining outcomes um you can look at this in terms of uh a generative model when equipped with factorial depth effectively entangles latent states to generate um outputs or outcomes which themselves can be prior constraints on the states and dynamics of of the level below such that the inversion of these models effectively disentangles the data or the content that is being observed to

explain it or um in terms of a disentangled representation which is the um the latent State and dynamics of the generative model so what we have here then is um a generalized discrete State space model with Paths of as random variables um just to um return to this um fundamental aspect that as one Ascend scales in real life in um you know in terms of uh the external States things slow down in virtue of the um the the the the synergetic um mapping from fast microscopic States uh to um slow macroscopic States um the same thing can now be implemented in these um Markoff decision processes that have a hierarchical depth um and the way that is done is effectively to discretize or quantize time such that there there are more updates at any given level than any um than the level Above So for every um update at this level there are uh say three updates or two updates at the level below and that um affords a temporal scaling um in accord with the scal inv variance um implied by the synergetic view of self-organization I've tried to cartoon that here in terms of if you imagine lots of these um little markof decision processes populating this uh circumference here and that as we go deeper and deeper and deeper into a model with hierarchical depth as we consider updates in Universal or clock time there are more updates at the peripheral level exposed say to the know the um external states of entire brain

um relative to the active to the updates
that as you get deeper and deeper and
deeper in into the
model so that's where the temporal depth
um becomes if you like intimately tied
to the hierarchical depth the higher the
hierarchy the slower the updates which
is why um we generally parameterize the
initial conditions um or generate the
initial conditions and the initial path
from the States above and that path can
pursue um um a trajectory over multiple
time steps until a certain um Horizon uh
and which case it then sends messages
back to the level above likelihood
messages um that then cause belief
updating um and the subsequent Step at
the level above and then generating
priors about the subsequent path the
next PATH at the level below and then so
on recursively uh to the depth of the
model um so that's the time bit what
about the spatial bit what about the S
spatial core scening the lumping
together of various States um and of
course this depends upon the particular
partition so in the General Physics
formulation You' be looking at uh
Markoff blankets and Markoff blankets in
Practical applications one can take a
shortcut um and that shortcut effective
rests upon um the simplifying assumption
for most systems that are um composed of
um coupled um coupled uh dynamical
systems um the interactions are usually
local so if we um just think about um
some
um um system that um can be described in

terms of local interactions so for example um in a um a ukan world um that comprises some ma massive objects that bounce around and if we ignore gravity for the moment then they're only going to influence each other or depend upon each other when they're touching when they're um when they're proximate um effectively what that means is the the the mark the particular partition that defines all the Markoff blankets in um in this particular State space um is composed of lots of local little tiles um so what we can do is we can use a um a device um very prevalent or indeed ubiquitous uh applications of the renormalization group which is called a uh a block or a spin operator that just groups together local tiles or quadrants of a say two-dimensional arrangement of tile of of States so I've Illustrated that here just by um in terms of um How We Do the spatial core graining in these um what will refer to as a renormalizing generative uh model with spin block uh Transformations um in terms of um this little um um this hidden state or hidden factor with its states and paths here um is responsible for generating predictions about the Dynamics of these two um factors um at the um at the lower level um and in so doing what we are effectively doing is successively grouping together sets of sets of sets of sets so that any set or group at one level is accountable for or trying to generate the Dynamics

of a small local group at the at the
lower
level um this is quite fundamental
that's kind of Architecture is quite
fundamental because what that means
is anything at the lower level only has
one
parent and if it only has one parent and
if remember the mark of blanket is
constituted by the the the um the
parents and the children the parents of
the children because there are no
co-parents there are no parents of the
children and what this means practically
is that we can now ignore dependences
between or effectively resolve or
dissolve any dependences between the
factors at any given level because each
factor is only responsible for its
children and it doesn't share any
children in virtue of this um particular
spin block transformation so this leads
to um a very efficient kind of
generative model um or certainly an
efficient kind of inversion of the gened
model um where we've got all these
converging streams but that the streams
don't have to um uh talk to each other
until at the point that they come
together through these blocking or spin
block um um uh Transformations so at
some point all the tiles or the groups
of the lowest level will actually come
together at in um uh at the top or very
deep in the model at which point we're
now encoding quite long trajectories
technically 2 to the $^$ of the depth minus
one um in terms of updates um but at the

lower level um there are no interactions between the blocks as we Ascend them and this is just to um remind you we can treat these d and e the um the initial States and initial paths effectively as likelihoods that couple um one scale or level to the scale below so how uh with if that's the kind of model that we aspire to because we think that's appropriate for any world that shows this scale uh invariant aspect in space and in time or in state space and in time um how are we going to build one of these things um and this is the sort of second um if you like um thing that we have been working on um um because it's going to be very difficult to specify by hand these um scale-free or re normalizing gered models we really want the model to learn itself so basically how does one build a renormalizing generative model and what we're going to do we're going to use very fast structure learning and this particular kind of structure learning is necessarily recursive and uh in the sense that we're going to build these models through the recursive application of these um RG operators or renormalizing operators so what's fast structur leing so fast structureal basically equips the model with unique states and transitions as they are encountered so to put that simply what we're going to do is grow a model from scratch from nothing and how are we going to do that well we're effectively

going to take one observation and say
okay that was generated by some latent
State and then we're going to take the
next

observation um and say okay I haven't
seen that particular observation before
um so I'm going to induce a second
latent State and furthermore I've seen
these two observations in sequence
therefore I'm going to um encode that
with a transition from the first to the
second state and then I just repeat I
take the the third one and if I haven't
seen that before I induce a third latent
State and a third a second transition um
accumulating all the Unseen or unique
instances of um

um of

observations in a likelihood M mapping
that just grows and grows and grows
until we have seen something before so
we don't induce a new state or
implicitly column of the likelihood
Matrix if we've seen uh the previous
state that's just a um um an instance of
something that has been seen before um
it may well now come along with a
path so the next state may not be the
same as a the subsequent State following
the previously seen instance of this
state and therefore I'm going to induce
A New Path and slowly grow my B tensors
and my probability transition matrices
until I can um basically encode all the
dynamical structure of the effectively
the training sequence of observations
summarized in a a maximally efficient
way and I mean maximally efficient in a

technical sense um in the sense that the implicit mapping that we are growing from from scratch um has the highest Mutual information between the latent States and the observations that are at hand um the way that you can think of that in terms of um free energies that the expected free energy um is just the mutual information in the absence of any expected cost cost so the expected free energy is just the mutual information of any mapping um um um plus um or minus an expected Cost U where cost is defined the usual way in terms of constraints or prior preferences

so one can actually look at this uh fast structure learning um as a particular kind of active inference but not about um it's not active inference Over States or Active Learning over parameters it's active selection of the right model in terms of the size of the likelihood tensors um and the transition uh tensors um and the criteria for whether we accept a new Latent state or a new transition is simply does it optimize or improve the expected free energy namely does it preserve or conserve or maximize the the the mutual information so we can actually look at this structure learning as just another instance of active inference but here deployed in terms of um um in terms of basic model selection or so some people call this uh structure learning that has exactly the same rules as um planning um and um uh and um accepting um parameter updates provided they they maximize expected

free energy as well um the application of this kind of uh uh fast structure learning to um these renormalizing models um looks complicated here's the algorithmic description of it um again please don't worry about the details here it's relatively simple you all we're doing basically is ingesting um observations but we're just doing it for local groups and um and then we are combining the initial conditions and the paths into new outcomes for the next level um and so because we are doing this for local groups the number of groups just get smaller and smaller and smaller as you as you get higher and higher and higher but obviously the number of combinations of paths and state initial States increases as you get deeper and deeper into the model and you may be asking well you know how do you ban that you know how can you um possibly assimilate all the data without having an extremely large number of latent States very deep in the model that are now encoding effectively evolving patterns or paths over quite a large number of time points into the future well it's trivial to do that because the size of the highest level I'll call them generalized states that basically generate paths and initial conditions for the lower levels um is upper bounded by the number of elements in your training set um so that you can control and upper bound the size of these renormalizing operators or likelihood

mappings and transition um um tensors
simply by giving it canonical training
data the kind of data that you want this
um agent to generate or this um um agent
to recognize and thereby
predict um so here's a a particular
example of that um what we've done
here is just ask can we very very
quickly learn a Jed model that will
generate a um a simple movie and this
movie is simple because um
it it begins and ends in the same state
so it just Loops over and over and over
again um so we only have to supply one
Loop um to this fast structure learning
so that the model builds itself and can
then be used in generative mode to
generate the movie time and time and
time again um so here's one depiction of
um the results of the learning the the F
structure learning um and subsequent
infants when exposed to um this kind of
image so the course graining here at the
very bottom in in fact um uses um um a
slightly finessed version of the spin
block Operator by um harvesting um local
igen modes or IG uh IG images um of
little patches of of the video to get it
into a reduced or coar grained rep
resentation that then is just passed up
using these spin block operators and in
this instance we just need uh two levels
to the model to uh have the highest
level see the entire image and then if
we present the image um um after
discretization using the singular Val
decomposition or the sequences of images
to a model it learns the model and then

we can generate um we can generate um
little movies simply by moving through
these paths or episodes I use the word
episode um because remember these
generalized states of the highest level
encode paths or trajectories into over
um 2 to the N minus one time steps in
the future when n is the um is is the uh
the depth of of the model and these
episodes themselves have Dynamics so we
just go through a little Loop here go
back to the beginning and thereby
generate um this image at the
bottom what else can we do with these
kinds of models
well one thing we can
do is
um present not the entire data but just
u a little prompt um
so in this example what we've done is
actually show it just the upper right
quadrant of the
movie and the and see what this
um this generative model um believes is
going on in terms of the um the
predictive posterior in output space in
image space in this
instance now because during training
it's only L to recognize a dove flapping
her
wings um that's all it can believe come
is the cause of its Sensations even if
those Sensations are very partial so
basically what we have here is a kind of
pattern
completion that inherits from the fact
that this generative model only knows
about latent causes that constitute an

entire Dove flapping her wings so um
even though this is what the dove is
actually um exposed to it's sensing this
is the predicted after a couple of
frames it very quickly um is able to
predict um what it would what it should
be seeing or what it would have seen had
it been given the entire uh data and
another way of thinking about this um
remarkable ability to sort of fill in
the gaps um is from the point of view of
compression so just a little digression
here
um a lot of the basing mechan mechanics
and um variational inference that uh
ensues from variation mechanics can also
be likened to um
the to
um the same fundamentals underwrite um um
efficient information transfer so we've
talked about maximizing Mutual
information in the context of uh um
algorithmic complexity and um the
associated Universal computation um then
what the objective function looks like
is basically how much can I compress
this content um to represent it in the
most efficient way so you can look at
this past structure learning under these
renormalizing or scale-free generative
models as the most efficient way of
compressing whatever your data you've
been exposed to um and because we've
compressed it in such a way that it can
only now recognize a dove flapping her
wings um then it can if you like fill in
the gaps in a very efficient way when
it's given partial or noisy data and I

use the word noisy here because you can regard the missing three quadrants here as extremely imprecise data data that has extremely low signal to noise for example so um here's another example in the example we we um took a um effectively um a movie or a damic or um a a system that had a periodic orbit um um which was each flap of the Wings will this work for um a periodic orbits and and generally for um stochastic um uh chaos um yes it works it works relatively efficiently so this example um we um used training data that was generated um um through creating images of a white ball that moved with a chaotic trajectory governed by a uh a len system so this is the kind of system that governs or is used to describe uh chaotic events um in meteorological systems or in fluid convection um so by applying exactly the same procedures as in Pre in the previous example um we now uh generating this as a three-level um um hierarchal or renormalizing model um that now accommodates the um stochastic or Randomness induced by not only random fluctuations on the Dynamics of this stochastic uh random dynamical system uh but also if there were any determin IC aspects the deterministic uh um um chaos that ensues from exponential Divergence of trajectories that now has been summarized stochastically in terms of probalistic um uh transition matrices here which means that um we can having compressed the training data we can now

withhold not a quadrant but just simply
withhold the data to stop the data uh
but the model will keep on generating um
what it predicts would have happened
given the statistics of the Dynamics
that have been installed in this
renormalizing generative model um and
what we've done here is um um for the
first um few hundred time bins at the
lowest level we basically rendered the
images or the input very very imprecise
so that everything is governed now or
everything is generated from this top-
down Prize or posterior um um posterior
um predicted posteriors that inherit
from from precise prior because there's
no precise likelihood um and then we've
actually uh removed the U um removed the
um imprecise input completely and just
allowed the um the agent to generate its
own beliefs about what's uh about what's
going on um so here this is fairly
precise in the first instance and then
we remove completely the um stimulus but
it still uh imagines or generates its or
predicts um what's going on in terms of
this chaotic um Dynamics um and then um
with a degree of uncertainty then um
continues to then generate um the um the
Dynamics associated with this kind of
stochastic
chaos here's a preter example um exactly
the same uh technology and approach um
but here explicitly um talking about um
video compression um so we literally
just presented a video of naturalistic
Dynamics in this instance a little bird
a robin um feeding um itself um and um

compressed that using these this
renormalizing um generative model um
such that we can again simply remove
stimuli at certain points um in the
video um and yet the uh the in in the
agent's mind nothing has really happened
uh because all it knows is that if it
was like this um previously then this
must have happened um and so on through
this in this instance um orbit that
doesn't have a closure it's just a
trajectory through from the beginning to
the end of the sequence and here's the
posterior prediction after just being
exposed to a couple of frames right at
the beginning um in the first um six uh
time frames of
um of about 12 sorry of um 18 um this
example moves applies exactly the same
um approach um as
previously but now we're not talking
about um ingesting or learning or
compressing images we're talking now
about compressing or ingesting sand
files um and that is if you like a
simple or limiting case of this
two-dimensional uh compression that we
talked about before so in this instance
now what we're doing is compressing um
one-dimensional frequency summaries of a
particular um wave file or uh um file um
that um describes flu situations and the
intensity of uh of a sound that then is
um summarized in terms of a continuous
wavelet transform um and then the
succession of different trajectories in
this um time frequency space is then
ingested or assimilated or learned by

the generative model um now
unfortunately you can't hear this um so
I'm not going to bother playing it but
um it um this is what the agent was
trained on it's basically Jaz
piano um and these are the um uing
predictions in the absence of any inputs
it's been used to generate music um
using um the stochastic
transitions um through various bars of
Music um that it has learned um
returning to one or other bars of the
music of the music when it reaches uh
the end of the orbit to generate
something that um sounds at least Jazzy
if
not oh started it but you can't hear
that so I'm not going to pursue that so
to summarize what we've done so far um
um we've gone beyond generative AI in
the sense of um
um having a a true gened model um uh
that entails active inference in agency
um and in the setting of these um
well sorry what we are going to do now
is go beyond just simply generating uh
content um having been uh trained on the
basis of some training data um and we're
going to now consider um agency by
putting action into the mix um and in
the setting of um the simplified setting
of optimal State action policies where
we can effectively um ignore the
imperative to maximize expected
Information Gain or um um explore in the
sense of uh responding to epistemic
affordances um we can
um do one of two things we can either

use some Universal function approximator
say deep learning to optimize the
mapping from sensory to active states to
maximize the um expected cost or the
expected utility part the expected free
um or we can use active inference to
realize predictions under a normalizing
generative model of rewarded events so
what I'm doing here is deliberately
setting up a dichotomy that tries to
distinguish the active inference
approach to purposeful Behavior where
the purpose is provided purely by the
prior preferences or the um um or the
the uh the constraints usually encoded
in the c um uh tensor um against a sort
of reinforcement learning approach and
this is a rather busy slide but um I
think it's probably worth just um uh
briefly uh rehearsing the distinctions
between active inference and
reinforcement learning so the you know
the story with respect to action as a
gradient flow on free energy which
basically reduces to um pick those
actions that maximize the predictive
accuracy um in the sense that if you
look at the expression for the variation
free energy the only thing that you can
change um is the accuracy in the sense
that free energy can be written as
accuracy minus complexity minus accuracy
and the complexity um is does not depend
upon action whereas the accuracy does so
that picture basically says um I can
describe behavior in terms of a
generative model that um is optimized
through um perception uh in the usual

way um cast in terms of say variational inference um that basically generates predictions about what should happen next and then I can pick my actions to fulfill those predictions in continuous state for formulations this would be start minimizing the prediction error usually the proceptive domain but you can generalize this to any any consequence of action I'm just going to pick that action that brings about or is most likely to bring about what I predict will happen next and what I predict will happen next um is optimized through perception through variational inference based upon my learned and selected um generative model so this is very similar to sort of um model predictive control um um and neurobiology would could be cast in terms of um um perceptual control theory um which puts action in the game basically of realizing evidencing um predicted States so I've tried to summarize that here just in terms of um you know our markov blanket where we're optimizing our beliefs about the world um noting that planning in this instance so con um planning as inference uh is all about the paths it does not have to be about action it's just about how does this world unfold um and from the point of view of these scale-free generative models we're really thinking about how do episodes follow each other which episode is going to follow this going to U um this

episode and we can say well you know you are in control of that um you know imagine a um a benevolent World in which everything happens according to um your beliefs about how to be an expert in this particular world and the agent can plan a series of episodes generating predictions all the way down to the bottom which then slavishly action come fulfill just by fulfilling those predictions um so one could in some sense

think of this in terms of control as inference at the bottom level in the spirit of model predictive control and planning as inference on the top level so belt and braces in terms of realizing active inference in this particular example where I repeat um we are really just worried about um State action policies um why are we worried about State action policies well these are the kind of policies that reward learning schemes um are aimed at um and have a similar architecture um but um to my mind a slightly more um over engineered architecture first of all you actually need explicit inputs which are rewards um there are some proxy for um the constraints uh you know um um under which um the basic mechanics would normally operate um but more importantly you're you're you're basically sending um your inference scheme is basically about selecting the right policy I've used Q learning here um um to maximize the discounted reward um so similarities and differences but we're going to focus

on uh active inference um uh in this in
the particular setting of these
renormalizing models where repeat
planning and thinking or imagining a
future can proceed at the very highest
level that has no notion of action
they're just patterns that unfold they
may or not be caused by me I don't know
I don't care um but the action at the
lowest level the reflexes as it were um
do um are in a position to make my um
post predictive posters come true simply
by picking those actions that realize
the predicted outcomes

um

so how are we going to leverage that
kind of merely reflexive active
inference um in the context of building
or learning from scratch um
renormalizing or scale free generative
models that would be apt for doing
something implementing a state action
policy um and I'm going to um appeal
here again to this notion of um active
selection as one way of um looking at um
action which can always be at least phys
mathematically expressed in terms of
um of um doing something if it minimizes
expected free energy um and in this
instance um what we can do is actually
apply it not to the selection of the
model but the selection of the training
data so if we just um uh apply
the the dictat that you only select
something if it um um minimizes expected
free energy that
includes um the expected cost what that
basically means is I'm only going to

select those data features or those training sequences that don't come along with a high cost namely those that are rewarded um and that's what we're going to do here we're basically going to um learn to play um a very simplified um game of pong where where this ball bounces around inside a circle um and we can move the bat um in a way to uh return the ball so that we can avoid penalties when the ball hits the upper boundary um and secure a reward um whenever um there's contact of the ball and the bat and how can we learn this kind of dynamic these episode in exactly the same way we learned the video of the robin or the the um the dove flapping her wings well what we can do is we can just um select um instances of random play that conform with um our um dictat that we cannot um include those costly events and so what we've done here is just generate training data training sequences using random play but omitted anything that does not that contains a costly outcome and furthermore just to make things more efficient we've restarted the random play after a reward is secured and then just use the subsequent play if it actually attained a subsequent reward so basically chaining together bootstrapping together or splicing together sequences of rewarded uh play um and then use that to um uh to um or submitting that for fast structure learning and now we have a generative model of expert play effectively or a

generative model that has that entails a state action policy that can be realized through um um reflexive active influence so active data selection um and um um sort of cartoon that here in terms of applying a certain kind of Maxwell's demon that only lets through non-costly episodes um basically yes admit this sequence for fast structure learning if there is a reward if not ignore this ignore this sequence um and this effectively learns um or furnishes a generative model of events that lead to a reward and nothing else so now this agent can only recognize expert play therefore all of its predictions have to be uh the predictions of an expert player and because action is fulfilling the predictions it will look as if it's an it is an expert player and here are the um the Learned trajectories um at the highest level of different episodes that we'll see uh um shown in in an alternative format uh in the next slide so here's the initial starting episodes little events here uh the red dots correspond to reward so you know first hit second hit and then it falls into a nice little orbit and this com completes um this orbit indefinitely by in it's the way that it has learned so remember it's only been trained on random data so this particular style of play is unique to this agent but it is sufficient now to play with the 100% um um uh performance um simply because it's now going round and round its orbit very much like the wing flapping her dug

flapping her wings um these are just different ways of portraying the same um the same Dynamics and the same the things that have been learned um the first eight frames and then subsequent frames um of expert play um this is the usual format showing the posteriors at different levels the posteriors over the states and the posteriors or predicted posteriors over the paths and then rewarded play as a function of the uh free energy at different levels here illustrated by the by the red dots every time there's a hit up until 512 time frames here and here's this rather long orbit through 50 episodes um each four uh time steps in in length that subtends this kind of play this is an interesting one and speaks to the fact that we've actually switched on inductive inference at the highest level um so what does that mean well what we've done is just um um basically take each of these highest level

States worked out the sequence that it encodes as by um um Computing or projecting down through the model and if that sequence includes a hit um we said that's an intended state that's the kind of state that I want to work towards through the successive um four um time step

epochs um

and if I want to be at some point in the future in one of these intended States in this instance rewarded States or episodes um what states must I avoid in

order to preserve or conserve the possibility of getting in or getting to one of these intended States um basically that's the technology behind inductive inference and it's sort of summarized here in terms of the latent States of uh you know 50 or so of these latent States in terms of how many time steps would it would it what's the shortest number of time steps until the intended State um and basically um we um as we move from state to state to state we want to keep in the white areas here avoiding these dead ends if you like from which there is no possibility of reaching an intended State and it's really trivial to evaluate these um areas that you have to avoid because you know the probability transition matrices so you just multiply it by itself a given number of times and you can work out can I get from this state to this state and if you can't um then you get this um black area here and you know you have to move from here to here to here to here to to get to um your um your intended state so that's inductive inference in brief and we've applied at the top level so we just the you just tell the agent this world um is under your control imagine the you know imagine what you want um noting that these particular episodes or events contain you are things that you should pass through or should experience um and then deploying inductive inference to basically predict what will happen all the way down at the bottom and then

using reflexes to make that happen um we can apply exactly the same um technology or the same procedure to different kinds of games there a slightly more um complicated game um SM to emulate breakout um where we've introduced uh certain stochasticity in terms of the initial uh conditions and uh sticky action but the the overall behavior is the same is basically choosing paths through episodes into the future given what it knows that can be achieved in order to get to uh rewarded uh States so effectively what we've done is create um an attracting set of episodes at the highest level of these scale-free um um generative models where all the aable paths and paths now in this generalized State space or event space lead to reward um we can go a little bit further and just um add in uh various episodes by merging uh or assimilating new data new training data if and only if um the path actually leads to an existing um part of this attracting set and I've just Illustrated that here um in this final example so this is the path or the orbit or the attracting set that was learned at the highest level by the um the ping pong playing uh agent but now we can um fill in um fill in this um effectively um allowable Dynamics to include all states that lead to a path on the orbit um and or thereby all paths will eventually lead to reward or to roam so we can augment the model of expert play with paths that lead to an orbit namely this attracting set um and

paths that lead to event that lead to
the orbit um recursively and get quite a
dense set of allowable transitions that
we can be guaranteed always keep within
the attracting set that will contain
these preferred States or um states that
we have um constraints uh where um we
have constrain those states that are not
visited simply because they cannot be
recognized and that's it thank you very
much

indeed amazing Carl thank
you

awesome okay Arun if you'd like to lead
in with a
first question and Lance too thank you
but go for run

first uh yeah thanks very much Carl um
so my first question is um based on a
sentence in the paper which talks about
any model being

Spar must be renormalizable could you
say a little bit more about why that
must

be

oh I can't remember give me a
clue uh I I'm presuming it has to do
with Markoff blankets and yes yes yeah
that's as far as I got I see yes no I I
remember now um I

it may have been in reference to some
work done by Dalton um showing that with
probability one provided the system is
large enough there will be Markoff
blankets and as soon as there are
Markoff blankets there is now an
opportunity to take Markoff blankets or
Markoff blankets in the spirit of um

that applying the renormalization group
that that I started off with um so as
soon as you've got um as as soon as any
um sparsely coupled random dynamical
system um can now support a um a
particular partition into Markoff
blankets you've now got the opportunity
to take Markoff blankets of Markoff
blankets and then apply that
renormalizing procedure that I said
before so all you need to posit is any
system um that has a Markoff blanket at
any given level and as soon as you've
got that you've got Markoff blankets all
the way up
um not necessarily all the way down but
um you've certainly got them all the way
up um the the physics approach would
would actually um also actually um
require for for there to be Markoff
blankets all the way down as well but
that's that's that's another issue um so
um what that basically means is that um
all you require is a sufficient degree
of
sparsity that would um lead to the
conditional Independence
that are definitive of Markoff blankets
and in one sense much of the sort of um
the um the maths behind the free energy
Principle as it inherits from random
dynamical systems or uh in physics say
Langevin equations is simply
showing that a sparse coupling a sparse
causal coupling in a physics or control
theoretic sense not a philosophical or
Granger um or um cause effect structure
sense but in the physics sense of a

coupling so that the change in this state influences the rate of change of that state so that kind of which could be written as an influence graph so if there's a sparse coupling on the influence graph the equivalent Bayesian graph or probabilistic graphical model induced by that sparse coupling can now be interpreted in the sense of Pearl's Markoff blankets so all you need to get to the Markoff blankets is sparse causal coupling hence sparse random dynamical system that is sparsely coupled just out of interest the spin block the motivation for the spin block applying the spin block or blocking Transformations that basically comes from the assumption that many systems that people deal with both in computer vision and in idealized models in physics has an exquisitely sparse distribution and that's because the causal inferences are all local so as soon as you look at a coupled map it is or a globally coupled map you'll notice let's take the lattice for example let's take a sort of idealized crystal so what you're saying when you write down a lattice is I'm only connected to my neighbors to my neighborhood or to my Markoff blanket that means that in a large lattice where I could be connected to N^2 neighbors I'm only connected to 1 2 3 4 5 6 7 8 so a fantastically small almost empty set of possible connections

and I use that phrase because there I can't remember who said it now but I really liked it the notion that if one looks at the connectome in the brain in terms of the connections it's almost empty you know the number of actual connections you know white matter connections external connections and indeed intrinsic um um external connections are almost zero in comparison to the total number of possible connections between the 10 to the 11 or whatever uh neurons in the brain so that's not surprising and you it's one of those nice examples that the the anatomy the functional architecture of the brain recapitulates the anatomy of the lived World in it has this incredible sparsity um um and that just means um the universes in which there is um the action at a distance is quite rare um um means that that you you're just interacting you're just coupled to your neighbors and that's and that gives you if you then look at the mark of blanket structure or the particular partition for these Itis structures you get to the spin block um blocking Transformations as a way of carving up nature you know in terms of little patches or Markoff blankets

I do have a follow-up question on that but then I'll hand it back to Daniel or Lance um so the Itis example is really interesting because I think there's a symmetry there that maybe we lose when we talk about blankets that only have one parent so in a normal Ice as you say

you're just connected to your eight
Neighbors in 3D
space when we looked at the spin block
Transformations before
we were going let's look at this square
and everything in this square has one
parent but then the square along is
connected to a separate parent so now if
you're at the
boundary we are breaking some symmetry
there yes technically or practically a
very good point um it doesn't really you
it doesn't really um
uh have any material impact upon the the
the algorithms or the partitioning um so
just practically what happens is you you
you you you just take your tiles and
then you have a half tile or a partial
tile um until you get to the boundary so
these these groups in the grouping
operation or the or the blocking
operation are not necessarily the same
um you know they're not the same size um
and indeed in uh um current um work
we're actually dealing with sort of
multiple streams so you can have sort of
audio files and video files separately
tiled and then converging right right at
the top um ideally what you'd want to do
is is
um not use this um um blocking
transformation but to try and identify
the Markoff blanket structure at the
particular scale in question and
actually use the Markoff blankets based
upon um well there are a number of ways
of doing that you can either use a
markof blanket Discovery algorithm of

the kind that Jeff Beck um Works upon
or you can use um empirical um analysis
the the dynamical coupling um uh by
estimating effective the
Jacobians you know for those people are
interested there's a worked example of
that in the particles and Parcels in the
brain paper in network Neuroscience um
it's really framed in terms of data
analysis but you can use the same
principles to um in to identify by a
particular partition from a large number
of um um States or boxes or or or or
pixels um um you know in three and 2D um
respectively um there's another thing
though that I think your question speaks
to
um which is um you know you've got you
got you know one block of four then
another block of four and then um
because of the sparsity they're
separately uh modeled at at the level
above so I thought you were going to ask
how in Earth now do you start to repair
the obviously rather false assumption
that you've got just local action
because of course in our world there's
lots of action to distance you we are um
talking to each other over over over
thousands of miles or Kilometers and
you know we have Vision uh so all the
and we have uh linguistic communication
via sound so you know it is interesting
that we have actually
violated much of the local action if I
was a virus this would be perfectly okay
because the only thing that's really
going to affect me as a virus is the

things that Arch molecular contact with
me but for you and me um and the worlds
in which we live in um we can't there is
no such thing as just uh the action
distance now starts to really matter um
and effectively because the two um
latent States generalized latent States
generating stuff um initial conditions
and paths for the two blocks below
because they themselves now constitute a
single uh contribute to a single um
latent State at the at the level above
as you move deeper and deeper you start
to now

repair the simplifying Assumption of
conditional Independence between the
groups so it is at the higher levels
that you now become you're able to see
these action this action at a distance
is coupling your what what happens over
here really matters in terms of what's
going on over here but you only ever see
that you only have that as part of the
generative model right at the top which
is why all the inductive inference is
implemented right at the top level where
where you can see everything um so that
it becomes you know if you like you are
now you are now in a position to um um
model and thereby generate appropriate
predictions that rest upon um these long
range dependencies um and of course you
know using the word long range I mean in
the sense of the the coupling distance
uh you you in a dynamical sense is that
what you had in mind uh that was exactly
what I had in mind thank
you thank you

Lance do you have any remarks or want to
add a question or I can yes please go
for it yeah I have lots of questions uh
thanks a lot Cal for the talk so I'm
just wondering about this all Road lead
to Rome slide that you had at the end
and I'm wondering whether you have some
sort of a curse of
dimensionality as you increase the
dimensionality of the system so it
sounds to me like the manifold of expert
play because
um because you're taking temporal
trajectories that's going to be a
onedimensional
manifold now as the state space of the
game um increases in
dimensionality you're going to have an
ambient space of trajectory that's much
much higher
dimensional and so it sounds like the
amount of
trajectories they need to you know
construe to bring me back to your um
onedimensional expert manifold is is
just going to grow and grow and grow and
it sounds like it might be hard to fill
that space so could you could you
comment a little bit on that and whether
it's a
problem right yeah in general in my
world you know these things aren't
problems but they're challenges um and I
think that challenge is yet to be met to
be met um so what I imagine people will
do um is that you will um
ingest
extra episodes or

sequences that bring you onto your
attracting set up until some bound um
and then you will switch off active
selection by which I mean the structure
learning um so you stop growing the
model and then you switch on parameter
learning so that now you're forcing the
Transitions and the um expression of
various generalized states to
accommodate variations around the
training data that you have been
previously exposed to so that fixes the
size of the tenses and hence the
dimensionality of the state Bas State
spaces in question um um but will in
principle now accommodate by preserving
by using sort of um Act
learning in the technical sense that you
only update a parameter if it um does
not uh increase expected free energy or
decrease Mutual information
um then um you you you're now in a
position to become more robust um and
perhaps I should qualify this and the
whole point of this adding in extra if
you like insets onto an attracting set
is just to equip the agent with a
certain robustness should itself find
find itself outside or off that
attracting set so um I know Lance knows
this but I'll just make this explicit
for for everybody else um this kind of
learning um and state action policy um
implementation is very much like
learning to ride a
bike um so you you basically um just
accumulate and remember those
experiences that were successful so for

example I you know managed to um rotate the pedals by one cycle and I don't fall off and I remember that and then I have another go and I wait until I've actually done two cycles and I don't fall off and I remember that I forget everything every time I I I fall off and then I keep on going until at some point I connect back my last cycle was identical to the first cycle and then I've learned to ride the bike but because I've forgotten all the falling off I can't recognize or know where I am when I when I've fallen off so I've got a very brittle expertise that will not allow me to recover if I fall off the edge so another example which makes it slightly clear I think is like mountain climbing you know if I want to be an expert mountain climber I cannot learn by my mistakes provided I'm climbing a sufficiently high mountain should I fall off it then I will die another example example here is learning how to cross the road as a child you know you cannot learn by um um experiencing being run over but the cost you um you pay or the price that you pay is you can't recognize when you are being run over um and that I repeat lends a certain brittleness and a lack of robustness to agents who may be exposed to Danger that doesn't actually terminate them so putting this robustness back in by growing the um by growing the attracting set or basically putting these insets on um in this sort of generalized um phase

space um you know it's just to make it more robust um in numerical studies it it does seem to um converge it doesn't seem to grow indefinitely yeah so those are likely violations are moving off the attracting set are themselves quite small relatively small in number um you know practically speaking uh if if you've controlled the sources of stochasticity or unpredictability so it may not be always the case that you need to have an explicit bound on the upper the size of the um the very high level tensors if if you do then one has to now think of a principled way of making that upper bound an emergent property of um um expected free energy minimization it may be that you know you violate now the mutual information that certain rare events are just so rare you're actually compromising the mutual information by including them as a particular option um or a particular column of of a likely likelihood mapping or yeah but I don't know but these are all really interesting questions to pursue

thank you Carl all right Arun go for it um yeah so following on from that I think it's an interesting question on the uniqueness part of ingesting data and the question is are there many more ways to fail than there are to succeed so if you're passing your data into sort of Max demon Maxwell's demon uh with a gate saying accept this data reject this data if this uh improves my Mutual

information if there are many more ways to fail but then get back onto the uh successful path but there only a few paths that actually keep you there would it not make sense to just continually accumulate these sort of recovery States because the mutual information will keep increasing and then you stop there then you parameter you learn parameter wise because we're not including the sort of Prior distribution of How likely you are to be succeeding at any given moment because with the uniqueness thing you don't know oh 10 times out of 10 I'm cycling or like nine times out of 10 I'm cycling and one times out of 10 I've falling off the bike yeah I think it depends um again I don't have any explicit numerical analyses or analytic analys mathematical analyses of this but intuitively um I think during learning there may well be um um good mileage in ingesting failures um or roots to success if you like or to the attracting set that that actually Traverse regimes that would normally be excluded by my constraints or my PRI preferences um you know just taking the view of the role of of the C tenses or prior preferences from the point of view of physics what that basically is saying is it's actually carving out vast regimes of State space latent State space that you should not be in because the it would be uncharacteristic of you to be in these things so it it's really not so much about the rewards and the preferences uh

it's really what you where you shouldn't be and that's what defines the attracting set um however before you've learned that attracting set it may be useful to include constrained regions of State space that offer a route into or onto that attracting set so I can certainly see that that would be a useful device but notice that once you're on the attracting set you will never visit the constrained regions because you now keep yourself on the attracting set um um and therefore if you've encoded those but you never visit them your Mutual information will actually fall so what that means operationally is what you'll probably do is have this sort of slightly over inclusive structure learning learn how to cope with all the necessary sources of Randomness so that your attracting set is sufficiently robust and you do actually visit from time to time but not um not never um s of excursions from the attracting set drawn back onto the attracting set um and then use basing model reduction to eliminate all the states that you don't want to recognize simply because you never go there um you know once in a while you know when I was younger I would certainly be able to recognize what it's like to fall off my bike but now as an expert bike rider I really don't want or need that so you know and that's just basically getting old and wise and you

know implementing basing model reduction
to remove the Redundant parameters which
would in this instance just be the
columns u_m and u_m

Associated slices or or rows of the of
the transition m_{ij}

so I think you could perhaps another way
of then of answering L's question is
yeah you just keep

accumulating u_m but u_m start to engage
Bas in model reduction at some point you
should reach an equilibrium which will
be an attracting set that allows you to
survive in the character states that

Define you in this particular
environment with this particular
volatility

thank you

awesome okay I'm going to bring in a
question from the live

chat okay Javier wrote so stochastic
plus dynamical probabilities learned by
the agent itself are scale free will
this hierarchical learning also be
applied to the goals of the agent
different abstractions granularities of
goals self-learned

[Music]

my answer would be yes but I'm going to
I'm G to ask Lance to to to to answer
one oh that that's

cheeky uh yeah I would say for sure u_m
because typically when we have when we
oper operation operationalize sorry the
goals in our gen model they need to have
the same structure at our gen of model
itself so imagine then we would have a
hierarchical raliz gos which kind of

make

sense cool okay I'll bring in oh yeah go

for it Carl no I just GNA say um you

know just to um unpack what Lance just

said in terms of practical things you

know if you're trying to um you're

trying to say um what my goal is is to

go uh is to have dinner

um that naturally unpacks hierarchically

into I've got to move from the living

room to the kitchen then I have to find

the you open the fridge and sorry I have

to find the food which then unpacks into

I have now open the and all the way down

to to the finest finest muscle movement

so almost by definition um your intended

narratives can always be um decomposed

in this scale-free way you um there

there's a this is very similar to sort

of um motor chunking and hierarchal

decomposition in in um um theoretical uh

motor control for example and you

probably find it a lot in in robotics

that there's a global long-term plan

that that provides empirical priz of

constraints on shorter little chunks

that themselves provide constraints on

shorter and shorter chunks and that's

exactly the um if you like the the

architecture implicit in these scale

free

models awesome and totally ties back to

the

synergetics okay I'll ask two questions

from ml Dawn he wrote how can we inject

the same separation of time scales by

imposing smoothness constraints on say

the parameters as opposed to playing

with proportional update rates at different levels and this relates to something Arun and I talked a lot about in the zero and leading up to it is identifying which Windows to use you could imagine picking the right separation of time scales if the classes on campus really do shift on the hour then the hour course scening is going to be great and effective but the 59 or the 61 minute my Alias you with the real variability of the system and so there's not necessarily even a smooth landscape for determining the windowing or the binning of coar graining and coar graining may have dependences on each other so how do you go about playing and finding viable approaches and robust ones amidst all these different levels of of meta optimization that are opening up right I'll take this one and I could take take the next one um then that's a really good question because there is an implicit separation of time scales between um so vanilla implementations of inference and learning however in these schemes there isn't um so the the know the um the Matlab version or implementation um of um active inflece under these and all uh Markoff decision process models um unlike um the um procedures that you might find in data analysis where you acquire you basically do some filtering on a on a an Epoch of data and then at the end you

update your parameters and then you rinse wash and repeat um in um in active inference um simulations and also in some um applications of what's called General filtering you have continual learning so the update of the parameters actually occurs at every time step um at each level of the model so the likelihood parameters at the lowest level are updated every time that routine is called um so there is continual learning the learning um is yes you're absolutely right is smooth in the sense that you're slowly accumulating drish parameters but the temple scale of that learning is the same as the inference um and you could also argue well could one also put Bas in model reduction um into play at the same time scale so basically can you at every time point at each level in the model um identify redundant parameters and remove them uh using Bas and model reduction in effect that's what's done uh using Active Learning well I said before Active Learning will only proceed um if the expected free energy of mutual information improves that's actually also implemented so this the kind the the the rate the rate at which it looks as though things change um between inference learning and model selection is certainly distinct I don't think that's quite the same kind of separation of time scales that is implied in the renormalization group that's as you were intimating uh Daniel

exactly specified by the number of ticks
or updates at level n per single update
at the u_h level n plus one u_m and that
is I think you know the you the key
question you're asking how do you get
right u_m and of course the answer to
that is the the model that has the
greatest model evidence or smallest free
energy or greatest marginal likelihood
for the kind of data generated by this
Environ in this world so u_m are there
any u_m sort of useful PRI you can put
over this u_m well one one useful prior
is u_m that u_m things are changing as u_h
to to well one prior which is actually
used in operation is you take the lower
limit on on What's called the u_h the
temporal RG flow which is basically the
the length of the window at each and
every level u_m at the moment that's two
what does that mean
how can you motivate that well it's the
most
expressive in the following
sense that if I u_m just imagine u_m what
imagine what this discrete
discretization of time means from the
point of view of classical Notions in
continuous State space like position and
momentum or position and
velocity if u_m if I use a
a u_m a scaling of two which is basically
means there are two time steps or every
time step of the u_m the level above what
I'm effectively doing is u_m u_m modeling
two time steps for every state at the
level above and that basically means I
can model the initial condition and the

subsequent condition basically in terms of the change which of course is the velocity u_m and what does that mean for the level above that well it now means that I'm modeling the chain in the velocity so I'm now modeling the acceleration and then so on up until u_m you know the highest order of motion so what that means is by taking this limiting case of T the temp temporal horizon or the the you the ratio of time time steps from one level to the other level u_m as two means that I got a very expressive model that can handle u_m can recognize u_m things that change very very quickly in the ex ation their jerk and all and the implicit u_m context sensitivity that that that gives you if I didn't do that I'd have to assume that I C grain over long periods of time and therefore the path is consistent during that time now that sometimes is a useful approximation and indeed there's a whole industry called you know called linear switching linear dynamical systems that makes that approximation and it means you can u_m approximate all the nonlinear complicated high order motions in some evolving pattern or system with a series of u_m u_m linear trajectories that basically go you know that intercede between a series of points at which you which you which you switch the problem with that is that the length of the linear trajectories for as for example I'm

committed to one Path U so one slice of
my B Matrix for the next eight time
steps the problem with this is that you
don't know when you when you deviate
from that transition Dynamics so then
you get into the game or perhaps I could
make my scheme reactive so sometimes I
do three steps sometimes I do eight
steps sometimes I do one step and I have
some criteria for saying your
predictions about my trajectory have now
been met please um I'm going to tell you
give you the evidence that I've been
pursuing this path you update your priv
level above and tell me what you expect
me to do next so this now speaks to um a
particular kind of reactive message
passing um that starts to have
implications for the computer science
and the sort of CR the crud um um or um
as um as I think we've rehearsed before
um The Institute you the the original
conception uh behind
um the actor model in computer science
from the 1990s so it's all about
reactive message passing you I I will
send you a message when you send me I
will return a message to you if you send
me a message um and what this um what
these um renormalizing time models have
and I should say that this also applies
in the absence of the the spatial
renormalization so any deep temporal
model will have this aspect what that
means is that from the point of view of
Any Given
level I will um be responding to
messages more quickly from my

subordinates than from my
superiors so once in a while I'll get a
request from a
superior my response to that is to ask
my juniors to do a series of things they
report back to me and when they report I
give them the next bit bit of request or
guidance uh until they complete their
job and when all my juniors have
completed their job of course in the re
normalizing spal re normalizing context
I'll have a whole group of of Juniors to
worry about um then um I have gathered
enough evidence for me to then return
the the message to my to my Superior so
you get the in terms of the
reactive message passing in this sort of
actor model of um of computer message
passing um you get the the separation of
temporal scales inherits from the number
of messages or re um requests and uh
responses um um exchange with the people
lower down in the hierarchy relative to
the number immediately subordinate um
and in principle they don't have to be
fixed so there could be a
criteria um before I pass the message
back back up to the to the higher level
I think that's Daniel probably what you
were getting at um I'm not there may be
somebody who's done that um um but
I'm I haven't done that and I I I don't
know I can't give you an example of
somebody who has done that you may find
you may find somebody um in Bert de's
group um well so of you know some of the
world experts um in reactive message
passing who have considered these kinds

of things um you know you in in
principle um you know there should be a
way of
um each level testing whether I have
fulfilled the predictions of my
superiors um and now it's time to get
more guidance you know uh based upon the
free energy
awesome the organizational technology I
didn't know that my PhD
needed okay I'll ask a question as we
get close to the end and and this will
be for for
anyone ml Dawn wrote how does the
training time of active inference
compared to that of reinforcement
learning and how does their scalability
compare and it would be interesting just
to hear anyone's very
takes where are we at with that entire
Arc from problem
conceptualization on through the
training and the inference how do we
even grapple and
describe with similarities and
differences in the training time and
resources and all of this
that's definitely Lance for for at least
five minutes I
think I'm I'm a bit incapacitated in
terms of talking um I think this
actually a good question for Aron um but
yeah I mean I can say a few
things um so uh scaleability in active
inference I would say it has been a
challenge and there has been um much
less man work hours that have been put
into that than in reinforcement learning

now for the past few years we we've had a lot of work um a lot of it led by C actually um trying to scale the structure learning in active inference and and I think we've done a lot of mileage on that so there's been the supervised structure learning paper there's this paper uh there was also some interesting work um presented at the international workshop on AC inference um I personally think there are still some uh challenges ahead and and I think uh one of them is is the one that we just talked um a few questions ago about you know if you if you learn an expert orbit how do you get back on the orbit as the dimensionality of the system increases so this is definitely um one challenge um now I know much less about reinforcement learning so um yeah maybe Arun has more things to say there um but but what I will say is a lot of reinforcement learning is powered by Deep neural networks when we use deep neural networks we sacrifice uh the interpretability of our generative model this is one of the strength of active inference is having more interpret um Dynamic clal models where we can actually here into them and see okay well this representation enclos this this other one enclos that and so forth um the fact though that reinforcement learning typically uses deep n networks means that it has been much more scalable uh just because there has been so much work done in like

training deep n networks this is a bit
my uh my perspective I don't think it
fully answers the question
though I can come in with a little bit
on that but I must admit that I've not
trained any reinforcement learning
agents myself I've worked with people
who have uh I'd agree Lance that the
infrastructure generally is much more
advanced for training RL agents for
doing stuff uh and completely agree with
as you say the explainability and the
interpretability of why did an RL agent
pick a particular action is very take uh
and I think the deeper those networks
become the harder it is to understand
what's going on in the hidden layers and
uh generally why they do what they do I
think with active inference yes we do
have more
explainability but I do wonder if we go
the far structure learning approach I
think KL had a particular uh comment
earlier about looking at the highest
level um identifying the desirable
States learned by the agent
from what I understand the
identification has to be done by a human
they have to look at the model that's
been learned by this uh renormalization
approach look at those top levels and
then say ah what do these levels mean so
an example in the paper is the Mist uh
classification it's the case of a human
being looking at those top levels and
going ah these top levels each
correspond to a digit in the handwriting
task

so we are uh sacrificing some explainability I think by doing this fast structure learning compared to what might be the oldfashioned way of creating uh G models By Hand by thinking sitting in my armchair and thinking this is how I might model the world these structures these likelihoods these states and when you do that I think just by hand you have 100% explainability that model just might not perform very well so I think there's always going to be that tension there um but in terms of how quickly these things learn um having not trained either i' would be having a reckon and at the moment I would say probably an RG um active inference agent would train faster simply because it's able to use the structure in the data much more effectively that would that would be my take and I think as well really interesting thinking about the active data selection part of this too I know that there is some stuff in the paper about the efficiency being both a curse and a burn but I'll hand over to Lance and Carl to give more details on that yeah

Lance yeah just to um echo echo what Arun said um also so so the structure learning in active inference and in reinforcement learning is is also very different um in reinforcement learning we typically have a a model with deep known networks that's massively overparameterized but has some fixed

representational capacity from the outside because the structure of that of that model will be specified and then typically the model will learn and then probably you would prune it using some techniques like Dropout or or other things if we looked at that from the perspective of active reference it would be like specifying a massively overparameterized model and including it with Bas model

reduction the um but but now in active inference we have this Bas and model expansion for example in this paper and what this Basin model expansion does is it learns from the Geto the minimal model for explaining the data and so in that sense it's extremely efficient and also provably because it extremes the um the marginal likelihood or the mutual

yeah I just wanted to Echo that last Point that's a really important Point um so Lance is drawing the distinction and there are many distinctions you can draw with um deep RL um and um deep active inference read in a more in in its um generic sense um and that that distinction at the level of structure learning or model selection I I think is absolutely crucial so you know deep RL starts with an overly expressive model with a known functional form um that is deliberately overparameterized and then shrinks it or reduces it to um maximize the mutual information the expected free energy sometimes they um refer to as a cross

entropy in machine learning so the reward function now becomes uh there's no explicit reward in unsupervised learning it's just replaced by the mutual information or the cross entropy um which should be distinguished between supervisory enforcement learning such as you know digit classification for example um whereas the you know active inference um under these re under fast structure learning is exactly as Lance says it starts from nothing and stops when it's minimally complex which means it's maximally efficient and that efficiency translates in many guises through to sample efficiency through to um um statistical efficiency through to thermodynamic efficiency and this matters because from the point of view of climate change this is the right way to do it as opposed to building power stations to to power large language models and Transformers that need big data so um the efficiency um on all levels whether it's the the sample efficiency or using the right data uh or whether it's right through to how much electricity do you need to actually train your model and how many millions of dollars do you need to train a foundation model um all of these are reflections of the failure of um machine learning to write in the efficiency into the ultimate objective function which as Lance said is just the marginal likelihood of the evidence for this model of of this kind of content or or

World awesome okay final question and a very apt one as we head into the rest of the Symposium so just give any short thoughts Tintin wrote hi all thanks for the incredible event does the sequence does the sequence in which the training data is fed to the AI structure learning model impact what the resulting learned structure would be so the learning curriculum AKA Symposium program challenge well the answer is sorry I'm mindful of lances he was recovering from his um jaw surgery um yeah absolutely Ely um and this is um again just in relation to to to to the last question something I think that eludes uh conventional RL certainly that rests upon um um classification procedures um because um the active inference as an application of the free energy principle is all about the Dynamics of self-organization and Dynamics means time and time means sequence matters so inevitably at some level the order in which you see things is absolutely crucial for getting the right kind of generative model um that is even in the case um actually of static image recognition because you're assuming that there's no ordinal structure or correlation structure to the presentation say of M digits you're assuming that they are selected at random as opposed to seeing all the tens and then all the all the ones and then all the all the sevens um so even in these edge cases um um which are you know the focus of um 20th century RL um

you know static image classification for
example um you know order does matter
but they they it matters absolutely
acutely because of course this the order
in which you um see uh
um content in
um this infrast structure learning
determines how you populate the um the
transition tensors and it's the
transition tensors that enable
separation temporal scales that enables
the ability to um encode narratives and
plans that far transcend the actual rate
at which you samp sample the environment
you could also to argue that you know
this issue has emerged um in a in a
strange kind of guise in Transformer
models so the attention heads operate on
the past so the the where you select the
from the past in terms of which token is
going to uh predict the next token most
efficiently um again is exquisitly
dependent upon and sensitive to the
order in which you train train train
models So Daniel have you thought
carefully about your ordinal approach
today I I let the structure I let the
structure of the niche and others'
constraints provide the first pass and
make requests from there so thank you
Carl thank you Lance for joining again
thank you run for all the contributions
with azero Carl great to see you
so next up we'll take a 30se second
break and we'll be back with Michael
Lennon so thanks fellows see you
soon e
okay welcome back from that

brief brief

break we're here with Michael lenon and
the session is co- inferencing
organizational symbiosis so Michael
thank you for joining take it
away hello yes um I am uh so grateful to
be here I would like to um approach
today's conversation in um in a
conversational style where um I share
what led me to active inference and how
the discovery of the principles of free
energy principle and and active interest
changed how I understood as a
practitioner and and how that has led
both to both personal professional and
organizational um shifts in
conversations um one of which will
conclude in an invitation to this
community to um experiment with an uh an
organizational structure uh designed for
um building both the global n knowledge
comments um as well as um um uh building
individual Ventures so similar to the
way in which uh for example Linux is
open source uh and then you have
Ventures building off of Linux to do
things in the world that that
experimenting with such a an
organizational model uh for uh the
active inference comments uh is some an
experiment worth taking on so I I I
don't want to let the the final invite
get away but but I'm going to backtrack
and do this very much in a storytelling
fashion so um so um I have some slides
but I'm not going to uh spend too much
time uh well in fact let me just
um skip that just just to recap what

this um

uh what this conversation is about is
different ways in which um not just
active and but um science has collided
with internal intuition and led to
transformational journeys of different
kinds and so um and so um I would
describe the um you know I I started out
as an individual uh as a uh

Economist um and um and as a
technologist working at um institutions
such as the World Bank and IBM and um
doing large scale um it implementation
projects and um and that work often
although there was a lot of um
engineering um it was often the human
factors that got in the way of success
or failure and so um that has been a
kind of

orienting um how to both be informed by
what the the the science and the math
provides and uh then build uh how does
the biologically based if you will way
of sense making um affect or
um alter the the the the evolution of of
successful project implementation so um
just um a little bit about um about
where this uh project um prior to coming
to active inference um I say project my
journey um uh led me through was was
um I had a uh I had as I said been
trained as a as an economist and um one
of the things that economists have is a
point of view about what it means to
grow the culture to grow the society and
um primarily denominated in dollars and
using um GDP as a way of um
of uh tracking whether or not one was

impacting the experience of flourishing and um I came across a uh a chart by um by an economist that showed that GDP had tripled in the last 60 years both in the UK and the us and that well-being had not changed at all and this economist who had been in office in the UK at the time when when he left office was kicked out um he went back to Academia and started looking into what does Science Show moves the needle of well-being and how do we make this socially more contagious and um and I both U loved the idea but I also was um uh shocked by that information because as someone who was trained in a particular lens a particular perspective what I had been trained to do and how to um cause effect and impact was in fact not moving the needle of wellbeing and so there was this kind of personal and professional crisis um that arose from this uh from this knowledge and um as I say that that led to this this journey of unlearning in some ways and um and in both personal interpersonal and organizational ways uh recalibrating the experience of what does it mean to pay attention to thriving and so um that Journey has been very satisfying um not only do I understand how I need to be to experience uh more goosebumps in my life uh I understand uh how I need to be to experience connection and belonging and to co-create with others uh a sense of um purpose and of meaning that um that I

had not had previously and it's uh one of the ways I sometimes uh describe this relationship between the subjective uh internal experience and scientifically objective let's call it was that I might know how to go to the gym and flex uh my arms to get a workout but by understanding Anatomy I could learn um the bicep had a particular placement in the body a particular Arrangement and I still had to do the workout but I could more effectively and um skillfully uh do the work that needed to be done of working out and so similarly the study of the science of well-being um and social uh social uh Sciences of of connections of groups and of um and of um how how for example um one of the surprising findings for me uh in in the realm of Happiness was um a finding by an epidemiologist and um basically he was studying how do emotions spread and um and he in studying this realized that your happiness um uh was not only more heavily INF uh influenced by the people that you're surrounded by but also by the next degree of separation the people to whom they are connected and even the third degree of separation and so that these three um Degrees of Separation predicted your happiness more than uh the way that I explained I had been explained what happiness was which was that uh it is an inalienable right like life and Liberty

the pursuit of happiness and that it's
subject to individual traits of my own
that if I just do me the best I can then
I'm going to be happier and so all of a
sudden there's this
disorienting um Insight in that it's not
about being my
best but it's just as important to inter
be my best and so it's like well what
the heck does that mean what does it
what does it look like to to inhabit
one's own experience and one's own skin
in in a completely different way and and
I know a conversation that uh you and I
Daniel once had about um about um the
impact of active inference you were
describing how you might have been
driving on the right hand side of the
road your old life and then all of a
sudden you um you uh go to a place where
everybody drives on the left and so some
intuitions still work exactly as as they
used to but others are completely
counterintuitive and so this developing
the capacity to to go against intuition
against the The Familiar is um is part
of what sometimes these these these new
forms of knowledge there's there's
knowledge that that adds to what we know
and then there's knowledge that changes
how we understand and and happiness
first did that for me intuitively at the
personal experiential level right it
first was like okay I I am playing the
game of my own life uh
inappropriately um do I how do I begin
to inter be my best instead of be my
best and this is where um uh some

subsequent work in working personally
and in with groups and and
organizationally led to um paying
attention to what was happening in in a
different way and uh so I uh as I say
this this um crisis LED to a very
fruitful um Learning Journey and um and
uh excuse me I
um I got distracted
um some of this led to writing different
ways of describing how to do work and so
for example one of the spaces that I've
been working in was in um government
Consulting and in government Consulting
top down
um uh uh is often the way to direct the
effectiveness of the
organization um you come up with the
right strategy you figure out all the
components and you then execute on the
plan and uh and so in a sense uh if
there are two forms of
of uh probabilistic prediction a basian
being one at the heart of of active
inference and another one being a fisher
fisher is one that that says it knows
where you need to go and the way that
you relate to to surprises and to
mistakes and to errors is you correct
back to what you felt your original plan
was whereas basian probabilism
probability obviously um can have
similar results but also can update
instead of reacting to surprise as
something to be fixed it can sometimes
be something to be adapted to and uh it
changes the way that uh manage you
manage um the the uh emergence of

differing information from what you originally predicted and so in the same way that I described that happiness uh had a disorienting effect on how to practice it active inference has a similar um disorienting effect about how co-regulate um um how to supervise the um the evolution of of differences between what you expect and what you what you get and um one way in which I I describe this uh to to folks is it changes your understanding or changed my understanding of how cause and effect works that um you know as as as I arrived to um to um active inference there were there were a couple of intuitions about uh how the world operated that active inference helped um evolve the first was that instead of instead of the world being composed of objects um that were fixed and stable that um that the mathematics of how to explain things was much more of a dynamic verb and so to look at the world as being not nouns but verbs and and and a way that I describe this for myself when talking to people about it is that instead of seeing myself as a self a noun that I'm a a be a being and so um that paying attention to the continuous verbing and the relating uh to to to others is is a is a different um uh a different game and so um the interesting thing about Act of inference was that there was a mathematics for describing this that um that related the internal States and the external States um

through uh mathematics of
information that the way in which um the
the way in which
um uh science

Works um is is is familiar and intuitive
to to to many of us um Let me let me let
me rephrase this in a different way that
you know when I say the inside and the
outside are informationally connected um
that that

um that the outside is all the
information all the stimulus of the
world and that the inside creates a map
of that world and that the mathematics
of creating that map not only um reduce
the world and is a mathematically um
efficient way of describing how to make
the how to get the most signal for for
all the information that's being
received but that it also
um it it also um

enables like I was describing earlier
with the understanding of the bicep a
more accurate way of of sensing the
world of um orienting to the world
and um and of responding to the
world um I

I should have uh shared my my slides
instead of trying to talk through them
but I'm going to take a a pause and ask
any any questions or shall I proceed
yeah uh I'll ask one question and then
looking forward to slides dsk
wrote there has been a lot of talk
around GDP not being a good enough
metric of national development has your
research informed you on a better metric
if so does it have a working

name so yes and it's a different um answer than a single metric so the metric a single metric was one of the problems and the fact that it was denominating currency was one of the problems so one of the things that uh has shifted um some of the work of uh the UN of of really a group led by Bill Boi and Ralph thurm on um on what are called contextual indicators are that um are that instead of an indicator being um GDP is both currency and it's um it's set to a single the denominator and the nominator of the indicator are both set in in the same thing so um let me describe this in a in a different way um there has been a diversification of what are the things to pay attention to so not only um dollars but also subjective experience of well-being for example better predicts the outcome of Elections than does Financial income and so um the idea that dollars are are predictive accurately predictive of um of outcomes has been often relied upon and turns out to be maladapted right and so that's um whether whether you're talking about biological phenomena whether you're talking about social phenomena there's a whole range of extra indicators that are now seen as um not a single one but but better visibility into these different nested levels of of life the second what I would call breakthrough type of indicator Beyond GDP that I was describing denominator and uh not numerator is that um most organizational

reporting for example is is um for example um Starbucks might say we are using 10,000 gallons less of water than we did last year so we are improving um our consumption and the way that we are related to um the water shed that we might be related to and and all that um reporting is if you will individual Centric and when you Center your just like as as with happiness you know where if you're just looking at the at the individual you're missing an important part of the picture when when measuring your relationship to resources to Commons to um natural Commons and social Commons um if you if you put your um uh individual Behavior within the context of the collective so let's say Starbucks says uh we used 10,000 gallons last year out of the total water shed then you can begin to draw comparisons about how everybody is using the Watershed as a whole and how the percentage the percentage behavior of of usage of the water has shifted that can be compared from one to the next to the next so if you will you're putting these indicators in a in a similar language in a similar uh they're speaking in a in a similar base than um than when they are um reported individually and so you know when when looking at at well-being it's it's important not only to to add additional things Beyond money but it's also important that these indicators be um set to the same um denominator or or you wind up not having the visibility that

you need and there's a huge cat fight about this because uh there have been a whole series of um of Privileges and a whole series of um social practices that are grounded in the way in which um uh individual corporations are um are structured and so this this kind of points to a different kind of natural Mee that I wanted to bring up that might be of of Interest so so um we talked about how with happiness um you know we we're doing it individually and that we're missing the relational um when you look at uh cells that um are part of an organism if they are traumatized uh they become more individualistic they become uh they consume more they reproduce more and they grow into what we call a cancer because their if you will relationality is traumatized and they become more and more selfish and so there's there's both a huge importance of paying attention to relationality at the level of the cell at the level of the individual human and some would say at the level of clusters of humans such as organizations and that with cancer part of the um New new form of treatment has been to call the cells informationally back into relation so instead of punitive burning those cells and killing them it's to remind them of their relatedness to restore them from a traumatized State back into relationship with other cells and they begin to behave again in better relation to the body around them and so that metaphor speaks to the importance of how do we

call each other back into relation if
the language of individualism is
permeating both our individual Behavior
and the language of Corporations is
equally um traumatized because
originally when when uh corporations
were first formed in the 1600s in uh in
the Netherlands there were really people
coming together to venture and do
together what they could not do alone so
trading with um the Far East was
was something that no individual uh
Capital uh no individual business owner
could could afford to venture but by
pulling their funds they were able to do
and they were able to become a great
trading power globally this model of
groups behaving as groups so a different
cluster of um of uh of existence groups
um and then behaving
um pro-socially with each other is what
grew the next level of
um economic performance and economic um
Effectiveness um however over time some
would say that organizations have become
traumatized and have become disconnected
both from the places where they
originated as well as from all the
stakeholders that they were serving and
so for example here in the US when when
the Supreme Court last century two
centuries ago now almost decided that
financially
Financial stakeholders were the foremost
important um stakeholders in
corporations they distorted the
relationality of of
Corporations and so it has become a kind

of cancer that that has become
extractivist both upon the its its
employees upon the the the ecosystems in
which it inhabits by architecture so
there are plenty of people that will
tell you in in an organization we want
to be more attentive to these um other
considerations Beyond Finance but but we
don't even have permission because the
agreements of the corporation have
evolved from
mutualism to um to selfishness if you
will to more consumerism more
productivity more Surplus and those all
have a place they're they're they're
important but they've become like like
the cells in the body that become
cancerous they've become
overly um overly uh they become
individually uh oriented alone and and
not sufficiently skillful at at
mutualism and so this all comes back to
a new basis for equipping organizations
to pay attention to how they're
interrelating within between and to
things beyond their current purview and
it is much of a disorienting and
challenging
um uh
um uh bit of information as I was just
as I was sharing the idea of learning
how to be more skillful at happiness was
to me personally you know it's like oh I
have to pay more attention to you to for
for us to be happier or to be the third
degree of separation as I was describing
earlier means that not only the people
to whom I'm directly connected but that

really that it's my dumb number the
dumb number being presumably the 150
to the 250 people that you have
meaningful relationships in your life
the dumb cubed so it's beyond your
direct sensory observation and it's like
how do you co- Haron eyesee across
groups how do you sing to use a
different kind of language songs in
common to which you're Co coordinated and
um and these are some of the things that
um active inference provides a
mathematics for how this works you know
that is fascinating of how the
synchronization of the individual to the
group and the nesting and and how the
signaling happens across the nested
groups and so it's it's um it's
fascinating the mathematics can be
daunting but the promise of it and the
ability to do these things more
skillfully and reliably is just it gives
me goosebumps it gives me goosebumps and
this is part of um why I'm trying to
tell this somewhat ineffectively I would
say today in a story Manner and through
experience so that people can pull on
the intuitions that they have the
memories that they have and say I dare
to tackle this
uncertainty building upon these
intuitions and feelings that I have and
instead of the classic hierarchical um
corporation that um that has as part of
its founding documents when you when you
plant a seed and you use the template
that says we are we are organizing
according to these rules even if you

want to change it's it's like it's a lot
harder to change the Constitution than
it is some minor policy in a back office
in a you know in your in your
um uh you know in some app that you
might be uh writing the code for you
know and and similarly it's important to
to incubate um uh organisms that have
relationality the mutualism and the
holistic
perspective codified into and and the
intering codified into how they behave
individually inter intercluster and then
in relation to the commons that they
might serve and
so before I take a turn into the case
for um the experiment that I was hoping
to invite people to uh do together any
any any more questions popping up sure I
can ask some short questions then slides
and invitation okay lightning round okay
Arun asked isn't there a cyclic
dependency of economic factors and
happiness for example High inflation
made everyone sad over
2024 so there there is um uh there are
um time lag factors uh that impact
um uh when we how the economy operates
and how individuals operate and these
are not necessarily in sync one of the
things that um that uh I found from the
um from the
science of Happiness was um more
effective ways to carry the suffering
and so um some have that this change
that just happened politically for
example um is a Bad Thing often when
things break open there are whole new

Realms of possibility to to be had and
so um happiness is not just about having
all positive emotions it's about taking
well-being is the word actually should
be using is about absorbing all of the
signal that the universe is is giving
all that's coming in and then optimizing
within that and so um the breaking down
can also be a breaking open for what
there is to to uh become and so that is
not to say that there is a lack of
struggle or pain or friction one of the
great intuitions for me um that that
arose from from studying people
explaining Act of inference was um how
things emerge and the person explaining
it was using water and uh they they were
describing still water as being the way
in which current conventions of how
things got done and how people calculate
it so if you're if you're in Still Water
if you're in a lake and you're trying to
get across um the way you Rod your boat
or the way you swim is all set by the
conditions of the water in um um in
flowing water um even though those that
same water has the same physical
properties uh as Still Water it weighs
the same no matter whether it's moving
or not um when when the friction of the
water over rocks um creates new dynamics
that emerge it creates whirlpools right
and that those structures um that
persist um you know the creation of
order out of disorder um allow for the
emergence of also meta patterns that are
again described by informational
mathematics rather than the physics of

the water itself and and this is this is
again trying to get it what is so hard
to Intuit it it's like oh these the
whether whether it's a single Whirlpool
coming into existing or whirlpools into
whirlpooling into into greater movements
that um that these frictions are also
creative and so to really um uh be
willing to settle into the into the dis
into the what you were calling the
cyclical disconnect in a symmetry um as
as something that is a signal also that
can be driving learning driving better
calibration um can be driving um new
learning uh because at the heart of all
of active inference is a mathematics of
continuous

learning

um yes I I love that and just flows like
water with our previous sessions okay
next question and then back into the
flow we go social Design Studio wrote so
curious about the transition away from
individualistic and mechanical into
Collective and organic where is the most
levered locality to

start well that's a a great question and
um um here's um a way in which I would
answer this in the in the world of
project

evaluation so there's a point of view um
in the world of systems thinking there's
a certain order of what to uh what of
the you know Donella Meadows is well
known for for uh the way in which she
created this categorization of ways to
intervene in a system and um uh in the
world of evaluating um projects and

programs complex projects and program
over time uh there's been some some meta
studies done about what are the things
that best
predict uh that if if the supervision is
done uh attending to certain factors
what are the things that seem to produce
effective and positive results because
we're talking about complexity you could
pick a thousand things but in in this
research they identified three things
and I'll add two the three things things
that they said you know when people look
at the following things they they tend
to not only manage what they thought
they were looking for but also what
emerges in other words managing the
surprise and the unexpected more
effectively and these three things were
looking at the inter relationality
surprise surprise right we're talking
paying attention to the interbeing and
the interbeing across not just the
individual or whatever whatever our our
individual unit of focus was originally
but but who
were seeing the system more
holistically second um
multiperspectivity um and the way that I
in my the way I uh describe this for
myself is um I say looking within
looking between and looking beyond
having one or more um disciplines and
perspectives and ways of knowing about
out uh those those lenses of of
attention on what is happening um uh if
we look to if we look within we often
become a better compass for our own

lives um uh if we um if we look between
we often develop and harness the
intelligence uh that collectively and
the Insight that we have and when we
look Beyond we often detect uh patterns
that are beyond our Collective purview
that might uh um break open and um
enable us to to transform in in powerful
ways so again to return to that second
Point

multiperspectivity um and I suggest with
those three uh lenses which is what I
call the me to we to all you know um and
it's it's kind of the label that I'm
putting on on this invitation I'm going
to bring up later um is um is a is a
good starting point is it enough no but
it's a great starting point and then
thirdly and here's where it gets messy
and is the non-answer to your original
question is um scope
definition um because the moment you
pick what are the boundaries what is my
mark of blanket you know what what what
is the what are the boundaries of the
map that I'm going to use as my lens you
begin to abstract everything else into
what fits in that lens so so if you say
we're going to have this conversation in
English the way that we everything we
describe will be denominated mostly in
nouns instead of in verbs and with
Concepts that exist in English but don't
exist in other languages and so um the
attending to what are the boundaries
that we will use to focus is is a
gigantic mechanism for seeing um how
you're holding your attention and so

again it's not a fixed answer but it's a way of attending to perception again repeating relationships diversity of perspectives and scope definition uh the other two that I like to add in that um were not a part of that research so that research by the way was um a 100 Years of developmental evaluation um you know hundreds of studies and and so it's very robust and you can you can go to the bank that that's a good place to start building your dashboard or not the dashboard your conversations about the dashboard um the other two that I would add and and these in part come from active inference and from other systemic view places and Views is looking at patterns how are we you know what are we clustering and looking at flows you know um what what are the core flows are we looking at energy if we're looking at biological systems are we looking at the water flows you know life organizes around water um so do do we understand what are the core flows of information of energy whatever uh in the system that we are observing that that that gives us a a key insight into the way in which causality unfolds um and causality uh organizes itself around and that it's not the mechanistic you know geometry that we we may have intuited in uh for for the longest time but that by looking at Source by looking at flow we we develop

a better compass for for the
world thank you it's it's awesome and
the live chat is super lit you'll see it
after um in the last 20 minutes do you
want to share your slides and open the
invitation

letter yes so uh let me um switch here
to share

I

share are you seeing my screen okay uh
me so

um the

invitation uh was

um was to test a new way of organizing
and a fair share Commons is a kind of um
a mutualistic Co-op that um first
originated in the UK and has been
endorsed by the

financial I I I should have read it down
here because I can't remember like the
finance trust accounting Board of of the
UK it's a spread through many countries
of the Commonwealth over the last uh 15
20 years and it's primarily an adapted
form of organizational structure that is
intended to both cultivate the practice
of mutualism within the group so it's
kind of like a co-op um in that
everybody has power uh to to participate
it's

self-governing um and that it attends to
the relationality of the founders the
workers and uh and external whether they
investors or they they self-organize
it's really uh there is a a pattern
language about forming co-ops and one
can use that pattern language to to um
to to form these Commons and that these

whether they are to create
Ventures um or to create um full
businesses that the pattern language is
useful for that effect um and so as I
described earlier a kinda Linux um you
know it's it's intended to grow the
commons but also to support the
application of those Commons and build
the organizational traits uh for
successful Ventures and what I am uh
hoping to find allies uh around um oops
let me um so it um the these are again
suggested and there are many ways to to
go about it but um you know there
there's certain core agreements that
differ from if you will the extractivism
and the foundation is uh
relationality um which drives mutualism
symbiosis um and uh the the invit ation
um is to um is to constitute one in
service of the active inference uh
Community for growing both the commons
and the capacity to um to prototype and
develop both the the experience of
mutualism but also projects and that uh
there' be two starting experiments um
the first around prototyping uh agents
for accelerating access to money
so often um folks incubating a new
project are looking to find funding for
that project um find uh funding for what
that project might serve and often
they're people willing to fund those
projects and the idea of writing up what
are the sources of different funding
what are the sources of what what are
the needs that they have that they're
looking to fund how do I um discover and

engage that um funding is an important agent that can be used in common after common after common and so possibly developing uh um an agent or uh there are plenty of methods but but agent toying if you will uh that component as one of the key flows you know that in business the flow of money is is an important element to to inactivate and and that the second is about um finding um resources of a different kind competence and complementarity and um what are the what are the components often for example a u uh the active inference Community has a wide diversity of competencies um the but but it is highly highly Technical and so to engage with people who who um have complimentary skills um uh and and um and to support the accelerated discovery of allies not only based on competencies but also on um uh other holistic considerations is um is the attending to the flow of mutualism um and so those those three um actions is kind of a starting point is um is something that I would like to um I'm I'm already in action on the uh with with other groups um and a variety of groups on the constituting the legal structure of a of a fair sharing Commons um and I would like to enroll um the active inference Community to become an ally with with kind of other competencies and other types of mutualistic eco-minded socially minded groups to grow um the capacity for mutualism and it's early stage I

literally um this this is germinating
you could say and really would like to
um would like to find allies who who
knowing that we don't have all the
answers but that we're willing to grow
not just the methods but the relational
soil um is um is what this is really
about so um yeah I'll stop
there awesome exciting I can ask some
questions from live chat and we can
explore just a little bit of
a shall I leave this on the screen or
keep going sure you can leave it up I'll
ask there's a lot of great I'll just
start with a direct question so people
can add more in this last 15 minutes
okay social Design Studio wrote does the
commons use emergent hierarchy so as to
accommodate contextual expertise and
thus relational
power so um great question and um there
are several definitions of Commons um
but I'm going to go with elrom she won
the Nobel Prize for
and um and she she basically defined it
as it's whatever Markov blanket people
choose and then self-organize around so
so um there are many variations and um
and in order she she identified eight
principles for common to not just be a
nice idea but to be able to persist
sustainably right um just to do a little
sidebar she she was trying to make a
business the reason reason she won the
Nobel Prize is because in the profession
um everybody was explaining most of the
economy's activities as either being
driven through hierarchies or through

marketplaces the Dynamics of those two Collective groups and she said there's a third kind of group a common and it has the following properties and um the socially mutualistic uh side of the world was just in love with her and this is why she won the Nobel Prize because she developed not only a body of Science and a series of eight principles that predictably describe those that succeed these eight include you know a sense of a boundaries that they have in common and whether they decide to self-organize around a fishery whether they decide to organize around um a data commons around uh a knowledge commons that is up for the group to to decide and then um the the way in which they co-regulate um she there there were many variations but but basically there was a um there had to be agreements on where to play shared attention and as people diverged from the Norms that there were ways to call them back into relationship and to have escalating um consequences if people deviated um now all of that sounds lovely but um but that was not enough either there had to be the ability to enforce one's boundaries not only internally but externally and so I'll give you an example um I I uh have some First Nation friends with whom I've been meeting every Friday for years to work on decolonizing our minds and they have a point of view about a piece of land that um that is their Garden of Eden and sacred and how it should be used or not be used and they use all

these principles but they cannot enforce their agreements on other people and so they really what they have is a wishlist I mean I know that's kind of rude and they'll be very pissed to hear me say this but without the capacity to enforce your your boundaries you um you don't have a Commons and so um this idea of uh how enforcement happens and whether that is done in a hierarchical manner or whether that's done by self organizing you know like when when uh in Valencia recently there was a a a a gigantic flash flood you know the government um did a very poor job for the first week and the response was done by self-organizing ant behavior of neighbors helping neighbors coming in doing whatever they and that happens in crisis after crisis after crisis and um and so there is a a capacity to self-organize if you will and get things done done um often what a government structure can do is help prioritize bring order to the what are the most pressing needs how to discern what will make the biggest difference collectively and so there is a value to having a a coordinating role but um but uh you know how that emerges or evolves or unfolds is um can either be by by cultural precedent of what the rules are or it can emerge through and from the group so um so that's my long comment about Commons and and commoning and how to be in relationship to the common awesome I I'll read a comment from Scott even before you mentioned Ostrum Scott had

written Commons Elanor Ostrom is a
co-management regime framing nicely
consistent with active inference in
human organizational contexts C Scott
Shackleford at all at U of Indiana for
awesoms Commons programs Commons
analyses have historically been focused
on quote asset based Commons this nicely
anticipates common co-management Beyond
Asset Management I.E risk management
skills coordination

Etc y i I should have made the point
that um one of the things that Commons
do that uh that the marketplaces and the
institutions have not been able to do is
for example it is financially rational
to fish to Extinction to deplete the
forest because the price just keeps on
going up so so so um the logic of
markets is insufficient for
sustainability and this is why
organizations that don't have a logic of
of of that friction of of um of
considering other factors other than
profit maximiz ation will inevitably um
deplete the resource um and so what she
created with this um case for Commons
was that there were sustainable
organizations and that these existed
both in indigenous you know in in
original people cultures they exist uh
throughout developed economies they
exist in a variety of places and are
unrecognized and what what you're
hearing this in this invitation
is like let's let's let's use commoning
even though it's unfamiliar even though
it's different it's actually much more

widespread than we ever have experienced
you know it's it's like believing that
there might be something other than
McDonald's you see McDonald's everywhere
so you think I have to eat burgers but
there are other ways to eat than
McDonald's and not trying to pick a
fight with h fast food
but how are we going to carry this
forward into 2025 and
Beyond yes so um so the invitation to
join um a conversation is is the
beginning one one of the Allies is um
that is both co-incubating other Commons
and that I wanted to use for if there's
energy here is um a a group called uh
evolute 6 uh based in Germany and and uh
South Africa and UK uh it's a small firm
dedicated to incubating uh and helping
Commons type organizations grow um it's
kind of the we're uh um and and so uh
trying to both invite the interested to
begin join the conversation sharpen uh a
vision and an understanding of what it
is that we think we can be in action on
uh what is it that how that active
inference can be applied to particular
problems the search for funding there
might be a better way I had to I had to
scramble to get something plausible but
if as we have conversations we might
very well identify much more um
important points of intervention with
the question that somebody else was
asking earlier we might decide that
there are certain flows and certain
dynamics that are in fact a great a
better starting point than what I

imagine right now but uh the invitation
is come let's convene
and do this in a systematic way and both
be transformed individually
interpersonally and ripple out into the
world awesome okay few more questions
while we're still here Frasier wrote I'm
not sure what this language of trauma
means I'd like to hear this
elaborated yes excellent sorry um I
forgot to explain that so um uh lately
in Psychology and in organizational
psychology uh there has become a
modified use of the word trauma and um
and what it refers to is the equivalent
of memory in in computation so you can
have a traumatic event and be
traumatized by it and then the trauma as
as I'm using it is the memory the scar
that's left either in you or the um if
it's it's if it's a collective trauma
that has a different um property I'll
describe it in a moment but but when I
say it's the memory the scar is what
remains in you and how you carry that
whether you heal fully or whether it
becomes a a debilitating Dynamic is is
when the word trauma can be used to
describe it Collective trauma is um and
this is an important distinction because
sometimes the same language can create
confusion
um uh when I'm acting as an individual
cognitive system I am all the trauma
happens within my body so I feel
something I fix something or I have a
story about something happening to me
all of that is personal trauma if I'm

part of a hive the the um hiving is
happening in the signals that we are
sending back and forth to each other and
so um the uh the this the the language
of how we signal and the meaning of what
that means is what can become
traumatized
and let me give you a simple little
example uh and and those are called
trauma agreements ways in which you
agree to carry the the significance the
pain the feeling about the situation
that you're looking at so so my father
took me fishing when I was eight years
old for the first time and we caught a
fish with fish and Hook and the fish was
flapping and resisting you know and I
was kind of shocked I'm like oh my God
that fish is suffering my father's like
no no no no no you just grab you know
whack them on the head knock them out
and they don't feel pain and in that
moment I entered into a trauma agreement
a different understanding of how to be
with what I was observing and as a group
we would talk about the our reality
assuming that shared trauma agreement
that shared way of of
understanding that same kind of dynamic
whether it is how what it means to be a
certain gender what it means to be an
organization that is um beneficial to
the world is it because it makes a
profit is it because it makes a
contribution there's a whole series of
assumed agreements that that it's like
oh we need to surface them that they're
not the truth they are an agreement into

which we stepped and we're now willing
to re-evaluate and this is both the the
the gateway to
disorientation but also transformative
adaptation um because it changes what
has been a habitual way when when when I
started considering that maybe animals
are still an omnivore but when I started
considering um that maybe animals had
feelings I had to start thinking
differently about what had been my
ordinary eating patterns and so this is
a language for
surfacing um anti- symbiotic ways of
being that we are now developing a
pattern and a way of recognizing them
and renegotiate aing
them awesome great answer last
question music minded wrote how does
Michael see the future of businesses or
more simply do you have Exemplar
businesses that are commons type in your
view yes there's a whole range of
Commons businesses and and and so even
though we're talking here about
incubating a new one in a sense um I I
believe that the the important work is
about what I was describing about
treating cancer calling back into
relation and so um some of us and this
you know if there's interest in this
I've been working with with global
standard setting groups about how to
measure and monitor well-being and then
how to so there's the sensing it and
then there's the what do I how do we
digest it you and this was another benef
another you know benefit from from

active inference was the way of framing
the cognitive process not just think but
sensing integrating predicting
responding adapting and and really
helping organizations think about in
that cognitive flow in that Collective
cognitive flow how to recalibrate them
their their attention so that they see
more of the world with with a denominate
a planetary denominator rather than an
organizational denominator and how do
they then more skillfully interrelate
with the different factors in which
they're they're they're engaging perfect
thank you Michael great presentation a
lot of people appreciated it in the chat
so see you next time thank you thank
e e

okay welcome back everyone we are here
in part one of the workshop that John
clippinger and our partners at first
principles first are coordinating so
with that for the
next session please John to you and take
it

away well thank you very much um I would
like to just to share a a deck with the
the group here if I could
um and uh and provide a sort of an
overview of of um what we'll be talking
about in this particular session can you
see that now is this is this good for
you okay

great um and we'll be anticipating with
Andrew is uh sure will be coming on and
providing the tutorial but what I would
like to do is in the first sense of this
is is being able to provide an overview

of why active inference uh is important
why active inference agents this whole
perspective that we're taking here is
worth people's attention and pay uh and
being concerned about um and what it has
to offer to address some of the the the
the limitations of the current uh AI
systems for example large language
models and some of the uh agenda
architectures have come out of them um
and and in that in that context what I
looking at is said what is the current
state of large language models and and
agent uh and agents and I think you know
there's a lot of activity that's been
talked about but there there is still no
Mission critical applications yet
there's a lot of an exploration you have
people like um Salesforce developing
Little Einstein that can do uh various
kinds of uh workflow automation you have
co-pilot and Microsoft but it's not it's
not it's not really core critical not to
say that it won't be but it is still
uh is still on the experimental phase
it's still a back box we do not know
rela how it works hence you have issues
of data Pro Providence you have a
question of how to you govern it
um and there's more recently there's a
concern about you've invested a trillion
dollars but there's an unknown business
model coming out so there's a lot of
money that's going into it but it's not
clear that the current framing of uh of
uh AI um is is now known as really is
going to be viable and part of the issue
here is is that we and I love this

example from The New Yorker is like who's modeling who we're actually putting data into the models we're drawing representations and we have that interaction but there's not necessarily an abstraction that's coming out so it's just it's kind of this uh relationship of of seeing ourselves mirroring ourselves and going into this sort of cycle um and what we have now is a certain hype cycle so it's around the peak of the this is 2023 from the uh Gartner group and it's the generative AI is at the top of the hype hype cycle and then if it's hard to read this particular uh graph but but you'll see that principle AI is is U basically um on the early cycle it's on the early phase of it but it's not seen as taking very long where something like uh you're going to have AGI is seen as being like 20 years out it's the top of the hyp cycle I think if we update this there now I think it's starting to be appreciated that the generative models and the expectations and the kind of problems and the costs associated with them and the lack of Improvement the the the lack of degree of improvement through successive models is starting to show through so the the Next Generation open AI shift strategies a rate of GPI Improvement slows in other words the new or Orion model is not did not have the level of improvements they were expecting I think that is something to take in account um and hence the cost of

of training and inferencing is extremely high and with that it the the M the effect you have this enormous consumption of energy uh that Microsoft is going to have to recommission Three Mile Island and drive one of their own data s this shows that impracticality of having to um have so computation um in order to have work make these models work and I I think that this is starting to be recognized within the the the limitations of large language models and sort of the the the parametric approach to to uh modeling uh AI modeling and I think to just start to step into what what we have now what we're looking at I love this is another cartoon from the the New Yorker it say says to think it all began with letting autocomplete finishing our sentences and well autocomplete is a sort of form of markof chain so we are we are having the sort of chains on top of chains that we've been able to to develop some pretty rich representations of of of human uh activity text and actually forms a cognition but there's not an underlying model there and so what we're looking at there may be some foundational issues that are coming up in in for large language models and that the fact that it is a black box is not explicable uh it locks internal causal models um there's not a way of reducing the complexity it's sort of a batch processing it's there's no real time updating and that batch processing we

see is extremely expensive and and so we
we're we're caught on on on that that
issue um it's not at all it has no no no
means of a spontaneous adaptation and
correction on its own um and it
certainly doesn't know what it it
doesn't know that has no way of
representing its completeness of its
knowledge and and adapting uh coherently
to that and and and also I would say
it's not it's not derived from really
sort of fundamental scientific
principles is really sort of an
engineering phenomena on a case-by casee
base uh basis now this is changing um
and I think the same issue if you start
to see foundational issues for
multi-agent uh llms in other words the
transition to to making agentic
architectures where you can um create a
particular agent that's optimized to do
a particular task or role a lot of the
success of that agent is depending on
the quality of the prompts of that and
sometimes those prompts are good
sometimes the prompts are bad but
there's not a really princip way of
doing that currently and so they have a
very fixed way of of of and specialized
kind of goals it's sort of a a a
mechanical it's it's so the
instantiation of an industrial model um
that we see um now what we we rather
than having a a a sort of biological
self-correcting model it is sort of
these fix static models now what I would
say the active inference difference here
um is is that it's really developed from

first principles and I think and and I think you're going to see that Andrew is going to go into this much more detail but it really C it cuts across different disciplines that themselves have been maturing fairly quickly in the last two or three decades and so it is going and basic basic physics Quantum information Theory the whole idea of computational Science Biology and cognitive science all these are intersecting into creating a new way of looking at what intelligence and life are and how they're interconnected um and it also builds in well established statistical methods including Markoff blankets blazi and belief models the stuff the work has been done with JD a Perl um and basically what you look at active infine does and and just sort to highlight is it codifies a predictive brain thesis that's been nurturing in in neuroscience and so the pr and then that in turn sort of mimics the processes of the scientific method and when I get so we think about what the method is that you're you're trying cify and and I turn to Richard Fan's comment well quoted comment that I can't what I can't create I can't understand so there's a new kind of criteria for what it means to have a um to understand something is be able to model to create a digital twin if I can recreate something then I understand what it is and if I can replicate that behavior and that that's sort of the the the principles that you find an active

inference from the very beginning is ah
how do I grant a generative model how
does a living agent have a generative
model of it s in its world and this this
really owes his work his Origins back to
to uh the work that's done with J Pearl
on his understanding causal reasoning
and also Markoff blankets you how does
some what is an autonomous thing and
this is where Carl fritzen an work how
how does some maintain its separation
from the world in which it is in and so
it is not governed by the things but
maintains its autonomy and there's a
certain kind of topology there that and
something has certain kinds of inter
internal States it has external States
observ states and action States and
there so there's a lot of deep Concepts
that's in it I'm sure Andrew will go
into that much more detail um and then
what we have is is the active invers
agents themselves as sentient live
things so you're creating something and
is very different than say a large
language model you're taking something
that has agency and it has a
representation of itself in the context
in which is trying to stay alive um and
so it's self-organizing is self-healing
is self-replicating and symbiotic in the
sense that it evolves complexity Mutual
synchrony with other living things in
its World in its Niche it's a joint
Niche construction uh hence sentience is
is sort of a is a kind of situational
intelligence and awareness that's
granted in the survival of that

particular agent and the person that's sort of the the context of the person who's really behind this is Carl Friston and and um my own experience of being someone who is very interested in sort of in artificial intelligence from from many many years ago and self-organizing system of cybernetics it really hasn't been until the last five or six 10 years that this has come together in a coherent uh not only Coan Theory but a whole Cohan formalism um and through what I what Carl would call a physics of living things and he's one of the most scied scientists in the world it's a neuroscientist that's done um basically developed a reputation neuroimaging and inferring the structure from the brain from Neuro Imaging uh and he applied the free energy principle which we'll go into more discussion to later um and so one of the I love this idea of a Trefoil not it's sort of like okay is it is sort of this autocatalytic biotic system it's really important it's a living system and it has certain kinds of characteristics that are all intertwined with itself one cycle synchronizes is exception or it's a representation of the external world to minimize uncertainty the other does it internally it has to align that with the external world and then you have another cycle that has expected free energy minimization it makes predictions about how to change itself in its World um and so what we have here is basically

a a a what I think active iners offers
is a really a soundness a grounding in
bi in biology and physics and so you
have a sort of a ground truth of
evidence base that do I make predictions
and do those predictions allow me to act
and survive in the world um so it can be
explicit it can be explainable and and
we'll see we'll talk about this later
but you can have since it has CA of
models and language and knowledge graphs
it can explain why it doing what is
doing and also it can talk about explain
why how much it doesn't there's degrees
of uncertainty associated with it and
there's also the potential of self-
policing self-correcting mechanism and I
think this this is really important in
other words you can design a active
inference agent to to live within
certain constraints and Bounds and you
can build in certain me me mechanisms
that have for self-governance that
cannot be captured uh so it's actions
with the limits of his designed Mission
and not to have generate negative
externalities so this I will sort of
leave with this one and then just to
quote William Blake I mean who talks
about this in a poetic form as a fractal
nature reality see the world in a grain
of sand and heaven and a wildflower and
hold Infinity in the palm of your hand
and turn an hour this just sort of pulls
it together in a more poetic sense so on
this I would hope to uh hand off to uh
to uh Andrew if uh

M thank

you I need

still

great thanks John

um and so hi everyone I'm Andrew um

Andrew pase I'm uh currently involved

with the active inference Institute it's

been great partnering with with uh Dr

clippinger with the director of the

Institute you know we've we've been

doing some really exciting work lately

um and I want to address a lot of the

points that that John has made although

most of that will come tomorrow when we

give some major updates on our biofirm

so today in the spirit of like an

initial workshop and kind of

introduction to active inference I'm

actually going to refer back to uh this

tutorial that I gave a couple months ago

uh at a International Conference for

computational social science meaning

folks who are involved in sociology

economics mix uh you know a broad range

of fields and uh the context was

applying active inference a lot of these

principles that John is referring to to

social science research uh and and I I

basically make the the claim and the

argument this is already being done so

I'm not necessarily providing a bunch of

new information but it's just to kind of

extend the dialogue to computational

social scientists like here's how we can

apply active inference to you know your

particular setting in different ways so

uh we just kind of cover some of of what

I went over and you know I apologize if

there's any Whiplash from how quickly I

go through these slides um but we're just going to move a little quickly um so the context again of this this uh this tutorial that I gave was it was split into three parts was a 3.5 hour Workshop where we actually we started with talking about agent based modeling you know how it's traditionally been done and then followed by that something that I would call a cognitive turn in agent based modeling that's happened over the past couple decades as you know there are these kind of calls for you know we have the computational resources now we have an immense amount more of you know Empirical research and validation of different kinds of principles related to human cognition coming from Neuroscience neurobiology uh you know cognitive modeling uh studies of neuronal Dynamics uh organisms and their environment all those kind of things so so there's a shift to that and then the most recent you know means of trying to establish that in the broader state-of-the-art area is uh using reinforcement learning agents so we'll touch on what those are and kind of how that works and you know anyone who's kind of been looking at the the agentic uh framework for for what's being done today they'll probably already be familiar with reinforcement learning then after that we'll go into AC conference and how that kind of kind of bridges together a lot of these different things um and apologies I want to just mention real quick uh you can

actually go to this this link uh the
quickest way of following that is just
by going to Google type you know my name
Andrew pashe and then you could type
GitHub or ic2 S2 in any case you'll come
up to uh the landing page for this
you'll see links to all previous slides
I've made uh slides that I'm making for
tomorrow uh I'm also making a lot of
this Open Access uh so so there's code
that people can you know play around
with and things like that um so
agent-based modeling uh this has been
done you know this goes back to roughly
70s 80s with a lot of you know uh
initial developments of computer
technology and being able to realize
that oh we can we can simulate uh people
right in some loose sense in in some
kind of simplistic sense uh how can we
simulate people and why would we want to
do that we would want to do that so that
we could maybe imagine uh hypothetical
scenarios imagine you're designing some
kind of policy or want to test a policy
uh and see you know oh if I recreate
that in a simulation and have people in
the environment where I'm implementing
that policy uh what happens what's the
collective outcome you know what occurs
so all kinds of fascinating work from uh
shelling and the the traditional uh way
of looking at uh housing segregation for
example th handling those kinds of
problems how where do people land where
how do they settle you know in in uh
like a residential environment uh even
more interesting work later after that

Santa Fe Institute recreating an artificial stock market uh where you where you have agents who are basically interacting with this kind of financial environment where they're looking at stock prices changing they make choices to you know buy or sell Etc see how that plays out Recreation of housing booms and bust excuse me not housing financial so this can be applied in many different ways point and so what what is Agent based modeling you know in it's fundamental principles and how does that relate back to active inference which we'll get to um key components of agent based modelings that we have agents kind of the catchword of the day um traditionally agents are you know that they just represent some entity uh a living organism you know this this can be used for for biological and Zo iCal studies trying to understand you know how ants or or insects you know interact in environment or at a broader scale how humans and entire you know societies and cultures might interact with their environment and and so what's really important about this with agents environments simulations that know we just kind of run a simulation over a series of time steps what and so we let Dynamics play out and then what is it that that composes those Dynamics it's the assumptions we make about agents what is their behavior what do we think that they do you know how do we program them in this way or in that way uh and

then also what are the aspects or characteristics of the environment you know and so again given that a lot of this work was done starting in the 80s going forward with with with very you know we're nearly archaic at this point uh forms of like computational resources it's like things were kept as simple as as possible and there are a variety of guidelines that need to be used to develop agent-based models for running these kinds of simulations uh you know and so so some of them even running up to to 2007 here this is the work of of drors laser and fredman or respectively uh have been involved with Northeastern University in Stanford um you know that they give some guidelines for how to do this um guideline one and two uh in some is you need realistic uh assumptions that you're making about your agents but not too realistic in the sense that they're too complex one because we don't have computers that are Advanced enough to run highly sophisticated complex agents and then also you know if we make if we make too many assumptions then the situation won't generalize well you might be making an agent based model for one very very particular scenario where it it's not adaptable across different real world circumstances right so we need some kind of balance in terms of the complexity of the assumptions we're making and the Simplicity of the assumptions we're making the rest are a bit more plain which are one uh reproducibility you should publish your

code you should document it very well
other researchers should be able to kind
of take up the torch and be able to
experiment with your code um you know we
at the Active Inference Institute we're
constantly putting out new GitHub
repositories and documenting our
processes so that other people can come
get involved so I just you know want to
mention that there's a big problem in
reproducibility in science today and
it's very well known you know so uh just
wanted to touch on that the
non-triviality it's like why do
agent-based modeling uh if it doesn't
help us uncover anything you know why
why not just sit down and have a
conversation about you know what what do
we expect to happen what are our
thoughts about what's going to happen
why you know do all of this um with with
setting up these programs and and agent
based models and signing them if we can
just come up to the same conclusions
through speaking so you you want your
results to be interesting you want to
make realistic assumptions that you know
we can kind of have all these things
synthesized um together so I made some
points about balancing Simplicity and
complexity this is just a representation
of uh you know this these These are
common problems in machine learning and
deep learning you know you you go and
get uh you know University education
these are the core comp components of
all of your courses you know this isn't
specific to active inference like wide

ranging you know concern how do you uh
create a model that is realistic enough
to capture uh and and explain the en the
phenomena you're looking at um but uh
simple enough to where it's adaptable to
different circumstances you know there
you know these are kind of in part uh
you know just as uh uh Dr clippinger
referred to Richard feeman you know just
some reminders from uh you know famous
statistician George EP box you know all
mod all models are wrong but some models
are useful right we're always trying to
approximate a reality and so the
question is how do we come closer to
approximating uh our reality and
so how do we actually you know using
modern computational resources using all
these new advances and all these fields
how do we actually make you know new
forms of what we'll call cognitive
agents who exhibit the kind of mental
Dynamics and social dynamics between
them that actually better approximate
real human
behavior so I give some definitions the
differences between traditional agent
based models and what agent uh
assumptions are are being made and uh
more often than not for the sake of
Simplicity an agent is pretty
one-dimensional if they observe X then
they do y otherwise they do Z these very
simple pre-program rules they're not
autonomous they're they're kind of like
you can think of them as like
thermostats or something like that right
the temperature is too high you you turn

it down if it's too low you turn it up
um that kind of reactive framework uh
there's no beliefs about what's going on
there's no internal goals per se aside
from a single homeostat that is you know
maintain the temperature at what you set
um cognitive agents however the
expectation for those is throughout the
the recent uh agent-based modeling
literature is like how do we achieve
cognitive agents to exhibit internal
cognitive mechanisms they exhibit
perception action they actually learn
and update their models I'll make a
point that you know as John reference
with in the case of llms which I'll
cover much more deeply tomorrow llms do
not I mean it's so expensive to train
them and you typically just train
them once right and so um they don't
learn in real time they're kind of stuck
with knowledge that they have they have
a lot of it to be fair they've been
trained on you know thousands thousands
thousands of documents uh and so um but
but that's it like that's what you have
and then and then we have to figure out
how to work with lenses from there with
cognitive active inference agents as
well as other kinds of cognitive agents
the ideas that they can learn in real
time meaning in the moment and adjust
their expectations and their beliefs to
what's actually going on now um and then
also planning so the capacity for being
able to infer or guess using evidence
that they take in like what's going to
happen in the future right and that's

immensely important in today's climate
right where we're that I mean that's in
part the reason we're doing H mod is to
attempt to plan to test things to to
come up with their own inference about
what's going to happen in the future and
then act in actionable ways to to
realize our own goals in the future
right so that's that's you know ideally
that's what an agent would do and be
capable of um and then with you know
this cognitive turn we we're also
looking at a broad synthesis of all
these fields you know as John mentioned
we're looking at things like cybernetics
and machine learning we're looking at
physics information Theory um people are
starting to it's great it's it's
fascinating that so many uh
interdisciplinary conversations can come
up and collaborations between
researchers of various backgrounds
because it's like oh you know once we
once we realize what it is we're trying
to do here we can start to pull from
maybe principles or relate principles
from other fields to this field what
works what makes sense what's
empirically you know what can be
validated so on so it's it's it's um you
know it's a very rich field right now
and then furthermore a lot of
traditional agent based modeling has
been done using uh a software called net
logo which is very specific uh it's it's
good if you want to do agent based
modeling like you just learn this one
programming language there's a GUI like

an interface for using it it's highly
simplistic it is not integratable with
other tools not integratable with you
know llms with natural language all
those kinds of things um and so it's
it's very narrow you know it's good at
what it does uh with cognitive agents we
would want to be able to pull from all
kinds of other resources that are you
know available including open source
ones uh being able to use apis being
able to pull in other kinds of real
world data not just purely simulations
you know so um there's just a lot of
richness behind this idea of moving to
cognitive agents then there's some basic
you know uh ideas of what makes a
cognitive agent again perception they
take in observations and they infer
what's going on in the world and kind of
like we what we do my stomach growls and
I infer that I'm hungry for it's a very
simplistic example um and then agents
are are generative the sense that they
actually come up with uh their own plans
or a you know choices of what actions to
take based upon what they believe such
as I observe my stomach is growling I
infer I'm hungry I will now choose to go
eat food as like an affordance that's
available to me in my environment so
this is a new way in a certain sense of
doing Age based modeling where before we
were looking at modeling an environment
with a bunch of onedimensional Agents
you know predictable agents on the
inside if they see X and they do y
otherwise they do Z but now we have

models in a model in the sense that we're actually able to internally inspect our agents right by having this beliefs based framework where agents observe real data or information and then infer beliefs about it we're able to then retroactively look inside the agent and see what its beliefs are as opposed to these kind of you know very difficult to to interpret uh blackbox models that we find very common and deep learning and and and you know very very highly performative yet very uh uh difficult to interpret like neural networks and architectures like that with with uh agents we would want to be able to see in a in almost like a human sense like what are their beliefs that they're holding and why are they doing what they're doing so we need to be able to accomplish that in order to actually realize all the additional benefit of doing this which is being able to look at you know internal cognition what are agents learning whenever I whenever I implement a new policy or something right I want to be able to know what's going on with them internally I want to know the full impact of of what it is you know this plan I I'm considering like actualizing so let's bring it back um you know how how do we address all these traditional criteria how do we make realistic assumptions how do we um you know how do we make our code reproducible that's an easy one because like well we're still coding the whole

time uh and then you know how do we how
do we derive useful insights from this
um and what are other problems that are
going on like now that we're getting
into to cognitive
architectures um and and where we are
starting to brush up against the
potential of recreating more blackbox
models right we we need to keep things
simple enough like how how do we
navigate all this common problem in in
social science research in general but
also in data science machine learning in
those kinds of fields uh reinforcement
learning common issue is what is uh how
do we handle What's called the explore
exploit tradeoff this this is famously
an issue uh if you have an agent who
comes up with a
suboptimal uh solution to a problem you
know let's say an agent who's you know
they're trying to figure something out
they find something that they deem to be
good they get a positive you know reward
signal or positive feedback from that
but it's suboptimal there there's
actually a better solution exists they
just don't know about it yet how do you
get an agent to actually be willing to
explore and same thing for humans right
we we don't always stick with the you
know the first decent looking you know
option available to us sometimes we we
explore in the sense that we seek better
solutions to problems and this is
especially important and something to be
addressed in agentic Frameworks where if
you're trying to use agents for say

software application or a use case like
you want them to be able to better
Solutions ideally you know at any given
moment especially in a changing context
that might call going forward for better
Solutions you know say agents who are
maintaining an environment that's
subject to uh natural disaster or or or
or climate change right like that that
might call for a whole new set of of of
policies or actions that need to be
taken uh you know that that couldn't be
relied upon in the past so we need
agents to seek new information
um unfortunately in reinforcement
learning you know that this is still an
issue um people are trying to figure out
how to kind of manually Define what what
defines exploratory behavior and it
tends to boil down to kind of simple
mathematic of of defining like okay well
we'll s we'll we'll modify these agents
so that 90% of the time they go with the
best solution they have they know about
and then 10% of the time they'll just
something random that way they're at
least exploring 10% of the time you know
you could put it that way active
inference again I'm sorry I'm kind of
zooming through a lot of this call this
is a over threeh hour tutorial um but uh
rein with active inference agents this
kind of gets naturally naturally
resolved in terms of What's called the
free energy principle which we'll be
getting into um but basically the idea
is if you're able to to have agents who
naturally seek information perhaps in

line with how we do right that when they come up against something they see oh there's value in seeking new information to find even better Solutions the ones I already have and being able to to do that in a way that isn't some kind of ad hoc rule of you know the 90% versus 10% as I mentioned that's that's um kind of a key component you know a key consideration designing agents um so this brings us to the active AC inference which I I describe here as the framework for doing agent-based modeling it uh begins with the free energy principle uh I I'll make a lot of mentions to this nice textbook that uh Thomas par Carl friston and uh you know their other co-authors published a couple of years ago um but you know there perhaps even some more updates to be made because it's such a lively field but uh that textbook really lays out a lot of the core principles and and ways of getting started with active inference um the the free energy principle everything that changes in the brain must minimize free free energy now despite all of this interdisciplinary communication dialogue between you know again cybernetics with psychology with behavioral economics um that actually sounds quite simple right to be able to boil everything down to oh we have a brain it's trying to minimize free energy that that that almost sounds simple enough to where we might be able to deoy that as a kind of you know simple assumption that we make in agent

based modeling it's
quantifiable uh it's it's a
presume presumed is the wrong word uh
but but something like an a principle
that can be applied in any given
circumstance whenever we're looking at
you know the these organisms who are
self-organizing such as humans such as
you know living organisms such as you
know plants in an environment that's
being taken care of uh you know that
they exhibit this this kind of uh
characteristic of minimizing free energy
as a as a form of quantifiable
uncertainty so we'll touch back on that
again in a moment but the the the what
active inference provides is a kind of
normative holistic framework for what I
argue should be applied to agent based
modeling it it itself addresses agent
based modeling that phrase is used in
the textbook this is not a foreign you
know out of left field kind of invasion
of active inference and agent based
model and proposing there's so much
overlap and I think it's just an
underutilized connection that's already
been made right um active inference is
based in you know it provides neural
process theories meaning that we're not
just you know assuming in the same way
with agent based modeling we might
assume oh all people are utility
maximizers um in this case we're not
just assuming it we're we're looking at
neuronal Dynamics we're looking at you
know Decades of search into using
neuroimaging studies and looking at how

you know how does the brain really
function how does it handle uncertainty
different kinds of Behavioral Studies
and tasks where you can you know record
uh fmri or EEG along the way and look at
the Dynamics of those
um uh of that data that's being
collected along the way and um and then
from there we can start to kind of model
how that
works so um as I already mentioned
uh we're still looking at a kind of
belief-based framework which is to say
uh you know we're looking at uh for for
agents who have beliefs and they update
their beliefs and they do that by way of
the free energy principle which is
minimizing uncertainty uh we we start we
start to ex we start to see these
organisms that we're modeling humans or
otherwise they start to have naturally
emergent adaptive Behavior right by by
trying to figure out what's going on in
your environment and then adapting your
actions in order to better understand it
to better realize your own goals you're
effectively minimizing your uncertainty
and uh and the free energy principle
basically uh is is a kind of free energy
is like a proxy for that uncertainty
that that renders a lot of these things
computationally tractable meaning that
we're able to quantify it we're able to
record it whenever we run agent based
modeling we're able to like actually
look on a chart and a graph how much a
free is being reduced uh over time you
know and and look at the agent's

internal beliefs about you know what
what is it that they're figuring out uh
within their environment over the course
of simulation um the view from from this
perspective is that uh another aspect I
I already addressed the explor
exploration exploitation balance and
tradeoff uh

self-evidence this means that agents in
order to minimize their uncertainty um
one way that they can do that is is um
to try and act to realize their goals
there's something they want something
that they expect right and so it's as if
you have this prior set of beliefs of
what you want to see in reality this is
the you know it's this is the kind of
framework that active inference deals
with but also referring back to you know
Jude Pearl as clippinger mentioned uh
you know this is a kind of basian idea
you know this is rooted in something
like like basian statistics or even
mechanics where the idea is that you you
have your prior knowledge you bring it
to the situation and then that will
impact how you update your beliefs based
on new information right so we're not
always starting from some blank State
some some starting from scratch we're
always coming to a situation with what
we thought before and then that gets
updated how does that get updated from
active inference that standpoint is
through free energy principal
um so you know it's I made some points
in this tutorial because I realized
we're we're we're really uh uh you know

I want to make sure we have time to to kind of bring this back um but uh with active inference that we already have so many aspects of uh reinforcement learning and active inference which overlap we're trying to create cognitive models you know active inference I don't want to say it's necessarily competing I don't know if that's the right word but it's you know it's it's it's making advances that reinforcement learning is trying to make itself um and uh and with active inference it's like well we with active inference we're actually basing this on you know a variety of of of principles you know that I'm again sorry zooming through you know we're able to like cognitively model you know what's what's going on in the mind in various ways that reinforcement learning is really lacking um today it's it's really lacking that it's it's uh you know these models are being built to be highly performative but they're missing the mark whenever it comes to something like realism and basing uh basing these reinforcement learnings and something like a you know science that extends Beyond just the mathematics of machine learning right we we want to be able to find biological physiological substrates for how all these things happen right um we want to be able from there to be able to apply everyday words and interpretability so what is it that defines what a habit is what what process is involved whenever a goal is formed and made and then from

there uh you know attempted to be achieved through action uh you know what defines planning what is prediction um and so I just want to see if there's anything else I might be able to touch on this is a nice um you know for for for anyone and maybe I should have started with this but this is a little bit more of an intuitive glance and this is actually these are not words from uh Carl friston surprisingly uh they're from uh Sarah Feldman Barrett who's at Northeastern she's uh you know nearly as rep reputable as Carl friston in the Neuroscience field and I just personally uh as someone who's also interested in in in neuroscience and cognition you know I find her work fascinating she's done a lot of work on um emotions and uh you know kind of affective neuroscience has to do a lot with looking at what's going on inside the mind of you know especially humans and so this is kind of trying to provide an intuitive glance that you know how can we think about basian Frameworks about um um inference about active inference as well so it's like uh this is a long quote from a podcast that she did your brain receives sense data from the world sights and sounds smells and so on receives sense data from the body so when there's a loud bang or a tug in your chest your brain receives that information as the outcomes of some set of causes but what caused that loud bang what caused that tug in your chest your brain actually doesn't know it only

receives the outcomes it has to guess at what the causes are this is called an inverse problem you know what the outcome is you don't know what the cause is so your brain has what uh you know another source of information available to it that is past experience memory um apologies for just like doing a slideshow karaoke here but you can read it for yourself I suppose but the point is and with some of this highlighting it's like we're actually taking account of a variety of things you know we're taking account of interoception meaning observations that arise from within the body or for an active inference agent uh you know it it it has to kind of model itself you know I mentioned earlier you know if I if I'm hungry this goes beyond the five senses in that kind of perspective right touch taste feel all of that um with this it's like well I need to understand what's going on within my body I need to understand what's going on outside of my body as clippinger you know referred to earlier the idea of a Markov blanket it's kind of like I need to figure out what defines the boundaries between myself and what's around me you know on the one hand I need to have a model of myself in order to maintain it uh otherwise I succumb to the environment around me to use some of the language used in uh you know the textbook it sounds a little bleak but you know we're referring to real world circumstances is if I you know if I don't take care of myself I will

naturally Decay like that is kind of the sort of the principle the world is entropy so to speak right it's like if I don't eat food you know I I I fall apart so I I need to have a model of myself what's going on within I need to have a sense of what's going on without uh and then and then I need to be able to kind of synchronize those things in such a way that I can continue to exist as an entity within this environment around me that's a that's a that's a very intuitive you know perhaps perhaps not very well technically described but for now an intuitive sense of of kind of what defines a markup like it so you know for this is a sort of foreshadowing for tomorrow we'll talk about this this really interesting project we're working on for the biofirm uh with with with uh you know Daniel and and and under guidance of of John um you know in that in that sense we're we're trying to develop homeostatic agents uh homeostatic meaning agents who you know their own sense of their survival is predicated on uh the the the homeostasis of their environments around them so we're trying to create agents who actually take care of and attend to the uh the environment around them as if their own Survival depended upon it you know this is a very interesting kind of uh kind of flip you know where where where you actually want to match and model the environments around you but also take care of it in a way that you can maintain M it within those bounds

just as you know I have to eat in order
to maintain my energy levels you know
that the agents need to take care of
their environment you know within some
kind of range to where it has you know
enough caloric energy so to speak um so
um I think for now John is there
anything that you think I have not
addressed that might be useful to to our
audience or if anyone else would like me
to touch on anything maybe something I
skipped or or otherwise would be happy
to to speak
more no I I think you you you you've uh
covered some of really key points that
um that are that what I I think is
you've done is what differentiates
active inference from the more
traditional agent-based modeling I think
is important because when people think
about agent-based modeling they they
sort of default to what the traditional
models are and I think that you've done
an excellent job as sort of uh
highlighting what the difference is and
what the the the having having that
broader perspective that more grounded
interdisciplinary perspective you you're
able to more not only represent the
complexity of it but also provide a a
way of modeling agents in within that
complexity um and I think that the the
the notion of a
the the homeostatic nature of of of an
agent trying to live in in conjunction
or really harmony with the world around
itself is depending upon its survival
and its ability to stay alive and

differentiate and have complexity but the synonym of the environment is really another key point because I uh I think a lot of people they think oh well you may say like navigating a fitness landscape you're just trying to optimize different levels but in fact the landscape itself is shaped doing joint Niche construction so it's a much more complex notion than traditionally understood General modeling and I think in in consideration what we be looking at tomorrow this whole idea of the btic character of model having it based upon principles and biology and nature and having being able to create an economic agent that acts that way I think is is is really quite important um and so maybe some people have some questions about that that we want to to focus on or not but that I I that's something that that I think we it's important to highlight yeah awesome presentation Andrew Ernesto if you have a question also anyone in the live chat is welcome to ask a question no questions thank you very much this is so inspiring well and as just to to put it in context I know this is something that you're very interested in in this whole idea of a btic economic system and you've been looking at this yourself so this is really trying to lay down the principles really the the scientific principles and computational principles of doing this and and and uh and we'll get in more to

that tomorrow but that that's the the
intent

here yeah

a lot more to say again in our remaining
40 minutes I'm happy to read questions
from the live chat I'm also happy to
screen share and look at some of the
registered participants backgrounds and
interests sure okay okay you can unshare
Andrew and

I'll pull up some word

clouds all right

okay okay

so

let's let it

load okay these are word clouds that are
generated from the participants
registration

information and we'll we'll zoom in on
them and and uh I I'll just kind of
mention them we'll we'll Zoom it in and
uh if anyone has a remark or something
that they want to uh like highlight or
call out about it just go for it okay so
this is a word cloud of the uh 150 or
200 registered presenters overall asking
about what is

your personal background so the biggest
and and and the this is scaled just by
number of uses of each term so one
person mentioning a term a ton of times
might make it large so just to be taken
as sort of a representative we see those
tiny two letters AI which could mean
almost anything at this point and a lot
of other terms like computational
research systems

Neuroscience cognitive

philosophy social

academic robotics technology

practice this kind of speaks to the

broad range of topics that that that

leads people

into joining and at least entering that

information on the uh form any thoughts

on this or we'll keep on exploring these

word clouds

okay okay all right again just just

raise your hand or or or go for

it interest

okay well there's I'm interested in the

notion of community and systems I mean

uh

applications the community seems very

strong yeah yeah and that that's

the the sensor of a of a later

one so here we asked what what would be

useful and by asking what would be

useful and what would be interesting we

alluded a little bit to the Dual

composition of the expected free energy

what would be useful what would be

utility what would be pragmatic and then

what would be interesting what would be

epistemic so heing what would be useful

you know here we are in in applied

active inference Symposium wondering

about how does active inference go from

being a Ying framework for ecosystems of

shared intelligences and all these

different kinds of things that we've

heard already and we're going to hear in

the Symposium to come so interesting

systems a way to

operationalize and really move forward

from framings of complex adaptive

systems to having the
implementations and also the community
and the people understanding research
models
collaborators and then a lot of
different topics that are being explored
in more
mainstream computational science Fields
like multi-agent algorithms Andrew
alluded to that and kind of brought that
up in the context of the the Goldilocks
parameterization
question the agent based model can be so
simple it gives you a great intuition it
shows at a a demonstration layer how
simple Behavior can lead to complex
outputs on the other hand the agent
based modeling can go the other extreme
where it's so sophisticated then it
loses its
transferability so that's what people
would find useful and then
here's what they put for what would they
would find
interesting
mhm
applications makes sense as that's what
this Symposium is all about and again a
lot of other ideas thinking about
different specific systems research
ideas architectures synergies toolkits
insights
implementations
support Collective directions
here's
asking what would help you in your
learning and applying active inference
and Community looms

large other people as well as some of
the more tool resource textbook
education examples Frameworks
interactive
tutorials different languages python
what does anyone think about
this I'm uh a little surprised that we
not seen the phrase llm anywhere
um I just uh kind of noticed that so one
of the one of the points that that I you
know I've been trying to work out and I
hope to speak more on tomorrow is um you
know integrating active inference with
llms and uh you know there have been
attempts to to do this and even even
with priston and Par and other folks you
know that there there's been a attempts
at trying to pull out an agent's beliefs
and then use an llm to process those
beliefs like kind of decode them and
re-encode them in natural language so to
speak it's as if the agent is speaking
and saying what it thinks and believes
and why it did what it did is you know
another layer of
interpretability and uh I don't know I'm
slightly surprised to not see LMS
anywhere um you know it maybe it's a
little too maybe that's a little too on
the frontier right now um I know it is
in many other areas you know it's
uh sorry in the couple minutes we have
it just uh you know something that I've
been working on personally and this is
feeding into the the biofirm is um how
how do we how do we use llms in a way
that we can actually integrate them with
you know all these major priorities that

we have uh where where we want to have agents who actually have defined goals right and and and ways of trying to stay alive you know in a way that they interact with their environment in a way that it's as if their own Survival depends upon it we don't want llms which you know again I'm just saying this quickly you know I don't to sound aring but llms in one sense can act as like functionally as Talking Heads where you just send in some text and they try to predict what would be the best text and response you're still subject to hallucination to confabulate making things up so to speak even though the answer looks structurally sound you know and and and so the my point is that rather than viewing LMS as uh as agents I don't I don't I don't think they are I think that uh you could have an active inference agent who is augmented with the capacity for natural language by having it interact with an llm right so yeah

sorrys my own yeah scan scanning across all the registration fields for llm mentions so H how does active inference apply to the design of llm systems AI outside of llms better ways of Designing internet skill skill protocols and governance systems and should converge with tiny ml to challenge lm's deep learning so it's it is interesting that that it's uh referenced with only a glancing blow and and uh We've even seen earlier like in the responses to similarities

and differences with reinforcement learning with frisen earlier about the increased amount of attention and Tool development and the the later and more professional maturity of training environments and this is like one of the key fosi for the open- source active inference ecosystem and for the Institute is to develop those kinds of methods and trainings and all onboarding all these different aspects of of increasing the professionalization so let's look at the challenges so this was what what what are the challenges facing your learning and applying active Influence People interesting the problem problem and the solution models the math software maths again mathematical understanding difficult physics scaling simple examples you know it's it's it's interesting interesting um because in the larger conversation about what AI is I mean it's it's that that conversation is so dominated by uh large language models and quote generative models within large language models um and and well at least I've been observing I tried to refer to my my initial slides is that they're starting to appreciate the limitations of of the large language models moved towards more agentic models and more towards causal models and hence creating aent architectures that do work with large language models but they're still there's not they're still not based upon biological principles that

we're discussing in active inference
there more of the mechanical
models being applied over and I I think
to me to from from a point of view of
sort of the
um the sociology of science soci there
seem to be these different worlds that
that I mean the number of people that i'
I've known sort of in the open AI world
they're they're very unaware of active
inference and some of the critiques that
are if you if you listen to the Gary
Marcus he's got a very pointed critique
of of um of of large language models and
not being able to achieve abstractions
but that doesn't extend to appreciation
appreciating what the the opportunities
are in active inference and so there are
there there sort of these islands of
understanding that they still
exist um and I think that's one of the
challenges and when you talk about
developing an infrastructure me you need
there's huge financial resources are
going into the the the large language
models infrastructure and where I think
that the totally bias on this I think
all the the opportunity is totally
within the active ERS of free energy
principle and I think that's where the
science is going and yet is it's still
it's sort of in in this isolate World um
and I think the only way to get out of
that is start to to really start to
showing the applications and showing the
with results and and uh because uh but
it it's it's it's it's interesting I me
to say they're they're different they're

different worlds um uh
but I I think that what's the there's a
peing and so the hype curve around
uh of the large language models and and
it's going to they're going to descend
my bet is they're going to descend into
a valley of doubt and you suddenly
you're going to see that showing up in
the in the Nvidia stock going down and
then people are saying you can't you
know destroy all the natural resources
or run these mods you don't have to you
know the other and Carl alluded to this
in his earlier comments today you know I
mean he was talking about climate change
and models compression I and I think
that's going to be a driver of this um
so but that that's some my my
observations on on these these clouds is
yeah I'll bring in a question from the
live chat also welcome Ma just thought
I'd bring you on if you wanted to
participate in this panel or just chill
so Steven select
wrote is the lack of llms
mentioned because large language models
are different to Dynamic agent belief
modeling and you want to take
that could you could you repeat that uh
yeah question lack of llms in the word
clouds here but just more generally
thinking about similarities and
differences between active inference
generative models and llms is is that
because large language models are
different than Dynamic agents or belief
modeling I I don't want to approach this
this question in the in the wrong way

but

I I will say that that that but as far as you know whenever it comes to llms and and dealing with you know their internal parameters are obviously highly quantitative and then they're dealing with with natural languages tokens and then switch back to looking at active inference agents which you know are highly quantitative in nature and uh typically you know if you're making continuous time models uh you're looking at again quantitative information and then even with discrete models you're looking at you know these kind of discrete values that you could potentially move in the direction of trying to you know make it I think this has already been done uh but the uh you know there there are agents who um you know have been used to try and understand like oh we can read individual tokens then move up to another layer and then we have entire sentences then move up to another layer we have entire you know paragraphs or ideas that kind of thing is done but I think this kind of like llm with quantitative uh uh grounded agents like that connection uh is still in my view that's kind of the frontier of where things are at and so it's difficult for people to conceive like how can we how can we translate encode decode re-encode natural language in a way that's readable by these quantitatively oriented agents and then vice versa how do we transition quantitative beliefs

into natural language that's the very thing that I've been uh you know looking at thinking about uh recently I made like a active inference like PDP agent who uh you know they what they do is that they use their beliefs about which action to take and then that action that they carry out is actually corresponding to a prompt to an LLM and then you can actually take those beliefs the quantitative beliefs find some way of meaningfully interpreting them and then putting that into these blank spaces in the prompt is as if you can have the agent say in real time I currently believe this with probability this give me you know a solution to this problem you know based on so so being able to find this kind of numeric versus the quantitative and the qualitative you know it's it's historical problem when working with with data that goes back centuries at this point so um it's it's really exciting though it's really exciting to think how to bridge that kind of Gap and I I still haven't seen anyone you know nail that down this this is the frontier this is where we're at and I think that's kind of the next thing for everyone to really be considering thinking about working on um and I'll give my view on how that could be done tomorrow well well um yeah I mean effectively LLMs don't really encode explicit beliefs or really maintain a persistent model of uh the world um even when we sort of give

them a sort of loop by which we communicate with them that Loop in itself doesn't actually factor into the predictions that it's going to do um so they they just lack a framework for recursively updating beliefs which is probably why it's not being thought of here um but also mainly that you know ILM agents are only agents by virtue of sometimes being connected to stuff and having um explicitly stated goal but I don't think that actually qualifies them to be agents you know yeah I I totally concur with that I from my perspective I I think that the reason the LMS are are are not looking that are are not looking at or that whole mindset is not looking at at belief structures in in subjective experience is I think is a product of a whole sort of objectivist reductionist methodology that um is has been tuned not to look at agentic systems I mean in my generation if if you if you were in biology you in and you talked about agentic systems you you were you were outed I mean that was that that's that was verboten and I think the the there's a whole school of thought there's a whole set of of in aonian sense a normal science built around um a sort of objectivist reductionist non- agentic notion and and and uh and so it's it's a new generation that's coming in in both in biology and in and modeling and cognition there's recognizing there this notion of an agentic perspective of

beliefs and updates and and and that's a huge shift I think in in sort of the epistemological perspective of of a lot of what has been traditional science that's really interesting there's also just operationally the difference between taking a very large fixed number of parameters in a given model 70 or 7 or 405 training on existing data and then finding patterns or eigen modes or activation um summaries that have secondary interpretability but there's not really an effort of trying to find interpretability at the level of activation of a given neuron it's about looking at patterns and trying to find Collective level interpretability like in in the AI interpretability field contrast that with designing from the ground up an example agents where the beliefs are explicitly encoded numerically and so that's a different mode of building and also one that's hinted at in in this 2023 paper with Ma and others that highlights that element of explicit semantic interpretability and also explicit semantic interpretability of uncertainty estimates whereas a string of tokens emitted by an LLM that says I'm not sure it's sure about saying that but that does mean that it has a given certainty about the referent so understanding what the communication semantics of LLMs are is incredible and what they do and I think when we look at this Cloud for people's

responses how can active inference make
a difference in 2025 where there there
is
llms

here many many kinds of systems
and that's sort of like a microcosm of
what the Symposium is different people
with different systems of
Interest asking H how do these
different attributes of the model that
we're talking about the first principles
grounding semantic
interpretability surprise and
uncertainty and its bounding as a
central component as opposed to reward
how do all those things really play out
what does that demonstration look like
in in different
systems yeah

here's go for it

Andrew no I was just gonna say on that
that previous one you were just looking
at I I I really appreciate the kind of a
broad-mindedness that the people seem to
be exercising I like seeing terms like a
symbolology scientist
governance um

industry I mean those are some
very I'm very excited to see people who
who want to you know so much work has
been put into kind of the theoretical
and then sort of like in in in vitro and
and in lab and and in simulation kind of
work that's been put into active
inference so I'm just really excited to
see you know with the the um thinner as
the the conference the the back of
inference conference that was given

recently and being able to see like a drone that's flying based upon the free energy principle and things like that I'm just really excited to see more that kind of real world applications right and and kind of creating uh you know systems and and applications for like real world use cases it's it's really cool to see some of those terms there yeah I think that's going to be really critical see see yeah

here's the cloud for how are you applying active inference again systems I think just reflecting people talking about variety of different systems types but and also reminding me of my own journey to active inference being excited about complex adaptive systems however not really having a software toolkit or a formal way to proceed past okay we think about nested multi-agent systems and then where's the software toolkit and then we're right back to where Andrew showed us on the slide it's like well we can do a really simple one in net logo or we can engage on this Enterprise scale engineering research program and yeah here's a more incremental interoperable way to have portfolios of motifs and generative model components and be assembling and standing across these different kinds of compositions like three ant nestmates in the desert three ant mate nestmates in the forest and these different kinds of environments being able to compose them and over the last several years seeing a lot of the advances in the category

theory in the first principles physics
surrounding all these sort of research
things and it's like that's kind of the
water drawing out and then that's the
question about how how it looks when it
applied and that's what I guess we
convene around in this Symposium and and
are seeing differently in
2024 than in previous
years that's curious so what what's the
change that you've seen in pre previous
years versus this year and then so
looking forward more of an application
Andrew or ma what what what do
you think I was
yeah but absolutely that that that is I
oh yeah please yeah I mean I feel like
um the work of Maxwell ramstead C Casper
HP I mean obviously Carl but um a lot of
people who perhaps came from more
Dynamics uh decided to take a real
interest in what we could do with
something that was so scale-free and I
find that it's been really really rich
in terms of the different kinds of
theories we've tried to sort of reforme
I know that's what most of my work is is
a Reformation of existing social
sciences um and to your point Daniel the
the problem we still have is that we're
missing like these key little plugs um
that yes you could couch as you you know
just different schemes of message
passing but you still need some
interesting tools to to have a a very
good description of the vast vast
variety of types of structures that we
have Which social sciences have done a

really great job at if not intuiting at least describing properly but we still don't have the best tools to make accurate predictions and that's part of the reason why most of our polls when we make them are are n useless and that should be the main thing we apply them to is just literally just understanding where we're at not even making predictions just understanding where we're at and we still fail at that so I'm very um I'm very encouraged by by seeing this and I hope that we will start valuing more and more interdisciplinary perspectives that are sometimes more technical but also sometimes uh you know more philosophical or um or social awesome also while we're in this in in the coming workshops tomorrow will be more of a Biore Regional biofirm focus and then day three will be entirely with clippinger at all on this topic and and I want to show one really operational open science Crossover with llms and active inference which is and I'll put it in the live chat as well here I used perplexity which does an internet search summarizes and translates information using llms so here we'll pick on Alexi there's a a profile a learning plan and a set of small medium and large project proposals about active inference so this is just summarizing expertise and experience here we have a learning plan it's just a starting position but this brings a new low bar

to the kinds of resources that everyone
can take on in their learning and use as
like a syllabus or a
curriculum and even as an interactive
tutor you know it will not get bored
with you asking to explain Concepts
again and again or using different
metaphors or making it applicable making
it like this poetry format putting it in
this narrative context and so even
asking how do I start with this or can
you help me understand what I don't know
that puts such a
centrality and and highlights
metacognition in
learning and then the project proposals
each of them structured according to
essentially what a grant officer would
be looking for and what a what a project
manager would be thinking about in terms
of small medium and large projects so
this
unprecedented High throughput Way
to fuse even scanty input from a
participant in a inclusive open science
setting into real projects that if
enough people think hey that looks kind
of interesting then we're well down the
road of actually making it happen using
different kinds of of funding which is
something Michael Lenin mentioned a
little bit earlier so it's sort of like
we can get really detailed and Technical
and Architectural thinking and and and
finessing the differences between
reinforcement learning deep learning
active inference and we saw slides to
that like when Carl showed the action

perception Loop similarities and differences okay active inference doesn't have a reward function it's not optimizing based upon this trained auxiliary reward concept it's just doing something direct with likelihoods like that is critical that is what makes the active inference generative models what they are but then looking at something like this which wasn't so accessible one to several years ago that just feels like opening up and unleashing what people can do and about using these as Tools in our Niche and I think it's a total open and unknown what happens when you can get to level of project and learning plan for for an arbitrarily complex field with the patience of an AI That's just going to accept questions all day long like that's almost the bigger applications question again just zooming out from how the architectures and the parameters are similar and different to how are we going to do what is authentic and right and best given the tools we have there's a bigger picture Beyond just holding the magnifying glass to different computational artifacts it's well said Daniel I really agree it's transformative if anyone has questions in the chat or here or any other things they want to mention

yeah what's going to happen in the next two sessions JN or what would be your goals or preferences for for this Arc of the Symposium for you yeah so I I I think as as Andrew alluded um this the session tomorrow is going to be really on the application of of U of active agents uh to a a design of a economic agent of new kind of firm biofirm this homeostatic um and the objective of that is to develop a new kind of incentive structures a point the collective action um that are based upon memetic principles that I think we'll see address a lot of the issues that people are having in the whole idea of B Regional Finance and how to deal with climate change and how to create those incentives so there one of the things I'll kick off with I've been talking to a number of people in the whole fields of of um um regenerative Finance by where they go bofi um this it ties into the work that was done in Elan ostrom's work on uh how common pole resource how do you manage a common P resource how do you incentivize people different scales I really think that what you identify some the requirements that people are looking for in order to address these problems to scale and I think what we're be talking about is to the implementation of active inference uh principles into an economic agent that it provides I think a a a reasonable way of of addressing these issues um and that uh um and and so that's part of the conversation that we

want to have and and and bring together
these two these two worlds um then the
third session is to really look at this
um from a a broader point of VI is
saying where is this going and they so
the people that we have in the panels
are very of very much involved with
Enterprises um applying it to nonprofits
to applying it to SAR problems to
looking at what they can do with LMS
currently and what the certain current
agentic architectures are and what
happens when when you s what happens
when you go to full autonomy what does
it mean to have autonomy um in in a in a
in a biological sense something keeping
alive and what does it mean to build
systems that have autonomy I this issue
of autonomy is sort of fundamental uh
when when when people think of of
autonomous organiz autonomous cars or
decentralized autonomous organizations
or the whole idea of U Central Bank the
the the the the principles behind
something like a Bitcoin is is like oh
that that that is autonomous but is it
really autonomous what is this new
notion of autonomy if we base it on
biological
principles um and I I think that's that
opens the door to a very different set
of discussions than than we're currently
having and it also opens the door to new
Notions of governance and how we can
build governance in to maintain the
homeotic states um and Andrew is
alluding to that so I mean that that
that that that's where I become a an

optimist and I'm not an optimist around the extension of LLMs and OpenAI and the tech giants but I am an optimist in being able to apply these principles to some of the fundamental issues we have around climate and social equity and economic equity. Awesome. Okay. In the closing minutes if anyone has last comments and David's question too if you want to address it Andrew or anyone else any thoughts about how we start to develop a multiscale approach to ethics. Um, I want to allow time for other people so I'll briefly say uh so so tomorrow um you know we'll we'll we'll present uh various aspects of the biofirm there will be a little bit of a Redux to some of the things I went through today just to kind of thread the Narrative um and I'll kind of talk about you know homo economicus and all these kinds of assumptions about human behavior and things like that but uh um I mean multiscale governance I what we're trying to do with our biofirm is to uh you know and Clippinger referred to um uh Elinor Ostrom's work on the commons and so I'm going to address that a little bit too but the idea is to be able to have a kind of like adaptable system of multiple agents uh you know and so we have a kind of small Society so to speak and uh you know they're kind of governing the commons around them and by being able to have something like the capacity for one shared goal uh and two adaptive learning such that they can

actually adapt what they do and what they believe and actions they carry out have those be localized to the current context which is an essential part of asam's like work is like it's not so simple to just give a singular top down instruction that applies in all places I mean to be able to have that kind of adaptivity and awareness of the local um you know while at the same time having agents who are kind of synced up in these various ways I think that's a one way of trying to approach governance and then from there you might be able to have kind of layered you know you could have individual hubs you know and then you could have some kind of like interconnecting system between them that that extends that to an even broader scale of multiple hubs and interaction between them so just some just some breadcrumbs there um thanks yeah Ma and then John with the last word on this

session yeah so I mean the question of Ethics is a bit it's obviously complicated but the the the key part about whether we coordinate around multiscale goal sharing Etc share potentiation that's intrinsic to power active inference right it just is the question that isn't going to be solved is what is good uh because you could you can always say well you know what is good is good by virtue of being contextually sort of stable but there may be other ways to be stable or that might prioritize other kinds of

individuals or entities Etc so I don't think we're going to solve that question however what we can solve is the is the global versus local coherence of certain kinds of beliefs relative to what might constitute goodness at given scales and find where the friction points might arise because this version of goodness is incompatible with this version of goodness and therefore Uh something's got to give somewhere an error is going to have to propagate so I think that's going to be answered but the the larger broader question you you'd have to look into different schools of thought around what constitutes uh good because we don't necessarily want to go towards um moral realism um because there's a lot of things that are baked into that uh but you might want to go into the um ethical implications of neomaterialism and how it is about interactions in locality and at that point it becomes about how do certain pockets of ethical framework sort of form and allow to just coexist away from other Pockets what happens when they start interacting uh which again we may be seeing right now in the US yes good point John any last comments on the session oh this is great uh um and I think U we we now surfacing some one of the issues that are really important to talk about and and um I think Le least what we're going to be presenting is is a framework um that at least we can have these discussions in a more

rigorous way and testable way um and uh
I I think that's part of the way moving
it forward so yeah awesome thank you
okay we'll take a one minute
break see you soon okay thank
you e
welcome
back welcome back hello son and Ma again
we are in the session physics and
metaphysics of social Powers so thank
you both for joining and take it
away thank you so much for having us I
was just asking if we should introduce
ourselves which uh which we will um so I
will introduce myself and then ma can
introduce herself so that I don't make
any mistakes uh because my brain is on a
hyper hyper abstractions at the moment
anyway I am Sonia I'm a PhD student uh
based in rdam at the araso school of
philosophy and uh we've been working on
this paper with ma recently and Ma
please uh yes so I'm a PhD candidate at
the University dubec amoral in cognitive
Computing and I'm also a director of
research strategy ad
versus thank you so yeah as Daniel said
we're going to be talking about the the
Spectrum the big spectrum of between the
physics and the metaphysics of uh social
powers
uh very inspired by the work of uh of
ramstead here hesp Hines and uh by mil's
work on on on power recently with Sarah
Grace mansky um and then we're trying to
combine that with some work that I'm
doing with two other colleagues uh
Wilkins and clav Vasquez on the concept

of active ignorance which we will be um
continue continuing to work on in the
coming
time um so yeah and then generally
inspired by by by ideas of semantic
attraction from Hinton and by Major
names in in philosophy and sociology who
have been treating these Concepts
through different Avenues uh less kind
of formal Avenues but still relevant
nonetheless now that we're actually
capable of joining um yeah the physical
with the metaphysical that's what we're
going to we're going to try here with
presentation so the concept of power we
we argue can be explored at at several
scales from physical action and process
effectuation all the way to complex
social dynamics so this kind of spectrum
wide analysis of power requires
attention to the fundamental principles
that constrain these
processes and in the social realm the
acquisition and maintenance of power is
intertwined with both social
interactions and cognitive processing
capacities so socially facilitated
empowerment grants empowered agents more
information processing capacities and
opportunities either because they're
relying on others to bring about desired
policies and or because they're able to
enjoy more information processing
possibilities as a result of relying on
others for the reproduction of material
tasks material is in Brackets because
these tasks need not be material so the
effects of social empow empowerment

imply an increased ability to compute
information thereby
augmenting the evolution of a specific
State space specific is key word here uh
really commenting back on what Ma just
uh just mentioned in at the end of the
previous
talk so empowered individuals attract
the attention of others attention is one
the central Focus uh of of our paper
they attract the attention of others who
contribute to increasing the scale of
their access to various policies
effectuating these these State spaces
that we mentioned so the presented
argument posits that social power in the
context of active inference is a
function of several
valuable
variables among them an individuals or
groups ability to attract the attention
of others with access to or the ability
to facilitate desired
policies and an individuals or group's
capacity to effectively process
information as a result of its power
amplifying effects this extended
computational ability also buffers
against potential loss or or possible
vulnerabilities semantic social
artifacts so narratives um ideologies
histories representation at large are
attentional scripts which ensure the
coordination of social patterns so you
can think of anything from psychological
schemata habits group think insulation
policies Etc and these are the
structures under and by which agents

mediate and leverage power and they're because of that the main focus of attention here so uh thinking about this the linking between the physical and metaphysical Concepts and the world gravitational metaphors abound in language this is like really prominent in the work of of Shephard Lake of and Johnson of course with special metaphors um I've been I've done some writing on it at the beginning of my PhD and also um laand Kent uh did some really nice work that's also up on the on the activ inference archive and then in his papers course um so the idea of power being pulled towards complex social attractors uh was also famously presented by Pier bord who linked various concepts of capital cultural capital political Capital Social Capital to power as this kind of social attraction uh and also traction and he borrowed the metaphor from from physical field theory in fact so there is already the connection is really present and obvious there uh only in in a metaphoric way um which we're kind of trying to expand on here uh so expanding also on his legacy to to you know take the epistemological and the empirical as totally Inseparable things that we're analyzing and his desire to actually ground sociological analysis in in a scientifically tractable manner we're trying to frame uh these kind of attentional attractors that move systems toward certain configurations through activ

inference and this framing allows for a new take on the dialectics between the individual in the collective again linking back to what was just mentioned at the end of the previous talk so in terms of um questions of information processing capacity not only our resource distribution and you know the division of labor a systemic fact but also the modulation and projection of future possibilities cannot be computed at the level of the individual clearly so this is something rather obvious in our context and our framing power this is how and why the minimization of uncertainty for for social systems is in fact social right so agents seek alliances and policies which ensure that cognition so information processing and future projection can be distributed and this allows for efficient practical traction because and here I quote from Steinberg at all 2023 epistemic confidence in knowing interpreting and acting together is that traction that happens when we when we're socially distributed uh and this in contrast with this kind of individualized agency where one's estimation of and here I quote again Divergence in mental States between self and others is what allows for the sort of prediction and interpretation of others Behavior which we might kind of want to frame as something that has a shorter range traction on on the future so now we're going to look at the theoretical framework and I pass it over

to Ma

so um obviously you have heard this 15 times but I'm still going to go through it because there are some Concepts in here that will become relatively silent after but to go over it briefly active inference explains how biological agents really minimize variational free energy to align their internal beliefs with external reality so effectively they've reduce discrepancies between expected and actual sensory inputs and we sort of Leverage The Descent on free energy to to to couch model uh perception cognition and action through bayesian belief updating and so free energy is the upper bound on surprise or the unexpectedness of sensory data relative to an agent's predictions and so direct computation of surprise is complex so agents use variational free energy uh to to approximate based on their internal in generative model and so um you can see that for example the Kullback-Leibler Divergence term measures how closely the internal beliefs match the true posterior of and so agents align their beliefs with actual data as they minimize it so the log likelihood term ensures predictions align with observations so to minimize free energy agents update their beliefs they allow for adaptive action selection that supports coherent goal directed behavior and so there are two strategies perceptual inference to adjust your beliefs to match the sensory inputs and active inference which is um acting to change the environments and align new

sensory data with internal models um active inference interestingly and as we saw earlier with the the whole word cloud spans or can span multiple scales from cellular processes maintaining homeostasis to cognitive systems using predictive coding to update beliefs based on hierarchical prediction errors and so this really gives us a broad biological principle where minimizing free energy promotes systemic efficiency and optimal homeostasis across scales and so um reducing free energy really means that you minimize entropy as well and you align your beliefs with reality to reduce this uncertainty important to note is that we also Factor um action and so expected free energy can be decomposed to extrinsic goal oriented and epistemic uncertainty reducing components that balance exploration and exploitations and so this dual strategy insert ensures that agents achieve immediate goals while gathering information to adapt effectively in un certain environments and so Central to active inference we really have this concept of a generative model which allows agents to predict sensory inputs and their hidden causes the generative process represents the real world states that produce observations and generative models reflect the agent's hypotheses about these processes uh the agent actions correct mismatches between its model on real sensory inputs uh to align

model with reality going quickly over uh
the the different terms in a
hierarchical structure each level
predicts the states of the ones below
and it corrects predictions based on
sensory feedback and there are
conditional dependencies that Cascade
between layers that support uh belief
updating and
adaptation we have observable states
that are the sensory inputs the agent
receives and tries to predict ch its
model the hidden States uh latent
variables that represent underlying
causes inferred by the agent to explain
sensory inputs there's a series of
parameters that Define the statistical
dependencies between hidden and
observable States likelihood function
which indicates how well the hidden
States and parameters predict observe
data we have a prior distribution which
is the initial beliefs about the hidden
States and the posterior distribution
which is the updated beliefs after
observing data and combining likelihood
and prior then obviously we use the
marginal likelihood which is the
probability of observed data under the
model to integrate over hidden States
and so we use this for model evaluation
and comparison to determine how well a
model explains sensory inputs so the
generative model really enables the
agents to align its beliefs with the
sensory data sometimes through
hierarchical predictions realistically
most often through hierarchical

predictions and feedback loops and they refine their understanding of the environments to improve action selection for adaptive behavior and so this picture is really uh very individual so um we can see though that within this picture it's really really Rife to uh to couch power dynamics so we start from Individual agents as predictive systems that minimize uncertainty through prediction uh through perception and action and so brinberg really made a salian point about this with his paper on the Primacy of I can where agents develop you know empowerment through embodied interactions which enable them to affect the environment so if we scale this up to Social Power agents use generative models for self- prediction but also to predict others intentions they form the basis of social understanding in Collective action Carl has lots of work about this um Connor Hines as well Axel constant maxel ramstead Casper hes Natalie Castell which I believe will give a talk as well and so attention uh is driven by Deep preferences and it shapes what becomes Salient which influences both individual and group Behavior at the group level shared predictive models facilitate coordinated actions so stigmergic processes as explored by our very own Daniel uh environmental modifications that guide Behavior allow distrib did control while deont cues as explored by

Axel constants enforce social norms and
so cultural practices further shape
Collective attention it creates shared
perceptual Landscapes which is something
that we explored with Toby Sinclair
Smith I think I said his name wrong I
call him Toby so I don't really know how
to say his name and finally we have to
know um how to couch joint attention and
this is work that and at the University
of Quebec in Montreal have also positive
that posited that this mechanism
basically enables efficient coordination
through mutual recognition of action
possibilities and
so power here power structures emerge
through self organizing subgroups based
on belief similarity so again this work
that has been explored by Natalie
Castell which we can call
decentralized influenced spreads within
Network structures and they amplify
certain narratives While marginalizing
others and so these these processes
create cultural capital where shared
norms and knowledge facilitate efficient
social
interaction
um attention here is thus shadeed by
those preferences but it shows that the
importance uh of critically constructing
shared narratives without homogenizing
scripts that stifle diversities uh is an
important thing to consider so it's a
point made by Anna AR simplistic
one-dimensional fuse like there is no
alternative constrain attention and
reduce complex systems to narrow goals

which is why fascism tends to be very attractive especially in times of high uncertainty and so distributed generative models over time including historical narratives and generational Dynamics give us the tools to Envision power as possibilistic which is a term that Sonia introduced here this this shifts the focus from isolated agency to Collective Dynamics and hierarchical organization we get a richer analysis of how agents and groups interact within constraints to shape future possibilities so active inference really gives us a frame for social power beyond very simplistic atomized ideas of agency and focus instead on this emergent distributed process that structure Collective behavior and potential thank

so yeah to return to the concept of narrative because it's sort of the thing that that um frames something that might go from from from the small scale to the large scale in different ways and something that we can all colloquially uh uh sort of latch onto narratives are are social mechanisms which have the power to make or break power you can say uh the cognitive exploration of of of the not yet engaged right of of scripts that are not yet there but one may kind of uh uh latch onto sort of effectuate anything you can think of here things like speculations fictions future possibilities of All Sorts um can be these can be understood as semantic attractors that enable kind of

relatively risk-free learning right so
you're you're projecting to imagine what
might be the case if and the simpler The
Narrative usually the easier it is to
sort of minimize like vast
uncertainty so these possibilistic
Explorations in the form of narratives
allow agents to explore counterfactual
scenarios and minimize uncertainty
without the sort of exposure to the to
the real world consequences of this um
this this safety mechanism frames both
The evolutionary advantage and
persistent appeal of narrative
engagement in human culture and in the
context of power you know leveraging
strong narratives as as Ma just pointed
out such as US versus them that's like
the the the one that you know the
biggest semantic attractor that we seem
to be keep returning to all the time
these types of reductionisms that
greatly minimizes uncertainty about an
agent or group's current predicaments
and they become because of this very
very appealing and sometimes
self-fulfilling unfortunately policies
that bring about their effects resulting
in Collective Collective behaviors which
can benefit sorry I'm so like which can
be Ben beneficial or detrimental to the
greater
whole the uh the effects of of
variational free energy minimization can
be observed of course at the level of
the basic structures of language in this
sense so semantic attractors like
redundancy rhyme alliteration Etc can be

understood as patterns which Aid memory
communication informational retrieval
policies because they permanently
reappear right so pulling attention to
the same phenomena and here I use scare
quotes very very strongly the the same
phenomenon uh thus greatly reducing the
need to interpret and organize
linguistic interaction and cognition a
new every time which is very
costly so we can understand for example
the possible emergence of rhythmic
patterning in communication as enabling
the storage and access of information
collectively by connecting a sound
pattern to a phenomenon we reduce
uncertainty about reality once that
sound pattern is witnessed again in in
sort of true pavlovian fashion
by way of the same uncertainty
minimizing mechanisms simple simple or
complex narrative constructs create
cognitive situations where agents reduce
complex complex aspects of the world down
to Preferred
scenarios and now we can look at how
active inference is a useful applied
framework for this pass it over too
again thank
you so I'm I'm here drawing on work that
is currently ongoing from other people
as well with who I'm collaborating and I
want to make sure that these ideas are
appropriately attributed to
them we're going to talk about
individual agency and empowerment and so
autonomy uh can be defined as a system's
capacity to self-govern using internal

Norms uh so again this is a topic explored by Jaclyn Hines Alex Kefir inesi poo in part adversive um and so empowerment here is measured by the maximum Mutual information between an agent actions and resulting States so this reflects how much influence an agent has over future States uh higher empowerment means greater potential for desired outcomes and so empowerment maximizes the entropy of actions and States while ensuring strong action State correlations autonomous agents are thus more driven by intrinsic motivation and this is a premise that again Alex Kefir is one of the Pioneers uh here it's thus connected to drives and control to systems that optimize future action State paths and so under active inference agents minimize expected free energy to balance exploration information seeking and exploitation goal Pursuit so this involves perceptual inference to update beliefs and generate model refinements through action outcom and so here power operates as an attractor in Social systems it it draws agents towards influential entities through access to resources and strategic advantages Happ it leads through it leads therefore to amplification through feedback loops um powerful agents help others minimize their expected free energy by providing some degree of stable predict Frameworks in reshaping the informational landscape so high power entities have

privileged access to information and can model their environment more effectively and they also influence what can be modeled and how impacting the collective predictive

landscape powerful agents can offload cognitive demands as well processing complex information more efficiently and directing narratives to manage error absorption and uncertainty so power here enables agents to shape others Focus directing attention to favorable States while diminishing access to Alternatives and this leads to feedback loop where control over the environment enhances influence and perpetuates power the whole story around

1% so power grants a broader range of operational capabilities and extends influence over others State space policies within its domain and so here active inference reveals power as this emerging property of the connection between perception action and social structures or powerful agents maintain and expand their control by shaping both informational and physical environments so let's go over three main components Precision modulation belief alignment and ultimately minimizing

Divergence first Precision modulation powerful agents shape expected outcomes by altering the social or informational environment as we saw they create narratives or scripts that effectively reduce the dimensionality of this state space forming stable

attractors such as social norms or shared stories and these attractors direct Collective attention and ensure that other agents align their actions with these preferred

States um but here basically the posterior belief about States uh has uh an α that modulates the impact of precision on these beliefs and so in belief alignments the power to direct attention and utilize Precision modulation really enables the leaders to make specific aspects of reality more Salient for others and so by establishing these attractors within the belief space the leaders guide others interpretations and behaviors um these attractors act as a focal points reducing ambiguity and encouraging agents to align their beliefs and actions around them thus fostering a shared understanding so the the alignment process naturally emerges from minimizing the expected free energy uh and so for example ex Le um where P of O given s and P_i is the likelihood of outcomes uh given States under policy p_i and Q of s given o and P_i is the posterior belief of about States and so this decomposition shows us the intrinsic and extrinsic components which balance uncertainty reduction with goal oriented behavior that is um shaped by this Precision that we mentioned earlier so finally uh minimizing Divergence uh can be represented just using a coolback liary Divergence uh which is a measure of the

difference between two agents belief
distributions and so a lower kale
Divergence indicates closer belief
alignments and so leaders influence this
alignment by modulating this Precision
which amplifies confidence in specific
interpretations and so the process
minimizes belief Divergence across
agents which promotes synchronized
understanding and cohesive group Behavior
without the necessity for direct
coercion although that is yet another
way to minimize the state space and
therefore is an available practice so
Precision modulation sets the framework
for control belief alignments ensures
Collective cohesion and minimizing
Divergence uh formalize as this
alignment so basically we capture the
Dynamics uh and influence uh the
Dynamics dynamics of power and influence
within social systems
and so finally we come to the conclusion
we have a lot more in the paper that
we're still working out so look out for
that um yeah so what we're we're
thinking about here in terms of this
kind of shift um attentional shift uh
from from ideas of territorial and
material power to this new realm of of
informational and computational power um
well this this for us requires attention
to attention uh for sure something that
we can we can look at in uh in ways that
we can formalize and of course we
propose that active inference provides
crucial theoretical tools for for making
power structures visible and therefore

analyzable and so allowing possibly for the potential to rethink current social asymmetries so again given this sort of apparent transition between the organization of life around the centrality of physical work and matter and things moving in space towards the centrality of attention and information uh especially considering the whole attention economy idea we kind of try to think about a possible sustainable social future which requires the addressing of both resource and information distributions and the latter is best analyzed through the power of attention um as we've tried to to show as a major factor or one major Vector influencing social dynamics and of course this is very important to mention hierarchies are inevitable in complex thermodynamic systems but we propose if we can begin thinking about this uh fostering transparent plastic hierarchies that remain contestable and thus open to novelty open to learning right this approach um has to acknowledge uh historic and thermodynamic constraints while also acknowledging space for systemic Evolution and while hierarchies are inevitable social scripts should be read writable by those with an interest in survival within them right and this helps reorient our understanding of an unfolding globalizing interconnected and interdependent life and participation in a system is only participatory if those

involved are able to track the
consequences of the variables and thus
future State spaces at stake a true
distribution of information processing
cap capabilities is essential for
systemic sustainability and proper
Equitable development and when thinking
through evolutionary biology lenses for
example let's try to polish them clearly
and not fall into social Darwinism
narratives and to quote from the xenop
feminist

Manifesto um lovely colleagues there
Patricia Reed for example if nature is
unjust then we change nature which is
what we are all in the business of doing
here a little

bit all right that was um that was our
presentation thank you
awesome okay well will'll be
back in a few
minutes

e

Carl hello welcome back
everyone I'm here with Alex
aoria we may have Carl join we may have
someone else join if you want to join
maybe write it in the YouTube live chat
and maybe I'll email you a link
otherwise write some
questions however to kick things off
Alex feel free to say hello and start in
and and I know that there's things that
we can explore together sure you just
picked it back to me and I was saying to
you all right well I I'll just introduce
myself because I'm not entirely sure how
most of the audience if they know of who

I am so I'm Alex roia I am a computational neuroscientist and cognitive scientist who has worked for active inactive inference and specifically predictive coding versions of it for quite some time I'm Carl's friend and collaborator uh some of you might be aware of more of our recent arguments of mortal computation which is a coraly or a new coraly or another coraly of the free energy principle and uh yeah and you know I'm all about what's wrong with deep learning and machine intelligence and I'm usually the guy you go to if you want to attack back propagation of Errors large language models Transformers kind of relates to a lot of the discussion I was able to catch Snippets of earlier today so if you want me to help dismantle what's what's the problem with deep learning you know that that can be fun too so that at least if nothing else everyone now knows who I am Carl needs no introduction maybe start off with that while I try to figure out why he cannot join oh okay uh what part of that do you want me to I mean I can go into Manifesto but I'm not sure that's what this panel should be about out maybe beginning with a five minute Manifesto and then there will be some questions and uh we'll see where we get well maybe to try to make it useful to the audience uh for the moment although I wasn't intending the panel to

be focused about that um maybe one thing
that we can talk about is there was this
discussion during the active inference
Workshop about metabolism and sort of in
steering a homeostatic metabolic drive a
survival almost orientation to machine
intelligence and so uh Carl and I have
now been arguing for probably now about
more than a year since uh our paper came
out like on yeah around November
actually start of November talking about
uh another way to redesign reformulate
to use Ma's earlier uh uh wording uh
intelligence itself uh particularly
machine intelligence
and so mortal computation without again
if for those that might not be familiar
actually my friend Jeff also two years
ago introduced this phrase the idea that
computation or intelligent calculations
and substrate uh in biological natural
systems are entangled and it's very hard
to divorce those two concepts uh whereas
computer science and a lot of like what
cat GPT is or large language models or
your favorite
GPT structure is Immortal which is the
idea that we can copy that program
including its weight tensors and it's
encoded knowledge and place it on GPU
server you want and it'll behave roughly
the same there's nothing inherent about
its physical
instantiation uh in this universe uh
that dictates or shapes or uh
necessarily embodies it right and so
there's this like move to uh embodiment
by the way active inference in the free

energy principle has all of these characteristics mortal computation which is again as the concept I explain earlier Carl and I sort of uh recast it we strongly redid it and said that um uh because while Jeff didn't want to necessarily directly explore it we put it as it's just basically thinking of the free energy principle and taking it to its natural extreme conclusion which is that machine intelligence systems uh because they are inherently B to their substrate must focus on this survival objective right and the free energy principle and one of its ways to talk about it uh basically is saying you know we have a markup blanket or a nested system of markup blankets and our goal in this universe is to protect those internal States uh yet interact vicariously with our external States and so that eventually gives us our ultimate goal which is to prevent ourselves from disintegration into a heat bath it's a very physics uh form of thinking about identity and identity construction um but this is kind of what's centrally missing in basically all of machine intelligence um is this the completely right path to go there's plenty of criticisms and I'm not one to say that there's only one direction we need to go if we ever want something like artificial general intelligence um but this might be a path worth exploring it also addresses the Energy Efficiency arguments I was hearing earlier discussed uh Transformers particular

particularly in large language models are uh grossly negligent of their energy costs they have often times have the carbon footprint of a large city uh and that's actually now out of date as you know Carl and I wrote that about a year ago um there's even discussions I find kind of humorous by individuals I'm not going to particularly name saying we should be powering now uh these Transformers and large language models with nuclear power plants and so we're kind of going down the rabbit hole of let's just burn as much of our resources as we can to just get this particular type of intelligence I argue very strongly narrow AI uh and we're cheating ourselves of looking and embracing actually Energy Efficiency uh which would lead us towards the real grand goal of green AI uh because clal GL global climate change is clearly and empirically a thing and we should be a little worried about that so uh mortal computation sort of helps steer everything into sort of what I would argue hopefully maybe in multiple years time the emergence of a whole new field because I consider at least my personal take I'd love to have you know heard Carl wherever he might be um but I Envision actually mortal computation is not just like the framework and definitions that Carl and I talked about which kind of take free energy uh and active inference to the strongest conclusion to a naturalistic machine

intelligence but maybe perhaps a new domain of thought of biomedic intelligence and where everyone else can come together and play in this really interesting space and there's a lot to unpack there but I don't think we want to make this about that uh there's plenty of talks i' given including active inference Institute we've had a podcast actually I think it was right when the paper came out so for those that are interested I just recommend you to go to the YouTube channel and watch wonderful videos that Dan LED oh is there is there a new link okay sure was was I just blabbing there to no one no that was perfect we we we just hot swapped okay I was trying to buy Carl welcome Carl we talking about okay thank you the overview on Mortal computation and and and it was a highlight of the year um Carl with the evening Sunset now how um goes it wait I don't hear you oh oh continue sorry it's all good Carl continue can you hear me now yep excellent yeah I'm sorry about that I got locked out I was uh paying the price for enjoying for the first two hours right wait just I really enjoyed the um the intervening sessions um and look forward to comenting some the cross cutting SE yes okay um Alex

mute it's a little noisier on jitsy and then uh Carl um also a little quiet but we hear you so um what were the what were the themes that that uh connected the presentations that you were in or otherwise one thing that um struck me can you hear me right now is this y excellent I mean there are lots of lots of little uh golden nuggets um throughout all the presentations um but one um one crosscutting theme was this notion of community that emerged in various guises such as the notion of Commons that was the subject of uh a lot of the exchanges in the first um in the first um uh presentation um and then the notion of community that was a Dumont of the the workshop um and then the of coherence and synchronization and narrative social science accounts of that and how that might be formalized in terms of um joint free energy minimization ensembles of Agents you know finding a coherent communal um and empowered free energy Minima so you know I thought I I thought that that was a common theme sort of focusing much less upon um individual agents but what happens when you have a an ecosystem of Agents um and what principles um can be brought to the table to understand interactions or ensembles of Agents so I I have a number of U you know lots of things came to mind I was trying as a physicist to sort

of um rewrite all of these wonderful
Concepts um in a mathematical way or or
a physics way um so for example I was
think about the importance of Commons
and it struck me that one way that we
could motivate the choice of capital c
for prior preferences uh would be that
these are the common goals and the and
the C characteristic States um that are
shared between um multiple agents uh and
of course um the last presentation took
us through some of the mechanisms and
the capital c circular capital c
causality um that leads to this Capital
C coherence so you know which originally
was meant to be constraints in the sense
of a constraint Maxim entry principle um
or cost in the you in a reinforcement
learning uh reinforcement learning
setting but coming back of course all
this started when I was hearing about
Commons and how it underwrote
communities it is very interesting
how development and the presentations
have
proceeded a lot more about the
communication and somewhat of the
multi-agent coordination as opposed to a
longer memory train for individuals or
deeper prospection for one individual
um and in a way the
renormalization method brought a a
Clarity or a rigor to what had already
been qualitatively
explored from 2013 and Life as We Know
It And all these other works that we're
talking about multiscale systems and
about blankets of blankets and within

this year to make it timely we start to
see some of the
ways in which that compositionality can
actually be brought into the generative
models that we've seen I mean it looks
like the textbook figure 4.3 and the
nested models and now that's
actually the
motif that gives a way of looking at a
multiplicity at the more inactive level
as having a Unity at a higher level
Alex in
your designing or work how how does the
single agent or
multi-agent design come into
play
now yeah yeah can hear okay good yeah I
was having mic troubles earlier um well
so it's kind of interesting I think I've
spent most of my career at the single
agent level in the sense that I want to
build I don't think we have right the
single agent and I and I think it's kind
of important to sort of scope that out
from an architectural learning
perspective I'm very much a fan of like
Carl's arguments and you know I Echo
them myself a lot which is the idea of
the inference learning and structural
time scales uh I still think we have a
lot to do but if I were to say how in my
research the multi-agent approach really
comes into play well I do think a lot in
terms of assemblies or ensembles right
putting together multiple circuits I'm a
I'm a huge proponent of another type of
this comes from cognitive science called
the common model of cognition uh I'm not

sure how many in the audience are familiar with it but it's sort of like a generic reprint that has emerged over the last several decades of cognitive science to say that we can think about and take our current Knowledge from neuro physiology and cognitive studies that brain has these rough levels of organization different types of memory we have the motor CeX we have phas of gangu we have all these very important structures and we can start appending to them some functionality and so in that sense that's where I think of multiple agents right because you can start breaking apart using again Carl and I've have argued this the mean field approximation uh there's natural mean field approximations that emerg through the result of Evolution where you have these little substructures that communicate with each other um obviously the sparse connectivity that Carl mentioned in his talk this morning is uh kind of indicative of how these little substructures emerge and then communicate and perform sort of like what uh Marvin Minsky would have said you know sort of a society of mind right uh so it's kind of a call back to that and I think in that regard I often uh one branch of my work often tries to highlight emphasize bring forth that notion of cognitive Blueprinting or cognitive structure um because then we can start filling in the pieces with what we know about

neurobiology like for example I use predictive coding quite a bit uh you know to fill in those gaps and so I think in that sense I think a lot of interacting subsystems you could think of those as agents and then my final comment uh without letting myself BL too much uh I have worked in neuroevolution so there is another scale Above This which is you know you do need to account for how multiple mortal computers or multiple agents interact with each other um and so I think in that regard I have done some work with very simplified agents where you can kind of abstract away the neuroscience and just say I want to look at these little tiny systems that interact and then we can you know modify them through genetic algorithms and neuroevolution so there is a little bit of that like higher level uh societal kind of learning I have looked at that and I think a big challenge just to bring it to the audience for the panel is uh merging those time scales in a in a coherent and efficient way I still think well there has been good work I know Daniel you work on M Colony optimization as well and other types of kind of aggregate algorithms uh I think bringing those time scales together and really interacting with what we currently know about single agents even like the common model of cognition and going back to the high level which is this multi-agent evolutionary time scale is a Grand Challenge and how do we of course

effectiv simulate those in a way that
anyone can play with this I think is a
big Challenge and yeah and I think I'll
kind of stop there before I go down more
radicals neuroevolution and neural
Darwinism something Carl and and others
have also explored it's like they're all
and then the multiple nested time scales
multiple nested spatial
scales it's like there's this attractor
to want to come back and find the
unifying
patterns and and it's and how those can
be applied is kind of an exciting
question I'll read one question in the
chat so feel free to give a thought or
take it any direction David Highland
wrote do you view mortal and IM do you
view mortal and Immortal computation as
binary categories or are they more like
two points along a
spectrum actually want Carl to answer
that I mean I could give something but I
really want to hear your thoughts on
that it's your favorite subject though
design I know but I'm actually really
curious if you think of it as a spe
because remember the conversation we've
had with Mike Levan and Chris fields and
we've been sort of debating you know how
important it is to lean into mortality
versus immortality do you think that
there could possibly be anything in
between because I honestly you know I
think of it kind of more rigidly I'll
just give my aners that and I firmly
think we need to like focus on Mortal
computation but do you think that there

can be a hybrid in between so sorry to
turn it on you but I'm very curious to
know your thoughts um
my my instinctual answer is no
um having said that I am sure that you
can present some um description of
mortal computation in terms of immortal
algorithms I mean I'm just thinking back
to sort of um um the trilogy due to
David Mah uh but does that mean that
these kinds of descriptions are apt um
for simulating reproducing
understanding um or describing the
actual process I I I suspect not they
may at one level of description they may
be but when it comes to
um applying the principles usually in
computer simulations I suspect that that
just would not work I
mean because Daniel and I were wrestling
with zoom and other um um communication
things I I I may we may have missed
whether you've taken people through the
the the key distinction between Mortal
and Immortal uh computation so AI do you
just want to briefly rehearse the sort
of the two Cardinal views of the of the
distinction um between Mortal and
Immortal uh computation I did I just
wasn't sure if anyone heard me so I
wasn't sure what was going on
Zoom yeah I mean certainly from the
point of view the free energy principle
the free energy principle is a
description of a random dynamical system
and as such um it is
um a system that constitutes the world
the you know the lived World um perhaps

not the lived world but certainly a
mortal world um that has um
characteristic time scales and therefore
um the free energy principle and
everything that ensues from that
including active inference um rests upon
that primary assumption that you are
describing a process a mortal process a
a a a process that could have um a
physics um although
um the actual free energy principle
itself is almost pre physics so you have
to derive basic mechanics and classical
mechanics and quantum mechanics from um
the existence that you the the the
mortality of this process cast in terms
of you know time evolving systems so in
that sense um you know there is no such
thing as Immortal computation
but I also I don't know if you said it
out loud but I also liked your um more
playful take on Mortal versus Immortal
that if if software was
Immortal uh and it had some autopoietic
aspect to it then it wouldn't care about
dying so there would be no wait there'
be no way there' be no incentive to have
characteristic States coming back to
another big CA um uh so you know in that
sense they cannot be immortal
computation simply because that is
removing the definitional uh stipulative
definitional aspect of the
characteristic states that we are trying
to describe through the F principle and
active inference and specific um VAR
variance of that such as predictive
coding but I just pick up on this

crosscutting CC theme uh you also um um celebrated that with common cognitive models so again with we've got this this notion commonality and uh um and you asked the question you know how How would how would one um apply these Notions and if you like sort of naturalize the common cognitive model under um a less anthropomorphic or functionalist active inference account and you gave the answer implicit in your question which is how would you simulate these things and I think Daniel asked the same you same question indeed the the whole point of this Symposium is about applications so I I just wanted to put a question to the two of you I was um impressed with the very um clear distinction between agent-based modeling and the common cognitive equivalent the co where the agents the members of The Ensemble were actually for example active inference agents that seems to me a very big move and a very difficult thing to actually do um most of the literature I know in this area um has at most taken baby steps into having ensembles of two agents with authentic agency in the sense that they can plan so I'm now limiting um the notion of an agent Beyond a thermostat or a virus to those kinds of systems that um have in mind or part of their Gena model the consequences of their their own actions so they can be future pointing and they can plan and select from alternative policies uh and of course the numerics and you know the the Practical

difficulties of actually
simulating doing agent-based modeling
for example um with authentic agents in
the mix in the ecosystem um I think is
you know um is a really important
Challenge and a really important
distinction between classical
agent-based models where you put lots of
theats or rents attractors together and
see what happens and of course they have
a synchronization manifold and you can
certainly box lyrical about um
generalized synchrony and and and Joint
free energy minimization of a generic
sort but I think that misses the point I
think Ma was making um
um that you know part of the the
multiple Loops of circular causality in
sort of your know cultural Niche
construction for example um you know
really does rest upon um the capacity of
any individual within an ensemble um to
um act given they have selected a
particular course of action particular
path into the future which they have
beliefs about the consequences of and
you know to me that that that that is an
authentic kind of agent so you mean
perhaps well both of you d mean your
your your ethology background um are you
aware of anybody's actually sort of
Bitten the bullet that was introduced to
US during the workshop and actually
started creating active inference agents
that actually plan and learn from each
other and carve out their own warington
Landscapes you know finding those sort
of joint joint

Minima I don't know specific uh citation
but certainly you've given a research
Direction and it reminded me of in the
shared protentions work where thinking
about what what does a shared Ensemble
what would it be for a planning Ensemble
or for a collective it could be a shared
protention as in there's the same plan
disseminated across the ant nestmates it
could be that they have different maybe
nonoverlapping but complimentary parts
that would be shared it could be having
a shared reference but they could have
different plans or it could even be a
shared consequence without even a shared
reference so there's so many ways in
which especially when we consider
ensembles that that are
heterogenous they have different kinds
of umelt different kinds of ways of
engaging with the niche and and possibly
some Comon
niche as well as some uniquely
accessible components there's so many
flavors of
Ensemble that I I can can only
hopelessly or helpfully try to think
about what tools and infrastructure and
coordination would even give us a grasp
to be able to sweep or design or
visualize across that otherwise we might
be locked in
way too early to what it means for The
Ensemble to have a collective protention
and then that's not going to apply to
every ant species and if it can't even
apply to all the ants out there then
it's not going to work for

aliens Alex can I can I just respond to that and then ask ask for your your view on that one so I think it's a really important point and um one of the big takeaways for me on our paper on the variational synthesis use you're applying the renormalization group to the the coupling of evolutionary to um sort of phenotypic uh time scales was that it was not uh feasible or appropriate to talk about consp specifics in isolation that it was you had to have the all the varieties of predators and praise and things that we um that not even animate not even agents in the mix to actually um derive the um the you the non-equilibrium steady States at particular at particular scales which you know speaks directly it is not um it is not U useful to um as you know if you like take the idealized gas mentality of 20th century physics and now apply it to agent-based modeling to assume that all the agents are exchangeable and are identical in some sense and then are converging on some common narrative um or some common um uh is not the right way to you know to simulate these things you can't do Predator prey relationships you know for example you can't have um even simple economic games uh but furthermore you you you when people talk about depletion of resources you actually have to model the resource you know whether it's oxygen food or water as things in in that ecosystem so I think that's a really important

observation which just speak to the challenge of using numerical studies to test out some of the hypotheses that uh that we' you know being being exposed to um you

know you know if one just pursues agent-based modeling without introducing that heterogeneity I think one is making exactly the same mistake um that people make when they apply uh equilibrium physics to non-e equilibrium systems um you know you know

a true gas is not an ideal gas and it only has the kinds of properties and breaks detailed balance and expresses Nona in virtue of the fact that there is no exchangeability you can't assume that every atom or every agent in our context um is exchangeable with with every other agent and to connect again to Commons and and that model of well within the firm we assume cooperativity and hierarchy in the market we have a flatness and a competitiveness and then there's this Commons that has sort of elements of each so then I think about doing ecosystem modeling it'd be like well we have within the ant colony we assume cooperativity possibly hierarchy among ant colonies you have red and tooth and Claw and competition and then it's like but can there be a unifying framework that would have the two genotypes and two phenotypes and two species and two adjacent niches and keep on adding in these levels of Distinction and seeing all those different variables

in their joint Evolution across multiple time scales rather than the concatenation of a Cooperative hierarchical game NE with in a lateral competitive game it's like capturing what those models get the the width and the height of but bringing it into a framework where it's being seen as just one joint evolving process so C is the letter of the day Alex can you speak to Cesar fell what what that kind of heterogeneity and structured diversity would look like from a point of view of tissue for example or an organ okay wait what was the c word you said I just didn't hear it it's any any word starting with a letter C you can use but you have to oh I have to use okay this is the game um Community I'm gonna pick that one just because it starts with c um so okay Carl threw in organs and organel but I'm going to push that aside for just a minute to address I think one of the questions I heard emerge from yours and Carl's discussion which was uh is there anything like what you are arguing for which is avoiding the mistakes and by the way Predator prey model came immediately to your speaking Carl um about these classical models that assumed interchangeable agents and is there anything that looks at these heterogenous systems or heterogenous agent interaction and a couple of thoughts I don't think there's anything like directly what we would want and I think that ties or centers around are

even the arguments you and I and many others in the community are making about uh why it's difficult to Showcase why active inference is so important even for places like machine intelligence um but I did write some thoughts quickly uh so I do think uh the first thing that came to mind weirdly enough was an application in the world of machine intelligence where they had some environment where they had a community of even though I say it with slight distaste large language models or Chad gpts or gpts that had some extra modality equipped to them and they had this like little Community now it was a virtual very simplified Niche if you will um but the idea is that they could observe this community um and so these different gpts would and I can try to dig up the paper and send it to Daniel later it's one of my students that actually pointed it out to me they would different roles they had different personalities if you will um and I know that they investigated that to some degree now of course that doesn't have any active inference in there and I'm not sure what degree of maybe perhaps light reinforcement learning they introduced but there is something like that and there is slight baby steps if you will uh in investigating some notion of embodied multi- embodied agent systems I think the AI Community is still sort of of barely starting to embrace embodiment let alone enactivism and let alone embeddedness which is the

idea of a community to go back to the
seword of multiple agents interacting
with each other um and investigating how
do social norms emerge or cultural norms
and cultural constraints and
relationships at this like more Global
level so I think there is a little bit
of those Sparks of that thought and
obviously I might not be aware of
further follow-ups another place that uh
I at least see the potential of no
longer having these interchangeable
particle like agents is in multi-agent
robotics um I know that they have done
things like soccer competitions and they
have these little and again each of
those robots they are the embod of what
you and I want which is a version of a
mortal computer so it's a little like
assembly of mortal computers and there's
a Cooperative nature so and then a
competitive nature so you get a little
bit of a very constrained real physical
real world Niche um obviously it's in
the constraints of it's a soccer game or
some very simple task so I know in
multi-agent robotics there is some
development along this path and it'd be
wonderful to have like a a a hardcore
because I'm more of a simulation
roboticist a hardcore roboticist to talk
to us about the state of multi-agent
Robotics I think that's the place where
you could begin to test uh hypotheses
and kind of or theories that are
constructed even from a active inference
point of view uh into these real world
robotic multi- assemblies you know they

have like we could analyze the competition Cooperative Dynamics I think that's our best bet in terms of what maybe exists at the closest to literal agents that aren't just very simplified even though they make simplifications in robotics too uh in order to have a little brain in there that you know does something meaningful usually learning is turned off which is my own criticism of Robotics um I think the last comment I wanted to make and again getting back to the Notions of what we need so again you and I have a fondness I think for some of cybernetics and cybernetics has this notion of complex emergent systems and we you know you have little uh building blocks at one level that follow their own physics or their own physical laws and then they maybe assemble or emerge into these higher level building blocks that are subject to their own laws then you have bottom up in top down causation which is a notion that you know comes from cybernetics and we always talk about that but then I think about like Santa you you might be familiar with the Santa Fe Institute and all their wonderful work like uh what was it not just Herbert Simon but uh uh John Holland and all the wonderful work on complex adaptive systems but they always did at the end of the day when they simulated it they did the particle thing this interchangeable particle simulation and then tried to analyze like those convergence properties or those emergent rhythms or patterns

that kind of emerge in those systems
with no to my knowledge and you know and
I did study complex adaptive systems a
while ago uh not with that heterogeneity
and let alone any of the complex agents
that we could begin to design today even
with deep learning regardless of its
limitations and so I think that that
maybe even turns into an answer to your
question highlights an open opportunity
Maybe revisiting and recasting Notions
of cybernetics and complex adaptive
systems in this idea of well can we
actually build more complex agents uh
for example that actually have neural
substrate structures or something that
we can simulate and then learn to
acquire their own characteristics in
their Niche I think the reason that
maybe a lot of people avoid it or maybe
why I don't perceive uh development in
that direction is the problem also of
Niche and you and I also had an an a not
to try to highlight this paper but in
the appendix we talked about the body
Niche problem uh and that is this issue
that uh you need a niche if you don't
have a niche and you're designing these
agents independently like deep learning
typically does to apply to any Niche uh
you're missing really half the problem
which is understanding the physical laws
and how do we simulate them such that
people can then play with those models
maybe you don't have access to a xenobot
like Michael Levin does or I don't have
access to organizes though I wish I did
um but you know maybe we can simulate

these and so I think that we need the community may this is a call to the community to develop uh viable niches virtual morphologies virtual uh areas that we can then connect these agents and then simulate many of them and try to analyze and understand do the things we predict even with these interchangeable particle models apply to these heterogenous communities that might emerge and again I guess that just to wrap up and I promise I'm done you brought up organel and organs and I think that highlights something that I wish I had more easy access to rather than uh being put into this position where I well may I need to build it which is you know these virtual organ systems or virtual organ organ LS and trying to understand how these dynamical systems interact and then I can kind of take that as that's my body or that's where my Mark and I can kind of start to put some instantiation of blankets and then okay I want to instill maybe a neural circuit maybe I put in a predictive coding model into this kind of virtual morphology so I can analyze and understand does do these interactions of the time skills actually come through when we actually build you know physical instantiations of these models and so I think that's going to be one of the biggest roadblocks to even my dreams of mortal computation which is this notion of phology something that we can actually work work with that it gets as close as we can to reality without

reality and in Niche um because once you have those problems solved then you can start building multi and if you have an efficient way to simulate them and I'm sure lots of people that are watching might be worried uh because it's very hard to simulate you know real physical systems let alone allow machine intelligence researchers to play with them so I blabbed a little bit there but does that sort of give you some something to bite on based on your questions something to chew on something to consider um in the last little bit I'll I'll ask some questions so I'm going to read one from the live chat and then anyone else we'll just do some quick fun random questions sort of like sampling from the uncertainties expressed uncertainties of our live chat colleagues okay Pablo FM wrote is playing a simulation acquiring knowledge SL experiences through simulation of scenarios and how can we use active inference to do games how do you fellows make your active inference work playful and fun if when let you go Carl you go I'm not sure if there are two there there are two questions there um a practical answer to um why you might want to instantiate um a simulation environment with active inference agents or indeed just a single agent in the in the spirit of computational phenotyping but I think probably more interesting in the um in

the context of um communities of Agents
um is in the spirit of the sort of
cognitive um agent modeling um is the
ability to um parameterize the prior of
the agents to best explain or replicate
Legacy data from a particular ecosystem
be it Financial epidemiological
meteorological
um purely ethological
um and having optimized the parameters
you then roll out into the future
and in rolling out into the future you
now have the opportunity to have a
grounded in principle base optimal Plus
in activism and embedded um um estimate
of what is likely to happen in the
future again in the spirit of weather
forecasting but of course this becomes
much more um if like useful um when you
have control of the system so Financial
Market forecasting or epidemiological
forecasting or uh climate change
forecasting in you're assuming that
there's a substantive um um
anthropomorphic um um anthropological um
contribution to
um climate and you all the other factors
that uh you know um pertain to um
resilience and uh resistance to things
you know health and the like um as soon
as you can
intervene then once you've got a digital
twin of this ecosystem that has been
optimized in relation to um the past
data in the sense that this now has the
highest model evidence or marginal
likelihood in relation to that Legacy
data you've then got an opportunity to

do scenario modeling with interventions
so you can now ask questions what would
happen if we um cut this Supply CH train
chain what would happen if we impose
this lockdown in this viral pandemic
what would happen and so on and so forth
furthermore because you've got um um
uncertainty in the mix because this is
effectively going to be um you're using
variation procedures that we use in
active inference to actually simulate um
the behavior that best matches the um
the Legacy data you've also got for free
uncertainty quantifications and not only
can you say if I inter been on this
system and did this um then we would uh
I can predict that in six months time or
in six years time that and I can give
you basing credible intervals o over
that moreover you can now apply the
principles of active inference to roll
out under different interventions which
now become policies and evaluate the
expected free energy of each policy and
select the one under the constraints the
the characteristic outcomes U that you
um commit to at this point in time and
you can now use this scheme to make
recommendations on the basis of the um
the relative probability of this
sequence of interventions versus that
sequence of interventions so I think
there are great practical importances of
being able to put these principles in
silico and basically do Evolution very
very quickly so that tomorrow you can
read the you can read the consequences
probabilistically uh of evolution in

this setting and that setting where the setting is actually something that we can intervene on or even if we can't just to know the consequences of our Collective Behavior Uh today so I think that would be um that's why you might want to make you know to invest in in these applications um with the fun and uh having uh and play I think that's a different issue I think that's basically um um you know one way of looking at the um the way that authentic agents behave under under the free energy principle which is basically to explore and gather the right kind of evidence that uh provides literally evidence for their generative models of of their world um which basically entails novelty seeking but now complement this with this um Bas red C we've been talking about also trying to find the minimal explanation for the data and the evidence that has been secured by exploratory playful Behavior arriving at particular insights which uh those are har moments that say oh I see it's just one of those or this one thing explains all of this playfully acqu acquired data and perhaps that kind of mechanism you know may be a really important aspect of the communal um um you know um cognitive agent based or yeah cognitive agent based I can't quite remember the acronym it was very nicely laid out in one of the I have to go back and look at the live stream a followup question from the live chat Arno wrote I agree with the approach to calibrate to pass data and

forecast but it's already quite
challenging for current models that only
have a few parameters to be calibrated
how then does one calibrate prior
beliefs of millions of
Agents I'm talking about epidemiology
and economic models sorry I'm going to
give a technical answer before Alex
gives a gives deted answer uh you do it
it's only difficult because 99% of
agent-based modeling uh uses the sample
distribution um you do not do that you
use variational you actually um simulate
and roll
the um the um probability distributions
themselves using variational calculus
and that means that what was yesterday
um really difficult agent-based modeling
and certainly epidemiological modeling
now becomes really trivial uh orders of
magnitude more uh more efficient what
you have to take care of is exactly the
thing that Daniel was was highlighting
before which is a heterogeneity you need
to put that into the mix and this
manifests in a number of ways uh so when
you apply um variational procedures to
effectively um roll out population
density uh probability density
distributions over various states of
being am I infected am I staying at home
um have I died um you have to make sure
that the heterogeneity is part of the
generative model so this is sometimes
called stratification so for example in
epidemiological models you'd have eight
bands for different age groups and if
you notice the way that the

epidemiological data was generated was usually age segregated so that people could stratify their models um another um way in which this heterogeneity becomes absolutely crucial um when you come to um agent-based modeling in the context of epidemiological or infectious disease modeling is called over dispersion now over dispersion is the fact that uh you get heavy tailed distributions that inherit from the heterogeneity so get this for free once you put heterogeneity into the model but it does mean um for every if you like subgroup of um the population you do induce more parameters but remember the number of parameters doesn't really matter it's the amount of time that you take to effectively invert your model and if you try and do that using sampling based approaches which is currently the the norm you could take literally weeks to uh do something um whereas if you use variational procedures then you could do that within within minutes so it's a really important question a very practical one yeah to restate direct variational inference on beliefs about ensembles themselves being variational belief inference on agent level semantic beliefs rather than sample from the sample what the agents are sampling yeah you're you're you're invoking the The Twist yes how on Earth can you model a variational density over things that

have variational densities you can do it under the f p room because the only observable things about an agent are on its Markoff blanket and of course they are not belief structures so there is a lawful relationship between you can um if you like simulate cognitive agents using traditional agent-based approaches but what you have to do is to amortize or to or to um um basically work out the um the mapping between the history of what something sees um and what it actually does but in principle you can do it by marginalizing out the um the beliefs of the agent in principle and and that's that's why I as that's why I'm so intrigued by this distinction because that will be a challenge Alex and then Carl just in our last minutes here what would you say was a favorite memory or moment in theory and practice for active inference this year and then what are you excited about for next year um I'll go I'll go first so a favorite memory of active inference let me think quickly well I I mean I'm going to be a little B EST just to give a highlight and a shout out and everyone will get to share in that memory if you stick around after the round table is actually some of the results from one of my PhD students um via and he's going on to give a talk soon and uh the idea is that uh one of the things I was proud of is he's been working in the area of Robotics which I think is a really good playground uh is difficult and uh hard

is it to democratize for everyone to work with robotics I think it really shows you how far we need to go and I think that's a good place to make strong advances an active inference um and one of my memories is that I remember him kind of showing me uh because he's working with partially observable Markoff decision processes and how difficult and unaddressed or lightly addressed it is in the field of working with raw sensory data to try to construct these continuous models and I remember him showing me some of his models rolled out trajectories in the future and I I'm always kind of in a fondest in my heart for generative models and I remember him playing out sort of these robotic memories of this robot trying to you know fantasize where it would want to go to kind of like move this block or to knock a ball towards a so that was one of those moments where we were really proud because the idea is we actually we actually made a breakthrough uh in actually showing that we can construct a very robust framework that applied across all these robotic control problems and so I was immensely proud of that and so I think my students drive a lot of some of my fond memories um and uh and then we got uh Chris Buckley shout out to him uh to you know uh kind of give us his insights into our work too and it was a really really nice work and so and I think the best way to say it is stick around if you want to know how that all works I was really

proud of that so one of my favorite
memories of the

Year awesome Carl with the last words a
fun memory and uh a direction or an
excitement for next year right I'm I'm
not going to tell you about my most fun
moments that's that's very personal but
my my my my excitement for next year is
to see whether any of your hairs go gray
after you do the master ceremony of
these of not only these uh Symposium all
the meetings because for those people
who don't know Daniel has to run this
almost

single-handedly nonstop for a marathon
sometimes 10 hours at a time I don't
know how he does it at is unless you're
using hair dye you're you're you're
standing up physically remarkably well
sir so I just see how long that
lasts thank you Carl I look forward to
the marathon

changing continuing and flourishing in
the coming

year we appreciate you Daniel thank you
guys it's been

awesome okay thanks a lot see you soon
nice talking to you guys bye

everyone

hello welcome

back now we have the much anticipated
student presentation that your adviser
Alex recently told us will be awesome by
Viet dwon on Dynamic prior preference
learning for scalable robust deep active
inference so thank you Viet take it
away hello uh hello everyone um my name
is vet I'm from uh the neuro adaptive

Computing Laboratory um Department in
computer science or CH technology um and
I am uh Alex advis uh and today I'll be
talking about Dynamic PR preference
learning to scale B plus active
INF so yeah first of all I want to talk
about the motivation for this so
nowadays you know from fun Shure work in
robotics um excels in implementing
robots behavior and planning for example
um creating um we can you know creating
autonomous robots that can actively
explore in the environment um
additionally um acquiring we can also uh
do a lot of Robotics applications for
example acquiring knowledge and learn
skills continuously um and um this also
envisioned um cognitive and
developmental robotics um so there just
a goal um for um robots robotics
research um that's we can try to build a
robots that can continuously develop
through interactions with the
environment and their learning processes
um and it is also also better um to have
interactions with their uh physical and
social world in the manner of human
learning and cognitive development so
there's a lot of Robotics applications
um around um and for example one
application can be to build ASA robots
and um building um social collaborative
robots and stuff like that um so
underlying um all these robot work um uh
we it's like active inference research
where we um often build a generative
um um it's so the developments of
Robotics actually depends a lot on um

active framework because um there's a great need for research um that integrates the dtive world model into robotics control system um as also deep learning is increasingly used in constructing complex action planning algorithms for solving control tasks um so um we can see that there's a need for scaling um active INF system spe especially and deep active INF system um to integrate into the real world applications so um resar in active inflence has led to many improvements um in terms of robustness uh of um model based agents that applies to both M processes um and partially absorbable MVPs or P MVPs so an example of an MVP environment should be Mountain car um uh for those who um knows or studies reinforcement learning already and um this m car environments um in mdp we only know POS velocity right um and that is the actual state of the environment um but when we get to PVP for example like robotic tasks um we as agents see only the image frame and we do not know the underlying states such as position and orientation of the block or um join states of the robots for example right um so basically active INF is basically a form of inference and learning that SS to balance first of all the um goal orienting objectives or highi preference um so this signal exploit um the instrumental signal um so that the agents the goal um and try to pursue its goal um and the other thing it needs to

balance the epistemic terms or epistemic signal where the foraging uh it's uh cares about foraging and exploration driving signals um so researcher often actually encounter great difficulty in providing the agent with effective prior preference as well as Computing useful instrumental signal from the agents the priors um especially in the real war applications where um the environment is very complex the robotic control task and um the trajectory or partway um to be desirable um and task specific goals um often requires uh a very very long um trajectory of preference um so it is not that ABS practical to provide uh a prior preference distributions especially the um stream or roll out of goal images so that is the reason why we need to find a way to also learn to generate the prior preference signal in active inference so on the right here is just our um preliminary results um but we'll get to that in a moment so um yes so we actually uh given with that uh all of this uh right we need to um uh we need to predicts a u uh drawout trajectory of a goal um stat dynamically right um it can be a goal state or goal observations but basically the goal is um if you look on the figure on top here um the top row is the real trajectory um where you see the mountain car right there's a car that it is to go backward and to build some kind of a momentum and then go full to the right to get to the

um to the flag if it is on only going to the right and it can it does not have enough momentum to to go to the top of the hill so um so in here uh is the um actual observations you can see that the trajectory is um a successful trajectory and the goal here is to um for each uh observation at each time step we predicts a um immediate predict predicted PR preference um image or goal at that single time step so that we have um the instrumental signal or idea of uh what we are trying to learn and then um from that we learn the actions planning uh on uh from the PRI preference signal so in this talk we we specifically analyze this issue um the issue of um uh prior preference learning um in the current literature um and discuss potential result solutions for scaling the instrumental signal inference to active inference by introducing the um contrasted recurrent State prior preference or crsb um this is the model um learning framework that um we also have um done in the past so uh this methodology frames active inference agents in term of progressively constructing and adapt adapting um a prior preference um at each time step so this also facilitate the Dynamics emission of a useful or dense very dense um instrumental signal at each Town step so we also um study how our prior preference um adaptation scheme uh when operating jointly with the Laten space driven epistemic for aging terms uh offers active inflation

um a very nice flexibility in
challenging problem context um such
as um sparse to reward um environments um
or and also composed of uh continuous
observation and action space for example
um for example robotic task um they only
we often like do not know the sparse um
do not know the dense reward um
functions in the actual real environment
and we we only know if the robot has
risked the goal or not right um and also
all these environments has you know
varying goal States so um we can also
see that providing the uh uh or manually
craft the uh distributions of goal is
very not very um
practical so uh yeah so here here's the
outline of this talk on the introduction
I've had just covered um so we so first
of I'll just assume that uh everyone is
NE active inference and I'm just GNA
briefly cover active inference
backgrounds um and deep active inference
um and also so cover a little bit about
um prior prior preference learning and
problems um uh of all those prior
preference learning framework and then
I'm just going go through the actual
methodology which is the dynamic Prim
preference learning um and then uh first
of all is the world model um next is the
self-revision mechanism and then next is
the um learning pre preference models
and then finally um putting all of them
together um to make agent um and then um
I'll show the results and then
discussion okay so first of all um I
want to um cover a little bit about

active INF um so just a brief overview
about how active inference work and what
is the underlying theories behind it um
so um it is a mathematical framework
that describes how intelligent systems
are human interact with the environment um
so it describe it also described the
environment as the generative process um
and the intelligence systems um or the
agent or um us as generative model um so
active INF built on top of the free
energy principles um this means that we
as agents or intelligent system
continuously observe the environment and
forming our and correcting our belief or
our own representation of the
world um so the generative process um
defines a real world environment here
right um this is treated as the data
Generation Um process where the
information P terms are sensed by the
agent so you can see that's the um
hidden State underly hidden States
denoted as here um it's generate a sense
of um some kind of um observations um
denoted o and this is also um sensed by
um our intelligence
system so on the right here is the
generative model uh is uh the actual
agent that we are trying to build on the
intelligent system right um the
intelligent system also has an internal
State um which is trying to um estimate
the process um States um so this is
called um the prior or the internal
belief prior to the observations of the
environment at a particular time stamp
um so this is denoted as um the random

variable s or state and P of s here u_m
 is the distribution over the random
 variable s u_m so this make the agent
 generative model u_m the generative model
 represents the agent approximation of of
 this generative process u_m in this world
 u_m by continuously seeing the
 observations or making the observation o
 u_m and modified or updating u_h its
 belief so we actually call the belief u_m
 as after u_h being updated after u_h
 making the observation or the posterior
 because it is the posterior after u_m
 making the observation right
 so in order to learn the model and
 estimate the posterior we first need to
 in introduce these terms u_m first of all
 is the prior over State u_m which is
 denoted as P_S which is also the
 distribution over state right u_m P of o
 here is the model evidence u_m it is the
 distribution over the current
 observations u_m denoting how well is our
 model at predicting the real data u_m and
 in order to compute this easily this is
 often written out the sum of probability
 u_h of observation given every single
 state and then next one is p of O given
 s this is the likelihoods u_m and the
 probability u_h this is the probability
 of U an observation given a States so
 this is a function it gives the
 probability given an input States and
 then the the last one is the posterior
 function functions u_h p as given o which
 is u_m the posterior function that Maps
 the observation to the states u_m so u_h
 normally in traditional inference u_m the

per function would be uh the the
transpose of the likelihood Matrix for
example um and deep deep learning
literature um it is often calls a
encoder and the likelihood here is a
decoder

so how do we estimate the post now um we
have to find a belief that maximize the
modal evidence right because um at the
current time points uh there there is
only one single observation o here and
it is real right so $P(o|p)$ of that
observation um it's going to be one
right so it should be the probability of
one um and then um we use the stat to
estimate um the observations and then it
has to be the uh the probability of one
two right so uh that that arrives us at
the conclusion that we have to um
maximize the model evidence P
ofo so maximizing p ofo is also
minimizing equal to minimizing minus $\ln p$
of O right because um it is is uh
minimizing the surprise
um so in the beginning we say that we
continuously update our belief after
making an observations of the
environment um we call the belief after
making an observation a posterior um
this is Q of s
right um so to compute posterior we want
to put in a um we just want to um call
this you know any kind of um
distributions um over the states um then
for example we can sample it we can
randomize it right um and then we try to
gradually trick this distribution so
that it's used the highest model

evidence or it's used the lowest

surprise so how do we choose U the

actual uh posterior um so that it has

the lowest

so in minus $\log P$ of O here is the actual

um uh minimizing surprise that objective

that we are wanting to minimize right

so uh we can uh analyze it like uh we

can uh do it with a base theorem this

not base theorem but base

formula or theories

yeah um and then uh

to evaluate the surprise minus $\log P$ of

O we need to sum out um the hidden

variables from The Joint distribution

$P(S)$ um

but well this summations there's a

problem right because that we do not

know like all the hidden States all the

hidden variables in the environment um

it is hidden from us right so it is

normally very very hard to uh to sum up

all these um State values so uh if we

cannot sum it up then we can try to

approximate it so first we introduce a

posterior um in into this uh sum right

we multiply both the denominator and the

numerator by the posterior um so this is

the belief that we are trying to tweak

right so um in here we're trying to um

take this posterior so that um we can um

minimize this

term so this gives us a we sum where s

uh the ratio um $P(O|s)$ and um over the

the distributions um $Q(s)$ is Red by

the $Q(s)$ this is like we of

so so according to the inequality of

convexity or Jensen's inequality says

that's the convex function of a weed
 means is always smaller um than the weed
 means of the same convex function um so
 you can see the convex function here is
 minus log of something right
 so minus L of something is just a hyper
 uh is just a parab something like here
 um and this is smaller than a what means
 of the complex functions also we put
 this S_u outside right um so this is also
 apply to the context of probability
 Theory um and then we can arrive at this
 uh yeah this
 equations so formally it's called an
 upper bounds uh this one right uh a
 quantity that approach from the above
 right so this is also means that this is
 a free energy um when we find a
 posterior that minimize this free energy
 term we will have a chance of
 approaching the lowest surprise points
 um which potent increasing the uh modor
 evidence so after we have this terms
 right here we can just uh put the minus
 inside and then it's become this form
 this is the conventional form that we
 call variation of re energy um and after
 decomposing this we will have these two
 form uh which is the K Divergence of
 between the posterior and the prior
 minus the weighted sum of the Lo of um
 the um P of or given s which is the
 posterior
 functions or now this is
 likelihood so basically um the K
 Divergence term is called the complexity
 and the minus I_p of our given s is
 called accuracy term where the

complexity terms compute how much surprise are State and two State distributions posterior and the prior are far away from each other or not right and the accuracy terms uh defines how accurate our model is predicting the actual observations right so this is also relatable to the variational auto encoder where we compute the loss of the variation Auto encoder to be the K Divergence of the posterior and the g distribution with a mean of zero and a standard deviation of one and then we also have the mean Square errors of the actual predictions of the images right and then VFE and MF is a little bit different done from variational out on coder VF is variational free energy and MF is marginal free energy as in the um course tutorial um the paper um so marginal free says that we compute the K Divergence between the posterior and the prior and the prior is computed from the transition model um not the um not the normal distributions of uh mean of zero and the standard deviation of one so uh here is the uh General model of trying to build a deep inference model right so here's first of all we're trying to encode right using the posterior functions So In traditional active inference this is the P given o is the transpose of the likelihood Matrix P of given s this is normally learned with the initial distributions and using doubly conjugate prior um and then uh in the Deep learning rerat um this is often known as

the encoder uh that's uh Often
parameterized by the neuron Network
um and then o here is often be an image
um and state here is just um some hidden
uh vectors that we are trying to
compute and then next one is the um lik
function right so normally we text in
the states um of um the agents and then
we're trying to estimate the uh
predictions the observation predictions
right um so in traditional active
inference this is known as the
likelihood Matrix or likelihoods um per
se um this is also learned with the
categor categorical distribution and
using the rich conjugate prior um and
deep learning lature um this is known
often known as the decoder um that is
often parameterized by a NE
Network so uh as We Know the variation
of free energies formula the complexity
terms K Divergence here right and the um
uh accuracy term is the minus L po right
even s um so here we compute the K
Divergence between the um encode States
and the actual prior States here and
then um next one we comput the um actual
L probability
the predicted observations and the
actual observations right
here um and that makes the um whole
objectives where we comput the two uh
objectives right here that's uh also be
the um variation of free energy
formula so we just talk about how
inference um inactive instrument work
but there's more we want to build an INF
system that can interact with the world

right so when it comes to interaction between the agent and the environment we also need to have actions and um also roll out in town step right so basically we have another elements here which is at and remember I also mentioned the transition model right so after we take an action in the world um we um the the world um all the environments evaluates one step right and it uh it change it own State and then emits a new observations um and we as agents or intelligence system also try to predict um given the action in the current states what is the next States and what and this States is the prior over um the states of the next step time step where we have not made the observation of $T + 1$ yet so after we have the s_{t+1} one as prior um we do the same process as I have mentioned before so uh we make the observations and try to change um our belief uh after making that observations and then learns from the uh new observations to take our prior into the posterior so here we learn it um using the variation of fery or marginal fery formula so as you can see from the previous slides uh we do needs somehow U to somehow plan the actions to interact with the world so if we're trying to minimize or surprise um with the current observations um so in the action planning we aim to take actions that minimize the similar free free energy term in the future um this is often called as the expected free energy by

the way um so in here uh we introduce some new terms right Theta um is the model parameters um the reason why this is here is because we want to also compute the um uh model the surprise over the modal parameters or the uncertainty of the modal parameters um so you can see that the expected free energy in the future it's kind of similar to the version of free energy formula it's only needs um another model parameters parameters and uh the given the π or the actual PL policy or Cent actions so after a lot of derivations right I'm I not getting into like the real derivation here but um here's the end results that's um something something minus um minus the weighted means of the uh lock P of OT right so I'll just focus on this um term is because this is the um goal of the talk today so basically um this term lock of P T is the expected free energy derivations um of the instrument uh instrumental term so basically uh this also means that um we are trying to maximize the now we're trying to take the action that maximize um the future observations that matches our um prior paraph distributions or what we want right so normally um to learn to take actions based um uh to learn to take actions um to align with what we want to see in the future um we of needs to craft and provides the priate preference manually in advance um and then we text the action based um we take the actions um that's maximize the

likelihoods um given that
distributions um but this is often not
really practical in the realistic World
um and because we can like craft manual
distributions like every single time we
encounter a new environments or
encounter a new task right so what we
can also do is we can also provide a
goal States or goal observations
directly to the agents which is more
practical um but in the end the prior
preference distribution is very sparse
um and this provides little to no
learning signal to the
agent as you can see on the figure on
the right and the left black right so
black is a prior preference distributions
and right is the single goal distrib go
St distributions or goal State
image so uh what can you do to
dynamically uh produce um the uh uh
prior preference right we can try to
learn a prior preference model right
that um produce a goal image at each
time
step so here is on top is the prior
preference model that predictions right
um and uh in the bottom here is the
model prediction of the future um given
the sequence of planned actions so we
are trying to say that we um we want to
take the actions um that have uh the
roll out in the Futures uh predictions
matching the prior preference uh Ro out
in the
future so how do we do that we first
have to learn the model first
right so in learning uh we can try and

say that each time we achieve a goal
then then we Mark the whole trajectory
as the prior preference trajectory or
the um positive um trajectory or expert
trajectory um and then for each
observation marks as goal we use a deep
learning like another deep learning
model to estimate startat observations
um for example right and then um in
inference time um we actually train when
we train the policy planner uh to
propose the actions right we make the
policy planner proposing the actions
first and then predicts the future
observation based off of that um actions
U that is planned by the policy planner
and then um we actually learn the policy
planner based on the signal that is
emitted from how a line me the predicted
future observation would be um to the
predicted goal um the predict the
predicted goal here is the actual prary
problems model outputs right
um then we train the policy planner with
the um mean Square Arrow or um some kind
of objectives um that's uh predicts uh
uh some kind of objectives between the
predicted goal image and the predicted
future observation
so there's this problem uh where uh uh
it exists between um all these learning
schemes because it is very computational
expensive um so it is generally uh
expensive because we have to work
directly with the images so for example
in learning we have to use a decoder of
the um of the PRI prence model um to
actually predict the go image at each

time step right in inference time is
 also computationally expensive is
 because um it has to learn from the
 images itself and compute it down from uh
 the um the back propagations will have
 to flow through the decoder um back to
 the action
 planner so here it comes to the actual
 methodologies uh we actually use Dynamic
 preference preference
 um in the state space instead of the
 observation space right so first of all
 I want to talk a little bit about the
 world model so uh this uh model
 architecture is um basically what I have
 just um drawn in the background sections
 but this looks with robots images so in
 here we have the observations O_t O_t and
 here's also the observation the same
 ations O_T minus one O_T um and we text in
 the encoder encoder is the encoder
 posterior text in the encoder and we
 have the transation prior um and then we
 have the decoder right encoder encodes
 the imin have the states and then decode
 the um the image and the state back to
 the image and then computes the accuracy
 loss right accuracy loss uh is minus L
 prop of the P given S as we have
 discussed in the variation of free
 energy sections and then the complexity
 is the k k Divergence term between the
 what the area and the prior right so how
 do we compute the priors we compute from
 um the previous steps uh States and then
 we use the transation model um to
 estimate the next States prior right so
 that's um that's how we can get the um

prior so

the current work um and popular Works uh regarding the world model or generative models um often integrates um the model called um RSS Lang which stands for um recurrent States space model um in the translation uh model so it's to zoom out a little bit um we are trying to uh integrate some kind of temporal information

Through Time right um so that's the state of the environment or the states that we're trying to predict be more accurate um and it carries more informations with time um so for example um we cannot um encod the uh enough information from just a single image because for example um there can be like uh position and velocity right the velocity is acquired um through multiple images um and cannot be acquired from a single image if does makes sense so what we often do is we try to integrate a recurrent neur networks that text in the previous States um or the previous recurrent States H_{t-1} my T minus one um and then we also text in the um encoded observations

um and then with in transition um we produce the recurrent n network we produce the current recurrent State um and also predicts the outputs to to put into the uh transition model um and also it's the recurrent stat will also uh be served as inputs to the posterior Network um so that's basically uh the idea

here and then uh the other way uh around

we could just uh do uh the model as normal so uh in active INF agent takes in the actions um that's it Bel uh that would lead to its preferred outcome uh using instrumental signal um to construct this prior preference passord has um for example provide a go statement observations directly to the agents for example mazaka in 2021 with contrastive active inference um we can also manually craft a PR prence distributions um we can also dynamically produces the preferred observation that each time set um but as I mentioned already um all these methods is one way or or another very expensive um or it's very not practical or in the reward applications or um it is very computationally expensive for dynamically producing the actual observations um for for the goals so we can also use Behavior cloning but it its trajectory diverges from the training trajectory due to um small mistakes made by agent as well as environmental stochasticity um and also it is because of ident IND independent and identically distributed assumptions of positive exper data distribution so we can um we we try we will kind of avoid it because it's it's not going to um be uh the same right the two distributions of the actual um actual trajectories in the environment should not be the same as um every single positive trajectory of the expert data so to tackle this we try to leverage a small quantity of seat

imitation data um to learn an uh a neur
network that's dynamically produced the
preference of State not observation but
State at each time step so this
effectively um produces an easily
generated dense instrumental goal signal
um and we also design the agent such as
that it moves according to the
trajectory that is is shaped toward its
own estimation of the future States or
future preferred States so it's trying
to Notch the its own trajectory to watch
the imitation or positive data
distribution so as you can see on the
figure on the left um so you can see
that in Behavior cloning it's trying uh
the actual trajectory is diverges away
from um the actual imitation data where
the prior preference over the states
trying to um get uh trying to be shipped
toward the imitation data itself uh
using the model signal um and ra is our
agent um is called robust active
entrance um and it learns from both
epistemic signal and instrumental
signals but in here we'll just focus on
the instrumental signals and how to
construct

so uh yeah so basically we use a decoder
um or the likelihood functions of the
generative model to only visualize on
the preferred observations um not that
in training or test time we do not use
the motor learning uh we do not use the
actual observation uh goal observations
by itself we only use the hidden States
so the goal here is to um get the states
the goal States at each time step be

predicted by the prior preference model
right um You can see that's uh on the
top here is the real trajectory and the
bottom here is the actual um dream that
the dream of the robots that it wants to
be right and then we are predicting that
dream

right so here is the actual model of the
uh um contrast recurrent State prior
preference learning um U so basically it
has

um it has also similar architecture to
the rssm to the world model um it also
has have an encoder um here it also have
transition um but it does not have the
decoder um so how do we actually uh
learn this

uh um PRI preference model so first of
all uh we uh learn the world generative
World model as normal right um but when
we say whenever um it's uh whenever the
trajectory is uh positive or a
successful trajectory um we try to
contract um the state of the uh crispy I
just call it crispy um we try to
construct the state of crispy to the um
world model States and then um whenever
the um trajectory is negative we try to
push away the states um from each other
so that the crispy model um can only
predict the um positive States so how do
we determine um if it is a positive
trajectory or the negative trajectory
without having a very sparse signal so
we will have a thing called brow or the
prior preference signal so how do we
compute this RW
signal so imagine um this is the real

trajectory um and it is a successful
trajectory for example right
so we're trying to compute um the prior
preference rate by uh having the um rate
of the successful States is at one and
it starts from one and then it decays
backward so decays um very fast and um
if so in this like really far away stat
from here um they do not have um enough
signal because there's no goal um
like there's no direct go from that
states and then there's real trajectory
that is failed right um You can see that
the mountain car cannot reach um the
flags on the right here so whenever it's
built um we can just PL PL the very very
last observation as minus one and then
we start to De Decay backward negatively
um and then uh we have the prior
preference rates uh signal as um very
negative so the reason why it is it has
to be last is because um we want to um
we want to blame the L observations um
hardest so um in order to uh in order
for the agents to just push away from
the failing States very very very strong
um in the very last
stage so in the real trajector that have
like multiple successes um we try to put
all the one in all the successful States
and then just Decay
backward um as similar to how we Decay
backward in a single success
case so with that s we now uh know how
to compute um the roow and then this
will serve as the dynamic signal to the
contrasted loss and it creates a dynamic
behavior of the contracted loss um

that's trying to either push away or um
contract um the two states of the first
speed State and the world model
state so for example in here um there's
if there's positive row um the
contracted log would be um cont
Contracting all the state together
and maximizing the similarity between
the preit states and the world model
States and then we also learn to U
minimize the K diversions between the um
complex uh uh the posterior and the
prior uh of over the states of the
crispy
model um this is because we want to
estimate the transations right so we
transition to the um next
goal State and then this goal state has
to match the actual goal um that is the
current state of the EnV of the rssm
model or the world model
um so in contrast right if row is
negative then we're trying to push away
um all these states away right so we're
trying to maximize this um distance or
minimizing the similarity between the um
world model States and the crispy
State and in here notes that there's no
complexity objective is because we don't
want to um predict the next goal that is
um the failing State that's basically
the the
idea so that arrives that's the uh
general formula for the objectives of
the contrasted State PR preference model
um basically
uh it is just masking out the sign of
the row and then multiply with the um

the K Divergence or the complexity uh between the posterior and the prior of the crispy model um and then we uh minus the row and seam of U the actual predicted State and the world model state so same here can be very much um uh any kind of similarity functions for example cosine similarity um and then in the in our work we actually use the actual cosine similarity so in train time we have we have just talked about training time right so we have successfully assuming that we have successfully um learned the prisy model that predicts the future goal um and in inference time we use this uh crispy transition model to estimate future rollout trajectory of the goal right so we can see that um on the top here is the uh observation um actual observations and in the second row is the actual prior preference roll outs or predictions of the future goal uh given the um given the current observation so given the uh PL actions we're also training the PL uh the plant actions right so the actor trying to predict an actions given um the observations the predicted of observation in the future um so we're trying to say uh this is uh we're trying to compare uh the actual PR prior preference roll outs with the actual predictions given the actions um and then we are trying to uh learn the action planner based on the signals

or how diverge um the future goal is to
our future predictions of the world
given the
actions so uh there can be so many um
there can be so many reinforcement
learning agents and active INF agents
that can utilize this technique um of
Prior preference learning so active INF
agents um I put my works here um it's ra
and then there's also Chris bakley's
work and baren M work about deep active
influence in varal um policy gradients
methods um and also Auto uh work is is
the scaling of the uh act active
inflence in PDP and not car environments
and so on right so re enforcement
learning agents there's all these sorts
of um after prcts algorithms like
dreamer series um soft a cretic dreaming
um ddpq tqc um and TD and so on so
there's a lot of agents that can utilize
the framework because there's no change
um in the original architecture there's
just an add-on um to how um to how an
agent would apply the priate preference
learning signal um into the
system so here the actual flowchart of U
of the agents basically um we start the
agents and for each episodes we evaluate
um the environment um using the world
model and the policy planner um actually
basically we uh uh evaluates in the
environment we let the robots play we
let the robots do the task and then we
collect the uh data and add them to the
replay buffer for example right and then
we get get the main training agent um
logic um and then first of all we sample

the training data from the buffer so in here we trying to have um we also have to use um the self- imited ation learning buffer um so uh yeah so self-imitation learning is very good technique um normally to balance the number of um successful and failing trajectories and we also learned um so first of all we learned the world model encoder decoder transation on rssm and then we uh train the epistemics model but in this case we don't need to care about it um and then next is the um constra recurrence State PR preference model um from the data and then um next we imagine the future trajectory using the world model and the policy planner right we try to estimate the future actions um to be um what is the optimized future actions for example we we estimates the future action based on how uh current state is and then we the world model will be continuing to BL out in the Futures and then we dream of the Futures and next is the part that we actually um infer on the uh crispy model we use a crispy model to imagine the future preference trajectory um given the goal um given the current observations and we predict the future goals roll outs goals right so after that we learn the policy planner right using the instrumental signal the instrumental signals here is computed from the um difference um between the future goal and the um predicted future dream of the

world right and then with that the policy planner can learn from uh from the uh instrumental signal so here's our um preliminary results on um active INF agent so basically um we can see um the two top rows is the mountain and car environments and the two bottom row is robotics meta World um push button uh environment so let's look at the first row for example the observation here is a mountain par that have reached the goal and for each um observation here we uh we predict a preference roll out or preference um or a goal State at that single time that so in here we can see that um if the car has not have the momentum yet um then we have to build the momentum by predicting the car going to the left for example right and then after it has built some kind of momentum we trying uh the actual prior preference model predicting the car would go to the right go forward right when it goes here and the prence model will predict the car to be be um going closer to the goal something like this so the same applies to the um robots environments where the actual observation is the third uh row and the uh prior preference roll out is on the fourth row so we can see that uh it is also trying to estimate the dream or the dream of the goal of the robots is to actually push the button so um there's different kind of experiment setup that we are trying to learn um basically we have uh four trials on each environment and then we

train the experts um using as soft critic
or um po um proximal policy optimization
um to collect um a small amount of state
data normally is about uh uh 20 episodes
something um 20 to 30 episodes um so we
test mostly on three Mains environments
or suits um first of all it's m car
there's only one task um metal wall is
13 task so here's um all R robots here
is metal W and then the robot suit we
have two task and then um the cam
regarding camera setups uh we use
um three views for met world and four
views for um the robo suit
so the green check mark here denotes uh
that the robots has successfully solved
task um here's the Robos suit door open
door and then um here's Robos suit lift
where it can successfully lifts the
block and here's more uh solving agent
tasks so basically agent with Crispy
exhibits and ability to perform
successfully early on um and uh so agent
with without crispy uh struggle to
actually succeed on solving the
environment um and we also test on other
environment as well for example genology
and Robotics or and risk task um you can
you can see in the middle image on over
here so actually go to the discussion
sections so actually um I want to make a
note that um with respect to the crispy
subm model um our model utilize
contrastive objectives to adapt its
parameters so basically it solely learns
on um estimating the world Moto in coded
States while pushing itself away from
undesired World model States or

trajectory so this does not require or
the learning of the of the decoder so we
we only make sure of the learned uh
decoder in the generative World model um
visualize the uh to visualize the
preferred observations um from the
estimated crispy States so
experimentally we remark that this is an
um that's uh we only use um this decoder
um to as an auxiliary tools um to prove
um that's the uh states of the contusive
recurrence prim preference model uh is
useful so theoretically Behavior
clothing will be unable to achieve a
great uh success in multiple multi goal
environments due to its Reliance on a
fix and finite size uh imitation sample
pool um the expert trajectory in this
fixed pool might further differ from the
actual trajectories um that the agent
needs to take to solve the problem at
hand
um for example there's a distributional
gap in in observations and therefore and
um it requires different sets of actions
to be taken for example right um
so on the other hand right crispy learns
to produce dynamically the goal States
and not the goal observation right um so
we can empirically confirm in simulation
that our agent um with Dynamic prior
preference learning does not suffer from
the goal Mis match problem u in the
training
phase so this is very applicable
applicable to the multi-goal
environments
so um we can see that um this PR

preference learning uh techniques has a lot of practical applications in different robotic tasks um with for example with potential dangers in the operations um and functioning for example when human safety must be considered right for example a trying to drive a autonomous driving car system where it has to have or it has to predict um the future U goal right the future goal is not to hit the human for example right um and we can also make a lot of advancement in imitation learning in research community and it can be um applied by the wide range of reinforcement learning and active influence agents um and then the last one is to scale the the active entrance agents on the real world robotics agents so Future Works words actually considered um importing or integrating the dynamic PR preference learning in latent State space model um for example variational Dynamics discret variable Auto encoders or tension waiting models um and also uh there's another models uh physical neuro robotic system um which entails an embodied and active and survival oriented formulations of active perception and World model learning so in this uh talk um uh we have discussed a lot of ideas about uh being able to estimate the particular goal or preferred States um at any particular time step is very useful um for cognitive control agents so as our results demonstrates the agent can

then learn to adapt the take actions
that leads to um a estimated goal um and
then um in contrast to the approach is
taken in the impementation learning and
behavior cloning literature um which
train the agent's policy based solely
based on the fixed collected expert data
sets in our framework the expert signal
is estimated dynamically um through an
Adaptive prior preference model right
closing this Gap the domain gap between
the actual trajectories and the
collected trading imitation
trajectory or preferred
trajectory um and then we also discussed
the generative model um and
encoder decod transations and how the
variation of free energy work and then
how to learn the model and then we
finally we discussed the contusive
recurrent State prior preference model
um with the self revision mechanism and
then discuss the model learning and then
inference um objective um and how to uh
integrate S_t into the agent planner
learning system and then we show the
results for the robotic task and then I
mention how it can be applied to
fields so that concludes my presentation
um and thank you very much for um
listening thank you that was amazing
really great job with the background
math done only like a PhD student can
but it was great to see how you brought
it all together and connected your
research to it all at the end so thank
you that's all for now see you later
thank you see

okay hi welcome back welcome back hi

Denise hi thank you for for joining and looking forward to your presentation so take it away oh thank you so much and thank you for having me today um today I am going to talk about the potential impact of centralized AI through active inference and spatial web

Technologies and what we're really going to be discussing is um the potential aspects of this next era of computing uh it'll be incorporating a lot of the research that you're hearing about here over the next three days um and decentral in my talk decentralized AI is kind of taking a main focal point here and there's a couple of aspects um in technologies that are evolving right now that uh I think it's really important to be aware of um and they factor into uh what I'm going to be talking about today so one of the main uh things that is occurring right now is our internet is evolving um we have a new protocol layer that's layering right on top of the other protocols that are part of the same internet that we're talking across today um the difference is is what these protocols hstp hyperspace transaction protocol and hyperspace modeling language is the programming language um what these are going to enable is for us to have programmable spaces so instead of just programming pages in domains with content we are going to be able to program digital twin spaces of anything and this is going to enable context to be baked into everything in every space

so all of the attributes about various things in various spaces and nested ecosystems and all of their inter relationships to each other so one of the things this is going to enable is uh active inference agents to be distributed across our internet and I see this as the next wave of innovation and through the through active inference these agents will be aware of each other and the network uh they'll be able to sense the world in real time and learn from each other and this is going to provide unprecedented levels of interoperability um so what we're talking about here is first principles uh automated systems and distributed collective intelligence so first let's look at the challenges in traditional centralized AI structures um or architectures uh centralized processing constraint you have high energy demands and scalability issues limited adaptability to realtime changes and you know we're all familiar with the limitations of uh these deep learning models they require massive amounts of labeled data and processing power they have limited ability to generalize across different tasks and environments AI models uh lack adaptability to real world complexity they're only good for one thing at a time uh like one uh optimized for one type of an output and they lack interpretability and transparency in decision-making they have high

computational resources leading to high costs and inefficiency and they struggle with generalization across different scales tasks now we need decentralized AI and decentralization really matters because it increases resilience in scalability for autonomous systems it reduces Reliance on Central servers and data hubs and it enables autonomy across distributed networks so this bridge is the gap in AI we move Beyond centralized AI for scalable autonomous systems that can operate independently and locally now we're all familiar with this guy here Dr Carl friston and uh he he gave a great talk uh first thing this morning on uh renormalizing General generative models um so what I'm going to be talking about are various components in in some emerging Technologies along with active inference that potentially realize this research paper designing ecosystems of intelligence from first principles now you know the this uh paper was published in December of 22 and um it really laid out a new framework for autonomous intelligence that mimics self-organized systems of nested intelligence found in nature and in in this paper it described a cyber physical ecosystem of natural and synthetic sens making in which humans are integral participants what we called shared intelligence and this in the paper it says this V this vision is premised on active inference a

formulation of adaptive behavior that can be read as a physics of intelligence and which inherits the physics of self-organization and in this context we understand intelligence as the capacity to accumulate evidence for a generative model of one's sensed World also known as self-evidencing and at the end of the abstract for this paper uh they also note we also consider the kinds of communication protocols that must be developed to enable such an ecosystem of intelligences and motivate the development of a shared hyperspatial modeling language and transaction protocol as a first and key steps towards such an ecology so what I'm proposing in this and what is proposed through all of this this uh technology together uh with all of the various parties that are involved in building these compon components is that the future of AI is shared distributed and multiscale an entirely new kind of autonomous intelligence able to overcome the limitations of machine learning Ai and as we know uh from understanding active inference this AI is knowable explainable and capable of human governance and it operates in a naturally efficient way there's no big data requirement and it's based on the same mechanics as biological intelligence it learns in the same way as humans so what I want to do is start with the spatial web protocol uh first

because this is kind of the um the framework for these agents to be able to have this distributed intelligence structure so what is the spatial web protocol it's just the evolution of our internet into spatial domains um so people refer to it as web 3.0 it's just the next next evolution of the internet but this protocol has become a global standard a new global standard for the last four years uh this this standard has been in development with the i e uh which is the largest core standards body uh in the world and back in 2020 they deemed the spatial web protocol a public imperative which is their highest designation and this summer it was voted and approved um as the new global standard and it's in the final administrative uh cycle with the i before it becomes uh public so there's a couple of things that are happening with this through hsm1 the programming language for this new protocol all emerging technology whether it's iot or ar VR distributed Ledger technology um you know all of these Technologies they're all going to be able to uh converge across the internet because they will have a common language for interoperability and in that you also have a language that AI can understand and uh communicate together and communicate with these various aspects of Technology as well so this is going to produce the next era of computing and deoe called it back in 2020 so it's the

next evolution of the internet and it's going to be empowered by active inference

AI now just to kind of understand what this means for the internet um let's take a look at what a website domain is today with the worldwide web a domain basically contains a website and these websites they serve up content so whether it's you know images or video or audio or you know information data um that is what websites are capable of doing now a spatial web domain is quite different because a spatial web domain could be anything in any space right and it can be programmable a spatial web domain is an entity with a persistent identity through time that has rights and credentials so you're talking about going from a library of pages in our internet to the internet of everything and hsmI becomes a common language for everything so it provides unprecedented levels of interoperability between all current emerging and Legacy Technologies uh it's a programming language that Bridges communication between people places and things laying the foundation for a realtime World model for autonomous systems and what you have are programmable spaces hsmI is a cipher for context anything inside of any space is uniquely identifiable and programmable within a digital twin of Earth producing a model for data normalization and from this we get adaptive intelligence automation

security through Geo encoded governance
multi Network interoperability and it
enables all Smart Technologies to
function together within a UniFi
system and then this produces a
Knowledge Graph and digital twins of
anything everything potentially our
planet so as as this knowledge graph
grows within the programming of all of
these spaces across the network you get
this Universal domain graph that enables
a digital twin of our planet and all
nested systems and entities within it we
get nested
ecosystems uh there there will be
intelligent agents both human and
synthetic across this network uh they
will be sensing and perceiving
continuously evolving environments
making sense of the changes updating
their internal model and what they know
to be true at any given moment and
acting on the new information that they
receive so why do we KN need a spatial
internet progr protol well the
increasingly graph-like nature of global
data the opportunity for Automation and
uh autonomic activities using context
aware cognitive AI the need for
composable systems and applications
including the governance of such systems
the int the intrinsic need for secure
transactions the rise of machine
learning and neural network computation
and Edge computing
the need for explainable AI and robotic
governance and the rise of iot and
sensor

mesh and some of the guiding principles of the spatial web standards are spatiality representing information as locations and relations in hyperspace ownership users own their data in digital property security secure data collection transmission and Storage privacy individual control over digital identity and personal Data Trust reliable permission driven validation of users assets and spaces interoperability seamless navigation and asset transfer across spaces and responsibility creating technology ethically for the benefit of

humanity and what what has become of these Global standards is a so soot technical standard so the spatial web is a soci technical system of systems uh takes into the account the um so the development of the spatial web must address both technological as well as sociological considerations socio technical standards integrate and balance the technical social physical and legal use of technology and the spatial web standard is is a soci technical

standard and what this results in is multiple scale cognitive Computing so the spatial web plays the role of a collective nervous system for smart infrastructure it hosts cognitive Computing uh inference perception learning and action selection by combining and abstracting distributed information the spatial web enables the formation of complex ideas and

facilitates decision-making
cognitive Computing occurs at multiple
scales in the spatial web individual
nodes host cognitive Computing functions
and networks of nodes collectively
perform emergent cognitive
Computing so hsml hyperspace modeling
language it's a human and machine read
readable modeling language and semantic
data ontology schema there are some key
elements to it there are entities
activities agents contracts channels
credentials domains and it includes
hyperspace and time so hsml enables
the expression and automated execution
of Legal Financial and physical
activities in the spatial web and all of
these entities are identifiable and
knowable by these intelligent agents
through this uh language throughout the
Network now let's look for a second at
the agent aspect agents and activities
so agents and that can be human agents
or synthetic agents um but these agents
are entities that sense respond maintain
a model of their environment and take
actions to achieve goals and the
activities that occur they're single
actions or partially ordered sets of
actions that are performed by the agents
and they unfold over time so agents
interact with domains and other agents
to perform activities in the spatial
web so what the spatial web facilitates
is a distributed ecosystem of
intelligence if you look at current
state-of-the-art AIS they're siloed
applications they're built for

optimizing specific outcomes and they're
they're really unable to communicate
their knowledge frictionless and
collaborate with other
AIS but this facilitates a a an
ecosystem of intelligence that is a
self-evolving system it's learning
moment to moment and upgrading its World
model and in that way it mimics biology
while also enabling uh general
intelligence so this is shared
intelligence at the edge of
everything so
how active inference works in the
spatial web it enables real-time
decision making you have agents that are
predicting and adapting to their
environment from their frame of
reference in the network uh they're
continuously learning from their
environment enabling realtime
adaptation and unlike traditional AI
which relies on pre-programmed rules and
static models active inference
continuously updates its models based on
new data allowing it to adapt in real
time so these active inference agents
would seamlessly communicate in real
time with each other sharing their
knowledge and perspectives and learning
from each other these active inference
agents excel in handling complex
uncertain environments by continuously
minimizing uncertainty in fact they can
quantify their level of uncertainty and
this makes them suitable for
applications where conditions are
constantly

changing so decentralized AI takes the processing to the edge to the edge devices you have a distribution of processing and power uh superpower gpus are not required the processing can take place on local devices throughout the network with infrastructure that already exists like your laptop or your mobile device or devices that haven't even been um invented yet um but we're already seeing a lot of those kinds of devices coming into play uh with you know um Apple Vision Pro and and you know the metal glasses and and different aspects of these

Technologies

so the processing takes place on these local devices throughout the network you have faster response times and reduced Network congestion and it uses real live data realtime data from iot sensors machines and everchanging context that's embedded to the uh in the network so this is the sensory information that's coming in for these active inference agents so that they can uh you know do the perform this action perception feedback loop and continually update their model of the world of their reference and it eliminates the need to tether to a giant database active inference agents throughout the Network learn and adapt from their own frame of reference within the network enhance data privacy by minimizing centralized storage and it minimizes complexity it uses the right data in the moment for the task at

hand and with hstp with hyperspace
transaction protocol you there's natural
guard rails uh included in this
framework so it creates a secure ethical
and user Centric environment for AI
applications fostering trust and
reliability in digital
interactions uh you have data privacy
and sovereignty transparency and
explainability security and
authentication bias mtig mitigation
ethical AI guidelines interoperability
standardization user empowerment and
control accountability and governance
safety and reliability Dynamic policy
enforcement and what this results in is
in intelligence at scale uh through this
network of distributed intelligence it
enables the cognitive architecture of
the collective intelligence of multiples
of agents that continuously communicate
coordinate and collaborate with each
other you have individual and
specialized intelligences together on a
common Network speaking a Common
Language
hsml efficient and Powerful cross
communication to perform tasks regulate
systems and address problems in real
time and it scales up and grows in
tandem with humans through hsml we can
take human laws and guidelines make them
programmable for these agents to
understand and abide by in real time so
we're talking about unprecedented levels
of cooperation between human and machine
intelligence
and what you get then is an Ever

evolving collective intelligence uh decentralization is an imper imperative for intelligence at scale these agents represent scores of people from all over the world sharing their individual and specialized knowledge all coming together on a common Network and it's based on each agent's own world model you have unique perspectives from whatever frame of reference uh within the network nested ecosystems and levels of self-organization and Collective shared intelligence so the idea is that anybody can impart their knowledge and their wisdom into these Agents from wherever they are in the world and this actually acts to preserve all of the uh cultural differences all of the um the belief differences the governing Styles um you know people whether they're in uh Kenya or Singapore or anywhere else in the world they should be able to create these agents to solve their local problems and then share that with the world because then these agents can share their intelligence with each other so the spatial web protocol enables a decentralized secure and interoperable web with the potential to profoundly transform Global Industries and everyday human life now let's dive into um kind of the the potential that exists for this idea of rgms these renormalizing generative models and the spatial web and we heard a great talk this morning uh by Dr friston he uh you know he talked a lot about the you know

the science behind these uh these rgms
so I'm going to kind of uh maybe maybe
break down some of the aspects of how
these work as leading up to then what
they can do within this uh potential
potentially within this architecture so
you know with
rgms one of the interesting things is
that it's a universal architecture um
they're a UniFi unified scale-free
framework for AI That's capable of
reasoning learning and
decision-making and they can generalize
across multiple tasks handling small and
large scale data efficiently they
combine perception learning and planning
into a single system and they can be
used to learn scalable architecture over
space and time similar to human thinking
they require significantly less training
training data than traditional models
while achieving Superior
results so what are these rgms
they're a new class of AI models based
on active inference they use
renormalization to simplify complex data
into hierarchical structures and they
operate on the free energy principle to
minimize uncertainty and adapt in real
time so the significance is that they
represent a Leap Forward from
traditional deep learning methods they
reduce data requirements and uh improve
efficiency and it's one single framework
for for all
applications so the science behind rgms
renormalization and multiscale learning
renormalization is a technique from

physics that's used to simplify complex systems to maintain consistency at different scales and rgms break down data into multiscale representations hierarchical structures rgms are actively learning from small details to large patterns simultaneously and they analyze like with an example would be like an rgm can analyze an image by recognizing edges then objects then scenes and understanding the the concepts at every at every scale so key innovations and advantages of rgms why they're superior to traditional models they learn efficiently with less data it's a unified system that handles perception and planning simultaneously and they understand conceptual relationships not just patterns and they can adapt to new environments with minimal training uh aspects of them are scale-free modeling renormalization discrete representation hierarchical structures fast structure learning unified perception and planning and explicit handling of uncertainty so if you look at how the brain processes data processes data um you know we we perform core screening all the time when we're learning right we break down sensory data into hierarchical patterns um that help us focus on the most relevant information um kind of this chunking of concepts to to grasp different aspects of um whatever it is that we're we're learning or focusing on in the moment

and rgms can mimic this approach by processing data hierarchically focusing on important patterns so it's a first principles approach uh using a way to simplify models without losing important information um they can model space and time Dimensions like we do and can learn the causal structure of information and it helps handle large scale problems efficiently using less data and less

energy now when you can consider uh scale-free modeling you know um scale-free modeling these rgms they operate across different scales so from very fine details to highlevel con uh complex systems so unlike deep learning which struggles when moving between small and large scale problems this scale-free architecture allows rgms to seamlessly work on a variety of problems including small data problems and large scale strategic planning planning so when you when you're thinking of the concept of

scale-free you know this is a great example here of a green apple you know you can you can see the green apple as a concept of just you know pixels in an image or a volumetric object or on a branch on a tree or in an orchard and you know the the way that these uh rgms can model these conceptual understandings of things is it can understand all different scales of what a green apple is so when you think of like um the supply chain or or factory operations or or anything like that

these agents can understand the what's happening across the manufacturing line on a very uh local level but also what's happening across the entire distribution line uh throughout the entire system so they can predict both short-term movements and long-term outcomes so you know the fine details as well as uh large larger uh outcomes with the highlevel complex systems all at the same time and they don't lose accuracy when switching between local and global data

scales so hierarchical modeling in rgms hierarchical data processing rgms process at multiple levels of abstraction from fine grain details to highlevel

Concepts uh they use recursive learning they continually Loop over data learning at each level continually refining their understanding of Concepts as they process new information and you get hierarchical

representations that they build models of the World by zooming in and out of different layers of detail and this is similar to how human cognition processes both micro and macro scales so you know think about when you're when you're thinking about the concept of driving a car if you're talking to somebody about driving a car or you know you're you're focused on the aspect of driving you're not thinking about the car as all of its various parts that make up that CL complex system like the brakes and the steering wheel and the seats and the

every aspect of the engine you just call it a car right because that's all you need to know in the moment for whatever whatever you're uh doing with that concept and so this is kind of a similar idea um and this layered approach allows rgms to understand complex multiscale problems without requiring vast data sets so conceptual modeling for decisionmaking um rgms then focus on conceptual modeling versus just mere pattern recognition um deep learning can recognize patterns but struggles with conceptual understanding it can recognize correlations without understanding relationships and it can recognize patterns but it doesn't understand the underlying meaning and rgms on the other hand focus on conceptual modeling they capture multiscale hierarchical structured Concepts that represent both fine grained and Abstract features it builds a hierarchical understanding of the world capturing cause and effect relationship ships so rgms understand the why behind patterns not just the what and the key difference rgms are not just a better generative AI but a transformative AI framework that builds world rgms can reason about cause and effect relationships making decisions based on conceptual understanding that are more aligned with real world outcomes so how do rgms Le aage active inference by incorporating active inference Active Learning and active

selection while incorporating realtime data to adapt its predictions and actions so what you get is realtime learning uh they adjust their beliefs based on new sensory input making decisions on the fly optimizing both learning and decision-making processes and they also are able to uh have unified perception and planning so deep learning typically separates perception like recognizing an object from planning like deciding an action rgms can combine both in a unified model they handle both within the single model and learn Concepts that guide both recognition and future decisions so for example you can have an autonomous robot that can recognize a door handle and also plan how to open it using the same model so how what does all this have to do with the spatial web architecture so the potential that I see with rgms is that um the nature of how they work fits very well with the nature of this architecture within the spatial web and just to kind of bring you back to what I was talking about earlier you have hyperspace transaction protocol this manages data flow between interconnected systems and it facilitates interoperability between different AI systems and the hyperspace modeling language provides a common language for these systems enabling uh computable context and decisionmaking and it provides a shared framework for context aware decisionmaking so this protocol aims to

create a standardized way for different Technologies and AI systems to interact and share information across programmable spaces within our global internet and this unified model this becomes a unified model for diverse applications handling complex multiscale systems through unified perception and planning it's a single framework that can handle multimodal tasks so you have versatility they can perform across multiple applications whether it's object recognition natural language processing strategic planning using a single Universal model and it reduces the complexity uh it integrates the data analysis the perception and the strategy execution which is the planning within one model so you have cost savings it reduces the need for multiple specialized AI models it streamlines system management and saves resources so this how the spatial web enhances the rgm capabilities the combination of this scale-free active inference with the spatial web protocol could enable seamless integration of physics of the physical and digital these protocols allow rgms to operate across distributed networks exchanging insights and information between physical and digital environments enabling more sophisticated iot uh applications uh context aware Computing through the spatial web rgms would operate in context aware environments

these agents uh adjust their behavior based on the physical and digital surroundings and interoperability multiple rgms these active inference agents and autonomous systems can operate in different environments whether it's your Home Smart City businesses and they could communicate collaborate and coordinate actions through through the hyperspace transaction protocol they uh energy efficient distributed intelligence instead of relying on centralized energy intensive data centers intelligence could be distributed across networks of smaller more efficient devices and this this creates a nested ecos system uh similar to a hallon architecture and if you're not familiar with a h what a hallon is it's something that can be both a whole in and of itself and also part of a larger system so the spatial web and rgms would form a nested ecosystem where each node in the network whether it's a domain device agent or subsystem functions as a hall on it's autonomous yet interconnected contributing to a larger system's goals as part of the larger interconnected Network like a smart City infrastructure um your smart home could contribute the contribute data to your neighborhood grid which informs the city infrastructure decisions all while functioning independently um if you if you look at this picture here of you know uh part of

a supply chain image you know each item packed inside of each of those boxes is a hole in and of itself but it's subject to and it's part of the larger system the Box the box is subject to the the trucks that are moving it and distributing it or you know whatever's happening within the factory with those boxes and then that factory itself is its own hole that is subject to a larger ecosystem of the entire supply chain as a whole so when you're looking at what this next era of computing really enables it's just nested ecosystems that can be empowered by this natural intelligence and active inference Ai and the spatial web protocol can work together to create sophisticated adaptive systems at various scales so rgms for autonomy and self-organization with recursivity the scale-free models naturally form a nested hierarchy each level of abstraction in the model is complete in and of itself and yet part of a larger whole uh autonomy and cooperation active inference agents exhibit both autonomy in their decision making and the ability to cooperate with other agents and self-organization the renormalization technique in introduced in the paper allows the system to dynamically organize information at different levels conceptualization similar to the self-organizing nature of uh honic systems or these in these nested ecosystems so this last part um I want to explore

the transformative potential of decentralized AI for the future of systems so the impact of distributed collective intelligence the impact on this next era of computing would be enhanced problem solving and Innovation increased efficiency and scalability democratization of knowledge and Technology resilience and robustness enhanced decisionmaking personalized and context aware Serv Services ethical and responsible AI collaborative platforms and ecosystems enhanced learning and education empowered communities accelerated scientific research and environmental environmental and sustainability Solutions and the implications for future systems and Society the convergence of rgms in the spatial web could transform Industries enabling decentralized context Ware systems that operate with minimal human intervention industries that could be impacted smart cities you would have autonomous systems that improve efficiency at all levels Global Supply chains rgms could optimize processes from local shipping to Global networks personal to critical systems homostasis so whether it's personal devices or critical infrastructure a continuum of intelligent adaptive systems can maintain homeostasis at each level in our Health Care Systems personalized treatment by integrating real-time data from wearables and medical records education we can have

Aid driven adaptive learning systems
that evolve with student needs and
Environmental Management we can have
ecological systems that could be modeled
and managed more effectively from
Individual habitats to Global Systems
balancing local interventions with
global
goals and the future of AI and human
civilization the development of
renormalizing generative models within
the scope of active inference and the
spatial web promises to have farreach
far-reaching
implications more adaptive AI these
systems are highly adaptable capable of
operating in new environments with
minimal training efficient Edge
computing rgms enhance the power of AI
on devices like smartphones and iot
sensors reducing the need for large data
centers improved human AI interaction
with explicit uncertainty models these
systems interact naturally with humans
seeking clarifications as needed safer
AI systems they recognize and handle
novel or ambiguous ambiguous situations
reliably preventing catastrophic fail
in unfamiliar scenarios so they can fail
gracefully and accelerated scientific
discovery rgms expedite research in
fields like genomics and climate science
by rapidly identifying patterns and
formulating plausible hypotheses thereby
guiding further experimentation so the
key takeaway here is that uh all of the
all of this research and all of these
Technologies together can provide us

with a future where AI systems can interact seamlessly with the physical world autonomously optimizing operations and improving human lives so um yeah there's my website I have a podcast and I'll stop sharing now thank you Denise awesome I don't know what time how how is how are we on we have uh 15 minutes and I'll read some of the questions in the live chat okay sure I'll start with Pablo who wrote how will all of this affect the gamification of processes in organizations well I mean the the interesting thing is is through hsmi you know all of these emerging Technologies are are going to converge together and become interoperable in ways that we've not experienced before you know we've had this um kind of vision of of this augmented reality existence but the framework hasn't been there and this is going to provide that framework so I think when you're talking about gamification in every aspect it'll probably become kind of a natural part of of our existence and how we move through our day how we interact with with our work with uh brands with you know anything um you know I think that there's going to be a lot of of really interesting ways to kind of perpetuate that cool there's a lot of great questions in the chat so okay I'll read one from Tintin so they wrote it's impressive the obvious question I have is of alignment of this system in terms of that alien message from elazer

gudowski for example and then they clarified what that exactly is so the key idea there is that simulated agents evolve and and came to think faster than their creators so at a moment they took control to ensure their own survivability it's the question of how to ensure that way more intelligent and faster systems keep aligned with our values right and so one of the things that uh is really interesting to me about this is through hsmi we can actually collaborate with these systems in the sense that we can take human laws and guidelines and make them programmable so these agents can understand and abide in real time and you know um versus uh who you know Dr Carl friston is is their Chief scientist um they were involved in a a program called Flying forward 2020 and and they were doing a lot of uh testing around this and because it was it involved a drone project that I I don't know there were multiple countries throughout Europe it was with the European Union that were involved and you know testing whether the they could program in laws around airspace laws and you know things like that they did all kinds of scenarios of like you know Medical Supply delivery uh perimeter security all kinds of things and what they found was through hsmi you could actually uh program these guidelines and these these agents abide by them so I think we are going to experience this unprecedented level of of cooperation

and collaboration so we can grow in tandem with the you know with these uh agents and their intelligence and I think that we can also you know take some um solace in the fact of how you know active inference works and you know how our human knowledge grows you know I think we're going to see a similar pattern with this knowledge among these agents and the only difference is and and is just me speaking from you know my own you know my own uh perception and and perspective of this

but you know they're going they're going to be dealing with real-time data and with real uh a real grounding of this programmed these programmed spaces and those are the things they're going to be operating off of to make their decisions right whereas if you look at a human you know we have so many other VAR going on and every when we learn through active inference it's a very subjective uh growth that's why you can have you know multiple children growing up in the same household and they all experience things differently right but when you have these agents and they're getting this they're getting their learning through actual tangible programmed information I think there's a stability there as well if that makes sense and and and and again that's just my own uh you know interpretation and feeling around this and I could be I could be wrong but you know just kind of throwing that out

there that's how we move forward
yeah okay next question from David
Highland who
wrote thank you for the informative
presentation Denise can you share any
thoughts or information about the
ownership model for agents is it a
decentralized market how we decide which
parts are public
goods yeah so with the protocol with a
with hyperspace transaction protocol um
because so it's interesting it really
shifts to this uh kind of self-
sovereign identity uh aspect of the
internet because right now think about
how web domains work you know if you
want to interact with a web domain all
the transactions are taking place under
the umbrella of whatever centralized
organization owns domain and you have
two choices you can either say yes you
can have all my data and you can decide
what to do with it all or I don't want
you to but then I'm opting out of being
able to engage with your domain well in
the spatial web because now transactions
can be at every touch point of every
entity within the network then
permissions can be programmed in a very
nuanced way right so it really puts that
back in the hands of the of the user of
of the entity within it you can you can
program you can program guard rails
around certain aspects of data while
allowing permissions and Freer like
Freer accessibility for other aspects of
data so whether you're talking about a
business you know they can guardrail off

their data they can guardrail off
certain aspects of it for certain
Departments of employees some for their
customer base some for you know um
Partnerships whatever right um and we're
going to be able to do that kind of
thing too and the thing with this is
this takes into account time the passing
of time so you can set expirations on
permissions as well so I think we're
going to have a lot more self- sovereign
control in this in this
area yeah that's a part that really
excites and Peaks me as well like with
the web domain or the email domain it'd
be like saying well on one hand we're
not going to look in your email let's
just go with that for now but we do or
don't accept letters from this area so
it's kind of all or nothing and it's
based upon like you pointed out the the
custodianship of the all or none
permission based around what boxes you
can send where
whereas having a more nuanced way to
talk directly about belief
sharing means that instead of just like
plugging the USB into the port there
could be even a active inference
negotiation around what elements to
share how to share when to expire all
these other features which can can help
like incrementally build rapport and and
add to the relationship instead of doing
all or none which can be just very prone
to false positive or false negative in
cyber physical interactions
yeah you're so right Daniel

yeah what do you think about learning trajectories and paths I know it's a interest of all of ours I guess like what what can we do to be learning and thinking about all these things well um so you know I I have I actually have put together a series of courses um to kind of give people a a real um kind of a prerequisite knowledge understanding of what this transition looks like and the different components and what it means from all aspects of like you know basic understanding of of active inference and uh you know the free energy principle and the spatial web Technologies to you know decentralized AI what does that mean convergence of all this Tech across the network what does that look like safe AI had do active inference benefit that and then rgms let's let's look a little closer at at that and what I try to do is I try to take the complex ideas but make them more understandable so that people can really wrap their heads around the these Concepts and then they can go dive into the more technical aspects of it and I know you with the Institute you have a lot of technical knowledge there so I think that you know between resources like you know some of my resources your resources and then I'm sure there are resources out there that maybe I'm not even aware of and you know I think there's going to be more and more resources available around this modeling these agents you know uh

building out you know these these spaces
uh within the network um I know that
versus is going to be launching a
platform that'll be the first interface
for the public around this technology
and I think it's coming to the public at
the end of first quarter next year and
with that there will be tools where
people will be able to build these
agents as applications within the
network and and you know I think at that
point it's going to get really exciting
um so yeah I I just think over the next
six months we're GNA see a lot coming
about with this and if anybody is
interested in the courses I've put
together you know you can you can visit
my uh my website and and sign up for
them uh they're not available yet but
they'll be available very shortly uh
within the next week or two so yeah cool
okay I'll I'll for last question ask the
same thing that I asked uh Alex and Carl
earlier what is a fun or a memorable
moment from learning and applying active
inference this year and then if you
haven't already mentioned it what's
something that you're excited for next
year um
this year honestly I I think that uh I
have been on this Learning Journey in
creating this course material around it
you know and at the same time I've been
building community and kind of having
them be my test guinea pigs around it
and stuff so I you know there's been
this whole learning aspect for me around
building this educational material so it

it has been you know a constant uh
feedback loop and next year I'm just
very excited about the potential of um
building these agents and what that
means and and how we can tackle some of
the the problems and the the issues and
the challenges that we have um through
this new exciting technology and this
this uh framework so yeah well thank you
for all of your great learnings and
digested and and sharings and and all
the ways that that what you talked about
today I think connected a lot of the
dots from hearing Carl's more technical
view on the rgm on through what what
does it mean to have multi-agent
decentralized as opposed to just
building up one agent so thank you again
Denise thank you Daniel thanks for
having me thanks everybody till next
bye e

okay the next talk is going to be by
Bradley Alisia and colleagues
purposefully non-standard cybernetics an
alternative to purpose-driven control
this is going to be a 1hour pre-recorded
talk so thank you

enjoy hello my name is Dr Bradley Elisa
and on behalf of Jesse parent Morgan
huff and Amanda Nelson I'm we're from
the orthogon research and education
laboratory and the cybernetics interest
group and I'm going to talk to you today
about per purposefully non-standard
cybernetics

an alternative to purpose-driven control
so we start with a question what is the
definition of behavior and this is

varied throughout history according to the behaviorist Behavior was identified as a reflex elicited by stimulus associations the consequence of reinforcement and the current motivational state if we go forward to the cognitivists which is more or less the modern view they would say that behavior is a set of internal mental states that are responsible for observable behaviors with goal directed end States now it's of note that both behaviorist and cognitivist Views involve feedback this leads to learning a notion of what constitutes a state and predictive capacity so we start with this paper Rosenzweig et al in 1943 Behavior purpose and ontology published in the philosophy of science and it's a formative paper in the field of cybernetics so in that paper they have this typology of Behavioral classifications so we're going to return to this typology throughout the talk and we're going to think about how we can reorganize this typology but it basically defines Behavior as a set of nested things the red text Non classes or they're basically all other things other than the uh black thing that we'll be talking about at each rung of this typology so behavior is active it's purposeful it involves feedback which implies some sort of kology or end goal it's also predictive which involves extrapolation and involves order of prediction so you might have first order second order or third order prediction

so to understand what teleology is and where it comes from we start at the Middle Ages where teleology was the end State or the end goal of some behavioral process was thought of as Divine causation we then move forward to teleology As Natural causation and then it branches off into two directions and the first was the vitalism of Bergson and evolution by natural selection made famous by Darwin leading from Darwin's work we had Lamarck who talked about teleonomy which is purpose versus history and we'll talk a little bit about that later in the talk kind of discontinuous to this are two different streams so the first at the top is where behavior is viewed as habit that's by William James and Thorndike that leads us to the animal molecular anti-teleology of Francis Crick and feedback is complexity which is summarized by control theory and dynamical systems at the bottom is our target paper that we just has talked about these are the views of Rosenzweig and Weiner and this is feedback relative to a goal this led us to cognitive science and reinforcement learning or at least the modern versions of this where people talk about goal directed behavior and variations on that in molecular biology such as molecular vitalism and the teleonomic and U methodologies of Leban Janca and Ginsburg so we'll talk about a couple papers here and frame some of this in context so if we think about Behavior it's it's essentially a form of

functional in the first paper here Rody and Germain they argue that functionalism is a pre-scientific relic so they actually argue that the whole idea of functionalism is something that comes from this era of divine causality biology is not designed so why should we expect to function and this is something that of course maybe is assumed about function that it comes from some sort of design or comes from some sort of divine intervention on the other hand economic systems are defined by spatial and temporal information and constant iterations Over States so we do have these economic systems but include modern components only select behaviors can be defined as selection of a goal or derived from signals from the goal so that means that we can't necessarily paint Behavior with a broad brush of theology only some behaviors may be theological and others may be not not all behaviors and this includes biological and machine behaviors are goal selecting and while goal directed behaviors seem to be purposeful and and it may be true that some of these behaviors that are known as goal directed are purposeful are there Developmental and evolutionary antecedents so are those Place those uh ancestral behaviors or those basal behaviors or behaviors that occur earlier in development are those also purposeful and the answer may be no so Rosen Luth at all Define behavior as any change in an agent with respect to its

surroundings this is much broader than the Neuroscience psychological definition

and it sets up a contrast between all behaviors and a subset of purposeful behaviors so if a behavior is put under some sort of selection or if a behavior is selection of something in the environment this might lead us to a subset of voluntary behaviors which then some of those may be purposeful and so purpose is actually purposeful behaviors is actually a subset of all

behaviors to demonstrate this we see that clocks both mechanic and biological are not purposeful as there is no final aspirational state mechanical clocks can keep time and you can set an alarm on them but that alarm is imposed by the outside and it's not a property of that behavior and of itself furthermore biological clock runs via physiological processes like a circadian rhythm and it can be entrained by external white but that again is not a final aspirational

State more generally teleology requires a signal from a goal so it's some sort of signal from a goal that's specific to that behavior and purposes progress to that goal so again with our uh example of circadian rhythms it's entrained by light but that light itself is not a goal it's just something that's out in the environment and the system entrains itself to

it we can also see this with the brain in that brains produced behavioral outputs but the exact mechanism is hard

to pinpoint and this is the rust and
tell article that we saw earlier and
this asks the question signal from what
to goal so there's a signal coming from
something to a goal but it's not clear
what that is this is something one of
our co-authors Jesse parent works on
this is a frontier map for teeology and
related complexity Theory Concepts so
the data from this is drawn from the
complex world book by by CAU this is
published in
2024 and it kind of goes over three
different concepts that are interrelated
so we have teeology organized complexity
and emergence and so you can see that
our paper Rosen blue 1943 is at the top
and this is sort of the first paper that
talks about teeology
in sort of a scientific and in a manner
of complexity
and so this is the first paper that
talks about teeology is a property of
complexity and of complex
systems so a lot of papers were
influenced by Rosen Bluth this uh
theology was inherited by Weiner's book
cybernetics and control in
1948 two other papers adaptive systems
in uh by John Holland in 1962 and
organizations in complexity also in 1962
the good regulator of course in 1970
inherited this concept but it also
inherited another concept from
organizations and complexity which is
organized complexity the good regulator
then passes down this concept of thology
to the book order and chaos in 1993 and

there's a direct link between Rosen
Bluth 1943 and the tame concept of Mike
11 in 2012 now we going back to the
concept of organized complexity this
comes from organizations and complex
the good regulator in 1970 and the
autopolis paper in 1974 both share this
intellectual lineage the autop polies
paper lends that organized complexity
concept to this paper UND deiv in 1982
to this book order and chaos in
1993 and then to the tame Concept in
2022 finally we have emergence which
starts with the paper by Anderson war is
different in 1972 inherited by the autop
polies paper in 1974 in turn inherited
by the dissipativity paper 1982 in turn
inherited by Order and Chaos 1993 so you
can see that we have not just theology
as a concept in which we've kind of
talked about already but also a number
of other related Concepts this is leevan
uh this is the tame paper and then janca
and Ginsburg this is the U paper and so
Lan proposes the tame or technological
approach to mind everyone Levan's
principle here is it's cognition all the
way down we can see cognition in terms
of Minds we can see cognition in terms
of molecular phenomena we can see
cognition everywhere in between and so
behaviors all those behaviors are
cognitive to some extent and as a result
we get these diverse Minds that have
these different goal directed
competencies that allow those systems to
achieve embodied agency so in Levan's
approach you know chology is a very

important concept not just for cognitive type behaviors but behaviors that go all the way down to the molecular scale and maybe even down to the quantum scale second theology is a means to explain why agents develop towards goals and so you know we can look at uh agents such as artificial agents simple organisms like bacteria and we can see that like teleology is a way to explain learn how to move towards goals the next paper is this U paper by Janca et al and in that paper they propose a concept called unlimited associative learning and this concept is a minimal marker for Consciousness so in this paper they also Champion teleology is an intrinsic reinforcement system you have this internally regulated model and service of attaining external goal and teleology plays a role in attaining goals so again in both cases there's a strong tendency to talk about goals a strong tendency towards too but we might ask the question is behavior at these different scales and any sort of behavior that's sort of motivated towards some sort of function teleological and purposeful or is it simply a product of the history of the system so you know we could talk about teleology and purpose and we've kind of set this up that you know this is something that has kind of come out of the natural study of behavior but sometimes you know we don't really fully you know sometimes we ascribe Teleology and purpose to things that maybe are just simply a product of

the history of a system first example I want to give is the Belousov-Zhabotinsky reaction or the BZ reaction and at the lower left you can see an image of the BZ reaction and this is a deterministic internal model that we can model with differential equations and we can solve those differential equations and get interesting solutions that produce a very chaotic behavioral output or the Dynamics of the system which are sort of the unfolding history of the system so if we take an example of some sort of chemical soup we can achieve Belousov-Zhabotinsky reactions or BZ reactions by setting up this sort of set of differential equations these ordinary differential equations setting up an initial condition and we get these spiral waves and strange patterns that result and so if we have slight variations in the initial condition we end up with very different systems very different types of patterns in these systems and we can describe this with ODEs and one might say okay there's a purpose to this but the purpose isn't necessarily clear when you run this simulation again and again with different types of initial conditions there's a chaotic behavior at a variational dynamic dynamical output so this is an example of the trajectories of BZ waves this is from a paper I brought an angle in 1993 and it shows how there's a lot of variation in the pattern but there's a sort of um you know constant aspect of

the pattern in terms of the tip trajectories so one might say okay the purpose is to produce these tip D trajectories instead of the actual pattern and the interactions which may be true but then of course you know that's something that I want you to think about as we go through the rest of talk we can also plug the bellow of Jaa tinsky reaction into was cybernetic model with higher level feedbacks and so I talk about this in this dimensions of morphogenesis lecture on the D.A Worm YouTube channel and what I'm getting at in that talk is that there are different dimensions of morphogenesis there's onedimensional morphogenesis two-dimensional morphogenesis par dimensional morphogenesis and so you see this kind of output is and so I'm asking the question here is this purposeful or is this a product of sort of a chemical a dynamical chemical system with a history and so we can also look at this as a cybernetic model where we have feedbacks and we have other sorts of complexity and you know we can fit it also into that behavioral uh typology of Rosen

lth okay another uh another example is this thing called an AI estasis machine so an AI estasis machine is is basically the output of an internal model and so an internal model of course as we talked about is something that has some sort of capacity

for memory and learning and maybe some sort of output that looks Behavioral or cognitive in some way so what we can do is we can take this internal model we can have an oscillator which is a stochastic process or a deterministic process depending on our oscillator and we can have a memory with capacity so we have this oscillator that produces an output and we have sensory information that can drive the oscillator forward this output creates a trajectory which is an interaction with between noisy perturbations and a historical trajectory that's shown in this red line and the historical trajectory is stored in the internal model as a memory with a capacity so as this red line unfolds there's a memory of it in the internal model and it responds to external perturbations which drive this uh Red Line in different directions and the oscillator tries to recover to the original state and so you get these dynamics that may look purposeful but really have this stochastic and adaptive component we get more into this aspect of dynamical control Control Systems I look at intrinsic motivation in dynamical control systems so this is this tiamin paper in PRX life and so in this paper they talk about how you might model maybe something like an alist stasis machine as a set of actions generated without a specific reward

signal so in this paper they talk about that as intrinsic motivation they model this with except States and as oft actions so an action maps to a number of states Ates and this results in a trajectory forced by a observation stochasticity this is entropy so you produce the trajectory there's observation that you have and that's the entropy of the system and so you maximize actions that increase susceptibility rather than entropy these are noisy observations that wead do in perfect observational States and you can compare arbitrary length sequences of such sequences in the future this gives your internal model the ability to sort of integrate information and to find sort of solutions to problems the idea is that you have the system that can work towards a goal but it isn't driven by any sort of teeology it's just driven by this memory component and the stochastic component and that's all you need a number of other papers here that I'm going to talk about I'm going to put this in the framework of anticipatory systems from our paper on tonomy and Benny and friston so to get to this we need to re kind of reformulate our behavioral typology so remember we talked about behaviors having these different comp components we have these non classes so in this case we're going from Behavior to active Behavior to purposeful Behavior but that's contrasted with historical or maravian behavior then we have feedback

prediction and Order of prediction and
so er Mar refined the Rosen blue F all
theological category to two parts
purposeful agentic and historical
maravian so if we go back to the Benny
and friston paper they talk about the
free energy principle is managing
different forms of of goal
directedness and how you need negative
feedback to attain homeostasis there's
purpose in in cases where the goal is
known which is a marvian system and
surprisal is minimized which results
uncertainty then we can expect purpose
purpose isn't really just some sort of
magical thing or it's not some sort of
thing that requires a strict goal it's
just certain conditions under which our
behavioral trajectory exists we can also
look at anticipatory systems it's the
Louie paper and this anticipatory
systems come from a generative model
with depth so this is where we have
temporal aspects of the system that lead
us to teleology so it's more that the
teeology is the history of the system
and this depth comes from an internal
model so basically if we go back to the
internal model we have some process that
generates Behavior we have this memory
the memory is basically something that
stores everything that the system is
experienced and that's how we get
toyology to kind of understand this a
little bit more we want to distinguish
between a marvian system and a
non-markovian system so a Marian system
is where you have in arrival times so

there's a series of things coming in in time this is exponentially distributed in this diagram we have sensory information coming in we have some sort of encoding this encoding as a binary Channel with noise where we have these lines that represent some sort of sensory input or perturbation of the system and in arrival time is the area between these lines so the arrival time is obviously not equidistant in this example but it's in between the red and blue lines or the red and red lines and so we can take those in our arrival times and if they're exponentially distributed we have a Markovian system if they're not then we have a non-markovian system and so we've kind of developed this uh distinction between Markovian and non-markovian sensory information and how that relates to an information measure in our Gibsonian information work so Gibsonian information is a Markovian but can also be non-markovian information processing system and the citation is here so going back to that behavioral typology we have active purposeful feedback prediction and Order of prediction feedback is everything that is circular non-transitive this is the present past and future your behavior is controlled by a margin of error Corrections of agent location the time relative to a goal position or state and so if we think about this in terms of a feed forward process feed forward process is where the past is

your input present is your process and
the future is your output in feedback
however it's a little bit different your
input comes in the past and then you
have this present State here and then
you have feedback which is the past and
that Loops back to your input which
means that your input is now the
future and so the future is actually
present with past feedback so you see
the loop here you have input to the
present you have this feedback which is
the past and it goes to the Future so
now you have this future State before
the present State and if that keeps
going out you get this all output with
alternative Futures so your feedback
loop can be positive or negative and in
this case it doesn't matter we're
interested in is sort of the order of
past present and future so we end up
with this input we go to the present we
go to the past we go to the future we go
back to the present so we're sort of
interdigitating different types of
causality here instead of just having
this past present and future aspect to
things we have this sort of convoluted
aspect of past present and future if we
go back to our historical map we recall
that non- chological things can be
characterized as feedback as complexity
and so there are two papers here that
talk about pit control that kind of get
us at that idea
so PID control is proportional integral
derivative
control and if we think back to our

example of chemotaxis and phototaxis we find that PID control can be applied to chemotaxis and other types of biological control so this is The Balter and Bucky paper they talk about this in terms of variational free energy minimization given a linear generative model they talk about pit control resembling a reflex Arcic mechanism so this is The Reflex Arc we find in sensory motor control and the free energy principle provides a more direct account of control this is where we don't have any hidden states in the process of proprioception so they're focusing on proprioception AB but they also talk about chemotaxis and so they kind of give us this sort of uh variational free energy interpretation of this um in go all which is a more engineering take on this the pi and PD components or the proportional integral and the proportional derivative components and directed Network synchronization are all sort of uh components of this control mechanism in particular the pi and PD components of PID Define present past and future components in an internal model control system we can then also start to think of feedback is complexity in terms nonlinearity and complexity as well so this paper by Westerberg at all stimulus history not expectation drives sensory prediction errors in a melean cortex

this is from the bio archive so the author has framed this in terms of predictive coding so predictive coding explains cortical responses in two ways a feed forward model where an internal model of expected offense occurs or in terms of an internal model of expected events or a feed forward approach and a calculation of prediction ER for unexpected stimuli or feedback approach the latter are actually Global oddballs which defy expectations in a local context so in Mouse and macak which are their two model organisms they see two types of signal they see a normal response where the internal model is matched by no stimuli and a Global odbo Response which is Emer which emerges as a higher order response to this normal response we also can look at this nonlinearity in terms of Behavioral output in general and then burstiness or nonnormality of so this paper burstiness of human physical activities and their characterization and quantifying postural sway Dynamics using burstiness and interent time distributions kind of gets at both this aspect of maravian systems and a behavioral system that exhibits sort of nonlinear complexity both papers focus on the burstiness of postural sway in postural sway we have this paradigm where someone stands on a plate and that plate is a forest plate that measures or the center mass of the individual and then we can measure this and we get this nice dynamical trajectory that has different features

that we can analyze so it's a behavioral output that exhibits a lot of nonlinear characteristics and dynamical characteristics. Teuchi and Andano focus on burstiness and postural sway in terms of adult housework and children's play behaviors. They studied physical activities of children twice, ages 2 and five, and they found that burstiness is not an acquired feature; it's something that's innate. Burstiness is present in healthy individuals, not in people with movement disorders or impairments. We also looked at inter-event time distributions and associated fluctuations. So we talked about inter-event times as being key to Markovian and non-Markovian systems. They found that they're bursty, so they're the stochastic signals that have a noise component, and sometimes this is a function of someone trying to maintain balance, and sometimes it's just noise that's generated by the nervous system that's added to the stochastic signal. In both cases, this is not directed; it's generated by the physiological system. It's a very complex behavioral signal and it's not explainable by either a Poisson or Gaussian model, which means that it has this very complex set of features. So if we do some experimental cybernetics on this, we can find that if we have the right internal model, we can produce very interesting output signals or behavioral signals. Some of these T kind of look like they could be chological, but they're not. This is just simply running some stochastic

one example is just a pure stochastic signal the other one is having a stochastic signal with feedback that then takes that signal and integrates it at a later date so what we have here are two feed forward processes one is a spiky stochastic signal in Blue on the left and then the other is a smooth signal that is a stochastic delay of that signal the signal on the right so on the right we have the blue signal you can kind of see it under the red signal so the red signal is the one that's bursty and it's just basically where at random intervals we sum previous output and we add it to the present output so we end up with these bursts and so the delay which is this uh sum signal that we uh that represent our bursts is the buildup of the input signal it produces these bursts or what we might call avalanches and so we can see that we can produce these nonlinear signals from a simple internal model so this is the classic cooper paper from 1982 it actually has a couple of interesting things to say of cybernetic oriented things so we have this feed forward which is where you have a process that goes from A to B to C to D and we also have these systems that have strong back coupling so we have a process that goes from A to B to C to D then we have these back couplings from D to C C to B B to A and so what's more likely to be the case of like you know actual sort of complex behavior is not the feed forward signal but the strong back coupling and this is

where we have ubiquitous feedback strong back coupling can be intractable to analysis but produces the most interesting aspect of network interactions and strong back coupling produces things like Collective Behavior emergent properties stochastic outputs and nonlinear outputs are Avalanche which we saw in our experimental cybernetics a few slides back we can think about feedback and we talked about this relationship between past present and future a few slides back and we also talked about maybe some of these types of behaviors that are non-random but maybe structured in a certain way and we can think about feedback just sort of revisit the idea of feedback that Rosen blue fed all gave us which is that feedback was purposeful so we can ask the question is feedback purposeful or simply non-random and so we can think of past present and future relationships as a trip around the feedback Bove namely that we can take our input we can take the present and have a feedback loop that gives us past information into a new present when we get a new time point and that ends up giving us a future that can then be recycled so this has implications for prediction we can go past present future to prediction or we can have inputs that produce these sort of differently ordered past present future Futures and use that as the output so feedback mechanisms reorder processes systems with a lot of feedback aren't teeology per se maybe they

produce outputs that look to but they're not to logical and so going back to the different orders of prediction which were kind of like the final output of our

behavioral um typ typology what are the different orders of prediction well we have our first second third and fourth or higher order feedback so we can have the these feedbacks that have PR past present future but we can Nest them together so that we can have a past of a past of a past contributing to the present or to the Future and so that's really interesting because what it's suggesting is these higher level feedbacks maybe are also not te logical but there's some sort of memory or maybe there's some sort of um you know inter prent interval and the way it's structured so there's some structural aspect of feedback that might be really interesting that isn't chological per se but it given some of these other examples might have some structural significance so if we build a a cybernatic model that is really kind of complex and has a lot of interlocked feedbacks and things like that we can get some really interesting results so our different types of higher order feedback you know we can kind think of these in terms of the derivatives of position so the derivatives of position revisiting that from undergraduate physics you have velocity acceleration jerk snap crackle pop and these are just where you start

with velocity and you get a derivative of velocity which is acceleration you get a derivative of acceleration which is jerk and so on and these can be thought of in this way of having these higher order feedbacks or these reordering is up past present future to give you these really complex Dynamics in cases of derivatives of position and feedback feedback if it goes farther into the past the effects persist farther into the future so if we go back a couple of feedback loops nested on top of one another and we introduced that to the new present or the new future those effects persist further into the future so we get these really longterm effects in a dynamical system instead of just being local how does this translate into the neuroc correlat of behavior well we go back to our idea about Behavior sort of being naturally te logical and we have to think about like kind of how we kind of assume that behavior does magic things so if we get a behavior like this with higher order feedbacks it tends to resemble the sort of magical aspect it could be something like emergence it could be something thing like burstiness or something like a set of avalanches that occur and so these are things that you know you know give us really interesting Dynamics but are no more sort of magical orological than any other kind of behavior and so more fully connect systems are perspective or might be sentient or

conscious that's sort of the idea of
does magic things so you know we don't
want to us misattribute sentients or
conscious to something that isn't and of
course this is something that we've have
a problem with with modern AI systems
that people think they're STI Ander
conscious when in fact they just have
this higher order feedback and this sort
of um you know behavior that isn't
necessarily what people think it is
magic things or you know what we talk
about like complexity results from
embodied function a good example of this
is kinematic chains which we'll talk
about in a little bit and now we
introduce this concept of convolutional
Cy cybernetic architectures or ccas and
so we go back to our behavioral typology
and we ask the question can we better
understand the role of purpose or
meaning and feedback or temporal order
so here we have active Behavior
purposeful Behavior theological Behavior
and the non- classes of those are
nonpurposeful or random behavior and
non-feedback or non- theological
Behavior perspectivevely but we're going
to reorder this behavioral
classification where now we have non-
prediction leading to the order of
complexity rather than prediction
leading to the order of prediction so we
start off with behavior that's active
instead of passive then we have
nonpurposeful or random Behavior instead
of purposeful and non nonactive or
passive and purposeful now non- classes

then we have feedback which is non-teleological versus feedback which is teleological. Feedback is now the non-class non-teleological feedback is now the past to the order of complexity and then instead of prediction being extrapolation we'll have non-prediction which is also extrapolation and then we have our order of complexity first second and third order so the more feedback we have the more complexity we have so this is an example of a convolutional cybernetic architecture we basically have this toy example where we have an input and an output the input is 00 Z the output is 11 one and we have these boxes which process the present to the future and then our feedbacks are give us like some past state and it feeds it back to the future or the present so we have a ubiquitous feedbacks here we have some feedforward elements we have elements of feedback that do certain things to the sequence and the question we ask here is what is the maximal number of steps needed to transform the sequence from 000 to 111 so we start with 0 z z we put that in we kind of move to our feedback loop our feedback loop does certain things to the sequence and then we end up with 111 and we actually want to know what the maximal number of steps is because we want to know you know how far we can go through these different uh feedback loops and get our answer this can also be restated in terms of

temporal phases past present and future
so we also have more explicitly this
input of the present we have the future
as an output and then we have these
different uh Loops that give us the P
give us past information and again we're
interested in getting the most diverse
information possible and so this is
again this maximal Loop that we're
looking for so we're trying to get the
most for iety out of these uh
convolutional cybernetic architectures
instead of trying to find the most
efficient path or the tightest path or
the most go directed path we can further
understand this or restate this in terms
of semantic components so we can talk
about this in terms of symbols reference
and understanding symbol being the input
reference being the feedback or part of
the feedback loop and then understanding
being the output and so we want to
cumulate as many reference as possible
for a symbol so that we get an output of
understanding and again it's the variety
that counts not optimizing the process
or trying to find a goal directed aspect
to this so if we think about how these
ccas are assembled uh we can look at the
different components of feedback and
then we can start to assemble these more
complex
structures so let's look at first order
feedback so CCA has input
and in this case we're looking at past
present future and maybe alternative
Futures and things like that and they
all produce a predictive output so first

order feedback would be an input going to a future going to the present and naturally the future is just like the present plus the past so the present comes in as input it's passed to this feedback which is the past the present and past are added to the Future and then the output is some sort of predictive input so again we have this reordering a past present future in the first order

feedback if we have a feedforward only case we have input which leads the present which leads to some complex output but of course as a feed forward only system we just only get the benefit of having the present we don't get to talk about the past and the future it's just like every input results in a an output and that output then is complex output within any context then if we put together a larger set of feedback loops we can see this sort of convolved mixed feedback where we have first order feedback second order feedback and a feed forward component so these are all different components producing two different complex outputs so we have an input that gives us the present we get a first order feedback that gives us some past and we add those together to a future state that produces complex output we can also have a feed forward part of that

where the input goes to the present and then the present is a part of that state and then we might have a a a second order feedback where we have an input

which is the present it passes A first order feed Loop uh feedback loop to the past and another first order or second order feedback loop to the past and those can both be combined into the present which is then informing a future state which involves a longer memory which has more information and that gives us another complex output but we also have this path from the second order feedback up to an alternative future which then has passed out as a complex output so you see what we have here is we have an input that goes to the present we Sub sub reference the past we sub reference the past again and that gives us an alternative future so happens for every cycle so you end up with these three different types of complex output and given this topology we can end up with these different types of characterizations of the system remember again with this convolution what we're introducing is this variety which is actually a concept from uh the ever good regulator theorem and a lot of the work of w ashb so we're doing with producing the maximum amount of variety so we you know we need variety to control systems so this is actually overall the appearance of purposes replaced with an onsemble of positive and negative feedback elements with diffuse effects so remember that we need to have not only a single Clos Loop feedback but we need to have multiple feedbacks and maybe some open loop feedbacks in there

so we can have uh maximize a variety so the imperatives of such a network are to gain processes where to maximize the number of processes and this leads us to maximizing variety and of course variety was a concept that was introduced by W Ashby in the ever good regulator theorem and he talked about how variety is necessary control but it also creates the conditions for adaptability modularity modifiability so we have systems that are both not purpose ful but they also have this sort of imperative for control but also for other types of conditions in a in a system that allow it to adapt be modular and be modifiable and so we can also add in polarity we can have different types of effects resulting from feedback as we talked about if we reorder past present and future that's all well and good but this also has relevant to different types of processes that result in systems so in in complex systems we have these different effects like Avalanches runaway feedbacks and dampen feedback these are nonlinearities and we saw an example of avalanches earlier in our experimental cybernetics and that can result from a number of feedbacks of different polarity and this can produce unpredictable and stability we can have things like runaway feedbacks which basically just the product of positive feedback and that can lead to mass instability and we can also have

something like dampened feedback which is a negative overly negative feedback which sometimes makes things more controllable and so we can have these effects that result from these processes that have a polarity that lead to these different complex systems phenomena that then lead to things that are maybe look they look to logical or purposeful that they're actually just the products of complexity and so if we kind of go back and look at like how we can model things like connectionist learning and reinforcement learning with cybernetics we can actually take the equations of connectionism and reinforcement learning and put them into this sort of cybernetic system so this is where you know we want to take modern learning systems or artificial Learning Systems in understand them using cybernetic approaches and maybe apply them into these kind of cases and so this this is congruent with hopefield and Sutton's alternative Viewpoint which is that the weights of a network are reinforced by the activity and selection of previous inputs so this is where we have we're looking at Q learning here and we're looking at connectionist learning and we're looking at sort of the way equations or and we're thinking about how you know we can get learning from these systems of equations and you know this is a dynamical system you know as much as a learning system so we want to

understand and of how this is how this works and so we start at a random set of weights and lack structure due to selection we started a random set of weights and lack of structure due to selection we proceed to a structured set of weights and optimal policies unwittingly we kind of move from a marvian system to a non-markovian deterministic system and so there's a tar question here which is how do we go from something that does not resemble purpose to something that is purposeful the answer is is that purpose is a post talk determination so we can model this as habits with weights so we have this input we have these outputs and we have this uh CC C that's that has polarity and we can have habits that are the serve as the inputs and the weights are polarity and we have non-teleological behavior in a directed graph so this directed graph kind of moves towards the end points the outputs and so the lesson here is that purpose is basically whatever we determine these outputs to be so when we observe a behavior a behavior is generated by these CCA graphs we interpret the output the way we' like to interpret it and as I said before we can have these different outputs that are that look purposeful or they look peely logical but that's just really a post talk determination then we move to wier's Alternative Viewpoint so Wier is more of a cybernetics guy he also has an

alternative Viewpoint and his Viewpoint focuses on communication and information which have both a control layer and a semantic layer so again we have to go back to the past present and future formulation so instead of thinking about this in terms of polarity we want to think about this in terms of not just the control but the semantics as well and so we have this past present and future we have an input which is the past we have a future which is the present with past feedback we have the present so the present goes to the Past goes to the future but the future also moves to a present where there's a present with past and future components and then an output so if we run this in time if we run this as a dynamical system we end up with this the present being sort of you know it's not really additive but it's basically you're concatenating things onto the present with information from the past and you're informing a future trajectory and you're you can actually generate alternate futures of a system from this and so this there's a control layer a semantic layer and this gives us information about the system both in terms of contingencies and synergies so connectivity and ordering provides both contingencies which is that nodes depend on other nodes and synergies for nodes working together so in this model we can't knock out any one node and have the system work we can't knock out the the feedback and have this sort of

Rich self-

reference we can only have you know a
feed forward model and we've seen that
that's much less

interesting we also can't knock out the
future and we can't knock out the
present we have to kind of continue with
all of our

nodes but also these nodes working
together produce the synergistic
um trajectory that gives us rich
information about time it gives us this
present it gives us a future but it
also gives us these intermediate states
where we get past feedback and we get
integration of

those this stands in contrast to the
imperative of purpose andology
is cybernetic control and behavior
purposeful or stochastic activity
tempered by recursively recombined
temporal ordering and so this is again
one of the

ccas and CCA is also a series of steps
that add recursive functionality so
sometimes if we think about this in
terms of our internal model memory this
is memory with sub referential power so
it allows us to have a memory again
where we can reference the past of a
past of a past and it informs the future
of the present and adding present into
future and things like that this creates
a hierarchical system and high degrees
of intentional order so again we can add
these things together where we can run
these ccas and we can
produce this memory with sub referential

power and the output can look very organized and hierarchical but the thing that produced that output is not necessarily ordered and it's not necessarily uh there's no overarching goal or any other sort of information going into it so once again thinking about habit and purposeful Behavior we have this paper creatures of habit the Neuroscience of habit and purposeful Behavior this was published in biological Psychiatry so this is where they talk about animal behaviors being composed of Habitual and goal directed categories habitual is non- theological but goal directed is theological they provide this sort of breakdown where you have teleic versus non-teleological behaviors and deterministic versus stochastic Behavior if we think about teleological and deterministic behaviors we get goal directed behaviors those are things reaching for a Target or moving towards a goal but of course we know that some of these types of goal-seeking behaviors are chemo like chemotaxis or phototaxis are stochastic and so if we have theological and stochasticity we might have something like chemotaxis or phototaxis we have something that's non-teleological and deterministic we have habitual work so we have something that's habitual it's not moving towards a goal but it's deterministic something like uh scratching or like a reflex we have non-teleic on stochastic behavior however we know that that looks something like a

random walk so we can actually characterize these different behaviors in this way the problem with chemotaxis and I put a question mark on the table so chemotaxis for example which is a movement along chemical gradient we already established that that was not toic the go directedness in that case is due to the environment not the internal model and so we can look to gibsonian information or we can look towards sort of thinking about why it's not te olical and so but it doesn't also doesn't resemble a random walk so that provides us with a problem which is where does this fit into this typology so a clue to kind of where chemotaxis and phototaxis fits into that

typology comes from this paper brain like neurodynamics for Behavioral control develop the reinforcement learning and this is from the bioarchive from this year cot at all test two scenarios supervised or extent and reinforcement or developmental learning so we have extant and developmental learning only reinforcement learning produces gold directed behavior and neural activity similar to biology it provides robust short-term behavioral adaptations to perturbations and it produces an emergent emergence of policies that are not purposeful in an of themselves but lead to things that look the part so this is reminiscent of our ccas it's also reminiscent of our or discussion of chemotaxis and phototaxis building a CCA

from a small Motif is a form of
purposelessness or purposelessness
reinforcement learning build up you
build up to function and sometimes
useless structures and that's okay
another example of nonpurposeful
behavior that kind of fits into our CCA
model comes from fetal substrate cycles
and biochemistry and so this is the
paper stochastic amplification
and signaling enzymatic feedback Cycles
their noise induced by stability with
oscillations this is from
pnas so this is where we describe these
feedback Cycles as these feedback loops
where you have an input of energy
there's an enzymatic
reaction and you get nothing that comes
out except entropy or heat so you
basically don't have an output your
output is entropy it's heat and of
course this is a problem because we
expect there to be an output we expect
this reaction to actually do something
and in fact it's creating heat but it
seems like that's a lot of work for just
heat so we have a biochemical pathway
which is inclusive feedback it does
thermodynamic work for no
reason heat is produced from an
energetic input but we don't get any
biochemical products
and this seems like it's a waste of
energy that has no
purpose but it's actually important for
regulation and other processes it
produces complex higher order Dynamics
or amplification effects throughout a

greater biochemical system so if we go from convolution to order we can see that the evolution of cybernetic control is a causally recursive chain of contingencies and so here we look at the contingencies in a CCA we go from 1 to two to three and then that's a feed forward system and then we can go from 1 to two to three prime to four which it feeds back to two and gives us this output four prime or we can go from one to two to 3 to four or 1 to two to three prime to four to five to six to 7 which is an output so we have these different outputs from these different numbered boxes and we have these uh resulting processes so let's take three prime as an example three prime is dependent on two and one and once feedback is operational four and five so four and five actually do feedback to three prime which of course is a dependency so if you knock that dependency out you lose your entire functionality of that part of the network seven on the other hand which is another output at the top is dependent upon six 5 4 3 Prime 2 and one there's no feedback from delay so you can see the different types of order and how there's a lot of recursive chains that lead to contingencies so again if we think about our feudal cycle our feudal cycle is like something at the bottom our her or feedback is like something at the top of this diagram and our alternate outputs alternate outputs are at the right and

they are of course maximizing for variety and so so if we think about the evolution of cybernetic control as a causal recursive chain of contingencies we can think about the kinematic chain which is where we have these dependencies so D is affected by the activities of ab and C we get this chain it's this is this robotic arm that extends out across two joints and the activity in D with kinematic is constrained by a b and c and so we can describe it in that way if we go back to these kinds of contingent gencies we know that seven is constrained by its intermediates and thus it is with the kinematic chain in in the kinematic chain we can do this in a linear fashion we can add on conditionals as convoluted cyber as the convoluted cybernetic Network evolves this restricts the system to a deterministic path behaviorally this leads to interesting consequences for a developing sensory motor Loop and a kinematic chain is basically sensory motor physics with contingency so here are examples of a kinematic chain in robotics and we can think about uh embodied robotics here where we have these control systems for these kinematic chains maybe the CCA where we have different types of morphologies and we can design them appropriately we can design and we can design them appropriately we can design behaviors that aren't purposeful

aren't teleological but still give us behaviors that we can that look intelligent we can also think about purposeful behavior and fore cognition so our approach to purposeful Behavior can be applied to Fore cognition so we have these different types of cognition some of them being embodied some of them being embedded some of them being extended which means outside the brain and some of them being inactive if we put this in the framework of Rosen glue that all's behavioral classification does for 4E Force us to rethink that behavioral classification as we did in a number of instances in this talk now there's one Insight of 4E that's interesting to us which is that the morphology and environment of a system is more important than its internal model so we've been talking about internal models and how these ccas model the internal state of behavior and how that shows that in a lot of cases teleology in per se are sort of elusive but if morphology and environment are more important than the internal model then maybe we need to incorporate ccas uh or or recast our ccas in terms of morphology and environment as well as an internal state in fact our CCA do that to some extent but it's uh sort of a way to put this behavioral classification on its head and so in the case of 4E purpose is not ascribed to a mind feedback is diffuse and complex prediction its highest orders are not teleological they must be open-ended we

talk about this in a paper from 2023 and Royal Society interface Focus so what is behavior and what are its properties so yeah we we're kind of revisiting what roseen Bluth talked about we're looking at a number of examples from neuroscience and from psychology and from other areas animal behavior

Robotics and we kind of break down kind of what we're looking at here so

Behavior can be intentionality it can be go directed and supervised and so these are things like control systems p word cognition and of course some of those are go directed some of those are regulated

complexity then we have Rich higher order complex Dynamics and those are things like Dy reactions which we talked about and embryogenesis which we didn't really talk about but also qualifies in that category so these are outputs of dynamical

systems we also have connectivity and convolution which are things like connectionism avalanches and open-ended learning and then finally we have behaviors that are hardw but recursive so with our ccas we're modeling some hardw Behavior behaviors in a recursive fashion we also talked about this in terms of non-markovian systems and habitual behaviors we want to go beyond the standard of teologico and purposeful Behavior so a standard way of thinking about behavioral regulation is that inputs or deterministic and outputs or

Collective output should be closed into so if we expand the basic closed feedback model we get an interconnected network of retrocausal ordering so this is something that we're seeing in the CCA where we talk about the order of causality. In the present future we have this basic Lo the feedback model but if we expand that out we get much richer retrocausal ordering which leads to of course to variety but it also leads to these interesting paradoxes and interesting effects of high feedback behavior is often open-ended and this can be characterized by embodied Dynamics unpredictable which can be characterized by chaotic Dynamics non-normal which can be characterized by non-markovian behaviors and systems Avalanches of different types power law Dynamics and so forth and variational which means that it's characterized by multiple goals and redundancies and so this requires a reassessment of Rosen blue theal typology and has wider consequences for the practice of active inference thank you for your attention uh one of our co-authors Jesse parent uh produced a lengthy blog post on the Rosen blue th all paper talks about its enduring insights and its con uh its context in cybernetics and then we have two papers we talked about in the talk from our group alist stasis machines and a primer on gibsoni information so I thank you for your

all right welcome

back here with Alexi Tolinski a friend
and colleague and collaborator on this
paper temporal depth in a coherent self
and in depersonalization theoretical
model so go for

it well uh I'd like to say hello to
everyone and thank you for being with us
after such a long and fruitful day at uh
6 pm. on the east coast and uh this
is I appreciate your interest and thank
you Daniel thank you active inference
Institute for this fantastic Symposium
um so

uh uh we'll be working talking today
about this paper on dissociative States
and it's a collective effort uh this
papers in peer review right now uh early
uh preprint was posted uh on the site
and it was uh uh work by Michael Levan
Chris Fields Lan D Costa Rachel Murphy
Daniel who's here uh David Pinkos and
myself and um this is the link to the
paper if you'd like to take a look at
the early preprint

uh perhaps some of you would like maybe
to get a feel for what we're talking
about and this is not the patient's
account this is a poem uh from uh Kurt V
and his song Pretty
pumpkin I woke up this morning didn't
recognize the man in the mirror then I
laughed and I said oh silly me that's
just me then I proceeded to brush some
stranger teeth but they were my teeth
and I was weightless just quivering like
some leaf in the window of a
restroom um and this uh poem was cited

by Paul moleno he's a specialist on did
in England on the Psychiatry and
Psychotherapy podcast you can also find
patients account on YouTube I couldn't
post the video here for copyright
reasons um so dissociations are a
spectrum and they range from mild to
severe transient uh to Chronic uh they
can be benign and pathological and
certainly they're not a bright line
category uh the causes vary for them as
well uh we call it iology in the
clinical World they can be caused by
psychological trauma neurological
conditions substances seizures
anesthesia and uh different sleep phases
there's a subset of conditions here we
call dissociative
disorders typically in the history of
these patients we see prolonged
elevation of stress accompanied by
repeated traumatic experiences in
circumstances where where they feel
There Is No Escape typically childhood
abuse and neglect and we call that
complex trauma which has a different
presentation than acute trauma and we'll
touch upon why this specific pathway of
uh iology uh is is important for
dissociative
disorders
um well I'd like to say why uh this
matters what's at stake what's the
motivation for this paper and we have
many authors and they have variable
motivations but I'm a clinician I see
patients all day and um uh for me the
main reason for that is these conditions

are hard to diagnose many patients go undiagnosed for a long time they're very hard to treat uh and we're sometimes content with partial results and the risk of self harm or suicide is higher for these patients and there are many many reasons for that uh we think that perhaps one of the reasons is that there is no Unified theory of dissociations we have kind of um neurobiological ideas the corticol limic inhibition hypothesis we have uh psychodynamic ideas we have uh attempts to moral dissociations from first principles by anona and other colleagues but uh these different attempts are not tied together and certainly not tied together with the patients uh subjective experiences so we are trying to make steps in that direction and and we don't claim that we've built a unified theory of dissociations but what we're doing here is we're focusing on one aspect of dissociations one focal point that is a collapse of the temporal depth in any kind of dissociation regardless of the cause by temporal depth we mean how far into the future the person can plan what we're hoping for is that this uh uh focused approach has the potential to uh allow us to to to um perhaps develop uh more diagnostic uh uh procedures and and to possibly um help increase efficacy in in therapy with full respect to how difficult complex and messy this work is um uh additional point of convergence is we explore multiple theoretical

perspectives in the paper and they all converge on the same um point and here is the main thesis um the collapse of the temporal depth is necessary and sufficient for the onset of a dissociative episode so we're going a little bit further than colleagues who already mentioned a relationship between temporal depth and associations Dean and and an unic and others and Carl friston were saying it's causal that there is no dissociation without the temporal depth collapse and the collapse of the temporal depth will reliably lead to dissociative episode though we are um uh saying that temporal depth collapse can be conceptualized as a common pathway for the onset of dissociative symptoms regardless of iology and finally we suggest that the focus on restoring the temporal depth can be an effective strategy in therapeutic interventions one of the theoretical perspectives we explored in the paper is Micha 11's te framework technological approach to mind everywhere which I think is very useful uh and unfortunately I haven't seen many uh attempts to uh use it in clinical applications this is the title and the link to the paper uh there's so much in Mikel Len's work that is useful uh his uh paper on cancer which is based on his experiments is also one of The Inspirations for our paper here but here's another inspiration if you dive into Mike 11's work you'll probably see

uh what he claims based on experiments
that all intelligences are collective
intelligences however at our subjective
level at the level of the conscious mind
we don't feel that way uh we perceive
unity all our percepts Sensations
thoughts feelings come together in a
cohesive and coherent ho a single
conscious experience in the here and now
and uh we will talk about this seeming
uh Paradox uh if you wish in in our
paper um some of the core tenants of the
team framework not all of them but some
key points uh commitment to graduality
there is no bright lines uh Michael
suggests that intelligence is built
gradually and uh by accumulation of
complexity material Independence uh our
model is not based on neurons or central
nervous system we will talk certainly
about a subset you know I'm a clinician
I work with human beings so we'll talk
about brain-based models but generally
the model applies to synthetic agents
and cells that are not neurons
commitment to empirical uh research as
opposed to just exchange of opinions on
the subject um Dynamics the C are not
fixed they're malleable and plastic and
C can be nested and over lapping
cooperating and competing both laterally
and across
levels uh this is an illustration of one
of the key Concepts in Leven framework
um uh called a light con and what you
see here depicted is various organisms
in the middle of the slide you see a
tick and a dog a human being possible AI

compound intelligence such as an ant colony and you see the sort of nested and embedded structure on the picture on the right and L con has two uh Dimensions there's a spatial Dimension and a on the horizontal plane and the vertical Dimension is temporal and tick for example is not particularly concerned uh with things happening a mile away uh it cannot recall far from the past and plan far into the future while human beings are capable of planning things until uh that happen after we die a living will and uh with the use of language and culture and books we can uh use things that happened before we were born so we're capable of huge light cones and I'm highlighting this word capable of it doesn't mean that in our wakeful hours and alert hours we are walking around with the significant light codes they contract and expand depending on various circumstances but we're certainly capable of them well ticks or not and the vertical dimension of a lone is exactly what we mentioned before that is the temporal depth there is of course a perspective on the same thing in FB and I like this very sucking phrase by Carl Friston who said because consequences follow causes the generative model acquires a temporal depth and more formally uh some agents are only capable of modeling their immediate environment and live in the Here and Now While others are capable of planning actions into the future and the

generative models capable of planning
are referred to as temporally deep
generative models
so temporal depth under fap is the
length of the temporal Horizon that is
considered during
planning uh we're moving on to
describing the if you wish healthy self
um and we do mention that it is a
materially independent U structure where
you know in principle we can talk about
the S of synthetic agents and you know
cell
collectives uh uh s depend on an
organism's ability to infer and here we
will uh provide uh a viewpoint on the
two clinical terms familiar to
clinicians such as derealization and
depersonalization so the organism's
beliefs about the world are the
inferences about the external
environment and the Continuum of issues
in this process we will call
derealization while the organism's
beliefs about its own internal milieu its
body and mind we will call the self and
the Continuum of issues of this process
we will call depersonalization
the selves are hierarchical composite
nested and with boundaries uh so we
consider self to be a system consisting
of hierarchically structured beliefs
which effectively makes it a dynamic
hierarchical generative model and taking
at its entirety the self is
informationally separated from the
external environment by a mark of
blanket or a near Mark of blanket by

boundary would like to emphasize I
apologize if this is Trivial to the
audience of the active inference
Institute that the boundary is not
Material like skin it is
informational uh through which the self
and not self interact so to emphasize
the self as we have defined it is a
model it is not an organism the organism
constructs and implements it so the self
is quite literally the construction of
the mind this is an illustration of that
from early work by Michael Leven I think
this was his doctoral dissertation
materials and what you see here is um
dark embryos that started from oite and
then you have blastoderm you after cell
division you have a collection of cells
and at the top left if nothing is done
then this blastoderm will develop into a
single ducky you see this green shrimpy
looking thing that's a single
duck however if you introduce a scratch
in this blastoderm and the wound
heals the same exact cell collected May
develop
into twins or triplets you see the red
yellow and green duckies now think about
that for a moment what happened
here um this Collective of cells
computed what is me and what is not me
and the hardware did not change but in
one case it made a decision that this is
all me and the rest of it is not me and
another one it says there's three Mees
so you can see that this is not
Hardware um s are also as I mentioned
hierarchical composite nested and they

have boundaries so now we're moving on to components of the self uh that are each component is separated by their own boundary and Mark of blanket and because we're using the word hierarchical uh we will uh Define that there is a core self a core component if you'd like a visual metaphor that Mark SS uses we can think about an onion and there's different levels of the layers of the onion and there's a core uh in the middle we will talk about that as apply to humans and mammals and rodents so now we are in organic uh in organisms uh this term is from effective Neuroscience from Yak bip and Lucy Bin's book um and also you know I believe SS and teral used it uh so core self is non-reflexive non-reflexive and dominated by raw feelings uh and part of purely effective core form of Consciousness so the effective homeostatic mechanisms allow an animal to problem solve in novel environments and I'll give you an illustration from Mark s um to maybe just um bring it home uh imagine that I'm in the room full of carbon monoxide and I've never been in before uh so I don't really know what to do but I feel badly I feel badly in a very specific way there's air hunger Suffocation distress uh and let's say I'm moving around the room but if I move to the left and I feel a little bit better and that's information and if I move to the right and I feel worse

that's also information perhaps there's
a door on the left and there's a draft
underneath the door and I feel better so
this Compass this internal mechanism the
homeostatic mechanism allowed me to find
out a possible
solution and um we have a whole
collection of these in know to mention
you know the actual hunger uh for food
and thirst and social homat such as rage
and fear and panic separation distress
Etc so the collection of that is what
pip and s's conceptual as the core self
uh uh therefore the set of predictions
in the core self that all the life
sustaining affects will be at or near
their settling points which means in
emotional terms contentment all my needs
are met I am a okay thank you very much
right and that is the minimum of the
variational free energy if there are
deviations from it such as hunger or
thirst or uh air hunger then that would
be a prediction error accompanied by a
negative feeling like in the example
that I used
above uh in mammals we're not putting
the equal sign but we're describing the
anatomical structures that are uh
participating in in uh serving this
function we're talking about the upper
brain stem regions including the
reticular activating system and the
periductal gray they mediate the
functioning of the core self and you can
see a lot more details in Mark's book
Hidden Spring a journey to the source of
Consciousness so um oror self is thought

to be present in animals without the cortex such as decorticated cats that are actually more effective and not less and the cases of human children born without a cortex hydron calic children they have emotions when you know uh you put a Brother on other girl's you know uh chest she smiles and she cries so the core self does not require a functional neocortex the higher levels of the brain structures in humans including the The Vortex mediate the higher levels of both Consciousness and the self and sometimes it's referred to as extended

Consciousness which allows us to reflect on what's going on such as to say I feel anxious right

now uh there is an asymmetry in this uh um model where the higher levels the peripheral levels of the self and and Consciousness cannot exist without the core um well the reverse is not true the core can't exist without the higher

Levits one of the other illustrations uh of that which PM cites this is based on Fisher's work in 2016 is that a tiny lesion a 2 cubic millimeter lesion in a parabal nucleus will reliably and IR irreversibly reduce a coma the patient will never become conscious ever again while you will never have anything like that happen in the cortex you know this small lesion in the cortex will not result in catastrophic outcomes like that this is one of the reasons why neurosurgeons don't usually like to operate around the brain stem this is too consequential too

risky so everything we said about
Consciousness the core and extended is
applied to self as well that peripheral
self components cannot function without
the operational core self and we will
illustrate that with specific examples
in humans later on in addition to this
one where the thing is completely
operational or not we can talk about the
influence and the core self has the
ability to change the regime of function
in other more peripheral self components
by inducing phase transitions in them
and the Brain mechanism for that is
generalized
arousal um when the higher levels of
consciousness are present and functional
then we have representations such as
fruit and vegetables and you know in the
core self we have raw effects such as I
don't know separation distress or hunger
but with the higher levels of
consciousness presence we are capable of
uh uh thinking things like I want an
apple or the affect the drive is in
Freudian language and in S's language is
attached to the
object and similarly we can have these
constructions about the meta object such
as I uh such as I look pale or I am a
pessimist and I here is a mental object
and therefore I is a metacognitive
construct an abstraction and it only
exists at the higher levels of the self
such as autobiographical self it is not
used or even needed in the core self to
reiterate core self as a system has the
capacity to

detect uh the effective prediction errors and uh attempt to minimize the variational free energy through action without any

eye we're moving on to peripheral self components and we'll describe only a few uh and the terminology here is different from what you may read in wonderful book by Neil Seth being you so the first one is autobiographic iCal self and that's a system dynamically representing our life's history if you for example remember your 10th birthday and the cake is on the table and Mom is there and your friends are there and the sun is shining and everyone's smiling and it's Tuesday that's an episode that's an episodic memory and a collection of them organized through time is our sort of Life

narrative um there is a bodily self which is a system dynamically building inferences about our body and various representations and interceptive processing there's a social self which is a system representing inferences about how we're seen by others and how we present ourselves and act in the environment each peripheral self component such as bodily self for example has its own Mark of blanket and each component is embedded into the whole self and also contains subcomponents creating a nested architecture that you saw on the T light cone picture

before about the unity that in humans at the higher levels of the self such as at

the autobiographical self level uh uh uh
these these levels create an impression
of a unified experience and this is just
the experience of the autobiographical
self not the entire system the
autobiographical self claims to itself
and others to be the entire self well it
is not so Anil Seth refers to these
things as delusion we have a delusion of
being a unified I don't know Alex right
but you know what we're saying is that
it is that system that is diluted um uh
the autobiographical self subsystem I'm
ambivalent about using the term diluted
I think it it's not the optimal one
because we use it in the clinical world
and I prefer it uh called inference or
belief that we're unified stated
differently we can say that a stable
belief that I'm whole in the uh I'm
complete in the order biographical self
um its generative model contributes to
us feeling as a Mantic self
therefore the subjectively perceived
Unity the coherence of the self is an
inference related to coherence another
thing we can talk about is continuity
and nearly as a taty stemming from the
definition of Mark of blanket ourself
remain the same persistent time while
all the processes or Communications
across its Mark of blanket remain
functional and we can um offer you
another viewpoint on the same exact
phenomenon from if you more
neurobiological perspective the
psychological
perspective if you recall maybe you're

at the train station or you know
anywhere in the city and you see someone
and you think you recognize them but
you're not sure you know this sensation
of recognition of familiar uh so the
familiarity novelty calculations is what
we're talking about here and
depersonalization is a breakdown in that
something that is familiar myself
suddenly abruptly becomes very novel and
not recognized
but actually these calculations
neurobiologically are not uh bright line
dichotomus categories they are
probabilistic and they are Spectrum but
we course grain them in the mind often
into yes or no familiar or novel so that
subjective impression of yes I know this
person or not is just the co grading of
the continuous
variable representational capacity is
important uh the Deep generative model
implies that the agent is capable of
planning into the future which in turn
requires an ability to generate store
and retrieve counterfactual data
generate store and retrieve is
memory so therefore when the
representational capacity is completely
or partially impaired for any reason the
agent completely or partially loses an
ability to infer leading to
depersonalization and
derealization if we talk about health
and pathology briefly the two key terms
here are voluntary and involuntary you
know again all of US during our day we
can either sit down to plan things you

know for you know 10 year temporal
Horizon which expands our temporal uh
depth or when we uh meditate or uh rock
climb you know then we can be more in
the moment in the here and
now but uh trauma doesn't ask permission
I mean perpetrators are you know there's
nothing voluntary about it and so that's
an involuntary collapse of the temporal
depth uh which usually is leading to
pathological
conditions this is a perspective from
the nonlinear dynamical systems you know
if again I am in uh you know adult
healthy alert functioning and I may have
small deviations from ime say I'm
daydreaming or something like that and
the collection of these points from
which I reliably return to ime can be
described as the attractor Basin and the
whole thing in terms of attractor
landscape is the set point attractor
regime I and ime would be the
variational free energy minimum and that
is a fairly stable system perhaps in the
autobiographical s right now in order to
for that system to change some energy uh
needs to be applied and with external
interference of sufficient power this
attractor landscape can change when that
power is applied temporarily there's a
phase transition the attractor landscape
evolves into something else perhaps
during that transition you have a chotic
regime and if we are in the case of
trauma then the functioning can be
chaotic for a while but then it may
settle down so to speak into one of the

possible posttraumatic presentations
there may be multiple posttraumatic
presentations and one of the possible
options is depicted here this is not
again Universal but you know it's one of
the possible options what you see here
is a multiple coexisting uh point
attractors represented by local Minima
of the variation free
energy and they can correspond to Alters
in
their uh and in this example you also
see that they're surrounded by a global
minimum of variation of free energy now
also if you look at the uh area the
region between the local Minima then in
dynamical systems that's called a
repeller and uh repellers are uh
notoriously chaotic so uh that suggests
that you may not have very good
predictive ability of how you will
transition from state X to state Y or state
Z to say a few more words about
this when we switch between the alters
then we are in regime of itinerancy this
is the term Carl Frist likes to use a
lot and you know justifiably so also
when we have these transitions from X
State X to Y that the time is disrupted
you know in the idea you don't have this
flowing continuity of time there's a
series of disruptions effectively then
your uh coherent large light cone is
broken into uh pieces
here now imagine that this patient with
with this condition reached out for help
and they want to get better but this uh
attractor landscape already acquired

some stability and so again we need some interference to destabilize the regime the difference here would be that in therapy in the process of healing the this uh destabilization of the pathological attract landscape is voluntary controlled and guided ideally by a licensed and competent mental health professional and uh that's that's the uh the difference from the onset of uh acute trauma so Psychotherapy can be one such intervention and it's the treatment of choice we have very limited advocacy with medications and now it's very uh uh uh kind of um fash able to talk about augmentation with psychedelic treatment or perhaps neuros stimulation it's important to say that just destabilizing that landscape doesn't guarantee anything at all so when people present this idea that you just go to like I don't know and have some iasa or anything ketamine infusion and you're better is a myth because you know the important part is that the patient settles back into health and just the destabilization of this landscape doesn't lead to that you still need uh therapy you still need you know to process the trauma not just to destabilize the pathological regime um and that therapeutic process when you destabilize the the pathological attractive landscape is another phase transition temporarily temporarily chaotic with the idea that eventually it leads you to a healthier functioning either a trative landscape

corresponding to cohesive uh system or
at least a little more coherent system
with less Alters partial
Improvement let's talk a little bit more
about the phase transition so before
things happen in in in health we have a
nested and embedded uh
self and that can be described in in FB
as a hierarchy of coupled interconnected
attractors Carl friston you know uh
wrote that slower attractor can possibly
control the phase transition of the fast
attractor and in our system the core
self is is feeling BAS it's effective
and it's one of the components uh of uh
effects of feelings is generalized
arousal and we know that the CH changes
in generalized arousal level can lead to
the phase transitions of the entire
neocortex from periodic to chaotic State
and back so this is one of the possible
uh kind of you know uh explanations in
our model how this works the phase
transitions um you know a few more uh um
thoughts about it trauma is not a um
guided transition it's an effect of
storm it's a flood of cortisol that is
abrupt and severe and that uh abrupt
increaseing generalized arousal leading
to the destabilization of a healthy
attractor landscape in the
autobiographical self but then when
there's an onset of a dissociative
syndrome after trauma then uh the regime
settles down to a lower energy uh
perhaps effective numbness that patients
talk about and in a a Psychotherapy not
immediately but when the is ready which

can take a year or more you know when there's active trauma work then there is again an increase in energy level with the idea eventually to settle down to a healthier regime of functioning a few clinical observations starting from dissociative disorders which are conceptualized traditionally as uh defensive response to prolonged and inescapable trauma clinicians talk about uh and patients report this this qual of emotional flatness or deadness or numbing sometimes they call it freeze so in our model it corresponds to issues with Communications between the core self and the peripheral self components for example that autobiographical self in this regime operates as if it is uninformed by the vital emotional informational flows that originate in the core self one of the possible ways it can be achieved not the only way but one of them is the frontal prefrontal cortex inhibiting the limbic structures that's the cortical limbic inhibition hypothesis and in FB terms that means lowering of precision associated with effective prediction error

messages um we're moving on to also severe dissociations but temporary such as general anesthesia propofol and uh uh uh it's a very profound change in the body that influences the Gap Junctions between the cells and recovery from anesthesia is a very interesting process Michael Lan is

fascinated with that it's actually a little bit of a miracle that we come back to be the same person we have been before and some people don't uh you may wake up a pirate uh and uh these conditions are called postoperative delirium dissociative Amnesia post-operative cognitive dysfunction this is also an illustration itself self is computed in plastic and malleable it's not a hard um you know Hardware um and perhaps we get oriented with the use of our memory systems including the episodic memories another cases a seizure where not only autobiographical self is impaired but also the social and the bodily self and one of the reasons for that is the necessary resources for the functioning of the self components are not operational in seizure uh moderate dissociations uh in neurological cases but permanent ones are for example the perhaps one of the most famous neurological patients hm who had a bilateral hippocampus damage uh and uh in Britain there's Clyde wearing and an Neil Seth describes Clyde wearing in his book being you these patients do not have episodic memory encoding and effectively they leave in about 10 minutes of the here and now uh there is no past and no future beyond that and if you read Clyde wearing's diaries that he talks about the series of Awakenings he said oh I just woke up and 10 minutes later oh I woke up and then I woke up and if you show this DVD to a clinician who is an expert in did they

will which I've done you know I talked to Roxy wolf one of the fellows of the international Society for the study of f associations and she said that you know this patient looks perpetually depersonalized there is no continuity in so Coro patients they don't recall the past they can fabricate the past and in Alzheimer's in early phases there's issue with recent autobiographical memories but some access to the remote autobiographical memories all of these cases are discontinuity in autobiographical self um uh more frequent perhaps cases or substances ketamine does of THC or iasas lead to depersonalization and dualization and importantly if you read the studies then these patients actually report they tell you that their sense of time is impaired not only that the planning is impaired as well um if you have an alcohol blackout and the person tells you on Wednesday that they have no idea whatsoever what happened on Tuesday after 6 then uh this is a discontinuity this is a gap in episodic memory uh there's also something called traumatic Amnesia this very sad cases of high school shootings where you know the person literally doesn't remember you know 15 minutes of what was happening in cafeteria what happens in acute PTSD is that you know uh hippocampus is releasing cortisol receptors and there's a flood of cortisol and the effectively the episodic memory of trauma the cohesive context contextual time stamp memory is

not laid down in a stable fashion you have shards of glass you have phobic memory semantic memories procedural memories but not the episodic one which is also discontinuity in autobiographical self and that is involuntary uh and we talked about mild benign everyday uh dissociations uh next time you wait up you know maybe tomorrow the day after you know try to monitor how what's going on uh and you may catch that pH of what met Syer calls you know a non-ic self where things are just flowing through your mind but your eye is not fully out borous here uh to summarize the level of Consciousness regime of Consciousness pathological States and all the underlying resources necessary for the successful operation of each self component collectively influence the complex dynamics of the system um a neurobiological perspective briefly the historically most famous one is uh cortical limic inhibition hypothesis by Sierra and barus who postulated that uh prefrontal cortex is uh inhibiting the lyic structures including the amygdala and the interior singulate cortex ACC the subsequent studies don't fully support that hypothesis but partially support it I like the work by Ruth lenius and her colleagues and guess what you know her uh revision her refinement of the cortical limic inhibition adds Dynamics transience context and itinerancy all the things were

like uh uh they talk about two regimes
even in patients with dissociative
syndromes over modulation and under
modulation again itinerancy in not one
regime but multiple regimes over
modulation is indeed hypoactive amigdala
and hyperactive
prefrontal cortex which is described
above but also these patients less
frequently but but you can see it are in
a reverse regime where amygdala is active
and and PFC is less
active uh uh over modulation dominates
in dissociative uh conditions they
refined the anatomical Network and
talked about different sub regions of
amygdala the central medial amygdala based
lateral amygdala dorsal lateral periductal
gray ventral lateral periductal gray
Thalamus and other Reg
and this network of regions collectively
Works to create specific posttraumatic
regimes they also described uh the
regime where patient has access to
painful memories usually uh is
accompanied by under modulation very
active state with amigdala active and uh
when they don't have access to painful
memories they're more in a dissociative
State we can also briefly talk about
what are the anatomical substrates and
the neuro psychological substrates of
the temporal depth collapse and again
acute trauma hippocampus complete damage and
dementia are affect the episodic memory
system uh if there's an insulin damage
then you can expect the disruption in
the bodily memories and perhaps in

ketamine use you may have a dysfunction of prefrontal Cortex and frontoparietal network which renders the work in memory uh of the patient dysfunctional you can imagine other circumstances when you have PFC in front of paral network impaired that usually leads to a psychotic state where you know it's very hard to think you know we need work in memory to be able to think um about dissociative disorders again the point made at the very beginning that prolonged inescapable exposure to highly stressful environments can be seen as lasting stress beyond the agent's ability to manage which results in the breakdown of um tame light going into the uh components fragmentation decoherent in discontinuity when the landscape uh attractor and repell Landscape corresponding to a pathological condition takes root and in contrast to that if you look at the single uh episode of canine intoxication then you have transient working memory disruption leading to transient dissociation again to summarize lasting or temporary severe or mild dissociative experiences are accompanied by the collapse of the self's temporal depth and we are ready for a discussion which I very much look forward to thank you Lexi great presentation thank okay where should we begin what would be interesting or or given where the work has gotten over the last few

weeks where what do you feel like is relevant well perhaps any colleagues have comments I know hus is with us and he's a esteemed neuros psychiatrist and he sees a lot of patients and perhaps neurological patients and he can comment and some of the things we've mentioned I had a uh clinical meeting with with uh colleagues who do a lot of dissociative works and I mean I'm seeking feedback and critique and so whatever people have whatever thoughts people have yeah if there's anyone in live chat they can write a question a few pieces that stuck out differently in how you presented it uh in comparison from working on the paper was the narrative self's role in maintenance of

The Narrative of

unity and about what is the experiencer that or is that all it can do why does it do

that why does the autobiographical self have that kind of a voice or a cognitive function yeah is audio working okay can we hear each other I I think why is a tricky question you know it might also be tell logical question I would say that it's probably functional you know uh for us to have coherent selves uh and um I don't know why uh but um we are observing it so we have descriptive level we have observation that I feel like the same elixi and it's a little bit of a miracle that as we know that the cells in the organisms they are they die and they're born but I still feel like the the same person even

so uh we change developmentally I'm not the same person as I have been at age 5 or age 15 but I still maintain this story if you wish this believe that I am the same and um I also protect the internal milieu from external threats you know of the same organism and if we look at other species say an oak tree that you know uh grows from an acorn it is still the same Oak it's the same organism uh so um I I perhaps in biology as well there's a need uh for the system inside the mark of blanket to maintain that saying whatever is inside the mark of blanket is the same me but it is a belief you know and uh um I don't know if I'm answering your question maybe I'm just free associating to it

so is that a clinically valid method um well it it reminds me of mortal Computing and there's a lot of focus on how behavioral activities in a niche relate to the material stability and I think you really drew carefully on Mike Levan all's work to talk about the analogous

Dynamics on a landscape of a narrative self and that's just kind of that was like a trampoline but then you also complicated that with multiple selves like the social and the narrative and so then there are these multiple disassociative or like it's kind of they each have their own unity and yet they can be discordant and

unified yes and so then is there any
general framework for different
possibilities open-endedly of different
regimes these are great questions or
even maybe challenges to develop it
further but I agree with you if we look
in the bodily self right I don't
perceive my hand to be a separate entity
I believe I I feel very to be part of
the whole organism myself my body right
and so in the bodily self there's also a
level of unity I'm just saying that this
is again a computation within that
specific subsystem that my body is one
whole self and we can see how that gets
uh perhaps distorted in
somatoparaphrenia or even rubber handed
illusion where uh this hand is perceived
as not you know so uh and so we see that
the system is plastic and malleable and
and prone to change but and when when no
uh interference is happening it
continues to be cohesive as one body and
one mind you know this autobiographical
I I I've noticed you Ed the term
narrative uh self I think that's Anil
Seth's term and I prefer
autobiographical for the reason that uh
rats they have episodic memories and
they don't have our verbal abilities
um uh and so they they also have
episodic memories which are what we when
the contextual memories right and I
think that uh there's the same eye in
all autobiographical memories and that's
unity uh yeah yeah thanks yeah a few
comments it's it's it it it probably is
more accurate to say it's a narrative

cognitive capacity that encompasses
autobiographical or aloe biograph
graphical type narratives um and I'm
reminded of a NPR radio discussion from
a few days ago that I listened about
Menoir and about what Menoir was and
could it only be written when the person
did or didn't change a lot but what or
who doesn't
change in every single one of these ways
so that what would that mean like what
so that that that was kind of a funny
connection and then and then um do you
have any thoughts on that or have active
comment no I'm just enjoying your
comment I don't know if I have
a a thought on that
yeah you you've see many clinical
Memoirs and presentations of self-
understandings and all of this so
there's just so
many and it gets brought
out so
sparsely like and
indirectly and it makes a lot of sense
to the person who might be saying but
that that could you could be too or not
confident enough about
that interesting you brought it up thank
you for that Daniel I think that you
know when we theorize saying Dynamic
theories then we are usually think about
different sub agencies of the mind like
Freud for example in his you know
structural Mo model he had the eat egoo
and super egoo where I want pistachio
ice cream is the E and pistachio ice
cream is fat and calories that's the

super ego response and stash for ES is \$4 and you have three and a half is the ego uh but there's a whole bunch of them we can name a sort of a sub agency of an internal judge you know which is sort of super eish or a neutral Observer or uh an accountant who's running Itali or and all of this so we all have sort of a jury there of different kind of regimes of operation if you wish but uh the patient is often surprised because when they for example self- criticize and you talk about that in therapy and say do you notice the voice of your judge it's kind of you know uh not very fair I mean there's no misdemeanor everything's a felony and judge is present all the time like no it's not it's me I'm doing it you know I screwed up you know so again you know the default regime is is me myself and I that I am you know I am bad and when we we discuss this things and we open it up and we show the different kind of uh we show a theater with different voices or a jewelry with different representations it's usually kind of an Insight it's a surprise but that allows a bit more control um for the patient in the clinical work so okay I'll read a question in the live chat Pablo FM wrote thank you very much I find it super interesting I wonder why humans seek for drug experiences having been so decisive in human history coffee could be a great example tea tobacco alcohol have also been very popular among civilizations it's a great question I

don't know if I have a good answer it's
a huge question and there's a good paper
you can find it on Google scholar a paper
by me, Zeller, Mark, and other
colleagues. I think Yakovlev was alive
it's called why depression feels bad and
what do addicts really want and it's
based on effective neuroscience and what
it postulates is that
uh of course there are individuals who
have multiple drugs that they are you
know using all the time so there's an
upper and the downer and other things
but there are usually some innate kind
of predispositions where people who are
wound up in tents they sometimes crave
alcohol because it's uh binding to the
GABA receptors and it's sort of you
know sedating us a little bit it makes
you a little more less kind of wired up
people who are very painfully lonely and
uh uh detached and they crave human uh
comfort and you know to feel together
and loved and accepted they may if they
discover opioids that would be a you
know lock and key match that would be a
dangerous
combination um and uh perhaps when
somebody's oriented to Performance and
they're very bored all the time and they
want to do something you know to do better
and more they may gravitate toward uh
stimulants and so there's these things
uh and there's some common factors among
different agents uh such as you know I
believe that multiple substances they
have the dopaminergic effect and you
know a lot of motivations are subjective

in individuals I've had patients say
that I'm smoking po to slow down time
I've had patients tell me that you know
PO is my oldest friend I've been doing
it longer than any person that I've been
with uh some people seek this state of
numbness and they want to kind of not
feel as acutely they want to sort of
feel desensitized

so these states of dissociation can be
desired um and that's briefly I guess
what I what I wanted to say about it so
well thank

you I'm reminded of a comment from ma in
the live chat earlier this morning and
she was kind of pointing to
how one mode of how we've seen active
inference applied in these presentations
is looking like athology at the
sequences of behavior but not really
looking at the self-maintenance as a
behavior like especially talking about
digital

agents or or the it's just about what do
they perceive what do they do and it's
taken as a granted or a separate
question how they
exist yet also there's this self
maintenance self

reconstruction character that doesn't
even need to have anything to do with
the athology

its Dynamics can just be like
morphologically playing
out not necessarily doing some cognitive
task out in the world in any
particularly effective
way yeah it's a very interesting thought

and for some reason I'll allow myself
the Liberty to reassociate again it
reminded me of yal Harari who when
interviewed he said you guys go to the
gym You observe your diet and you take
care of your body and he said I take
care of my mind I meditate an hour a day
and uh he goes to silent Retreat a month
and a year and the interviewer
challenged him I think he was from The
Daily Show he said how do you manage
that and harar said I got no kids you
know so but in addition to that he's
also in therapy so there's a uh that
idea of really maintaining the mind and
not just the body with behaviors we're
all oriented to what do I do do do what
do I think think think which are sort of
verbal virtual behaviors but how do you
feel as my great colleague rajie uh
Warrior says we're human beings not
human doings we have a very tough time
being we're all about doing more that's
true and so uh yeah and also Harari
talks about informational hygiene where
a lot of stuff on the social media is
noise and trash and a lot of it is
consumed and we have pollution but you
know so he doesn't do you know he reads
books and he thinks in Millennia but I
don't know I mean I went sideways from
your comment but it just that's what
where my thought went so
yeah it it seemed like a very relevance
when people are discussing the the the
unity or the articulation of mind and
body which is kind of what this unified
promise on the cover of the textbook

suggests is what what would it be to
have a unified model of brain mind
behavior and so
on then the the analogy or the
connection with the physical and the
cognitive is
just it has to be taken as a starting
point not like a hot
take yeah
yeah but what you just said reminded me
when I read Michael Ians's work and got
more familiar with it I actually had
regret I'm like why were we not taught
that that was a bit of a revolution in
my mind it was a big pivot because if
you go through graduate program in
psychology it's all about the unity you
know I am the same you know cohesive and
coherent and the fact that we have
intelligence at the level of individual
cells and then the cell Collective and
then tissue and organ and all of this
thing that I think is one monadic Alexi
is a pyramid with Communications bottom
up and top down and laterally it's is a
revolution I think that you know I I
wonder what models of psychology can be
built with that and they don't exist yet
it's all about one big
block what happens to it I am depressed
you know there's no collectives this is
one big block
right yeah it's a really important in I
think that's the kind
of frictions and and experiences we'll
start to learn more about and see
produce in different ways like just
seeing how you connected the topics to

specific quotes translated the ontology
re
reworded descriptions using kind of like
a
translation sometimes really just as
simple as a lookup of words just authors
using one word and being able to see an
ontim ontology
term equivalent or roughly
related it's uh and
and what what does that look like or I
guess as your last comments what do you
think after the paper after this now
like what are the
directions towards which people would
say okay that's applying active
inference in a clinical
setting I my goal eventually I think is
to get to Therapeutics and for example
you know um actually I probably will not
use that example it's it's a good
material for the second paper that we'll
follow up with uh but I I do think that
you know the question is what do we do
in therapy differently how do we
understand how do we formulate cases
differently with this information I I
haven't heard it clearly said out loud
in the did literature there's a lot of
perspectives there's a lot of endless
complexity and there's a lot of nuance
but if we can highlight a central point
and say there is a temporal depth
collapse I think it allows us to do
something with it in clinical work and
in diagnostic work and that's my hope it
may not materialize but this is what I'm
hoping for so well even from a

researchers

ethological perspective it's kind of
like that's the transferable and the
interesting research question how could
you be

observing not on the inside and make
some kind of inference or assessment
that would matter clinically with no map
territory pretense not that it's 100%
reliability just taking a clear look at
Diagnostics available rethinking it from
dynamical systems perspective
maybe different uh

Biometrics maybe pupil it's like it's
not it's just kind of that's the open
question and then where it provides
clinical utility

that's where it

matters yeah thank you perfectly said

yeah I agree okay thank you

Alexi see you thank you Daniel see you
byebye

for e

all right that has concluded the live
presentations section of the
Symposium I'm now going to play the
pre-recorded videos so one second to
rearrange get it in

position now we'll continue with a
pre-recording

videos hopefully all of them

concatenated into the next few hours

okay this talk is by Leonardo MOA

Rodriguez Groove as the musical glue for
bonding ladies and gentlemen hello

welcome to today's talk on Groove as its
musical glue for social bonding my name
is Leonardo mul Rodriguez I'm a PhD

student at Sim and this is a
neuroscientific and crosscultural
perspective I've now I'm a bit
self-conscious by the fact that the only
audio you'll be hearing today is my
voice no music but do books sing No
we're doing a deep dive into the
scientific literature not musical
theater maybe musical theater in the
future of uh my active
inference well my my name is Leonardo
Müller Rodriguez I'm a PhD student at
the Cambridge Institute for music
therapy Research Center and my focus is
on sleep music four dimensions and if
you're curious how does this tie in with
danceability where relates to musical
entrainment U more specifically slow
beats I'm also a composer music producer
producer and musician so how does groove
Foster social bonding there's a little
outline of today's presentation Groove
and its relationship to Spotify
stability the Neuroscience of it the
social bonding the crosscultural aspect
flow States in music what musical
features Define Groove an active
inference application of GrooVe and a
later update to spot
uh suggestion so what is GR in essence
it's the Apex of an inverted u-shape
relationship between rhythmic complexity
and grooviness so and pleasure if you
wish uh
grooviness is at a moderate level of um
rhythmic complexity which is um rhythmic
complexity is if you use the musical
term syncopation is the amount to which

the Rhythm

deviates from the meter or the beat or alternatively you can think of the tempo the beat is a Tempo and it relates to how certain you have your predictions of that Rhythm so Groove is a pleasurable sens sensation of wanting to move the body to music driven by rhythmic patterns that balance predictability and surprise here's another study looking at the same thing and we'll see into the neuroscience in a bit that proves this now spotify's

danceability is a measure they use and they Define it as so we don't have the the actual computation that they do it's the suitability of a track for dancing based on a combination unknown combination unless you work at Spotify of tempo Rhythm stability beat strength or pulse Clarity and overall regularity and this data set um someone put this on on Reddit and they've looked at a million tracks and was like okay you beat me in terms of data sets um and so they've categorized all the musical genres several musical genres in terms of uh as a function of dance ability and based on the discussion and it makes sense um on Reddit it's more about rhythmic Simplicity seems to analyze because it's more more danceable music is techno deep house and then you have salsa music which is lower and people in the comments who are salsa daners like what what is going on and especially metal heavy rock or hard you know metal fans that were thinking

like wait is comedy more dancable than
my favorite runner so they're a bit
offended with that um I least sleep
music makes sense that it's kind of low
um yeah so it lacks this relationship
between the complexity and
predictability fun fact if you listen to
all the million tracks it will take you
seven years and nine months if you listen
to it on 24/7 basis here's my data set I
compared sleep soundtrack sleep
playlists um that are either very
popular on YouTube or that I've created
my own self or playlist that I have
millions of subscribers on Spotify to
high energy music such as Jim music that
has this beast mode playlist has a million
11 million uh subscribers or this region
uh music with region is a Latin popular
dance music very popular 11 million and
this has about five million and you see
the dance ability is in general higher
for this higher energy music compared to
sleep music and here compare vigilance
and high energy music is also more
positive apparently uh sleep music is
more negative anyway let's dive into the
Neuroscience of it and here the main
thing I want you to focus on is the fact
that beat
perception is present in the brain and
uh frequencies increase their strength
at the onset of the pulse here there are
different tempos that were used or
different speeds uh between uh pulses
and as you see there's always an
increase just after the uh pulse
regardless of the tempo it happens all

the time and it happens at the level of beta and Alpha brain waves and gamma brain waves these higher brain uh frequencies higher frequencies and um it activates these uh brain regions here and we'll go a bit more in detail that seem it's quite wide range and it turns out that we can call these the musical Rhythm brain Network because it makes use of the dopaminergic and rhythmic sorry the dopaminergic reward system in the brain um for beat perception and positive effect of groovy music and this involves a supplementary motor area premotor area nearby the pman as well as for pleasure and the urge to move to the music is a nucleus acaban and this connection to the medial orbital frontal cortex because we're talking about an auditory stimulus um auditor regions must be involved and uh the cereum also for fine motor movements turns out that musical Rhythm predictions and surprises are measurable in the bra in the brain on in two forms so the first one is about short-term habituation to a rhythm via repetition of a certain Rhythm and then the deviation from that the surprising change to that rhythm is seen here in this paper where this's this measure called the mismatch negativity which measures prediction errors in relation to Rhythm deviations changes or Omission you don't have that note that used to be in that Rhythm and the other form um of surprises and predictions is by the recruitment of long-term memory or uh we

can talk about it in terms of implicit music theory syntactic knowledge where here regardless of you're being told in advance that there will be a non-traditional chord sequence um that will be played you will have this sort of mismatch negativity which is more specific because it recruits long-term memory here is called the electric response in the auditory nerve area and yes that's an interesting aspect um by traditionally I mean traditional music western music theory um for harmonies here again we're just looking a bit deeper into this whereby beta waves we saw earlier and gamma waves The High Frequency uh waves us for early in terms of beat perception are responsible for top down so the beliefs the predictions better and the updating the surprises um through the gamma high frequency uh um activity in the brain so another aspect of music is that it's usually done in groups such as musical performance um appreciation music therapy or learning about music and it turns out that music in groups synchronizes brains and this is called interpersonal brain synchrony and in addition it recruits the same musical Rhythm Network that we just saw with a you know a bit more broadening of it it has the amygdala insula which is usually involved in visceral U activity and more generally the prefrontal cortex the Motor areas uh you know could involve actually moving if you're performing and the basal ganglia in general which

incises all the dopaminergic reward system which nuclear accents uh and dopaminergic system Etc put in cor we saw that earlier now music the real function of music beyond the auditory cheesecake that Stephen Pinker says it's not just pleasure not just report systems it has to do with social bonding and building trust music is a participatory um activity and dancing actually is a way to participate without any musical virtuosity or even familiarity to the music this is why I call it embodied Democratic music participation open tool and fosters joy and social bonding and music strengthens this interact uh interactive rituals you can talk about it in terms of ritual and it's a social tool for enhancing uh groups so as as you can see here you start to predict you the beat you see repetitive motions this induces Groove induces dancing to it and this enhances social bonding within this sort of you can frame it within an agent environment system moreover uh seeing others dance and move encourages dancing so this reinforces the social bonding aspect and this was exemplified here where eyes open uh with low Tempo High Tempo High roof actually increases the collective movement um in a group of people listening to music and this is what they were measuring here with this cross wavelet

and uh dance promotes also here
interpersonal movement interaction
promotes self other integration it
enhances social cohesion through
synchronized movements and more
specifically
moderate uh syncopation and low as well
for also there an implication for parkon
disease low and moderate syncopation
which is groovy
music um does so as well he
encourages self other
inclusion with social bonding and and
we're talking about music The Groove and
is it applicable to a crosscultural uh
definition of Music well first there has
been this is with Japanese music
actually there has been an optimal
Groove Tempo range identified um which
seems to be a u inverted u-shape
relationship non linear relationship
with Tempo is this seems to be from 100
107 to
126 BPM beats per minute uh Tempo and
The evolutionary interpretation for this
is the fact that it's near 120 BPM which
is the optimal tempo for walking and
dancing so this is where Groove might
come into sensory Moto synchronization
is most optimal in that around 120
BPM um and there are also universals
cross-cultural universals of music which
has been identified using statistics um
and the main aspects that are
crosscultural is Beats dancing making
music in group percussion repetitive
melodic sequences these are musical
features that are Universal just going a

bit deeper into this they used a music
for all around these spots it's quite
geographically varied data set although
only 304 recordings you know there's
millions of music but I think it's nice
General generalization we can make based
on the Black Box the black box here are
musical features that are at least
50% uh shared in globally and the most
uh which are the most you know the
whiteness is is shared um the most
shared beats subdivisions by binary tary
subdivisions or scales that have up to
seven notes so this is interesting and
this was control phenetically meaning
that this ensure that the these musical
patterns and associations are not solely
due to Shared cultural or evolutionary
histories another aspect that is
crosscultural musical bodily Sensations
and emotions what I want you to focus on
here is this correlation they asked um
Asian participants 5 500 people and
Western participants 500 people to
associate emotional labels to musical
features and there's a very strong
correlation
.94 uh between these so there's a sort
of crosscultural aspect here obviously
it's not all cultures maybe African
Latin and uh Middle Eastern cultures do
uh feel differently but here's just a
Latin um influence that's indeed
instrumental sinking in syncopated music
which is uh groovy music which here was
Samba um encourages the urge to dance
so Groove at least is present in Latin
American music now musical flow is not

restricted to groove what is flow flow
is a state of absorption total
absorption in an activity in other words
the feeling of losing yourself in the
moment uh flow state is usually
characterized as as a state of emotional
resilience against trauma and here
emotional expressivity is one of those
not

Groove uh uh musical features that
induces musical flow and here the two uh
main things I want you to focus on is
the blue um histograms which are states
of flow here were performers piano
performance and this is a musical
feature and so there was higher flow
during swelling Dynamics so crescendos
de crescendos which is volume um
fluctuations as well as um
repetition of um

Melodies and not only just any Melodies
but stepwise Melodies that are not too
far apart in terms of distance on the
keyboard um or in terms of notes so the
closer they are blue is again flates the
closer they are the The Melodies the
more flow inducing they
provoke um another musical element that
provokes flow that is not Groove is the
narrative Arc storytelling emotional
Journeys in music and this interesting
paper by colleague Annie hesit
identified stages emotional stages of
the hero's journey which is a a
narrative Arc found in myths
films popular films yeah and stories
narratives in general they've identified
emotional stages of the heroes journey

in patients and clients specifically in
this case was eating disorders but
applying to just more
generally undergoing music therapy so
most they were look they were doing
guided imagery with music which is a
type of music therapy
and depending on what they're saying the
clients

these themes were identifiable within
this uh the hero's journey narrative Arc
so that's very interesting um and this
is very important for music therapists
because this uh informs further the
scanning aspect that they usually do in
gim which is that they try to identify
using the music where is the emotional
stage that the client and the therapist
should start their um work emotional
work and so this is actually a emotional
journey is are actually a concept that
is applied by DJs they usually talk
about energy arcs for their DJ sets um
and it's also very prevalent in film
soundtracks um you know film
soundtracks are using so music is being
used to guide the emotional journey of
audience now

flow as neural entrainment I think it's
neural entrainment is evident for beat
perception we've seen it several times
here's more evidence like accurate beat
perception increases neural entrainment
here is just saying that beat
sensitivity uh there's a sort of tempo
range where this sensory motor
synchronization decreases uh with very
slow Tempe um but here again beat

perception this the beat is being um
being identified in a neural activity
and even if you imagine certain rhythms
these frequencies appear also in the
brain activity so neural train with beat
perception is clear but how about for
emotional expressivity and um emotional
Journeys well this

study um reconstructed the music that
was listened to by the uh participants
from their neural patterns it was a
combination of EG and fmri technique
that made use of EG fmri and deep neural
networks to reconstruct the music based
on the neural

activity and you can see that not only
very impressively resembles quite a lot
in terms of spectrogram information but
obviously there's not just Rhythm
activity or yeah Rhythm information
that's being displayed there's more this
frequency activity

so this takes us to this study that
demonstrated that spectral flux is
actually the most

um accurate measure for neural
entrainment to music and you can see
this with this red line is the highest
um and with a neural

entrainment and in addition they've uh
demonstrated that it is a metric that
encompasses amplitude Envelope as well
as beats they they have mutual
information here and spectral flux or
spectral novelty is called is a measure
of Novel energy increases in the
frequency

spectrum um so that's an interesting

aspect now nevertheless as you can see
other strong neural synchronization
musical
features um in addition to spectral flux
are Tempo range so uh roughly as you can
see here with the orange uh line 60 to
120 BPM uh High familiarity with music
and pulse Clarity we we to perceive the
Beats so let's keep it simple what are
the groove's musical features two things
moderate rhythmic complexity within an
optimal range of around the walking
tempo of humans
um this these two Encompass most of
these characteristics syncopation Tempo
micro timing event density this has to
do with and even beat sence has to do
with rhythmic complex complexity soap of
attack is a variant of beat saliency a
low frequency band is the base enhances
dancing uh volume and then this is the
core progression and that now keep in
mind that sensory motor synchronization
is not limited to this uh Groove Tempo
range the sensory motor uh
synchronization Tempo range is shown to
be between 60 to to 180 BPM below and
above this the sensory motor
synchronization which is usually finger
tapping or dancing to the Beats
decreases um which is probably linked to
biomechanical
reasons um however probably training
increases your capacity to do so uh to
synchronize however the beat perception
still can occur Beyond those uh temp
this Tempo
range uh I recommend watching this

YouTube video by Adan NE which uh goes into this uh the extremes of be perception let's apply active inference into two Groove but first before we do that we need to understand that the brain is constantly generating predictions about how the world Works in order to minimize the uncertainty about how the world Works two the brain is a layer thing and it balances top down down and bottom up influences where higher layers send beliefs about the world down to lower sensory layers which these sensory layers as we can see here um actually send back up sensory stimuli to adjust the uh beliefs through the mechanism of minimization of prediction errors or minimization of free energy now talk about Groove roof balances confident prediction uh confident predictions with surprise about the Rhythm uh just reminding you it is an optimal playing with belief updating about musical rhythms now listening to groove is accessible to groups of people and therefore can become a shared stimulus for motor and emotional synchrony and therefore under the inference uh framework um generates generalized synchrony uh which means that people have shared World models the listeners have shared World and this is depicted here um uh with this quote it says neither a model of my behavior or your behavior but a model of our Behavior

quote by

friston um

and generalized synchrony actually

requires flexibility in terms of your

higher uh

beliefs um and because you're trying to

model uh flexibly your Collective

identity and Collective behavior and

because of this and the social bonding

aspect I place musical groove on the

spectrum of the Rebus hypoth hypothesis

which is a relaxed belief under

psychedelics meaning that

it's not as uh the the flattening of the

free energy landscape is not as Extreme

as psychedelics are supposed to do but

musical Groove can achieve this through

the the social

bonding so dancing to the same music not

only increases trust and social bonding

but it also allows for individual

creativity compare an expert dancer and

a novel dancer um a novice dancer and an

expert won't be having m mimicry but

they will be sharing emotional synchrony

and this uh provides the mechanism for

trust and

bonding while considering uh creativity

of individuals and you can make use of a

Federated belief multiperson system um

model

Groove as the shared marginal likelihood

that people nonverbally communicate

um so because idiosyncratic dance moves

have been shown to even increase more uh

the social bonding and Trust I encourage

you next time to take the risk and bust

on some moves on the on the dance Pro

all right so what are what I want you to consider right now

is the current models of the context that encourages dancing yes they say musical properties they have temporal predictions that you can do induces pleasure induces arousal and this together with regularity chis the urg to move to the

music and um this will generate motor planning or policy generation under the active influence uh framework and and this desire to move and in training um body movements create social connection

I like to add a bit more to this in terms of contextual factors BAS on the literature review we're presenting today this is a hypothetical contextual Factor um framework for

danceability um basically Groove is the mechanism for uh

dancing but there are several contextual layers that people individuals might need to cover um before starting to dance for instance I have two example individuals a white individual here this white Arrow who

requires um seeing other people seeing their friends to before starting to move whereas this red individual just requires to feel a change in emotions to start wanting to move and these layers uh have a nonlinear relationship with the groove meaning that uh your entry level contextual entry level to start dancing um might be seeing a friend starting to dance and you might not need familiarity with the music to start

dancing to start connecting to the groove uh whereas DJs sometimes make us a familiar music in their DJ sets to encourage people to start dancing they say getting those first three people to dance is always the most difficult part and uh once you have them the crowd joins and you have a crowd dancing so this was modeled with a reduction of entropy to do relatively simply uh whereby depending on the distance to the groove there's a higher um entropy level that has a nonlinear relationship uh with grooving and K has is based on personality and the relationship between each layer there's another way of conceptualizing this as a the gravity of Groove the layers uh are done like this so implications for Spotify's danceability we saw that is more likely has to do with rhythmic complexity comedy is more danceable than uh Blues apparently or or heavy metal head banging is not as you know dancing as as laughing veence as well the most uh joyful music seems to be ASA and the less joyful music is sleep maybe Spotify is inspired by uh the rapper Nas who has a lyric saying sleep is a cousin of death so let's go beyond this uh and uh uh Beyond Rhythm and Tempo and incorporate what we've been seeing today into Danceability so if you want to generate a formula that goes beyond the ability unknown equation which is most lik

rhythmic Simplicity we
convert uh we Sorry first uh want to
point you to the study which asked
musicians to perform groovy music uh
with simple and complex M Melodies and
they show that
there's about 30% occurrence of fast
notes specifically 16th notes um when
people perform groovy music and this was
also confirmed in terms of listeners
saying yeah that's groovy um and so here
I've just uh used the optimal Groove
range and transformed it into seconds so
you can have an idea and ah too quick
for Spotify they can't no I'm just
kidding uh you get to see the the the
equation um so what I've done is I made
use of uh spectral flux which is this
spectral novelty makes use of for
transform and then logarithmic
compression to enhance uh weakened
spectral components and then you compute
the difference between time frames in
terms of positive energy
increases and so it measures positive
increases in Energy Event density is
another measure that's been used in
these studies and it's a number of
events onsets per window so here's a
Groove uh formula I suggest which is
open to to your feedback um which is 30%
of energy
onsets uh that is in the 16th of a note
um time
window um divided by a
measure that's dependent on the tempo
and it measures four times the beat
because most music is on 4x4 meter music

so I want to give you this quote dance
is an energetic mode of of participation
for a large population regardless of age
familiarity with music or instrumental
singing

virtuosity to summarize we found that
Groove is a musical mechanism for social
bonding through movement and therefore
through movement is danceability right
and we can make use of the active
inference to model group as a musical
mechanism for generalized synchrony even
uh as Federated beliefs to um model
individual creativity further Works um
would encourage the usage of uh um other
social bonding metrics in music such as
group

singing uh chanting is seen around
rituals around the world and this has
implications for group music therapy
life performances and

team building music recommend systems
Etc so please do comp contact me I'm
open to collaborate for this uh Groove
um project and discuss any of your ideas
uh we could discuss about musical uh uh
you know exchange music if you want and
I just want to show here my YouTube
channel which explores thanks to the
YouTube channel I've actually learned
how active inference works you know they
say teaching is the best way of
memorizing

and in here for instance

I made explored Chris Fields um work in
active inference with the holographic
boundary um for social interactions
which is which could also be a a further

uh direction of of trying to to
conceptualize musical Groove in that
manner but shared marginal likelihood is
probably very effective as well uh thank
you so much please do ask asked me for
the references um I will send them they
have been sent or yeah I should be on
the chat by the way asking answering
some questions thank you very much for
your time see you very soon byebye
thanks

all right I'm going to play next a short
video from Anna Pereira and active
inference Institute research

fellow hey an A Prayer

here I was disappointed uh that there
was an

resolvable conflict uh during the time
of the Symposium but wanted to share uh
or contribute um a small update
nonetheless uh for those of you um who
haven't had a chance to hear from me
before um we've been looking at
accelerating and disseminating active
inference for the

Humanities uh we're actually just
starting to call it

aih uh and we continue to make progress
um and quietly inate in the background
we're not really at a point to do um a
large more solid update uh but we wanted
to quickly say hello and and say that we
are incubating uh we continue to be
available uh for collaborations or
conversations um others find
useful uh and also to provide an update
that uh we've been really grateful for
the opportunity to incubate within the

larger architecture has provided
several mutualistic affordances that have
been a really great construct within
the larger
a uh
space so we look forward to being in
touch uh at Future conferences and
symposiums and available for
conversations um in the meantime uh
wishing you a great Symposium and uh be
well

all
right the next talk is by aswin Paul
bridging biology and AI active inference
for biologically plausible
decision-making models this is a
15-minute

video hello everyone my name is ashwin
Paul I recently finished my PhD from
monash University in Australia and is
currently working as a machine learning
researcher in deep learning so today I'm
here to talk about using active
inference to model biologically
plausible decision making uh and thank
you active inference Institute for
organizing this

Symposium let us start uh
by mentioning a recent Cutting Edge
experiment that was developed by C labs
in Melbourne so what they managed to do
is they developed a chip a silicon chip
and they cultured real biological
neurons like mice cortical culture and
human cortical culture and they managed
to couple this chip with a computer game
like the game of pong that you see on
the screen right now so the experimental

result is that given meaningful feedback these neural cultures actually demonstrate relative improvement over time in playing this game as you can see from this graph right like the MCC and secc neurons actually demonstrate that they can actually improve in game uh play when compared to Baseline groups like the silico um agent or the control groups so given such advancements in experiments from a theoretical perspective active inference is a great candidate to model such experiments and biologically plausible decision making so we have emerging literature and a lot of evidence um suggesting this so when we actually try and do modeling there is a clear need of scalability in the active inference literature where we are used to um modeling very small environments with very low number of states uh and uh very low number of actions Etc so in order to actually um use active inference in such scenarios we have to scale up the existing active inference algorithms and in the first part of the talk I will be talking about some algorithm and techniques that we can use to scale up modeling decision- making and active inference and in the second half I will talk about using these algorithms uh to actually uh model the particular experiment that we saw earlier so when we talk about um scaling up active inference or scaling up decision- making in active inference the first technique I want to talk about is

evaluating the decision making backwards in time and not forward in time as we usually see in the active inference literature so when we define the expected free energy and compute which action should be taken given a particular observation rather than evaluating expected free energy or computing it forward in time as we see in techniques like sophisticated inference we can actually use dynamic programming and model decision making backwards in time and this result is actually published in paper in this particular paper if you are interested in this particular algorithm and a more general version of this algorithm was recently published as the inductive inference algorithm by our colleagues in versus AI and in that paper we note that the inductive inference algorithm scales up decision making drastically and can be applied to bigger generative models another technique that we can employ in this regard is to actually learn prior preferences and in order to do that we actually take inspiration from the optimal control literature and this particular paper and this particular paper introduces an algorithm called said learning that improves scalability and efficiently computes the optimal action when compared to cutting edge algorithms like the Q learning algorithm and a particular Matrix in

this AI called the said Matrix um is actually very similar to uh the prior preferences in active inference and we can actually learn this set matrix orders of magnitude more efficient than um classic reinforcement learning algorithms and what we do in in um in this paper is that we show that we can actually learn prior preferences using a similar learning rule uh and once we learn this PR preferences then decision making is really easy because we don't really have to plan a lot forward in time and we also demonstrate that learning is robust in this particular algorithm and um talking about computational complexity both these methods one is DPF method that computes um expected feny backwards in time and the other one that um learns by preferences are actually orders of magnitude more efficient than the classical active inference we see in the literature or the recent um sophisticated inference and both these methods actually learn um um or scales linearly with time against the exponential computational complexity of decision making that we see in the active influence literature um so we tested these two algorithms uh in terms of its scalability and adaptability in um State spaces um as high as 900 uh and what we saw is that when we actually um introduced stochasticity into the grid these algorithms are more adaptable uh than um Cutting Edge reinforcement

learning algorithms like Dina Q uh and we demonstrate that these algorithms are scalable which gives us confidence to go ahead and model um experiments like this and when it comes to modeling neuronal Dynamics we already have existing active inference algorithms uh that model neuronal dynamics that is from um CBS ricken developed by Professor is and colleagues and in one of their Works they actually talk about um a similar experiment where um neural cultures are actually performing a perception task uh which is called B blind Source separation and in a more recent work they also talk about decision- making in neural cultures uh and um develop a decision- making model uh that can be equated to neuronal Dynamics so going into details um we have this algorithm called factual learning algorithm uh and uh they develop a version of the variation free energy uh that is term to term equivalent to a neural network cost function uh and this is valid for a particular class of neural network and more details can be found in this particular paper and there are two terms of interest uh in this algorithm one is a Time dependent risk parameter um and a Time dependent State action mapping that this particular agent um uses for decision making and because of um how it is developed it is biologically plausible and can be shown that every term is oneon-one related with the cost function that a particular class of neuronal networks actually minimize so

in one of our recent Works uh we um generalized this algorithm to have memory and we tested it in a similar um control task like the card P game that we see on the screen and we can actually draw comparison with the card Pole game and the pong game because both of them are reactive task where are where where the agent is supposed to actually react to the environment uh and take decisions quickly uh to balance this pole or to deflect the ball back um so we observe that um compared to planning based algorithms um like DQ and DPFE the counterfactual learning algorithms perform really well in this particular um task and gives us confidence to go ahead and model uh the biological um neurons or the or the kind of experiments that we saw earlier right and in addition to being able to model this uh we also have um the advantage of explainability in these algorithms unlike deep learning algorithms that are black boxes so if we actually putb the environment say in a particular episode we can actually see that the risk term is shooting up in that ex episode and um this um particular term captures um changes in behavior of the counterfactual learning agent and we can actually try to explain how decisions are being taken at every time Point uh unlike blackbox approaches so given these um several algorithms that we have can we use them to actually model um the manifestation of n natural intelligence uh in experiments like disin right so we

for that we actually develop uh an experimental based generative model that is based on this particular experiment so for example um this is the pong game environment that they use in the experiment um um precisely so for example um we have the x coordinate of the ball that being communicated to the dish brain chip uh through 38 distinct frequencies in the experiment um and uh the y coordinate of the paddle and the ball being communicated through a different voltage levels in the experiment um so for example like we have three observation modalities uh in this experiment and we note that these are the number of states and the ways of communicating that to the agent and we um uh enumerate the state space of our generative model um using this experimental details uh and match our generative model to be exactly like um the experimental generative process and we also have like the active inference parameters um in our generative model like uh the transition dynamics of a pom DP uh and counterfactual learning based State action mapping the risk term and the prior preference distribution Etc that we discussed in the talk earlier and given that we have this experimental based generative model we can model decision- making um of inico active inference agent and hope to compare them with the experimental result to draw insights and so on right so first of all we uh perform simulations using the see that counterfactual learning agent

um with memory Horizon um as low as two is actually uh performing at par with the HCC group in terms of performance so here CFL um T stands for um CFL memory Horizon T So as expected with more memory Horizon it performs um better uh and uh we can actually look at which memory Horizon is performing similar to the experimental group and so on and in one of our recent works we knowe that the active inference agents are incredibly data efficient compared to deep learning methods and it is only the active inference based model that is counterfactual learning uh learning method that demonstrate data efficiency at par with the biological neurons when compared to deep reinforcement learning algorithms like a2c and poo if you are familiar with the Deep reinforcement learning literature um so we also compare the planning based methods that we mentioned earlier like the dynamic programming method so in this particular task or the pong game um planning is not really useful so with increasing Horizon of planning um we don't really see um Improvement um and um planning Horizon of performs as good as HCC neurons and planning Beyond a point is not useful in this particular task uh because of the nature of the task and we also not um the explainability of our models for example if we look at the average risk um in the counterfactual learning method we see that the risk is actually only reducing um significantly uh with a

memory Horizon of four and which and this can also be correlated with the performance

Improvement that we saw um in the results graph where with memory Horizon 4 there was a significant jump in performance and these kind of explainable insights um that can be drawn from our uh models is a clear advantage over black Mo methods like um deep reinforcement learning uh and we we can also look at um the entropy of the CL parameters and we see that the entropy is decreasing over time that is the evidence in learning and parameters reveal um differences in behavior of different agents and that is a clear Advantage here and to summarize um we actually developed an experimental um informed generative model and simulated decision making in the particular um uh environment uh we tested different active inference based algorithms in this setting and uh we also noted the similarity of memory augmented counterfactual learning method um to the biological neurons and we are interested in investigating similarities um of the memory based CFL method more to the biological neurons and also insights like this can Inspire more experiments where um aspects like memory can be studied well uh there are several um limitations to this work where we haven't conducted um a really um involved statistical anal is in comparing these agents we are right now qualitatively comparing them um with um

average parameters but we would like to do more statistical analysis uh and we would also like to um Inspire um experiments from our theoretical predictions in our future work so I thank um all my collaborators in this work and if you're interested in this particular um work and the algorithms that I discussed uh you can actually U get access to all the papers here and uh feel free to contact me if you have suggestions or more questions about this work so thank you for your time uh and um I look forward to meeting you all when when I have a chance next thank you so much

all right the next talk is by Bert

Burgers towards autonomous Urban digital twins this is a 20 minute

video for the next 20 minutes I will talk about transport policy planning and how active inference could be applied through digital

twins urban planning is a very broad field encompassing everything that occurs in urban regions from the design and operation of infrastructure to social programs to scope this I will focus on the transport system giving a rough sketch of what active inference May entail for this domain modeling the transport system relies on operationalizing decision making many individual choices lead to busy roads therefore the main story line focuses on the difference between active and passive decision making the presentation is structured in three

parts first is the context give an outline of urban digital twins I will focus on livability as it is an objective of Transport policy next I will present some domain specific building blocks which are then combined in the final part finally I hypothesized that renormalizing generative models can be used as transport models concluding that active influence based modeling might require A New Perspective on trans transport planning first I will give some context in order to frame how active inference transport planning and digital twins come together the field of Transport planning is moving towards the development of digital twins these twins enable quick calculation of effect give an intervention hard than performing a comprehensive study per project which can take months a digital twin can be reused many times an example application would be the construction of a new neighborhood what are the effects on congestion air quality and inundation risk there are roughly five levels the first introduces a compelling visual interface often involving a 3D representation of the real urban region next is the addition of information on top of the visualization and the geographic data works the third step not only Maps data but incorporate different physics based models hence it is called

predictive transport models are one example others are airflow or water simulations high performance computing speeds up the scope of parameters which can be

tested the fourth level Builds on top of the predictive models by adding an interpretive layer the added value is scenario based planning where a range of outcomes is calculated for a hypothetical intervention instead of configuring parameters exactly you could propose a preferred outcome to the computer and that would only simulate what is necessary as it understands relations between models the last level introduces autonomy not only are physics based models configured intelligently as in step four if the T actively goes out to construct its

Niche transport policy evaluation is going to somewhat of a change and that before the development of Transport models allowed the Paradigm of predict and provide we could forecast traffic flows and accordingly invest in infrastructure to prevent congestion more recently the field has acknowledged that models trained on past data are inherently flawed in predicting in certain

Futures as such it is moving towards scenario based planning sometimes described as desired and provide along with this change comes a reflection on the philosophical context for Simplicity I rely on two Frameworks the first is

utilitarianism it relies on the
Assumption of Rational Choice enabling the
quantification of benefits and costs in
terms of utility and
disutility between every origin and
destination there are several
alternative Routes each has attributes
such as the distance traveled and the
cost combined with preferences we get
generalized travel cost
the rational Choice assumption enables
visibility this allows us to take the
ratio of estimated B parameters in
utility
functions these ratios can then be used
to calculate for example the monetary
value of travel
time in practice we always use monetary
value as the final measure so that we
can compare the cost of implementation
against the benefits
experienced finally utilitarianism leads to
aggregate increases in welfare at the
ignorance of the
distribution not only does this apply to
wealth but accessibility too take for
example the discrepancy in
infrastructure spending between core
cities and the
periphery to address these concerns
contemporary policy evaluation attempts
to move away from
utilitarianism instead of attempting to
relate all preferences to monetary value
we leave them
separate not only do we reject the
visibility of indicators but we also
consider many more than before shown in

this wheel spending from living environment accessibility health and safety finally the distribution of costs and benefits is explicitly considered and much work is now spent on reworking existing modeling Frameworks to make these distributions explicit there are three main objectives in planning infrastructure accessibility safety and livability capturing liveability is much more difficult than accessibility and safety livability is generally defined as the fit in an ecological relationship between resident and the living environment with regard to the spatial temporal scale of this relationship livability is local and sustainability global for practical reasons fit is operationalized through valuation the more an environment can satisfy one's preferences the greater the fit theoretically it has been long known that fit could also be operationalized as a continuous transactional process however little progress has been made and this is where active inference comes in these two approaches to livability are best defined by contrasting them to other on the one hand we have the static approach which is concerned with measurable comes on the other hand there is a dynamic approach in which livability is viewed as the hidden driving force behind that which is measurable by Framing liability is comprised of three components indicators percepts and needs desires I can show

that the location of fit determines which is which either it is a weighted function between percepts and our preferences through valuation or liability is a weighted function between indicators and percepts which are biased towards our preferences finally these two approaches might be complementary and this might be useful for modeling purposes there are many different transport models when it comes to those used for urban planning these can be distinguished into micro and macro simulations the micro scale simulates individual vehicles and their interactions meanwhile the macro scale applies constrained optimization over a network preferences constrain this optimization here I will focus on macro models which combine transport and land use patterns fundamentally we rely on seeing the transport system as a market Travelers generate demand while the Network provides Supply the four-step model forms the backbone of microscopic transport modeling and I will go over each step first is trip generation this step determines how many trips start and end in each Zone this is about determining the row and column sums see also the top right figure for trip production we look at residential zones and choose social demographic factors like household size car

ownership and income levels for trip
attraction we look at the number of jobs
shops or school
capacity trip distribution iteratively
balances row and column sums to fill the
origin destination

Matrix balancing is done using a gravity
model where the number of trips
increases with the mass but decreases
with the travel

resistance with the full origin
destination Matrix we can split these
flows per travel mode
used the utility of each mode can be
weighed against our order in the choice
set

logit models can be estimated for
different segments of the population u_h
that is the figure in the the equation
in the bottom

right finally we assign trips to Links
network according to wardrop's First
principle Travelers select routes to
minimize the travel time leading to an
equilibrium where no user can reduce the
travel time by

changing the entire workflow is
iterative because assignment affects
travel resistances due to
congestion not shown here is the
modeling of land use which affects
zonal

data next I will Outline Three
conceptual building blocks which will be
combined in the
synthesis as we just saw modeling
transport networks requires a
representation for example the travel

time per

link additionally we would like to

approximate human process ception for

example by using images or

sounds the field of urban representation

learning attempts to automatically

capture the content and structure of

cities luckily there is a strong

inductive prior from geography similar

things are closer together as such major

design choices revolve around which

measures of distance to

include examples are ukian distance

travel volumes or accessibility

generally an application of urb

representation learning relies on three

choices first is the need to discretize

the Uber region into spatial units each

acts as a clause the figures on The

Right Use

hexagons secondly is the type of data

data used where traditional transfer

models only only use tabular data we can

now also use street view images and

text last but not least is the way in

which parameters are structured and

learned craft neural networks are widely

used as they naturally represent the

transport Network Additionally the loss

function can be used to steer

training accessibility is a major field

of study in transport planning that

connects well with active inference

Concepts in urban an urban

representation learning we aim to

capture spatial similarity in edings

accessibility provides a theoretic iCal

foundation for which similarities to

capture intuitively accessibility is akin to Landscapes of affordances which are shared by everybody living in an urban region location based accessibility is theise from constrained Maximum entropy and can be configured to equal utility based accessibility the calculation is straightforward for each Zone we Sim the product of destination attractiveness and the deterrence function based on generalized travel cost the deterrence function captures how travel cost relates to similarity as shown in the maps accessibility patterns differ significantly between bike and car networks Hy is a wellestablished element of Transport networks think of how local streets feed into high into arterials and highways each operating at different speeds and distances where local streets facilitate access main roads enable throughput to capture this hierar key in urban representations we start by defining the study area creating Columns of course hexagons and then mapping them to the final resolutions I have used a unet to capture both bottom up and top down information each block in this net is a a graph convolutional neuron Network a precomputed accessibility graph sparely connects centroids of hexagons where the edges are the accessibility

value reconstruction loss ensures we capture input data while consistency loss ensures Columns of hexagons are integrated combining context and building blocks I will now go over the synthesis we started by discussing digital Twins and two approaches to livability static and dynamic the dynamic approach frames livability as Niche construction when it comes to policy the problem boils down to the objective function used good free energy acts as an objective function for policy evaluation unlike utility free cannot be sum the cost decision makers perception is itself is subjective meanwhile with utility we all live in the same fully observed world if Urban regions engage in Niche construction perhaps we can create digital twins that embody the n construction process autonomy then becomes about influencing an existing process rather than creating something from scratch there are three potential approaches agent based models on a hex crit simulating individual Travelers simulating each hexagon as a cell with the expectation of emergent Behavior or use a renormalizing generative model with a fixed hierarchy the third option is the most promising as it requires a single model than the combination of many smaller ones in line with this third approach I want

to consider a fixed hierarchy and renormalizing generative models are perfectly suited to this purpose the idea is that these might already be transport models as they upap Origins to destinations using cost functions sticking with the notion of a market equilibrium there's a demand and a supply side to the operation of renormalizing generative models just like in a four-step transfer model the demand side determines the landscape of afforded embeddings and these embeddings capture the spatial similarity the supply side aims to calculate the distribution of flows through the network it is just that here the network is a hierarchical structure not the physical transport Network on the demand side the landscape of a forness is or the landscape of afforded embeddings is perceived through the combination of top down prior and bottom up data from the top down we impose our cost function on traversing the lateral graph attributes on edges are valued against each other for example travel time and the price of petrol we also impose our preferences in term of the content through which to see so the content of the embeddings for example the concentration of emissions air from the bottom up we receive information for this lateral

accessibility graph like the travel time

at this

moment as well as the content of

discrete spatial unit and this content

is in the form of embeddings Ander from

high dimensional data using neural

networks so that they can so that this

process can be automated

on the supply side we have a landscape

of affordances at each layer in the

hierarchy and we need to pass messages

them supply is about narratives of

attention either about errors flowing up

or predictions moving

down to move laterally you have to go up

and over in doing so carve the transport

system at its joints

for example when you need to do

groceries you probably do not zoom out

to the courses layer instead you will

find CP which more quickly and walk a

few

blocks using active inference as a

framework to understand decision- making

in urban regions is still

speculative the standard workflow of

xante evaluation to study the benefits

of a project might not be applicable

anymore as the digital twin goes out to

construct its Niche on its

own the separation between technical

calculations and political motivations

in planning becomes blurred how do we

capture both travel choice and policy

objectives and one

model accessibility is already a central

topic yet it does now gained additional

consideration due to the reframing into

Landscapes of performances

these affordances may not only be in terms of T tabular data but can be high dimensional like

images accessibility heavily draws upon constrained maximum entropy basan mechanics treats this the same as free energy

minimization it therefore seems worthwhile to reframe accessibility in this way

too active inference is useful not only for autonomous Urban digital twins but also for level four twins to refresh level four twins use physics based models with the addition of an interpretive layer and this additional layer contains an understanding of the sensitivities between models which allows an urban planner to quickly Converge on a set of designs rather than simulate thousands of

Alternatives therefore a practical starting point would be to take level three digital Twins and upgrade them with an inference module to explore the parameter space

efficiently the apparent complementarity of static and dynamic approaches of liability suggests that we could use representations learned through valuation so a supervised um way of learning Urban representations to then use them in learning the structure of an active inference model

in s it is fair to say that autonomous Urban Digital Trends still need a lot of

work regardless considering free energy
as an alternative to utility is a
valuable exercise in understanding
decision- making in urban
regions lastly I would like to thank the
team of Symposium organizers for the
time and effort

all right the next video is 23 minutes
long iness hippolito dysfunctional
markof blankets from stuck states to
adaptation

hi everyone I'll be presenting today uh
dysfunctional Mark of blankets from
stock states to adaptation this is a
paper that has been um in progress for
quite some time and it is um a
collaboration with ma alasa Alex Kea and
Daniel

fredman so I want to start by uh
highlighting that there are stuck
systems or stuck St

all around us and that we find that
example in

neurocognitive um levels such as in OCD
patterns in PTSD trauma Cycles or in
addiction circuits then on the social or
ecological scale uh we find it for
example in poverty traps in eamers and
Market uh monopolies and climate tipping
points then in AI we also find them for
example in a large language models Loops
who isn't been stuck in a I apologize
kind of cycle then uh in reinforcement
learning local Minima in recommendation
filter Bubbles and in neural networks
training fails so what um these kinds of
systems uh share is that uh they are
some sort of critical threshold

uh they have um or present um self reinforcing loops and uh they have this persistence to uh change so they tend to remain in the same state hence why they are stuck so then the core problem is going to be a sort of a paradox so um the view is that systems adaptive or complex systems um coupled with the environment they should adapt but sometimes they don't so then uh from this perspective a few questions um become Salient why do flexible systems become so rigid and do not have that flexibility to adapt to the environment how do these adaptive boundaries uh between the system and the environment break down and what maintains that dysfunctional State what is there that must be so strong that um the system cannot uh change to a different uh or look for or act for uh to attain a different more preferable state so then of course that this presents um these are presented as Challengers because there is an everchanging world and the system that needs to adapt to that everchanging World um and uh it seems like um existing approaches um such as dynamical systems theory and network uh science um still have some limitations to quantify and address this kind of phenomenon that we call a dysfunction so then from our perspective uh we need um a framework that unifies that uh unifies this explanation so we need three things first we need to identify and characterize this functional States then

we need to provide a mechanistic explanation for the Stuck States and finally we need to offer a principled approach to designing these kinds of interventions and that's what we set up to do uh in this paper and that I'll be presenting you so um our framework Bridges um theoretical Frameworks uh from complex systems theory and the free energy principle the main or core um Concepts uh they are going to be useful are Notions such as emergence self-organization and adaptation from complex systems theory and coupling self-maintenance and free energy from the free energy principle by uh utilizing these two Frameworks uh we developed uh what a construct that we uh defined as dysfunctional Mark of blankets and with that uh we hope that we will offer a formal theory of a systems dysfunction and we hope to develop quantifiable measures for this phenomenon of stuckness and finally practical recovery strategy so that we can act upon uh the system that is stuck in a therapeutic way and we uh uh hope to kick the system out of that stuck state into a healthy State okay so as systems we all know that um they are comprised by components which interact and from that interaction we observe some sort of emergence um for example in the case of an immune response we have different immune cells each one with a specific

role and they interact in the form of self signaling Rec recognition of events and that that interaction then gives rise to emergence that we observe in the form of coordinated defense or system memory then uh from the free energy principle which we uh are very acquainted with uh in this conference uh we learn that systems that exist act to minimize their free energy and um further to that um the understanding or the Insight is that system are coupled with the environment and they act upon the environment to precisely uh attain or look for more preferable States for their self-maintenance and we can explain um that behavior through um active inference then another very useful uh construct is Mark of blankets of course the framework is a dysfunctional Mark of blankets and the mark of blanket uh as we all know is a statistical boundary between what is inside of the system and what is outside but further to that uh an important uh Insight is the conditional Independence between internal and external states which give um the system sort of a very important boundary um to have a distinction between the self and the no self and that can be seen also through the the notion of it's an operationally closed system A system that self organizes and maintains itself for that maintenance it's also quite important that it is thermodynamically open it is an open system it's not a closed system and

that's what allows them to remain alive
by adjusting to the environment and here
you see for example we have a a neurons
in the form of internal states that
could be part of a network
so a key Innovation concept that we
bring in is the notion of selectability
and selectability is a systems capacity
to explore navigate and select among
possible States these can also be seen
as the system having multistability many
points um that are afforded to the
system in the environment such that uh
the system can be healthy in that
navigation so you can see that from
example that comes represented here in
the form of state A B C or D different
points that the system can explore
within the environment and select the
most preferable ones and um the notion
of selectability allows us to um think
about State space accessibility or
affordances the adaptation capacity and
the boundary flexibility to adapt rather
than being rigid and stuck in a
particular state
so then uh within the free energy
principle we can think through that
Paradox it becomes if systems minimize
free energy if they Accord to the
principle why would they get stuck does
the principle not work or is it the
system that is uh not in a stuck state
so then uh what we realize is that um
what is not what is really within the
Paradox is that we cannot explain this
persistent maladaptive States and there
is not really a formal treatment of the

dysfunction so then there is a missing link between Theory and intervention and our solution is the dysfunctional Mark blankets there are two types of boundary dysfunction that we can observe uh in these systems that are not adjusted attuned adapt to the environment so they can come in the form of stuck States or blanket breakdown and stock States I mentioned this in the first slide and uh for example in the case of depression what we observe is rigid kinds of thought patterns reduced emotional flexibility impaired Social uh information processing and shortcuts bypassing healthy emotional regulation and in a more General way we can Define these stck states as the sil reinforcing kind of Cycles being trapped in deep energy Wells and being resistant to environmental perturbation then on the other hand we have blanket breakdown and an example of this is the immune system when the immune system attacks its own tissue and what happens here is a very interesting phenomenon of excessive permeability uh on the boundary of the blanket right loss of system Integrity so the self no self distinction the system doesn't know what's self and what is not self that's the loss of of system Integrity because of the boundary dissolution okay so then we arrive at uh the definition of the dysfunctional Mark of blankets and this this is a statistical boundary that um contrastingly has lost its adaptive

capacity for Effective mediation between internal and external States and the key characteristics of um the dysfunctional Mark of blankets are that the system is persistent in High free energy despite attempts and what we observe is the breakdown in information processing and the loss of effective environmental coupling I will unpack this further but let me just situate us um for a moment so uh we will go from the framework to practical tools to cross scale applications that's where we would like to go uh so to do that we need to First Define quantifiable measures then um identify a dysfunction index and finally the recovery metrics then as to practical tools we need adaptive pations in order to kick um the system out of the valley and Recovery strategies and these on more therapeutic kind of site and hopefully we'll be able to apply these cross across scales so now looking at the framework itself the first thing that we need to do is to um find that function index and what that is going to help us do is to measure and identify and delineate the this function it will measure the information flowing through the wrong channels like a diagnostic kind of tool that um spots the system bypass and the higher the value the more dysfunction for example like a cell with a compromised membrane then uh we will need also a recovery metric and this will be a pro a progress kind of indication indicator for us so

it shows a dysfunction reduction and the thought here is that um we ask ourselves are we fixing the bypass problem so then uh we will need uh to uh in order to advance without we need the landscape uh reshaping kind of like bringing out a new GE graphy of points uh that are afforded to the system making sure that we don't get stuck again and hopefully we attain the systems adaptability where we find the question can it be explore can it explore and respond flexibly to uh the environment so why are these two measures relevant well they are they will uh serve the role of being Universal tools across scales we will be able to apply them uh to uh all scales from cells to ecosystems and they will help us diagnose and track recovery and guide our next stage which is the intervention design okay so now moving to practical tools adaptive perturbations and Recovery strategies uh we have adaptive pations here playing a crucial role and um an adap itive perturbation is going to be a controlled intervention that temporarily disrupts a system state to facilitate adaptive change and what we mean by this or our goal is uh not simply to push a system out of being stuck but it is to help it to become more resilient precisely to make sure that it doesn't come back to the stuck State as you can see an illustration here on the graph below so uh then we Define the Adaptive perturbation design as in the first term

as traditional intervention approach
where we would like to nudge the system
towards uh these better states such that
it adapts based on the system's response
then with the middle term we can reshape
the energy landscape have a different
sort of more um flexible uh geography
that makes it harder to get stuck again
like smoothing out the deep valleys and
then finally uh we can promote
exploration and that means help the
system uh discover new affordances new
possibilities hence enhance
adaptability um there are kind of like
two examples of adaptive perturbations
that um I just want to mention and uh
one is by the parameter changes where
for example you can think of that as a
temperature modification in cellular
environments and the other one is the
external force or input an example of
that is transcranial uh magnetic uh
stimulation so why is this adaptive
perturbation relevant well it combines
three intervention strategies self
adjusting all parameter parameters will
adapt over time um and it uh includes a
balance between immediate improvement
longterm perturbation sorry longterm
prevention and future adaptability the
Practical impact of um an Adaptive
operation is that it will help us guide
for Designing interventions and work
across different scales and prevent the
kind of uh so to speak Band-Aid
Solutions
right so then why do perturbations matter in
the end well for the stuck States

they're going to be very relevant to help us overcome energy barriers um break self-reinforcing cycles and enable the system to have more States and the capacity to navigate um and explore uh the the environment and select the states that are going to be much more preferable to the continuation and maintenance and self-maintenance and self organization of the system itself in the case of the blanket breakdown which is this is the case of for example the immune system that has lost uh the capacity to differentiate between what is and what is no self and attacks um its own tissue then utilizing an Adaptive perturbation uh will be uh useful to restore boundary function to stabilize information flow and to reset System Dynamics and therefore uh repair uh the capacity of the system ultimately to recognize what's self and what is no self what is internal States what is external States okay so uh I will recap uh the journey now uh from the dysfunction to the recovery um so first we need to measure the dysfunction then we need to track the recovery then we will need to design interventions and we do that through the Adaptive perturbation to implement the recovery finally we need to monitor that kind of recovery such that we now have um been able to uh get the system to be adapted to the environment uh rather than uh just um in a very short

term so key relationships that come up within this journey include going from higher dysfunction to Stronger perturbation that is required going to better recovery to reduce intervention strength and going from more selectability to Greater adaptation capacity future directions are where we need to go is to extend uh these mathematical formalism to non-erotic systems to multiple time scale Dynamics and coupled systems interactions in order to do that we also need to uh develop measurement tools such as real-time dysfunction metrics early warning indicators uh that can predict a Tipping Point or a point of no return and then we can apply a preventive measure and cross scale validity tests some of the implemental implementation challenges that um we can see um include of course measurement intervention validation and scaling up and in order to address them we will need uh better tools more precise tools standard tests and infrastructure the potential impact of uh this framework um is like I said across skilles uh in healthcare for example we can look into uh Therapeutics for uh any sort of disease kind of cycle uh it includes mental health and aging patterns in climate action uh we can apply um this uh framework to tipping points to as a preventive measure and to Restoration in AI we can apply to uh develop more robust AI we can also apply

it to uh make more smoother ethical and governance uh measures for safety in AI as well as um can help us design bio inspired AI such that it's a system that is uh more adapted uh and hence more effective finally in Social systems we can also apply this framework for uh policy designed for economical stability and social cohesion and um coming to the end uh call for Action um so uh research priorities include developing standard standardized metrics and build shared data sets create open-source tools and establish cross-disciplinary teams so please join us in uh theory development tool creation application testing and knowledge sharing conclusion and takeaways so we went in this journey from uh these functional systems that we observe that are stucked in a local minimum and we want to develop a perturbation in order to help the system become more resilient and adapt to the environment out of the stuck State and hence a full recovery in the long term so then key takeaways uh are that we hope that we um have developed the foundations for a formal theory of system dysfunction quantifiable measures for the stuckness phenomenon and a practical recovery strategy that can be applied that can be further in the future applied to scales thank you so much for your time please do not hesitate to contact myself or any of the authors uh if you'd like

to uh engage or learn more about uh this work

all right keeping with the co-authorship theme next talk is a 25 minute recorded talk by Alex kefir resilience and active 2024 this is Alex

kefir bu this is Alex kefir I'm here to present some ideas on resilience and active inference um so this is largely just a unpacking in a careful way of some Concepts from this paper from 2022 that I was involved in um but also will explore some some novel directions um that this uh this thinking has taken since

then so the concept of resilience really if you look at the emology it's just it means like the capacity to recoil essentially or to or to to bounce back or just to survive through hardship and change um so obviously this concept is sort of a fundamental one if you're looking at uh life life forms um dynamical systems more generally um in the context of like um ecosystems and sustainability it's an important concept um so that's why we're interested in this um it's actually also sort of a fundamental Concept in Material Science so so since this paper has come out I've been approached a couple of times by by journals of Material Science for um for follow-ups and such um so as I said the the fundamental concept has to do with recoiling or bouncing back um but there's sort of there are some peripheral sort of different facets of this of this meaning and we explored all

of these in this paper um so roughly speaking you can think of resilience in sort of three ways um so first there's inertia um so inertia I mean I guess literally just means really uh sort of not changing um so persistence uh but you can also think of it as resistance in the sense that right unless you're unless you're exists in a vacuum uh then you're a system that's embedded in some broader system and inertia is basically continuing on your way despite changes uh that might that might occur or interactions with that broader system um so if you think of the red arrow as representing the path of a of a subsystem or an agent through its uh State space then and The Black Arrow is some kind of potential disturbance um then the top diagram here illustrates inertia um the central meaning again of resilience arguably is elasticity or the ability to bounce back so in that case you undergo some perturbation or change um and you end up recovering your initial doesn't have to be that you recover your exact initial position but the point is you make your way back to a reasonable state in terms of the kind of system that you are um then there's also this sort of third thing called we call plasticity or adaptation um so in this case case you begin in one configuration uh the red arrow at the beginning here and after the perturb uh the perturbation or disturbance you end up in a slightly different configuration so that should

be read not as just that you change your state like you update your um your beliefs on the time scale of inferences or something but you actually you actually wind up somewhat adjusting your model in response to this change um so this is more like something like AI stasis um whereas EI C is returning to the same state um so I'd say there are some close relationships just to to get into some philosophical conceptual analysis for a second here um the first thing uh inertia probably doesn't really exist in its pure form right in nature in a way or it if it does it it reduces to sort of elasticity so you know I mean presumably if you uh if you perturb a system um then you have to cause some change in the system however slight um otherwise there hasn't been a physical interaction right and inertia is going to look in practice just like the capacity to sort of instantly recover um from that tiny from that tiny change um there there is in the limit case there is sort of a different meaning here because again there might just not be a change but I think um I think in the context of realistic systems there will always be some some interaction um but AR ARA these two these two concepts are very close um plasticity seems like um on the face of it like a different thing because um well first of all uh the ability to be to be plastic in this sense sort of explains how you could come to have this property of the

elasticity um so if you look at the example of neuroplasticity for example um it's the ability to adapt to change and to basically update your synaptic weights to arrive at a new model that allows you to be uh in part allows you to be resilient in these first two senses um but there's another sense in which if you just assume that there's some property of the system that doesn't change through this um through this process which I think is a reasonable assumption in the case of uh you know biological systems um then there's going to be some kernel of the system that survives through change and then you're back to your sort of original

um okay so if we look at this concept the lens of the free energy principle um I mean first of all it turns out to be a very fundamental at you know as is often the case when you try to unpack um sort of uh fundamental properties of systems in nature in terms of the FP and active inference uh you find that there's a fairly direct mapping from the properties that the FP attributes to systems or um the types of systems that are well described by the FP and this uh and this concept of resilience um so I mean Carl Friston has said in discussions of this that essentially uh systems well described by the fdp are resilient um uh full stop so uh but we can look at at the three uh facets of the meaning of resilience here so um certainly systems described by the Fe

are they have this property of inertia
in the sense that um by definition right
they're things that have managed to
persist so they limit their exposure to
uh to unlikely States unlikely
observations um but they're also sort of
by definition elastic systems um because
because they have this attracting set so
um having an attracting set of States
means that um you know of course there
might be changes or perturbations that
are sort of catastrophic that destroy
the system but as long as the system is
not destroyed then by definition it's
going to make it make its way back to
this attracting set of States um so
resilience in the core sense is really a
um almost a defining feature of systems
that are described by the FP uh and then
finally plasticity you can think of as
something like Basi and model selection
um on some time scale and so that that
might seem like something extra but it's
only extra if the multiscale aspect of
the FP is extra and I would argue that
it really isn't because at its heart the
FP is a is a scale free description of
things and this multiscale and scale
free are sort of uh sort of um connected
Notions
um so that's
plasticity um okay but so most of the
interest I'd say of this lies in the the
mapping of resilience onto this this
formalism um of active inference so uh a
key concept uh in this analysis is the
notion of degeneracy and this is a uh
somewhat technical term um here we're

following we mainly follow on the definition of and in fact the analysis of degeneracy in terms of active inference in

uh SJ at all 2020 this is a great paper on this topic um so here we have just the expression for variational free energy um in the top row um is a expression in terms of basically a Helm Holtz free energy so it's an energy term minus an entropy term uh uh so if you just look at that for a second um the we've identified uh again following this paper from 2020 we've identified degeneracy with the entropy term um so we'll get into why that is in a second um but basically uh if the concept of degeneracy is useful redundancy um so this you can think of this as like complexity that that earns its keep or that's worth having um so complexity in this this bottom breakdown of variational free energy uh you uh you analyze it in terms of complexity which is a the the difference the K Divergence between the prior belief uh overstates the the generative density Over States States and the approximate posterior so this is kind of like how how difficult is it or how much change has to uh does the system have to undergo in order to represent the states that are um inferred from some observations um what's what's what's the departure between that posterior distribution and the prior distribution Over States uh that's complexity um minus accuracy um the the the expected uh log probability

of observations given

States um so what's the connection between complexity and this idea of useful redundancy uh sorry of just redundancy um there's there's a assumption here that complexity and redundancy are closely related um I think the basic idea is that so redundancy uh in this context is quite literally just duplication of States um so the I'd say the classic example is like an error correcting code so if you're trying to transmit some uh bit of information over for a channel uh you might you might send 10 copies of a bit in case in case some of those bits get flipped or corrupted uh and you want to make sure that the that the message gets transmitted um so that's redundancy it increases the complexity because um uh well one way of looking at it is that it uh it it increases the description length uh right this this K Divergence is like how how how much how much more do you have to say in order to communicate that an event drawn from this Q distribution has occurred um given that you assume this P distribution um so uh so you can think of it as well what these these extra bits are are the Redundant bits um so there there's a sort of conceptual connection between complexity and redundancy um and then degeneracy on this first definition is just the complexity that serves some purpose and we'll get into that in a moment um it's just also important to note that there's

this definition of um degeneracy as partial redundancy in the in the literature uh and this is partial overlap and function so um the idea is that um redundancy exists in a system uh sorry degeneracy resists uh exists to the extent that multiple states in the system serve some function in common but then those States also have um otherwise uh have distinct or disjoint sets of functions um and in practice I think these two almost always line up so there might be sort of uh limit edge cases in which um in which they're different but in but in practice um right if you have multiple States they'll always have some different properties and therefore um perform some different functions okay so so this is one of the core things that we pointed out in this paper um of ours so um this this actually sorry this breakdown is from the sjad all 2020 paper uh but we sort of have a different gloss on this C Center term here um so this is the complexity term in the variational free energy you can break this down in terms of um further break this down in terms of this uh expectation under Q of the negative modog probability of States under the generative model um so this is called a cost term in the 20120 paper um we describe this in epistemic terms as as an certainty term so this is U this is basically uh you can you can read this as the Shannon uh minimum description length of the of the states

given the generative density $p(y|x)$ but it's sort of the expected surprise $E[-\log p(y|x)]$ how surprising are these states that I that I infer $p(x|y)$ using the variational posterior under my generative model $p(y|x)$ so that's an uncertainty term $H(p(x|y))$ and then we subtract this degeneracy or the entropy over $p(x|y)$ over the variation of posterior Over States $p(x|y)$ so I mean first of all sorry I realized that I didn't quite make the the link here between $p(x|y)$ degeneracy and entropy so why is this the case uh why do we identify these uh it's it's because $p(x|y)$ this means if if the if if this $H(p(x|y))$ distribution Over States is higher in entropy it means that there's a $p(x|y)$ degenerate mapping in the likely Ood from states to observations right so meaning for multiple States uh you can get the same observation $p(y|x)$ and so then when you invert that likelihood $p(y|x)$ mapping in order to recover this posterior $p(x|y)$ this means that for one observation you might have several $p(x|y)$ several different states and the the spread of the of the distribution Over States of course is just the entropy so that's that's the link to to genery there uh so so this complexity term is uncertainty minus degeneracy $p(x|y)$ um one reason that's interesting is that when this complexity is zero $p(x|y)$ it's not as if the system is completely certain of what's going on $p(x|y)$ the uncertainty term actually remains it's just balanced by this degeneracy in other words all of the uncertainty in

this ideal case um is is useful um and so I think this really this really captures what what degeneracy in this context amounts to um it's uncertainty that useful and is it's useful precisely because there's some amount of uncertainty that's inherent in the source of the information you're trying to model right so um here p is the the in this context is the generative model that the system has of um of it's the sort of internal generative model of of the external States um and so there'll be some right there'll be some entropy some Source entropy associated with that system that that you're trying to model and and um so long as you match your variational posterior matches that degree of uncertainty um then you're in sort of an optimal um an optimal model so in other words for maximum model goodness try to ensure that um P equals Q um so uh this ties nicely into the idea of plasticity resilience is plasticity so your goal here is to adjust the posterior to match the generative model essentially um so you can do that um you know in in like perceptual inference uh you can try to infer a state uh a q distribution this is again an approximation to the posterior the true posterior over over those States under the generative model um but equally you can adjust the generative model in in a in a multiscale setting you can learn learn or adjust update the parameters of the generative density in order to match

what's in the external World um
so right this this complexity term um is
affected by changes to p as well as
changes in Q and so I think the
fundamental idea that I had here was
um in an ideal uh sort of optimal system
under the free energy principle you
should expect um first of all P will be
adjusted so as to model the environment
for sort of um basic you know um free
energy principle SLG good regulator
theorem types of reasons uh and then Q
also be adjusted to match to match P um
so this involves um this cybernetic
principle called The Law of requisite
variety which is sort of a um um it's
closely related to the good regul good
regulator theorem um but basically it
just says that you know if you're trying
to regulate or control some states then
you need uh at least as much uh variety
or uh in probabilistic terms that that
translates to variance um or or entropy
um you need as much Variety in the
controller as you have in the in the
environment and so it's a fairly
intuitive thing because you basically
need sufficiently many degrees of
freedom in the controller uh to be able
to map to to model what's in the
environment um so you can see this as
this degeneracy um property of resilient
systems or uh any system under the FP as
as an instance of this law of of
requisite variety it's also clearly by
the way a an obvious case of constrained
maximum entropy so there's work um
showing that that many many lines of

thought really showing that the active inference uh sorry the free energy principle is um is equivalent to a principle of constrained maximum entropy uh so here the constraint is this uh you can you can think of this as constraint by accuracy um but you you basically maximize this entropy term subject to that constraint uh subject to the generative model okay so um looking just a bit more closely at this notion of degeneracy um so I've said that it's Central um one pointed question you might ask here is well are all do all resilient systems uh have this degeneracy property uh and so I want to say the answer is yes um so the type of degeneracy that we've already talked about is this degeneracy and the likelihood mapping uh so this is multiple States mapping to the same observation um and again that's uh that can be cast as you'd expect to see that given um that you're performing maximum entropy inference um so that that is um crucial but equally important I'd say is degeneracy and the Dynamics uh so this characterizes just systems that again that have an attracting set of States so if these S_t are states at a given time and then S_{t+1} is the state or set of States at the next time step um so this in order for for a system to be resilient um it really has to be the case that this transition um uh that these transition probabilities or the Dynamics of the system are such that um that that

mapping itself is degenerate um so you know this means that basically you can perturb the system in several different ways and you'll still arrive uh will still end up at the same at the same uh state or or set of attracting States um obviously if the if the perturbation is too large then that might not be the case the system will diverge and you'll destroy you'll destroy it but by definition the systems that survive through change are the ones that have property um so there is a very deep link between resilience and degeneracy if you consider Dynamics as well um just a couple more points here one is um something that I've recently been thinking about it's it's it's actually a theme that's been on my mind for some time but I've recently written about it in some other work um and this is what what does the principle of requisite variety and um what do these concepts of degeneracy and resilience look like in in the context of multiscale active inference um but specifically here I don't I don't just mean um the fact that excuse me the fact that model parameters as well as states can change but I mean U multiscale active inference in the sense that you've got potentially higher level agents that are composed of lower level agents um so in that kind of setting uh requisite variety ends up looking like the the property of higher level agents that they're composed of a variety of different types of lower level agents um

so I mean I'd argue that if you combine this law of requisite variety with multiscale active inference in this sense then you really sort of automatically have something like a diverse ecosystem in which there are many different types of Agents that participate um so here in this diagram we're illustrating first of all just the same degenerate State observation mapping that we saw in the uh in the previous slides um but here the nodes are individual agents that compose this higher level

agent

um also as pointed out in this paper um by by um Carl friston Daniel fredman at all um the in this kind of context different agents can be expected to and I think empirically in some simulations are shown to take on the roles of these different types of States so of course the breakdown here is in terms of active and sensory and uh active sensory and internal States so here you've got active sensory and internal agents uh and that's one instance I would argue of this um application of the of the law of requisite

variety um to this higher level uh system um I also show here the the policy selection based on expected free energy for completeness um but that's sort of the main the main point is this this specialization and diversification of Agents um in this in this setting um in terms of the second definition of degeneracy you can also think of these

agents as playing um somewhat distinct but partially overlapping roles um and there's some some lots of interesting work on um on that sort of multi-agent perspective from the point of view of activating

um so that's really it I'm just going to conclude with some with sort of a philosophical question that I think is is kind of important for um for tying this this together and and sort of checking that it that it just makes sense um so I've said earlier that um any system well described by the FP is resilient so hold on a second uh not all systems are equally resilient though uh right teacups by are sort of the Paradigm of a fragile as opposed to resilient object so since the FP describes things in general doesn't it mean that all things are resilient um and doesn't that contradict this obvious fact so I think the answer here is that it's simply it's a matter of scale so yes all things are resilient all things that is things that are reidentified over time and that persist through change um all such things are resilient but it's a matter of scale so if you if you introduce a very slight perturbation to this teacup um very slight you know um relative to a certain a certain reference frame or a certain scale but large enough right relative to the scale on which that system is resilient then by definition it will survive right and so obviously again in a in a a real

Universe in which there are interactions and you're you're always interacting in some way dealing with some sort of um random fluctuations um then yes for the system to persist through time to be inert and elastic um it has some right the material has some uh coherence that allows it to to persist through those changes so it is a resilient system it's just that if you introduce a change that's um on a scale that's That's too great for the system to handle then it'll be destroyed um but sort of the just the basic coherence as a spatiotemporal object of this teacup is itself a form of resilience uh and then when we when we get into more complex forms of life more complex generative models that's when we when we when we begin to see more um impressive forms of resilience and shading it even into into plasticity and the ability to to really survive through more significant changes so thanks for for listening slw watching um I'm happy to participate in this Symposium um thanks to all the other uh the co-authors of this um paper by Mark Mol 2022 that was the departure point for this work uh yeah thank you all right the next video is going to be 28 minutes long it's from ma alasen sustainability under active inference hey everyone my name is mar baras a PhD candidate at the University dubec and director of research strategy adversus today I'm going to be talking

to you about budding work on sustainability under active inference which we had started with multiple authors that I listed down below uh starting from the core components and scaling up to a computational model we are pursuing this research in multiple collaborations ADV verses around sustainability ultimately this is meant to Charter a path to much larger projects so feel free to reach out if you want to

collaborate so let's begin with the most fundamental aspect of sustainability namely

resilience resilience refers to an agent or systems ability to persist and adapt in the face of challenges or perturbations under active inference resilience is a multifaceted concept involving different strategies that allow a system to maintain mainin or recover functional

stability resilience can be broken down into three core components inertia elasticity and

elasticity so let's start by defining inertia it describes an agent's capacity to resist change when subjected to disturbances it really is the stability and robustness of a system in active inference inertia is represented by high Precision beliefs which can be thought of as highly confident low entropy beliefs about the system's current state inertia is reflected in the Precision weighing of prediction errors

higher Precision implies stronger
resistance to updating beliefs
maintaining a stable State u means that
Precision is implemented as a scaling
Factor

so here higher Precision really implies
increased resistance to
change the problem is that inertia alone
makes systems brittle errors they cannot
adapt so then we move to the second
component

elasticity elasticity describes the
system's ability to return to
characteristic or retractor States after
perturbation it corresponds to the con
cept of homeostasis so the capacity
to adaptively

recover in active inference elasticity
is expressed in terms of an agent's
ability to minimize free energy and seek
stable

States it can be mapped onto the
relaxation back to attractor states
which is driven by minimizing free
energy as the system follows a
trajectory in doing so restores the
system to a preferred State analogous to
returning to an attracted plant so for
example in nature biological systems
really exhibit elasticity through
homeostatic feedback mechanisms like
temperature regulation or blood glucose
balance these systems have the ability
to change but they ultimately return to
where uh they were without necessarily
learning and this may yield the same
errors and ultimately can crap Under
Pressure much like a rubber band pulled

too

hard so let's move on to the third

plasticity it refers to the expansion of
an agent's repertoire of adaptive States

it allows systems to adapt not only by
returning to previous States but also by
exploring new States

enhancing

robustness so in active inference

plasticity is related to functional
redundancy and structural

degeneracy having multiple Pathways to
achieve the same

outcome structural degeneracy means that
you have different components perform
overlapping roles allowing the system to
adapt to novel challenges but we will
dig into this a little later

so here plasticity is captured through
posterior beliefs that can change
flexibly based on new sensory input or
structurally if you actually add a new
state we can use the variational density
such that the agent can represent a
diverse range of potential States
maintaining a dynamic model of its
environment think for example of neural
plasticity where different neural
circuits can adapt to perform similar
function after

damage to understand how plasticity
comes into play we really have to hone
in on certain key concept underlying it
so complexity refers to the uncertainty
or diversity of System

Dynamics in active

inference complexity is associated with
the range of possible States an agent

consider complexity is linked to entropy
of the system State

distribution higher entropy reflects
greater uncertainty and the number of
potential States the agent must account
for variational free energy is a way of
balancing two important things how
complex our beliefs are and how
accurately they match what we
observe and so we can U balance this in
a formula that goes $U = \Delta k - \frac{1}{2} \Delta^2$ equals the
difference between the complexity of
updating our beliefs and how accurately
these beliefs predict what we
observe so in this formula the first
part Δk measures the complexity or
the cost of updating our beliefs to it
line with reality

second part written as $\frac{1}{2} \Delta^2$ represents how
well our current beliefs can predict the
observations we

make so we said we were going to turn to
degeneracy and redundancy and I'm
obviously here leveraging work by no
and Carl

friston for degeneracy what we're
considering is that the presence of
different components are capable of
fulfilling similar roles with in a
system in her paper they talked about
for example left hand right hand and
redundancy means that you have similar
components performing the same function
which enhances robustness against
failure the redundancy here involves
overlapping functions in state
transition matrices B which indicates
similar transitions across different

states and

degeneracy can be Quantified by the entropy of the posterior distribution which represents the number of alternative configurations that lead to similar

outcomes and so managing complexity through diverse beliefs really means that the agents can adapt to changing conditions degeneracy means the agent can switch strategy when one pathway becomes unavailable which promotes adaptation and redundancy ensures the continuity of function when certain components fail which adds resilience to system so degeneracy here helps reduce the complexity cost by Distributing probabilities across multiple States thus maintaining low free energy and supporting robustness in diverse conditions

but while resilience is really critical for maintaining stability it doesn't fully capture the dynamic interaction between systems and their environments resilience focuses on an agent's ability to withstand or recover from

perturbations but it lacks insight into the broader System

Dynamics without considering systemic interdependencies focusing solely on resilience can lead to shortterm adaptation without promoting long-term stability this is work that Mark Miller and inish poo have uh written a lot about so for sustainability we really need that um systems not only adapt but

also ensure the long-term viability of their environment and resource base we can through active inference integrate resilience and sustainability and and show the need for systemwide coordination for harmonious interaction as Carl likes to say active inference really is just the story of resilient and sustainable systems so let's Define sustainability a little more sustainability is defined as the enduring capacity to meet needs without depleting crucial resources and here resources is taken a little more liberally we can expand Beyond material resources and Encompass social ecological and System Dynamics sustainable systems are those that persist over time and do so in a manner that promotes stability and health across all Dimensions environmental economic and social we achieve this under active inference by longterm adaptability and interconnectedness across multiple scales sustainability emphasizes the creation of conditions that allow for continued adaptation without exhausting resources effectively ensuring that all parts of the system work cohesively to support resource renewal and plasticity so sustainability can be framed as optimizing the expected free energy over deep temporal scales which is work that Casper Hesp uh has pioneered balancing immediate need needs with long-term system

viability so the optimization of expected free energy can be represented as $U(f)$ of expected U SMA equals the risk plus the ambiguity of X of f expected

s so risk quantifies the Divergence between the predicted paths I.E the future states of the system given an action sequence and past observations and the preferred paths which are ideally sustainable and align with long-term goals this encourages the system to pursue states that meet its sustainability preferences over time

ambiguity measures the expected uncertainty in mapping states to observations and it ensures that the system continuously learns about its own Dynamics and the environment this helps the system reduce uncertainty and improve its predictive Dynamics U over time

the components of expected free energy balance the drive to fulfill preferences with the need to minimize uncertainty and so this balance supports plastic responses that enable the system to adapt to change but also actively guide itself towards more sustainable and self-organizing configurations over extended time scales ensuring resilience at both individual and Collective levels but sustainability isn't just about survival or at least it's not a simple story it must incorporate UD dionic well-being focused on growth

fulfillment and development and hedonic well-being which captures immediate satisfaction

so in active inference this means that the system should balance present experiences with long term strategies that enhance overall capacity for adaptive growth so when we talk about a hedonic well-being we represent growth and the system uh the system's ability to find diverse beneficial States and it aligns with long-term sustainability by ensuring that it can survive and thrive through extended adaptability and resource

renewal it represents the growth and system's ability to achieve its long-term goals this this concept aligns with active inference because it emphasizes long-term

sustainability uh by optimizing internal models that allow agents to find those beneficial States and So within active inference agents act to minimize expected free energy which involves the balance as we saw between risk minimization which means that the outcomes are aligned with the agents preferences and ambiguity minimization which ensures that the learning uh continuously happens so that's what enables the agent to grow and Thrive and so UD hedonic wellbeing can be conceptualized as the successful maintenance of favorable States and the expansion of the system's potential allowing it to reach higher order goals over extended time

Horizons and so this Behavior reflects structure learning where in agents adapted changes in the environment by restructuring their internal models which lead to growth and a diverse set of beneficial States now honic well-being involves immediate moment to moment satisfaction which is necessary to maintain ongoing motivation and effective response to environmental demands it focuses on immediate satisfaction and so the agent's capacity to respond adaptively to current needs uh it can be seen in active inference as the system's ability to reduce momentary free energy and maximize instantaneous rewards so we can see this at the lower level say of higher equal models so you minimize prediction errors in the present you seek sensory states that satisfy immediate needs or preferences like finding food to satisfy hunger and this corresponds to reducing free energy associated with specific observations the internal reward or momentary satisfaction that the agent achieves after fulfilling immediate needs corresponds to the agent reaching a preferred State as predicted by its internal mod this is crucial for maintaining motivation and ensuring effective responses to environmental demand here we also have to factor interdependencies within the system these inter dependencies can be represented through mutual uh information equations which uh

ultimately um quantify the connections between different components the plasticity at different at individual levels translate to the sustainability of the entire system aligning short-term actions with uh long-term goals so sustainability is an emergent property that depends on the balance between all elements in a system this really involves presenting resource depletion maintaining well-being and promoting resilience through self-organization and learning so it's a continuous effort to increase resource redundancy and avoid over exploitation thereby ensuring sustainable practices over time you can think of it as avoiding a local minimum the sustainability of a system depends on harmonious interactions among its different components which range from Individual agents to the broader Collective this means that the system must align both internal and external goals to ensure stability over extended period periods enhancing resilience and sustainability so for example uh the resilience components of inertia elasticity and plasticity will work synergistically to support adaptive actions that contribute to sustainable outcomes the emergent stability is promoted through decentralized interactions where each component of the system self organizes and responds to environmental feedback which reduces the complexity and increases adaptive capacity and allows the system to

maintain a dynamic but stable equilibrium also sometimes referred to as homeostasis so we can think of attractor states which emerge as agents minimize their V variational free energy over time and this process supports the alignment of internal models with the environmental demands ensuring that agents remain adaptive while reinforcing the renewal of resources and overall system well-being

so these are all nice ideas and you know we think they hold up mathematically but we wanted to test them out and obviously this is the beginning of the journey so we built a small and very simple agent based models to illustrate

sustainability under active inference at the level of one single agent within one environment this is extremely simple and the following studies will cover more so the model here simulates an agent managing food consumption in different environments to ensure long-term availability our objective was to show that uh the agent's goal is to maximize well-being while ensuring resource sustainability over time and this involves balancing immediate needs with uh future availability of resources so we started with a validation case

um in this case the agent makes decisions uh based on current food availability without affecting future availability and so the agent observes food and decides whether to consume or not uh based on its current state and

PRI

preferences so we have the generative model uh we have hidden States representing food availability and saity and the preference uh C which encoded a strong preference for sa while maintaining food availability as as you can see these two preferences are technically contradicting and so in a static environment which was the first case the agents decisions are really straightforward they just maximize short-term well-being without needing to manage resource

Dynamics so as you can see uh in the tiny little plots here uh the we can see the the agent's Behavior specifically uh whether the agent chooses to eat or not over a series of time steps and so uh the middle plot shows that the amount of food over time which is represented as food left and the last bottom plot shows that uh the uh you represent the agent's level of SA or how satisfied or full the agent feels over time higher values indicate higher levels of sa so uh the agent uh has a strong preference for high sa uh since food is continuously available the agent keeps eating at the beginning which increases its saity level and since the environment is static the food remains available indefinitely and so the agent successfully increases its a uh we then wanted to show whether or not what we did was just an artifact of the Simplicity of the model so we

screwed with its matrices and so incorrect matrices caused the agent to misjudge the effect of eating which leads to an unstable sa level so we we showed how errors in perception and action models can significantly impact the ability of an agent to achieve stable and desirable States even in an extreme simple uh setup so then we tried something a little more complicated which is a dynamic environment where the strategies have to change so uh the environment changes based on the agent's actions so resources deplete when consumed but replenish over time when not used and so the agent's challenge was that it had to learn to balance consumption with replenishments uh which effectively had to demonstrate some degree of sustainable Behavior so the model includes the transition Matrix that Maps the state of food available based on the agent's actions and expected free energy here minimization uh guides the agent to consider future State and ensure availability um so this was the case as you can see it hasn't changed much the only thing that has changed is the Dynamics of the environment and the ability for the agent to ultimately learn those Dynamics so the agent had to adopt adaptive strategies that balanced immediate rewards with long-term stability uh and this was showing us how active inference can foster sustainable behaviors in fluctuating

environments um so when we look at this plot um we could see that the agent uh has learning turned on or not and we wanted to see whether an agent that didn't learn did just as well as an agent who learn just to see again whether it was an artifact of Simplicity of our model and also the agent can die if the resources are depleted and it has high sa it is not satiated over time it dies which you know ethically might not have been the best thing to do but whatever so agents that have learning quickly improve their survival Times by refining their internal models and strategies and so the rapid Improvement really shows that learning enables them to find efficient and sustainable ways to interact with their environment without learning there's no significant increase uh in survival time uh as they aren't able to really adapt their behavior based on past experiences so they end up uh in suboptimal decision Loops leading to a lower average survival so we can really show that learning is crucial for agents ability to adapt which in turn allows for a longer more sustainable uh presence in the environment and ultimately uh you know the also the environment's ability to actually be sustainable in itself the agent overcomes initial limitations and eventually converges to strategies that maximize their survival which really shows the power of dynamic adaptation in un certain environments so we saw that the agent's

behavior in both static and dynamic environments really shows how well-being can be optimized while maintaining resource availability and in the dynamic environment the agent learns to moderate consumption

it is able to understand the tradeoff between immediate satisfaction and future resource sustainability the agent has the ability to create adaptive strategies that adapt uh that align sorry with the principles of resilience and sustainability and so here our model illustrates in a very very minimal way how minimizing free energy can lead to behaviors that promote long-term viability rather than short-term gains and so these simple models are just laying the groundwork for modeling more complex systems such as ecosystems or Urban environments where sustainability requires coordinated interactions between multiple agents and resources so to

recap we started by understanding resilience with an active inference we broke it down to inertia elasticity and plasticity we then explored sustainability building on resilience and we incorporated well-being as a critical component we demonstrated how agents can balance immediate needs and future viability achieving sustainability again all through active inference and so it Maps systems that can adapt sustain and Thrive by balancing

short-term adaptation with long-term sustainability but we still need to show how active inference can effectively be scaled to real world applications involving multiple interacting agents and resources

uh involving the ability for the environment to fully deplete and therefore how to actually manage such possible um such possible patterns and how agents might learn over different areas um we have to transition from theoretical models to more practical implications that involve challenges in data acquisition model complexity computational scalability um we also need to factor Mutual interactions and the effects of each subsystem on each other and the larger system such that they promote more resilience and plasticity so all of this is extremely seminal we have to move further and actually demonstrate in much larger scale and more complex simulations this these ideas but it has critical potential applications achieving all this we can use active inference to model and predict adaptive strategies for mitigating climate impacts at different scales or improved decision-making processes in organizations uh enhancing resilience and sustainability in economic and contexts so active inference can be extended to other complex adaptive system and we can support initiatives aimed at fostering sustainability in

health care Agriculture and urban
planning which is work that again Mark
Miller and Ben White teni poo uh are
really pushing forward

so thank you for listening and I hope
enjoyed all right the next recorded talk
is 50 minutes long By shager Ramen
Humanity story a search for moral
Clarity hi

everybody so I'll be going over what I'm
calling Humanity's story a search for
moral Clarity and I'll get into kind of
what that cheeky title means um going
over kind of myth morality narrative uh
self- understanding and um the meaning
of life and no that one was too far but
I guess not not too much far than one
I'm claiming um so yeah before I get
started I just want to give a big thank
you to the active inference Institute um
you know for overall the work they do um
it's been super rewarding um and
engaging to to be a part of uh I put my
information here so feel free to reach
out if anything picks your interest or
you know bothers you or you have to
correct anything feel free to um give me
shout so we'll be discussing um you know
a few things as I mentioned so one of
the big roles um that I feel active
inference can um you know one of the
ways that I can at least intuitively
scale

into human culture first of all is
through our use of symbols of what makes
us unique as a species um can be
understood you semiotically or through
symbols um and the road there uh as I

make the case in myth of objectivity is through morality um largely through bodifying norms and expectations which again ties very nicely into active inference I believe and the last way is to narrative and specifically I'll be talking about uh kind of at a meta level

The Narrative of modern narrative and you know why are we why are we doing this you know it's you know epistemic reasons interesting as it is but also I think that kind of applying a rigorous framework to humanity specifically Bly as we get into narrative um we'll see it more broadly tackle um some other areas of humanity um this is necessary I feel for cultural cohesion um something generally lacking these days um you know with partisan um and in polarizing um the nature of our culture at the moment in in our politics uh and so it'll achieve hopefully or at least understand the outlines for cultural cohesion uh and things like social and personal uh integration and I'll do this by as I mentioned by expanding upon my myth of objectivity hypothesis right and so my own road to to um becoming interested in the free energy principle and active inference was being disillusioned with the economics Paradigm um finding it to be as as I'm sure other folks were um insufficient to describing human behavior even through the kind of quote unquote narrow lens of the marketplace conceiving people as a neoclassical

framework does as
utility maximizing um individual
agents who are well-versed in what their
own preferences and interest are um you
know it's kind of quite
unrealistic to say the least uh not only
that but behavioral economics didn't
really um do it for me either um it also
held up this idea this neoclassical idea
of rationality um as the kind of goal
standard and you can see that with some
of the recommendations you know the way
we should
um you know default constantly default
401ks um you know maybe that makes sense
but it does it comes up with that advice
it justifies that advice using this kind
of principle of aity meaning an agent um
preferences should not change over their
lifetime which of course is not not
realistic and so of speaking of um know
behavioral paradigms that are lacking um
you know at first I thought well the
neoclassical model on its own isn't
interesting um maybe we can marry that
with you know some views of moral
philosophy because I'm sure moral
philosophers have you know a much better
view of of
humanity that's what I thought uh so
here are a few uh few different moral
philosophies that maybe some of you
folks are aware of maybe others not um
but kind of briefly um deontology
utilitarianism are you know traditional
from the enlightenment ER going back to
Kant um and Stuart
Mill um paradigms of moral philosophy um

interpretivism is a bit more recent but
um I think still been around for for
quite uh some time um and the last one
is uh just a couple examples of some
Eastern philosophy and so when we think
about I guess the question I'll POs to
you know those listening is what type of
thermodynamic systems in terms of um
their adaptability um you know how do we
characterize these various different
systems you know are they single
equilibria are they steady States um are
they maybe chaotic maybe they don't
reach any equilibrium
um you know how do we think about these
um so firstly deontology and
utilitarianism um and you know if not
familiar with them um don't worry I'll
kind of explain it but um the onology
and utilitarianism um I view as single
Lake album even though they are uh
different um they seem to have one um
kind of one unifying uh mistake which is
you know the environment or kind of
context doesn't change or in other words
there is no uncertainty in these systems
that they assume you know by their
prescriptions of morality and moral
philosophy so deontology assumes there
are certain set of rules that you should
never violate that um never change over
time um so you maybe thou shall not kill
thou shall not steal etc etc um there's
no conditions under which those should
be violated um and
utilitarianism this is all of your
actions are based their morality at
least is um evaluated based upon the

impact that they'll have on the the
aggregate welfare of others right so um
this is kind of featured for instance in
uh the good place for those um watched
it but the problem with both of these is
that um you know in a word it's
uncertainty right there's no epistemic
foraging there's no model for how Agents
come to learn okay what are the rules um
the maybe the deontic rules that I
should follow what are the ways that my
actions impact others like what is that
process of um socialization that kind of
awareness those you know the empathy um
you know all the things that you know
you know should I read a book should I
read fiction um you know how should I
speak to people um you know all these
are essentially epistemic tasks um of
learning tasks that are not featur Ed
into you know these twin systems um and
and I'll get into perhaps what this
reveals about Western Society um um of
spoiler but this is the kind of
analytical bias of Western Society I
think revealing itself in the ontology
and and utilitarianism specifically so
interpretivism um this comes out of
postmodernism I believe or at least it
was um rebr into um the dialogue
following modernism into postmodernism
te Etc so it says essentially kind of
rejecting these uh analytical reduction
single equilibrium as I presented it
here um you know frames of moral
philosophy and says that well it's based
on the agents interpretation of their
behavior you know what are the context

for their actions also how do they interpret their actions etc etc well of course it's an important element of morality um the kind of classification that at least I would label is chaotic right so if we're just saying it's up to someone's interpretation well how should they interpret it um should everyone interpret it differently is everyone's interpretation equally accurate how should we you know yada y y the point is that it's not super helpful for society um as a approach to morality all right so final two confusion and dosm um uh I see them as being adap of it being steady state um focus on the kind of rituals and behaviors not content specifically not focused on particular deontics um and not kind of hyperfocused on the impacts of behaviors to others although that can be a component as can you know interpretation but um it's really about the emphasis of these two paradigms um kind of an inherently acknowledging um that the world is changing that Society is changing and yet the goal of um morality at both personal and social level um is to find coherence so it's kind of still privileging um some kind of equilibria um some kind of Harmony but not kind of how you get there and I'll get into how it it does um with active inference so you know before active inference what I would say is that of Daoism and um Confucianism they come from a tradition that um Carl

Yung would label as a the mytholog
mythological construct of the cosmic
human or Cosmic man um aka the symbolic
self which is the you know term I'll use
for kind of going forward kind of ties
into um you know my theme of symbolic
agents Etc so the point is that uh you
know Carl after serving um doing a
crosscultural analysis of various
different mythology IES and and
traditions found that
this particular what he called AR pipe
reemerged uh and essentially the cosmic
human was the perspective that we all
kind of part we all kind of connected to
one another um not only to other people
but to the cosmos and to the world um in
general and our actions do have
consequences because of that inherent
interconnections um moreover we do have
a certain responsibility to you know the
world as a whole to other people um
because of our place as a kind of a node
in this hierarchy this some signaling
where I'm going with this um
but it allows as you can see um an area
both for the kind of
characteristic uh state so the ways to
minimize uncertainty kind of using both
that you know epistemic and exploitative
um you know Paradigm of active inference
um but kind of does so specifically um
by conceiving this kind of broad view of
the self
right and so here of course is how it
connects to active inference so active
inference unlike the neoc classical
framework and unlike other

um single equilibrium um approaches to social science and to science in general it allows um nodes to communicate unicate um both with spatially and temporally and so kind of over time an individual um passes information to its or to kind of quote unquote its its future self and um in some ways it's its past self in terms of goals um but the self is not a stationary construct it is something that is constantly seeking um who what other nodes are kind of included in my hierarchy um what other are including kind of myself um and the symbolic self is an example of our ability to exate all these different nested nodes so our the result of us being a symbolic agent is that every time we meet someone say you know who says they're Canadian um well if if we Haven met this person before our model of what it means to be now include that person we now have a you know instance of the con the general concept of a Canadian of that kind of General archetype so that means that the model we have of Canadian whether or not we are Canadian um we have the ability to compute to constantly compute what the self is um and by virtue what the other is and so this the kind of Road the tra the trajectory that I outlined that kind of brought us to here and what it means and all that um I draw from our social bias so what are kind of some instinctual impulsive elements that first um started

to see Human Society expand kind of
outward laterally or even over time well
at least laterally at least kind of um
within a specific time uh

I make the case that it is social bias
something that we are inherently born
with um you know although we are not um
you know born racist per se um we are
born with a specific bias and that's
accent bias or shth it does seem to be
something that emerges in utero and we
have preference for

um pronunciations for vocal accents uh
over others those we are more familiar
with and while that's all negative it
does have clear biological benefits of
course that's why it exists and that's
why it you know persists today um and
secondly what what do we do with
a society that's based on people with
the same accents um well that's
inherently un instable one when that's
not um super adaptive AI is extensive
and allows for greater cooperation um
it's unstable because allows for deviant
free writing things like that okay so
enter morality enter

morals um and so here now we're
connected in this um symbolic node Way
by codifying those Expectations by
minimizing surprise using morality right
so this is my myth of objectivity um
you're supposed to behave cooperatively
you're supposed to follow these Norms by
virtue of where you where you've grown
up you know that is social bias that is
shth um evidently you are not doing that
um so how do I deal with that surprise

well over time of course that model you
have um the shared model you have of um
you know that's based on identifying
with those like you starts to wear thin
starts to break down and so in comes
morality as a way to stabilize um
to offer kind of a hood that you place
on um
deviant and that you can then you know
use to shame and guilt um into aligning
with the shared model um or of course
there's punishment noiz and other things
right but that's what the the tool that
morality probably had in the p and
continues to have with us today all
right okay that third leg
um okay so if this is the self um what
are we modeling off of right so we have
the internal model Act of inference we
have the Marco blanket that screen
between us and the world so then we have
the external States so what once we have
kind of symbolized itself what is the
world what exactly are we modeling off
sometimes it's the world you can say
okay we think about
um farming think about you know someone
who's born and raised as a farmer the
land will feature very prominently in
their model of the world think about a
hunter the land wol figure um you know
the world itself the physical world will
um you know will mean something um
however it won't mean everything so what
are some other things that will
constitute the world probably the other
right so social bias of course Works
into that there's those who sound like

you there's those who's not or those not
like you um so by virtue of having a
bias means that you're biased away from
other people and that's why we don't
like to admit that we have biases um and
that's why we were troubled when um
people will tell us that we're
biased so that's one example over the
trouble um with the other and you can
think about you know the Cold War think
about any time that you know a country
is um you know under attack the other
becomes prominent becomes Salient um the
in group binds together and you know for
a while um you whether it's 911 or
whether it's World War II um there's
kind of great social cohesion trouble of
Wars end um enemies become
salient um and then what
happens well this is you know how we
have this is you know enter kind of
Mythology not these things are entirely
separable um there's a lot of overlap
but um you know for our purposes if we
think of you know mythology and we start
to think of um narrative what we're
talking about is uh
in the language of Mel in fields we're
explicating
the reference frame we're extending the
measure the kind of yard stick of what's
good what's bad um where is you know
according to mik 11
the frame of reference that allows
agents whether you know bacteria or
complex organisms to judge whether they
should approach something whether they
should avoid it is it an internalized um

coordinates coordinate plane or frame of
reference um what we're doing eventually
once we get to myth once we get to um
explicit morality once we get to
narrative is we're exiting that frame of
reference we're saying okay um you know
these group of people say are evil um
you know we we're all descended from
this um ancestor you know from you know
Hercules from beol from this of half man
half god um we explicating these
reference frames to gain cultural
solidarity to gain cultural cohesion
right and so um I mentioned this
uh this is kind more going into my um
myth of objectivity hypothesis but I'll
just going to briefly show it because
you know I would love to get some
feedback from the folks listening to
this um so this is how I
have um I hope to formalize um you know
how we um you
know became symbolically literate um if
you will so that kind of passed from
shith to morals um to you know full
symbolic agent um and so I just have
kind of an agent here um assuming that
another Agent B has a fellow X and
therefore shares the model of the world
has some error know the error reaches
some threshold um and then suddenly you
know if then else um you end up um with
a new model eventually leads you
to um modeling the prediction error so
here I have um a somewhat familiar
equation but kind of modified but
instead of
um uh taking the where the posterior is

kind of conditional on um traditionally
conditional on the model um here in
place of that I have you know an
objectified error um so the point is
that you have these
categories of where you know typically
the prediction error will just be
whatever kind of noise that
you experience that are not in line with
the prediction um and either you'll
select a different prediction model um
different model at that point um or you
just assume it away as um as noise in
this case there is an explicit model of
so when someone decides to steal food
when someone decides to
um not help with uh childbearing when
someone decides
um I can't think of any um moral
deviations um but um the point is that
there is a specific set
of deviations from the typical
predictions versus just being um
surprised generically so it's a modeled
version of error um that then allows uh
an agent a to view a given agent um as
possessing the ability to select between
adherence to a higher level or
superordinate model um now an objective
model um of the group or a narrow one so
they can comply or they can deviate
that's a C and D um subscrib and the
point is that um or you know I guess one
one takeaway is that okay this isn't
just metacognition um that of upper
level superordinate G um and this is
transc meaning that it's not
um purely objective um it's framed as

objective from um the point of view of
the agent but that only means that
transcending their own subjectivity and
the subjectivity of the agent
by conceiving that they have the ability
to select
between a higher level or a lower level
model and so in this kind of simple
way at least at this level of
explanation we can see
how active inference point of view we
could have gotten into something like
objective morality and more importantly
something like Free
Will and of course the various different
significance and so just
um again a couple you know I view this
as um the kind
of hypothesis hiding in plain sight why
because morality is something that we're
so impulsive about um something we have
so many emotional um ties to um you know
for instance um
we're constantly looking for vict um
we're constantly claiming to be the
victim we're constantly looking for um
people who
victimize um and yet we can't really
agree on what the specific contents of
morality should be even though as I'm
when I mentioned deontology I'm sure
there was folks
listening who had ideas of what could be
deontic meaning forever forever there um
so went over the some of this um you
know there's you know social biasing uh
morality there's impulse to do this um
and it's some kind of research that I've

done um in the economics going back to that neoclassical Paradigm although I was kind of dumping on it earlier um what these single equilibrium analyses allow for uh is the ability to say okay where does this model break down where does this model fall so you might say it falls down everywhere well that's not quite true um people are uh selfish they do have um stationary preferences at least for um a long enough time for that model to be useful um however it does fall down in certain interesting ways one interesting failure of the single equilibrium economics Paradigm is the need for social trust um and so social trust means that we have certain expectations we have expectations that company selling something sells what they promise to sell um it means that you can you know trust the market system trust the institutions um in the market that you inhabit and engage in economic activity um so you think that this at all levels you essentially need it you essentially need to take medicine and understand that hope the people are um you know so ethically inclined um you know because of course if um they do something bad they might not get caught for it yada yada on point is that there needs to be some sort of um agreement on social trust so it's an example of how kind of morality seeps its way into something even as um even as intended to surgically remove all kind of humanity from the Paradigm

all all morality all essence of morality
from econ um the study of Economics
social trust and morality seeps back
into it um the same thing with um
Linguistics um won't get into this one
too much now but the de of
transcendental signify so it's a way of
structuring otherwise of elusive um set
of symbols linguistic systems um we
always have something there's always as
I mentioned there's always God there's
always some sort of myth that's tying
everything together um even in secular
society we have not um you know we don't
have kind of uppercasse God G God the way
you made might have in the past you know
there's a lot of different religions
there's a lot of Institutions and
probably agnosticism and secularism is
kind of increasingly dominant um so what
do we have in this case we have
democracy we have um science we have
kind of other um essentially kind of
mantras if you will um that require this
kind of shared belief um element which
kind of has very moral um
characteristics but is is kind of tied
back into
um a kind of shared belief system um
then Heathers our uh you know norms and
expectations too it allows for us um
they kind of have you know social and
political dialogue
um yeah right
um I yeah I spent a bit of time on the
hypothesis
um quickly um I don't know if I'll have
time for uh this entire run through um

but once we have established that
um we achieved a symbolic capacity um of
what is uh the next step do we just kind
of walk around shouting symbols um no we
have kind of narratives that uh attached
to our symbols it's the way we kind of
piece them together it's how we connect
one symbol to another at a very um kind
of crude way of describing this um it
was a recent paper on narrative Theory
and active inference um which I
encourage um you know folks to go out
there and and um and seek um it covers
episodic memory it covers narrative
identity as so some interesting topics
throughout on narrative um the one thing
that I wanted to add is um
it makes the claim that narrative is for
the future it's for prediction just as
many things in um active inference are
however I think that you could equally
say that in order to
establish um you know a productive
prediction um we write narratives about
our past so we can separate you know the
person we were in high school um the
person we were um you know in our 20s
our 30s so we can we just tell stories
about that about our last job in order
to draw a boundary between us um and
that person so it's a way of um kind of
inheriting in in um inhabiting various
selves um but not identifying not
overidentifying with the past one so I
just want to add that because I don't
think it was covered in the uh in that
paper um and so here I have a couple
different of the core elements of

narrative and why I think that it it SPS
so well in the The Narrative Paradigm so
I think the story describes as a set of
you know the underlying events um so
not explicitly what's in the narrative
so if we read a novel um and you know
the the novel is a short story about um
you know Harry Potter it's uh exactly
what's happening in that version of
Harry Potter maybe he's um you know
already at school doesn't kind of go
into his backstory necessarily um um but
the but the the story is all the things
all the underlying events that happen
that are related that are necessarily
depicted in that narrative um and so
this is essentially the the generative
process right so it's the outside world
that's um you know that's being modeled
in um The Narrative and in the kind of
causal chain that is the plot um so the
point is that um uh I kind kind of
quickly but the point is that when
discuss
or when of
analyzing
um analytically the core elements of
narrative um you know a lot of the
themes that emerge are kind of similar
to um at least some of the key elements
of um active inference
skip this one for now
um yeah um one thing I think it's
important to notice is that there is
implicit versus kind of explicit
um narratives um and
so there's this there's the idea of of
feature placing um so this is

essentially

how um

vocal vocalizations you know emerged
prior to symbolic CU so you know other
animals have

um verbal Expressions they're non
symbolic um a feature placement
is um kind of one single

verbalization um that might be some way
that we can think about how

um language began um and so essentially
the kind of going back okay Concepts
need uh context if we just s fire um

what does this mean does it mean that

I'm fired from a job does it mean that
something is burning um or does it mean

that you know I'm just um you know
someone said something goes really cool

and I'm just saying fire uh in that kind
of slang

context um

so explicating the explicit so um

here uh I just wanted to raise um the
instance

where creating narratives um although
everything can be narrated um not
everything um is

narrated um so there could be
unconscious events that happen in your
past or childhood or even during the day
that you hadn't really thought about so

um various different therapeutic
techniques um address that so internal
failing systems is one where you address

um know different emotional states or
different psychological States and try

to ify those and try to kind of give
them

um uh you know abstract structure in your mind by engaging with them you know through journaling or through uh dialogue with a therapist um so that's internal family systems there's also expressive writing similarly and so if we think about um one of the impacts that um narrative can have that that that's one and sometimes it's good to um you know not sometimes it's good or sometimes it's just um a fact that we don't have narratives like we don't have narratives on all of our feelings every day I don't think we would be able to kind of narrate everything that's that's happening to us um if we think about meditation and mantras the ideas there is that you know for a time at least we shouldn't narrate everything or at least we shouldn't accept the first kind of order narration um which we think about mindful minutes meditation and kind of focusing on their breath as opposed to fall following our thoughts um you know if for anyone just practice meditation you know you'll know that um sometimes we should just you know be quiet and not listen um to the narratives that that show up in our brain have a couple different obviously examples

um right so here what I'm talking about is and as I mentioned on the previous slide or I had it written down I don't know if I um spoke to it um but what we're talking about here is that the problem of in promise of cultural narratives as is most known as Master

narrative or grand narratives um
that traditionally would have been
the purview of Mythology and religion
more modern times
um is kind of constructed um in other
ways and so um as I
mentioned from our nomadic multi-band
past um to even agrarian societies um
and then kind of coming into the
universal age
these um started from these fully
integrated narratives where a particular
nomadic um HRA tribe would have a single
kind of Master cultural narrative um or
at least that's what we understand as
our kind of best guess is that this was
the case or wasn't um you know which
religion am I part of I'm part of my
tribe's religion um you know why did uh
we catch food today and and and not
yesterday there was you know an answer
from the shaman for that you know this
this it was fully integrated um the
second kind of epoch was the leading up
to kind of the the axial age when the
universalizing religion you know the
setting the stage for Christianity um
you know Hinduism um you know Daoism
that that I mentioned um so for a very
long time um this is kind of following
from you know 10,000 years ago when we
um started farming um perhaps can tell
recently we had these dominant religions
um that essentially kind of took over
the the cultural
sphere um those are the the universal
narratives religious Nar I'm sure folks
are aware of and most recently in um

weird Societies or Western societies
it's kind of Western educated
industrialized rich and Democratic um
that's the an acronym for for weird
there um and my case here is that these
are you know we're larg in a culturally
fragmented environment
so earlier when I mentioned single
equilibrium analysis neoc classical
economics but specifically um
deontology um and utilitarianism those
are all examples of from I think a
thermodynamic systems point of view from
active inference point of view this
would be kind of a um a very um single
equilibrium um you know kind of ways of
looking at the
world um and why are they well um you
know according to
Joseph enri and I think According to
some similar analysis from um Ian Mel
Christ um this is because Western people
are distinctly
individualistic um they're a bit more
left brain um centered than um at least
the axial age portion of um Society so
those um in traditionally of settled
societies um that are more justult and
more R brain oriented um
would think of things a bit more
holistically rather than as kind of
reductive so the good uh of course
there's um know many benefits for left
brain thinking it's um more individual
it's more analytical uh and it'll be a
bit more geared towards science um
versus mythology versus um stories and
narrative um but what is it missing well

it can make a lot of um mistakes and not random mistakes um a specific type of uh mistake which is um leads to kind of reductionism um and which is how we kind of ended up with which might be according to these views how we kind of ended up with um some of the paradigms that we did

and so the the fragmentation we can understand along kind of three different these three different dimensions right and so one is aesthetic uh the next is kind of institutional and the final the last is uh kind of objective and so um this is essentially how I see our Master narrative currently broken down um and of course it could be additional ways but here's of three ways as a starting point to understand the kind of cultural narrative um

and how specifically it's it's fragmented so aesthetic we have the kind of modern era the postmodern era and um now what some people are calling the metamodern era it's maybe a bit too soon to um turn on it however I do think it's super interesting especially in the light of the free Eng principle um which I'll hopefully have time at the end to kind of um touch on um the second one is our institutional um narrative so uh e economic framework um you know how we should distribute wealth um you know what does that say about us and and kind of what's the most quote unquote Vision Way um but what are the narratives around um our economics what's our narratives around our

politics um and finally there there's
these idea of uh of you know objectivity
still kind of creeping in so um what are
the kind of quotequote true narratives
out
there
um and so
so you know reasonable question is okay
this is you know we all have kind of our
own stories um perhaps there's still
vague senses of a master narrative and I
think you know there always um
are uh there always kind of might be
um at least some emergent
um Master
narratives by virtue of us kind of all
occupying the same at least you know
digital space the same kind of time um
well Converge on some things um however
without
a um kind of
coherent or
a more of like a religious story that we
um perspective that you know we had in
the past where um you the narrative is
not just um okay we're this you know
Third Rock From the Sun we're revolving
around um we revolving around space and
you know we're you know victim to two
laws and um this is what happens to be
happening with Society um what we lose
with the kind of grand Cosmic narratives
of the past um the type of psychological
um coherence um so where do we fit into
this world you know the idea going back
to kind of cosmic ban you know where is
our place in the universe um what is the
kind of personal meaning for me that's

where a
um a tighter narrative kind of more
intentional Master narrative um comes
into play so our current fragmentation
um leads to personal anxiety can lead to
social anxiety um and it can lead to
kind of the polarization uh and
partisanship and the results are
fragmentation institutional misalignment
whether the company you're working for
um shares your
values um whether they're working on
something that is you find deeply
meaningful that is deeply meaningful for
your identity um and your values um
that's um what you mean thereby
misalignment okay so in terms of
Aesthetics um as I mentioned um the per
own is is sometimes called metamodernism
um and so that kind of gets
into the idea of genres meaning that we
have kind of plenty of good um you know
television so we have plenty of good Nar
is available um for us to consume um
that does include um kind of classic
genres
um but the problem is that they're all
genres so we're all kind of consuming
um content we're all consuming it
through different media um even in times
past um not even too far past uh we
would watch TV at the same
time you watch the same movies because
because they're in the movie theater
that's where you could that's the only
place you could watch them um that b so
recently um now we're watching movies at
different times moving from movies from

um 20 30 years back um from 50 years
back and the essentially the
fragmentation um has gone up an
explanation as I mentioned there there's
an analytical um orientation
here um I think it's it's worth noting
that along with the fragmentation that
has occurred in weird societies um along
kind of side that that kind of
loneliness that happens um with the rise
of uh the analytical world is some great
some great
literature um and so one of the reasons
I argue that this came about was
specifically because um once theology
once
religion
stopped serving the role that it did um
that left in particular artists in
particular those kind of aesthetically
inclined to seek out um art to create
art in some cases to fill those gaps for
um offering a narrative for answering
the questions the big questions that
were typically answered by those
institutions
yeah um and
so kind of going back
to my quest what we're kind of after
here is we're looking for greater um
cultural cohesion we're aiming for
better personal and social integration
so I argue that these pieces are
connected um and I'd argue
that if the free if the freeable told us
one thing um it that agents minimize
uncertainty however it tells us another
thing is that one way to verify kind of

any Claim about um a self-organizing system it's to not look at one Paradigm assignments to look at is to synthesize various different views so what we can glean from that is essentially a at a modern approach um which is to not necessarily ensure that everyone in culture everyone in a particular society is consuming the same content or media but it's to highlight certain aspects of it that each person can find meaning and the right type of meaning in their own content and even if we achieve that through different areas we'll all be experiencing something similar with each of our traditions of and once we've identified the certain components or make the case for certain components um that answer maybe some important questions from acers point of view and allow greater visibility into that uncertainty minimizing process at a cultural scale um so here I think poetry is a great example because of it um you know it reaches everything from the senses from its sounds to its visuals um to its more deeple deeper symbolic meaning and I think great great example of that is TS Elliot um the great many references he makes in his work um also um the kind of poetic conventions he aderess to makes it

very uh in some cases accessible
um works that he offers and so whether
it's TS Elliot or Kendrick Lamar
um in each case what these artists
constructing is a explicated symbolic
self an
explicated um a number of different
nodes in a hierarchy um that are
connected with all these various
references in a particular song
um that are that can be made clear you
know for instance in a um this website
it's genius so you can see where the
different um lines in this case Kendra
Kar has and what the references
were and so in a sense what you
experience when we read poetry or we
read a work of fiction that has um deep
references deep um cultural references
like this it's constructing an extensive
self um bringing coherence to it and to
everyone who listens to it um and by
kind of calling it out you're able to
share in it almost as if you know it's
something you know it's a greater work
of literature in the past um as folks
would and so that I think is is of a few
examples of
um ways we
can start to apply a rigorous framework
to the humani that tells what kind of
literature does for us why it does what
it does for us um and parse through some
of the components that
make it
meaningful to us and kind of allow us to
um use literature um in ways that I
think it was originally intended to in

past times

has so leave it there and um you know

once again feel free

to um reach out to me with my contact

information and um and please you know

follow my um you know substack and and

let me know if um anyone has any

feedback and I just want to thank Daniel

Freedman again um and you know all the

folks at active inference Institute and

Carl priston for um the this content and

um the discussions here so uh thanks

again

all right the next recorded talk the

penultimate recorded talk

is by Bradley Alysia

earlier Bradley gave a talk on

purposefully non-standard

cybernetics and this talk is going to be

an overview of Open Source and open

science so that's really cool and thanks

for bringing this important

complimentary topic to the Symposium for

all all who are working on this okay

it's 51 minutes

long hello my name is Dr Bradley Alisa

and I'm a member of the active inference

Institute scientific Advisory Board

today I'm going to give you an overview

of Open Source and open science I'm my

home institution is the orthogonal

research and education lab I'm also

affiliated with the open worm foundation

and the University of Illinois champagne

or champagne I'm also with particular

interest in the orthogonal research and

education lab is our open source

interest group and I'll talk about that

throughout this this talk this starts
with a question what can we do with open
source and open
science and I pair these together
because sometimes we're building
software sometimes we're doing
scientific discovery which kind of
relies on the open source software and
so for doing both we can actually build
uh systems that can sort of help us do
better software engineering and science
so they're kind of grouped together so
in terms of open Source you know it it's
maybe a trism that open source is about
software and putting it out into the
world but that's not enough and you
really want to what you want to do
instead of just putting software into
the world is building a community around
technical standards and code and this is
true if you want to build a community of
users or an open source community that
helps build the software because you
need to have some interaction with the
software to make it
sustainable you also want to have good
technical standards and and with
academic softwares we see that's been
lacking this also allows us to open up
the organization's processes and
products so having these communities and
having these technical standards are not
only good from Just sh this stamping of
sharing software but it also opens up
our processes and products and allows us
to serve our communities better our
academic
communities open science is where we

share data share results share research artifacts of the public so we do a lot of sort of transparency related things we give people our results and our data and it's software and papers and presentations and we're very open about it we can give them access to these things so when I say open science I also kind of mean Open Access and it's they're kind of in the same bucket open science and general allows us to open up our organization's research capabilities so that people can benefit from our discoveries and we can benefit from their feedback so a lot of organizations have proposed different sort of road maps or statements on open science and what that involves so not every organization's view of this is going to be the same but I'm using the young European research University's Network statement on open science at the link below and to them you know they have a very strong emphasis on collab collaboration and citizen science so this is where you have collaboration around the world around the scientific community and also involving say citizen science where people just want to contribute their own um analyses sometimes it's data sometimes it's observations things like that there's also visibility and idea sharing so know being able to get to the outputs and read them and understand them is important open education is important idea sharing because you need to have you know uh focus on

collaboration alternative models of publishing you need to be able to publish the work to get it out there into people's hands incentives assessments and research Integrity so those are all kind of part of this open science tree now again every organization who Champions open science will have a different view of this we don't really have a formalized view in the orthogonal lab but it follows this kind of outline line close quite closely and so if you want to know more about the where the open or where the orthogonal research lab is coming from you can read this paper and I put this out in 2020 and he needs to be updated but it's a good overview and this is building and distributed virtual laboratory adjacent Academia um and and so there are a number of things in this paper it's just basically it's very similar to the model that the active inference lab uses were much more focused on sort of educational opportunities and opportunities for people to sort of pick up the load and build uh research artifacts and you know having students come in and do short-term projects but it's it's it's a very similar process so one of the processes or one of our goals is to integrate open access in our open source practices for advancing distributed research projects so we want to be able to help people make these things open help them build research

projects the other thing is that we want to encourage practice and open working through the creation of digital research artifacts it's like papers or presentations and when I say open working I mean I borrow this term from uh the Mozilla foundation and their approach to this whole Enterprise is basically working open so it's working in these collaborative distributed groups and how to make the most of that that kind of experience you know things like note taking and meetings and managing sort of the human capital so that's also part of it open source and open Sciences managing human capital managing these work groups so one of the big thing other big things about open source and open science is this idea of practice and so you know it's it's very practice driven as opposed to sort of theoretical or philosophically driven and so you know practice means that you need to establish practices procedures routines workflows and Technical Solutions for working open so this means you need to adopt open practices it could be Community review of code releasing code under a license pre-printing manuscripts and generating open data sets so these are all things that involve you know planning but also practice and being able to produce something at the end related to that are what we call open workflows so when we want to

release an open data set we can't just pop data out onto the internet we have to put open data sets into a production line model where we you know clean the data we analyze the data we produce metadata and then we publish that formally in a repository with metadata attached we also want to have Community standards for open participation so we can't just expect people to pick things up and run with them without giving them some standards for working so you know you know usually in academic work you're trained a certain number of years it becomes kind of intuitive to you what to do but a lot of people especially students undergraduates they come in and they don't know what to do they don't have like those standards ingrained and so you have to have Community standards but also because sometimes the incentives aren't there for people to follow long lengthy procedures and so they don't in any case this helps you standardize this e these efforts then we have open Technical Solutions so open source and open science are built or enabled by Technical Solutions you couldn't really have open source in open science without having open Technical Solutions like GitHub live streaming tools and graphic tools and of course those things don't make something open source or open science by themselves but they enable those practices so this is a nice graph where I kind of talk about vision of sustainable open science and

so I have a couple of categories here to the left and the right which are sort of suboptimal areas and then where we want to go is we want to go to this Zone this elusive Zone in the red box so the elusive Zone the red box are the set of tools that would allow us to do sustainable open science and Open Access and the stuff to the left and the right are kind of like things that we do now that are suboptimal so we start with the suboptimal and we talk about black Open Access which is kind of an fancy word for piracy or informal sharing and so each of these types of Open Access are colored black just means it's kind of related to under the table or you know piracy or something so one of the examples are is silab where you know it's I my position on piracy is it's Justified given the incentive structure in science but you know you have systems where you can just get access to things and then of course sometimes we share PDFs by sending emails to our colleagues and sending them the paper and it's all black Open Access because it's very informal and it's not really sustainable because you're relying on people to have you know a you know someone has to get access to these and then share it with someone else um then you go over to the other side you have two other areas of Open Access you have gold Open Access and diamond platinum Open Access so gold up and access is where typically an author will pay an APC fee which is a

publication fee it's usually a fairly decent sum of money to a publisher to make a paper open access to all lot of tend journals will restrict access to paying customers if you are a member of an academic Library you can get access but if you're not then you don't have access and that's one of the things about like places like the active inference Institute or the orthogonal lab is not everyone has that academic Library access and so it's really hard to get things get access to papers and things that you need so this becomes a very important set of points here gold Open Access then of course is is beneficial to the Publishers but not to the authors Di and platinum Open Access is more beneficial to the authors there's no APC fee but someone has to take that fee and you know they have to absorb that fee so that's not optimal either so this elusive zone is a way to sort of build on sort of the lowcost but still like sufficient resources we need to provide access to our work we need to open it up give everyone access to it and allow them to interact with the work in a way we couldn't even do with any of these other types of Open Access so the first thing we have is green Open Access which is pre-printed this is where it's like not the gold standard but it's also you know you can post a paper on a pre-print server there's a pre-print server that gets funded through like a foundation and you know that's that

makes it open de um there you know formal sort of you know there's a DOI which is a formal link it's a permanent link so your work is in a place that has you know some level of reputation with with a mark of approval on it you can also take that preprint and go through what's called open peer review where you open up your paper to peer review not to like three referees you've never met but to the entire community and the thing about pre thing about peer review and I won't get into this

uh any more than what I'll talk about here but if we do peer review with like three people on an editor you know you pick the people who are maybe the most available or maybe the most visible people in the field or maybe people with the most time and they have certain opinions about what you should do what you should add on to a paper to make it publishable and sometimes it's not in the best interest of the science

furthermore given that they still don't really have a very good handle on detecting things like mistakes or even fraud so you can have a paper that goes through a three reviewers and an editor and still be retracted for mistakes or fraud so that's not good either open peer review allows us to have people many eyes on the paper many comments you can incorporate the comments by generating a new version of the preprint and you have this continual feedback of your work and you can eventually Maybe

publish it somewhere but you can also
update the pre-print
continually and have that process going
um so that's you know Green Open Access
and open peer review now a lot of papers
also have data and we can include open
data management in this elusive Zone
because we also want to have a strategy
for managing and sharing data and this
goes along with the peer review and the
access of the preprint that that's the
link to it and that the data can then be
reviewed and worked with independently
of the preprint but also you know it
it's transparent and people can actually
work with it far beyond the LIF span of
a single
paper then we have overlay journals so
we can have preprints we can have but
that's sit on a pre-print server we can
have multiple versions but sometimes we
want to build our own journals because
we want to narrow the focus of the
preprints funnel them down into
things of interest for certain Fields so
say for example we had an active
inference overlayer it would take
preprints in the active inference area
only of those preprints and present them
as Journal content so what journals do
besides inferring status on people is
they give you this curation of the
literature of research so that's what
overlay journals can do but without the
overhead of a journal so you can have
like your preprints you put them on this
with the skin over over the top that
gives us this emperor

of uh you know what a journal would give you so that's the elusive zone of Open Access um and I just wanted to point that out because I think it's you know a good way to think about sort of the production of Science and the production of academic work I posted a blog post for Open Access week 2024 and it's kind of a critique of open accesses that exists and this is why I kind of went through this and basically I argue in this post that open access is essentially failed at least up to this point that we've don't have the right incentive structures we don't have the right intrinsic and extrinsic incentives to do open science correctly Open Access and so we need to change those and so you can read that post on my blog synthetic daisies so in that blog post I talk about the intrinsic and extrinsic motivators for doing Open Access sustain so this you think of this more generally as a reinforcement learning of working open or reinforcement learning of open so when I talk about intrinsic motivators I talk about the internal drive to do something so you know the idea is that you would do things that are meaningful to yourself that's an intrinsic motivator or what motivates me in terms of a scientific question what motivates me in terms of how I can contribute to the field and then how can I benefit from or contribute to a team these are all intrinsic drives these are things that sort of you would generate

and you interact with the group based on those things by contrast extrinsic motivation is the external encouragement to do something so this could be money paying people this could be cost and benefits so if there's a cost to opening my data set I do it if there's a benefit to opening my data set I will do it and that leads to an inequality within the scientific community and then of course Prestige which is of course if something is prestigious I'll do it but if something isn't prestigious I may not do it so it interacts with intrinsic motivation but it also is this external force that that sort of is encouraging people to like a you know an a reinforcement mechanism extrinsic motivations can also be where things tie into existing activities so one example is where a lot of funding agencies and journals will mandate that people share their data if you want to publish a paper you have to share your data so this is like tying data sharing and doing this to publishing the problem is there's no intrinsic motivation to do this it's not meaningful to everyone and so people will do it to differing degrees of completeness or of usefulness so if you're forcing people to do something versus it's something that people want to do and think is a good idea it becomes uneven and you don't have the same quality everywhere sometimes there's you know of course there other types of mandates and

other T and there's been General uh an overemphasis I think on mandates and sort of top down motivation to do things in open science and Open Access um trying to get people to adhere to certain standards or trying to enforce more uh

uh trying to enforce moras that may or may not fit the situation so this is a problem in open science Open Access and even open source

where where people you know you have to kind of figure out the psychology of the people doing and enabling a lot of these open

practices so if we look at this graph we see that their intrinsic motivations and extrinsic motivations that interact so at the middle of this in the gray box you have intrinsic motivations things like I want to do science I have Theory building goals I need to create tools these things feed into extrinsic motivations such as funding Community Building uh adhering to the standards of the field and adhering to rigor standards of rigor so this is all kind of these these intrinsic and extrinsic motivations interact and then contribute to how you carry out your open science Open Access and eventually open science so now I'm going to shift to sort of how we approach this in open source and so this paper from neuron it was published in

2017 it's called the commitment to open science and open source it's called the commitment to open source in

neuroscience and uh P gleon hrew Davis
R Angus silver and Georgio ESC were the
authors they're are involved in an
organization called
incf which is the international neuro
informatics Corp and they've all kind of
done work in this area and they're
contributing to a project I'll talk
about in a little bit um that's open
source sort of solves a lot of problems
in Neuro Imaging uh and how to do that
so one thing that they point out in this
paper is that code sharing is useful for
a number of things in
science but they make a claim that
releasing code can be used to replicate
results so one reason why we do code
sharing is so we can replicate our
results we can be confident that those
results can be replicated because we
have the code and everything is you know
perfect um which is not always the case
but this is the claim this is the
motivator that they're talking about and
so sometimes you have to be strategic
about how you release your code
sometimes you know you just release your
code as a matter of you know requirement
but sometimes you can actually benefit
from releasing your code and in this CA
in the paper they talk about releasing
core scripts and libraries before
publication it will offer early feedback
from the community so you can release
your code just as a matter of sort of
requirement or you can benefit from it
by releasing your scripts and libraries
early so people can debug them and help

you if there are any problems with them or whatever and so you end up with this nice um optimal situation another paper that I'd like to highlight here is this paper from nature technology a technology feature in the journal Nature about six tips going public with your lab software so you know it's not enough to just write the best quality programs and to sort of put it out there you need to really kind of think about a lot of other things going on under the hood so one of those things is that you need to sort of have carve out some time for maintenance so if you put software out into the world the reason you're one of the reasons you're doing so is to sort of serve your community you know sometimes you want to be the standard in the community other times you want to give this gift this to other researchers so that they do better research sometimes you want The Prestige of having like the top software in your field but whatever the reason you want to be able to maintain the software so that it has a longevity and so different sources that they talk about in this paper estimate anywhere from five to 15 hours a week to 20 30 20 to 30% of the time of the project to be spent on maintenance maintaining the software now as you might guess you know there isn't a lot of motivation for that unless you know your software is really successful and so it's it can be a

problem if you you know take the leap into putting your software out there but don't devote the time to maintenance it ends up hurting you

later another problem is that you need to simplify runtime so a lot of times you'll release software and people will try to install it and fail because they don't have the right dependencies they don't know how to run the software and especially with a lot of academic software there we kind of build a clu of plugins and other types of software on top of software so that's not ideal and that's a problem that can't be solved by necessarily by in incentives and motivations some of it is going to depend on how we implement the software and I'll talk about that in a little bit

you also maybe want to spend time in interface design automated testing a documentation to help users who you know when you're in your own research group you're sort of an bubble where you understand what you're doing you've used the software you understand the specific problem for which your software is written well when you release it into the public you might get people who are not as experienced or simply don't understand you know what some of the variables are or whatever and so there are a lot of ways that you know making this easy to use is a is an imperative and then a need for transparency in community building so if you release software you need to have users to make it worth

your

while and if you have users you want to build a community around that so that you can do things like have a more you know have a user base but also have a community where you know you can solve problems

collectively and address open scientific problems and really for everyone to benefit but also transparency because we need to understand every 's methods and we don't want to make it you know obtuse for people when they use the software to understand the methods that went into generating the result that they're getting so you need to think about all these things when you open up your software when you go open source with academic software

especially this brings us to another point which is the need for research software Engineers so in Academia we usually homebrew software we end up putting it out there and you know we maybe find time to maintain it we find time to build a community but the software is not up to the standards of say professionally designed software and so one of the things we don't have is our research software engineering skills in Academia a lot of academics know the technicals matter but they don't know how to build software so software needs to be maintained and needs to undergo quality control and so those things are you know separate

skills so the problem of course is that in Academia we tend to have graduate

students or open source contributors build packages or whole software platforms and their contributions are well appreciated but their involvement is itinerant so that means that they might build something and then leave the lab in two years where they may you know contribute to an open source project and you may never see them again and so they built something with no documentation and obvious viously no maintenance and it just kind of sits there and Withers so I've seen this a lot in my career where people will build software for one purpose one use or like for a single paper and then it goes away because it's not maintained I mean there a lot of code I've written when I was a graduate student say where that was the case and I wish that we could recover that software and maintained it but you know who has the time so really what you need to do is you need to include research software engineer roles in your organization in your academic organization and sometimes open- Source projects will have this expertise but sometimes they don't because there's no requirement for it it's also a different skill set than say like the academic skill set that people building the software might have so this is something where you can either cross Trin people and they're programs where you can cross Trin research software Engineers but more often than not you just have to help people acquire those

skills and so there all sorts of roles
for RSC in different you know academic
and open source
organizations things like intentional
software design continuous integration
release Cycles bug fixes and maintenance
strategies and those are all things that
are separate skills from the academic
world separate skills even from like you
know if you even if you're computer
science researcher you're not
necessarily a software engineer so you
have to have that set of skills or at
least you know have a way for people to
acquire

them okay now we talk about open data so
open data is also a part of open science
and as well as a part of Open Access
so Wikipedia defines open data as openly
accessible exploitable editable and
shared by anyone for any purpose so it's
an data set that you can access you can
exploit or change you know you do
analyses on or whatever you can edit it
and you share it and so anyone should be
able to do this now when they say any
purpose they mean that it's licensed
under an open license so an open license
is a permissive type of liability
protection and copyright for people who
put these things out there that they
work on and they want to benefit from in
some way but they also want others to
benefit from as well so it's it's a
legal regime to maintain you know to
manage liability but also to manage the
social relationship between the person
making the data set producing the data

set and the person using
it open data allows their data to be
easily shared and reused but requires
governance issues of ownership and
adherence to community
standards so this you know part of this
is you know open is in sort of producing
data sets for public consumption the
other part is working on what they call
an open data stack so an open data stack
is a management strategy that is needed
for sharing reusing And archiving data
so we typically start at the bottom with
some sort of interactive interface for
uh working with data we need a metadata
format we need a Cloud Server to host
our files on to host our data and we
also need standards so we can have these
higher level standards for data
production for open uh science
production in general things need to be
findable accessible interoperable and
reusable so we need to have those four
principles embedded in our workflow and
then we need to have a strategy for
archiving data and for that we might use
something with dryad which is a a data
repository so we have these techn this
technology layer a standards layer qu
security access layer and an archival
layer so put together that's called a
data stack and you want to build a data
stack that's you know with tools that
you know how to use and standards that
you know how to adhere to but also
things that last you don't want to pick
something that will last a year and you
have to migrate to something else so you

want a sustainable data stack with sustainable tools with things that are easy for your team to use and so this also holds true for our Open Access publishing strategy we don't want to pick a strategy that isn't sustainable because we'll stop doing it now in the sea elans community and this involves my work with the open roomm foundation you're fortunate in that you have many open databases so the neod seans is often used in biomedical research and there's some basic research on it as well and it's a tightnit community where there are many many open data sets meaning people produce data they share them uh with the community and they have tools that are curating a lot of these data sets and papers and and things like that so we have all sorts of data sets uh available for all sorts of modes of investigating seelan biology and behavior so we have a number of open databases covering Anatomy genetics gene expression regulatory elements Behavior conomics so this includes like the worm wearing site which includes data for anatot conomics so we know in seans we know all of the neurons in the conecto we know a lot of their connections either through Gap Junctions or through synapses and we have worm wiring diagrams for both the male and the hermaphrodite so we have very rich data on worm brains we also have worm Atlas which is an atlas of structural Anatomy

so on top of the connectome data you can fit structural Anatomy data and that's at worm atlas.org the seans database uh run by MDC in Berlin features a lot of data on proteomics and RNA and so this is another database that we have available senen is a database of gene expression and celegans and the nerve system so we can you know build a model of the connectome we can have the worm wiring data set senen data the worm Atlas the mdcc Elegance database and we can actually you know produce a study from that so we have all those layers of data that we can include in and say a model or an analysis we also have this uh resource wormbase which is where we can explore the world of celegans Def find mutants so cgans has a number of defined mutants for specific genes they specific mutations that we can backcross and so if we have a strain of say some mutant defin mutant we have this worm that has a certain mutation that might be expressed in some Behavioral or anatomical deficit and so we can compare that to the wild type and say things about mutations of function and so this this is a really useful resource both for experimental work and for looking at the celan genome and finding you know like alignments and other types of data like in one of papers we recently did we found uh different types of we aligned different uh open reading frames within genene so it's really a good resource and then of course uh nature methods

this paper uh by people in the open worm Foundation um and this is called an open source platform for analyzing and sharing worm Behavior data and this is um on something called the tpsy tracker which is a movement tracker so they can actually take microscopy images of celegans they can track the worm's movement and they can characterize different movements they stereotypical movements worms make and they can actually track the behavior of the worms and classified so that's you know that that's the kind of data that's available in seans and so if you think about the active inference Community you know what kinds of data sets and what kinds of resources could be built around things involving active inference so you might for example think about you know and and some of these resources are just more General resources that you could bring to bear so you might say for example you know go to a neuroimaging resource and then go to a behavioral resource or build a behavioral resource there are all sorts of ways that you can build these um open data repositories to help the community Aid the community in solving problems and answering questions so this is a talk uh by eof freed this is called open science is an antidote to poor practice cynicism so one of the things he advocates for on this video is to use open science as a way to combat poor practice like we said open science is about reinforcing

practice and when you reinforce practice of open science you reinforce good scientific practice in general because you're focusing on the process you're focusing on you know cross checks and and you know other things that involve teamwork and so there there's a lot of this sort of advocacy for helping science through open science but also combating cynicism where when we see open when we see poor science we often think well this is just the state of science that it's actually failing or it's bad and so you can fight disinformation as well with open science this is a nice set of views so in the orthogonal lab we have done a lot of work modeling open source communities and their sustainability so I talked earlier about Open Access sustainability now I'm going to talk about open source sustainability and so we've done a lot of work with modeling Frameworks to understand this problem so why is sustainability important well in the case of Open Source many projects fail for a host of reasons sometimes they're related to people's projects you know not being of good scope where they weren't asking very good questions sometimes the team falls apart the team is dysfunctional sometimes the project runs out of money so there are a lot of reasons why projects fail and sometimes open source projects especially have problems with maintaining containers problems with people contributing so

they fail for those reasons as well so you can have things like we're scoping of a project we're trying to do too many things we're trying to do too few things we might have bad management or a lack of participation or we might have no resources in terms of time or money so these are all things that we want to be able to figure out how to manage at a reasonable level and to do that we have to understand sustainability so in these you know in these models that we've built we use different methods including active inference reinforcement learning and large language models to model open source communities and understand what makes a sustainable so the first one is this preprint on that kind of includes a lot of the work that we've done uh especially in terms of reinforcement learning models and active influence models this is our research search gate and then there's this newer work on combining agent based models and large language models this is sarasta Wall's Google sumar code project from this past year so she actually presented in front of the Google sumar code um uh Symposium and and she presented her project there and so this work so this work on large language models involved using large language models to generate issues and then generate agents that could solve those issues and see which kinds of you know almost create fake projects and then create issues to solve and then

see how the agents were able to solve them and so you know we used the combination of agent based techniques and large language model techniques to sort of work this out a very exciting project another approach is to take the multi-agent reinforcement learning route and so you see this grid down here on this figure and all these dots and they represent agents that are doing different things sometimes they have different statuses open source Community sometimes they're of a different level of Competency and you can actually run these agent based models you can Implement them in Misa as a python package and run these agent based models and then use reinforcement learning algorithms to sort of optimize the strategies that they're using so this is you know again where are using these uh sets you're creating these projects you're using these sets of issues and you're measuring their ability or their competence to solve those problems and so uh

shomon RV Raja Galan and hu shle were all involved in this work over the past several years we've been doing these kind of reinforcement learning models for understanding open source sustainability then we've uh done this work on Collective cognition for AI ethics this is where I applied active inference and we modeled agent behavior in terms of active inference using a package called interactively and this is the GitHub repo here interactively this

applies a partially observable partially
observe Markov decision process or a
pmdp

model uh to the problem we also used a
morally arm banded approach to model
explore and exploit Behavior so explore
and exploit is sort of this tradeoff
that agents make between exploring a
project exploring issues to be solved
and exploiting specific issues solving
them so it's like making you know
coordinating agent

behaviors and then figuring out what
strategy they're going to use and
modeling all of that so it's a very
interesting project uh it needs more
work on it so if people in this group in
in the active inference Institute are
interested in following up on it would
welcome and so there are a number of
issues about maintenance that are very
important to sustainability and this is
a special in the academic sphere because
academic organizations and open source
typically don't have the resources that
private organizations might so
maintenance and sustainability issues
are multitude in open source so I'll
talk about a couple of them here so the
first one is Log forj and this happened
a couple years ago where there were a
number of Open Source projects that got
hit with the security threat and it was
because their security fell at a sync
with the current state-of-the-art so in
open source we don't maintain things as
well as say like commercial products so
you have these holes that can be

exploited by malicious actors so love 4J was so serious that they had to um consult with the US government to try to help fix this problem and so this was a big problem um with just kind of maintaining the security of software but also it exposed a big hole in the open source model and so this is still something that probably requires meta governance you know some organization between like the federal government and open- Source organizations to solve the second is called the Mak or taker problem which is basically where some organizations make open- Source software and other organizations fork or copy the software for their own purposes thus being a taker and the problem here is where that relationship is not reciprocal that you know the makers will develop systems that encourage forks and then the open source contributors or the takers are expected to contribute back to the makers some sort of you know maintenance or something like that but they failed to do so and so this is the problem where you need to coordinate this reciprocal relationship and so an example of this is the word press for controversy so WordPress is the maker in this case and like I think it's WP engine is the taker in this case WP engine was forked from WordPress and it resulted in a fight between the two organizations one being the you know the maker and one being the fork or the Aker and they've been fighting over

commercial issues and fair use issues that have served to break the Norms of Open Source so there's this primary issue of maintaining sort of a reciprocal relationship between organizations that rely on basically rely on each other to survive but you can have all sorts of you know fights and disagreements about different ways that the software can be used commercially and other types of use that you know are although are stated in the license sometimes evolve away from the license and their practical concerns that you know isn't going to be solved by just some sort of reciprocal relationship or even intrinsic and extrinsic motivations and then finally the final example is XZ backdoor autotools this was an exploit that was made in in an open source repository where it was it resulted from lack of maintainer vigilance so this maintainer in this particular case was burnt out and they didn't feel like really kind of closely inspecting every pull request which is where a contributor will submit a change or contribution to an open source organization and so malicious malicious actor was able to get exploits into the software and it became a huge security problem it's kind of like I forj and the problem here is it's you know the failure here is at the level of maintainer and it wasn't the maintainer fault necessarily it's that sometimes

people in open source organizations as well as Open Access organizations open sites have become burnt out especially when you have to do all these things all these practice oriented things to realize the open vision and you know also you have disincentives that lead to problem so if you're not paying your maintainers enough or they not Mo you know if they don't have that intrinsic motivation high enough they can't do the prop the job that they need to do to make things work or the extrinsic motivations aren't well good enough you don't really pay or maintain her very much and therefore disincentivized to sniff out exploits and so these are our problems with maintaining open source in an especially in an academic environment this is a second uh YouTube video I wanted to point out this is behind the scene behind the screen there's a word play here which is open source software is both a tool and topic for research this is from William Schuler at Santa Fe Institute and he talks about sort of the history of software and how you know you can research open source software but also use it as a tool it has a lot of interesting issues in here so I recommend watching this video now I'm going to talk about how to sort of implement open software or like a purpose or an organization so we talked about you know one of these points about Distributing open academic software is

being able to run it properly or
run it in a way that doesn't rely on too
many
dependencies and so this is where we can
use something like Linux which or you
know some flavor of Linux to package
data analysis tools so that people can
use them in an environment where they
don't have to worry about dependencies
they don't have to worry about having
the proper software requirements so
there are two examples of where we can
use a Linux dist to package data
analysis tools and present them to
people in the community sometimes with
fairly low technical
standards so the first is narof Fedor
and so this was created as free and open
source software this is where we have a
version of the fedora this Dr of Linux
and we in Fedora you package a bunch of
neuroimaging analysis tools and you put
them in and you just give people the
package they install it on their machine
and then they can run those tools
without worrying about dependencies or
getting thing you're run on Windows or
Mac or
whatever the other example is n Debian
this is run by incf this is the group
led by por gleon and and Angus
Angus um run by run by or Leon and and
colleagues and this is where they have a
Debbie in drro of Linux and they have
these neuroimaging tools in Debbie and
so you have Fedora and Debbie and you
can choose between the two dros and you
can run these neuro iming tools and

these environments and that's that makes it easier I think for neuroscientists to do this kind of work to work with these pack which is uh a similar thing could be done for active inference and especially in Fedora you have these different flavors of Fedora like for artists or for musicians where you can package tools for those professions and they can just you know use it out of the box so focusing on the neurop Fedora uh project this is a special interest group whose goal is to provide Neuroscience researchers and enthusiasts with a strong easy to use ready made free and open source software platform for their work and anchor syha has let the taken the lead under aladora Morgan Huff is part of the orthogonal lab has also worked on this as well then we have neurodes so neurodes is not an operating system but it's a virtual machine it's a virtualization or a containerized environment where you have a basically virtual machine on your existing computer and you run everything inside of that and it really paves over a lot of the operating system differences and system requirement differences that you might find trying to install open- Source software on your home computer so like you know this is where you've basically created a controlled environment for the software so it really helps people run it when they have very little technical expertise and just get it to run that's all you need to do so neurodes uh is a

containerized data analysis environment
specialized for neuroimaging research
and this is something that you could
easily do with the active
inference it's where you package the
software code with the operating system
libraries and dependencies required to
run the code in a single lightweight
executable so this is platform agnostic
it's just you open up the container you
run it and you can do everything inside
there so thank you for your attention um
I especially want to thank the members
of the open source interest group group
where Bradley alisah Jesse parent Morgan
Huff Sarah Bala Brian mccorkel hu shle
rivy Roger gopalan and Shu
pomom um we have other people who cycle
through the group from time to time and
you're welcome to join
us uh please join the worth AI research
lab and then be on the lookout for our
open source activities we usually do the
interest group in the summer if you're
interested get in touch other wise happy
infering all
right that was a great
presentation this will be the last of
the recorded presentations for the
Symposium and last video on this part
stream we have JF claer an experiment in
artificial
agency an experiment in artificial
agency I'm Jean FR CER I'm a research
fellow at the Active infin Institute and
I do research in in cognitive
robotics one of the principal question I
ask myself is what does it take at a

minimum for an autonomous robot to learn
to survive in a world it knows initially
almost nothing
about other questions which follow is
are can a robot act for its own
reasons does it have agency can its
agency be
programmed and would such a robot be an
active inference
agent but one way to find out is to
actually build an artificial
agent uh and by artificial agent I mean
a sense making adaptive and
self-directed
robot so this project is a continuation
of a personal project that I started
quite a few years ago and three years
ago here's where I was I had a Lego
robot uh with uh quite a a few
sensors um like ultrasonic uh sensors
infrared sensor um a color sensor
luminum sensor um Motors of different
types all this uh running uh from a
Raspberry Pi with a board that that
provides an interface between the
Raspberry Pi and the various Lego motors
sensors and the robot also had a 360
degree
Beacon that allowed one robot to see
another through uh the other robots 360
Dee Beacon
Seeker so let's look at them in
action I Nam them Carl and and D for I
think quite obvious
reasons the Carl had gone through quite
a lot of training runs so it was better
at finding the food which is the piece
of paper on the floor um with a a beacon

infer Beacon providing the scent the attractive scent we can see that Andy is uh having bit of a hard time and they see each other so that if one robot starts acting panicky which they do if they collide too often the other will notice the pattern of movement and if one robot makes a baseline then the other robot will perceive that and say aha that's where the food is and try to move in the same direction but we see them uh moving about and trying to get the food and uh eating the food and Paul is clearly uh more apt at this task than ND and what they've learned through various trials are the policies that um are most successful at finding the food so that's where I was um three years ago the way I implemented uh the cognitive capabilities quote unquote of these robots uh was based on a uh architectural model uh described by Marvin Minsky in his book The Society of Mind this architecture posits that interesting behaviors emerge from many simple actors that interact in simple ways so we're looking at cognition as a collective intelligence the implementation was a hierarchy of these simple processes I call them cognition actors and they formed an abstraction hierarchy with at the very bottom um

cognition actors that simply wrap
sensors in uh effectors
Motors and then at the very bottom we
have cognition actors that care about
very uh simple things like the location
food the then at a higher level
information obtained from the food
location cognition actor feeds into the
food
approach cognition actor and this one is
in tries to figure out am I getting
closer to the food and so forth and so
on other cognition actors uh are
concerned with avoiding
collisions and avoiding Dark Places now
each cognition actor as I said is a
separate
process and each such process uh does
very simple things it makes predictions
about what it will observe
next it then updates its beliefs based
on uh predictions that were successful
or prediction errors received and then
selects a
policy randomly in the beginning but
then based on uh what was successful in
the past and applies that policy Acts
and then decides and then predicts what
and what the next uh incoming Sensations
will be and so forth and so on
essentially implementing an UDA Loop
observe Orient decide and
act there's one big problem though that
this Society of Mind these this
hierarchy of cognition actors is
predefined it's hardcoded for each robot
it's the same the only thing that's
learned

is what policy is more effective over time based on experience so the robot acted for my reasons for the programmer's reasons not its own the robots had autonomy but no agency in 2022 I joined the active Institute and continued this project under the uh symbolic cognitive robotics um tag and I asked myself what if what if a robot started with only a rudimentary Society of Mind what if it didn't have that pre-built Society of Mind this this hierarchy of cognition actors it started only with the the the most basic ones the ones that wrap sensors and motors then the Society of Mind uh would grow from experiencing autonomous engagement so the robot would start and with all these very basic um prior cognition actors for sensing and affecting and then would create new cognition actors and these cognition actors uh would be very um very simple in the sense that they would take their uh Sensations from the the uh uh the sensors themselves and would act uh by uh engaging the motors and then uh observe how it changes its its its perceptions of say color or distance and whatnot

Etc so cognition actors would be added to the Society of Mind BT them up to make sense of observations and actions now each new cognition actor would select an umelt composed of cognition actors one below and how do we know that a a a a

particular level of cognition actors in
This growing hierarchy is complete it's
when all cognition actors one level
below uh belong to an
envel now it would not only grow but it
would also shrink
because whenever uh con actors were
recognized to be hopelessly ineffective
they would eventually be removed and new
cognition actors would be would take
their place and the hope would be that
uh as this cognition hierarchy grows uh
becomes more uh
competent we would have a robot that
starts acting uh intelligently and and
in survivable
ways so the Society of Mind would grow
shrink and evolve as needed to survive
now that's that's a key Point here um
this is an implementation of um a form
of mortal
Computing obviously the the robot is
physically the wheels uh the sensors
themselves uh are not mortal they're
they're like a a a a computer on Wheels
you turn it off you turn it on it's it's
still alive it's still it hasn't died by
from having been turned off however the
Society of Mind itself is Mortal it
grows it
evolves in such that the robot will be
competent at finding the food it needs
to keep the Society of Mind alive so
what we have here is a um Immortal
Society of life of of mind trying to um
achieve a level sufficient level of
Competency to maintain
itself this is work in progress this is

um system that I'm in the process of building and it's composed of these subsystems agency where uh the smarts are for growing shaping uh and evolving a Society of Mind

there's another subsystem that it runs on the robot itself which I call Body which provides a um an interface to uh activate uh Motors and to read sensors there's also a a a version of the body subsystem which will run in a simulated

world and the subsystem world implements that simulated environment will also um animate an a real

environment and then there's a subsystem that I call Observer which provides me or any uh other human being with a a window into what's going on with the within the Society of Mind of the robot the agency subsystem runs on a powerful computer the the body subsystem the physical one runs on the brick pie which is powerful enough to run the significantly complex programs but not two computationally intensive ones and um world and observer and and body simulated run on a um a separate computer the karma body subsystem as I said gives access to real and virtual sensors and effectors

and here I have all the um the sensors I want to use and the motors um removed from a a the robot just like connected directly to the uh rickp and they are uh they can be uh activated they can be uh reached uh via an um arrest API which is essentially um they exposed a a a web

interface uh to the agency subsystem
the karma world subsystem uh here we see
the simulated environment it's a grid
World
um the the gray tiles represent just the
the surface and with a brown tiles
representing obstacles so the the robot
will bump into these
obstacles and will perceive them with
its uh simulated distance
sensors the green patch is represents a
Food Patch and the robot needs to find
food in order to consume the food and
replenish its its resources if it runs
out then it
dies the uh we can see that some ts are
are uh darker others lighter and they
this gradient this luminous gradient
corresponds to um a a scent gradient so
the closer the robot is to the food mood
the lighter the tiles
are there will also be um poison patches
which will be red so detectable as a a
red surface on on the simulated floor
the robot must learn to avoid poison
food because that will harm it and if
it's uh armed uh if it's
um excessively harmed then it will also
die
when a robot spends a enough time over
the food the food is consumed and uh
another Food Patch appears somewhere
else so the the environment is
dynamic and it changes with the uh with
the its interactions with the robot
that's an important
uh element
here now the um the karma agency

subsystem is the most complex and in the
the key subsystem and it implements a um
generative cognitive
architecture uh for the robot
implementing agency so we're going to
it's going to implement um cognition
actors as we've mentioned
already it will implement something I
call metacognition actors a very limited
uh understanding of the the the concept
of meta cognition one metacognition
actor per level in the hierarchy and the
cognition actors are responsible for
adding removing uh cognition actors as
uh needed as required then very very
importantly we'll have well-being actors
and these actors are represent the um
homeostatic values that
the Society of Mind must keep in and
within a acceptable bounds in order to
survive we have fullness integrity and
engagement we'll talk in more details
about what they are
later and then very importantly uh
there's an app perception engine actor
uh last year at the Symposium I
presented the um my implementation of an
a perception engine and this is the um
very important part of uh component of
the Society of Mind it is responsible
for making sense of
observations in order to be able to make
predictions we'll go into more details
well now let's go and look at these
well-being actors they implement uh
homeostatic uh States and I have three
one is fullness which represents how
much energy the uh the agent as a whole

robot has available to to do things that could be

Computing doing some comp uh expensive computation or um just moving about so there's a an energy budget that gets depleted and also gets replenished when the robot finds food and spends enough time in a Food

Patch there's also the Integrity well-being actor and that represents uh the health of of the uh agent the if the agent the

robot suffers too many collisions then Integrity goes down and integrity is is recovered over time think think the think about lives in in a video game that's pretty much the same concept and finally a little bit more abstract uh well-being actor is engagement and it represents

how to what degree the robot is engaging its environment we want to avoid the dark room situation where the robot U stops doing anything uh stops moving stop engaging its environment it h it it has a it it feels quote unquote an imperative to engage its

environment now the agent as a whole meaning the Society of Mind of the agent strives to maximize well-being for the entire Society of mind now low well-being levels will signal risk to the survival of the Society of Mind so that information will be available to each individual cognition actor and to the metacognition actors and will modify uh constraint uh constraint what they do key concept here is that well-being

imparts normativity to
beliefs beliefs acquired in the context
of a um low fullness or low Integrity
becomes a uh unpleasant undesirable
belief now this this this normativity
Associated to beliefs will focus
attention and motivate action as we'll
see later
on so well-being actors aggregate
metrics
uh received from the various cognition
actors and then aggregate these metrics
and broadcast uh wellness events back to
the cognition actors creating a give
essentially informing them of the um
ostatic State uh of the agent as a whole
now these metrics include the work
that's been done the food that's been
consumed
these these two will impact the fullness
um values then uh poison consume
Collision count will impact the
Integrity overall Integrity of the agent
and then uh Pol predictions made
predictions received prediction errors
sent policies executed we'll see that in
more detail later all these are
indicative of the uh Society of mine
engaging its environment
ment now metal cognition
actors their role is pretty
straightforward they're responsible for
adding cognition actors that are missing
and to remove underperforming cognition
actors each cognition actor is assigned
one level in the abstraction
hierarchy and it looks at the conion
actors under its supervisions

supervision and determines if all the con actors below them are part of one or the other's umelt so the notion of umelt here a cognition actor can only observe or act within its umelt and what is the umelt of a cognition actor it's other cognition actors of a lower level abstraction so at the the first level of cation actors the prior level uh where we have the um where they wrap sensors and Motors um they don't they they don't have a novelt well they do it's the actual external environment of the robot they don't have an internal enel anything above this level uh the cognition actors have as part of their umel other cognition actors and they constitute essentially the internal umelt of of within the Society of Mind so if a a one layer of the um Society of Mind is complete it covers all the conion actors below then it's time to create a cognition actor at the level above create a new layer of the Society of Mind and when you create a cognition actor you give it or the metacognition actor gives it its umelt mostly randomly with um the constraint that there's going to be overlap between umvs of various cognition actors and making sure that they have access uh to the motors directly or

indirectly now when
when well-being is low within the
Society of Mind when the well-being
actors are reporting low well-being
levels this might
accelerate the removal of cognition
actors the metacognition actors being
aware that we're not doing well now
we'll say okay which cognition actors
are underperforming our expectations and
let's remove
them and in order to for them to be
replaced with new Cog actors which
hopefully will do
better and if uh well-being is low this
will also slow down adding new cognition
actors so there's an influence of
well-being on the the actions of the mid
factors now I want to say adding and
removing um it's important to uh to to
understand that adding is bottom up and
removing is top down for example if I
have a a meta if I have a cognition
actor that has let's say two other
cognition actors in itself I cannot remove
a cognition actor in its unv because it
depends on it but I can remove a
cognition actor that is not in any
others umelt so that removal will be
from the top to bottom and addition will
be from the bottom to
top what does a cognition actor
do well a cognition actor observes its
umelt of other cognition
actors it generates
beliefs from analyzing its obs
ervations and
then it generates

policies in order to modify or validate
its beliefs and it does this this
observe believe and act always within
itself and one time frame at a time so
each cognition actor goes through its
more or less UDA Loop one time frame at
a time now the the lower the lower a
conition actor is in the uh Society of
Mind the more concrete it is the less
abstract it is um the shorter this time
frame will be it will cycle
faster it's important also to realize
that there there's no there's
um there's no synchronization between
cognition actors each starts its time
frame independently of other cognition
actors
and so that the they they may overlap
but they don't coincide they don't
overlap completely and the more abstract
the cognition actor is the higher up it
is in the hierarchy of cognition actors
within the Society of Mind the longer
its time frame will
be which means that very abstract
cognition actors will update their
beliefs more
slowly and will take
actions
uh not as quickly as a a more concrete
low-level cognition
actor so we can see here the the
cognition actor the large one surrounded
by its lowlevel cognition
actors the what does it observe what
does the con cognition actor observe in
itsel well it observes it observes the
beliefs of its of the C actors in its

zelt so as a more abstract cognition
actor I observe the beliefs of the more
concrete cognition actors in my
envel then based on these beliefs and as
we'll see based on the normativity
associated to the beliefs are they good
beliefs are they bad beliefs the
cognition actor will
uh um decide on what its goals are in
this current time frame
the goals would be modifying a
particular belief or validating a
particular belief then it will generate
policies to attain this goal and then
emit this policy to its envel and say
okay do what I want and we'll see in
more details what do what I want
means So within each time frame a
cognition actor Will predict
itself after having made sense of it
that's a key uh aspect that we'll we'll
go into more
details it will update its beliefs by
analyzing past and present observations
so a a a cognition actor has a memory of
the current observations in this the
current time frame and observations made
in previous time frames and it will
update its beliefs but from analyzing
current observations and past
observations then it will select a goal
affecting a belief in it tends to impact
and construct a policy and recruit its
invel to realize
it so if you look at the little U
redesigned UDA Loop here first there's
predict make predictions
about beliefs of of other cognition

actors in uh my umelt in order to
predict I need to have a causal model of
my observations that's what a perception
will do we'll see in more
details then the as a cognition actor I
update my beliefs select goals act and
back to predicting in the next time
frame a perception is defined here as
the process of assimilating sensory
information into a coherent unified whole
so let's look at uh
predicting what are the observations of
a cognition actor observations are
predictions it makes about the beliefs
of other cognition actors which are in
its
umelt
plus that plus prediction errors so it
makes predictions of itself of contion
actors in sunv about their
beliefs and the
umel cognition actors receive that
prediction look at it does it
corresponds to my current beliefs if yes
not saying anything I'm not contesting
the prediction if it doesn't match then
I emit a prediction error and the
cognition actor accumulates these
predictions that are uncontested and the
prediction errors and together they
constitute the Cog actor's current
observation
ations
so we'll see that not all observations
receive the same attention that they're
not subject they're not equally subject
to prediction there's a prioritization
there's a notion of attention we'll see

that a bit

later attention attention prioritizes

which observations to update so the

cognition actor starts a time frame with

observ observations made in previous

frame and if there and it wants to make

to make those observations current to

the current time frame as many of them

possible but there's a limited amount of

time in which to um update these

observations so it the Cog actor will

need to prioritize which observations

from the previous time frame to update

into the more current observations so

which one will it try to update first

well it will look at

predictions that were

made by higher level predic uh cognition

actors about its beliefs and knowing

which observations support the beliefs

will say ah these is are the more these

are the observations that um more uh

abstract more important cognition actors

are making about me my state my beliefs

so I'm going to I'm going to try to make

sure that I have up-to-date

observations that are of interest to my

parent my my parent conition

actors so that's the first one I care

about the first observations I care

about the ones that are of interest to

my parent function actors actor sorry or

actors the second most important

observations in in terms of uh

updates are the observations that

support the strongest desirable and

undesirable beliefs remember that uh

beliefs are the result of analyzing

observations current and
past if I have very strong desirable or
very strong undesirable beliefs
then I will want to verify that those OB
observations are still current so there
will be the next uh observations that I
will try to uh update then finally the
the observations that uh support weak or
neutral beliefs and then whatever
remaining observations they are so the
cognition actor will take will
prioritize the observations it it
that existed from the previous time
frame and say okay I'm going to now make
predictions um that these observations
are still valid or not
valid and if those there are obser
prediction errors then then the the the
observations are updated to the
prediction error and if the observation
is not contested then the observation is
is is um the one that is
current this implements a predictive
processing uh
architecture because as you can imagine
a high level cognition actor will make
predictions about
beliefs uh of the the its children
actors this will trigger predictions
about the observation supporting dis
beliefs one level down to the conition
actors in the layer below and all the
way down so predictions flow down and
prediction errors flow up through the
entire hierarchy of uh the conition
actors in The Society of
mind now in order to make predictions we
need to have made sense of uh

observations that's what Apperception does so we need to make sense of observations in order to predict them now in order to to explain the concept of app perception in which in last year's presentation I went into quite a lot of details but just I want to give a an overall understanding here let's take the example of observing uh two lights light a and light B and the light can either be on or off and at different steps in time the lights may change their state from on to off off to on or not so here we have a set of observations of light A and B over time and sometimes we may not be able to observe a light like uh light a in in the fourth state is a gray rectangle we don't know but we have a good enough set of observations and now the goal of the app Apperception engine is to find a causal theory that explains how the observations change time why are we seeing some light a turn on after having been off or Etc in in order to formulate a causal Theory kind of a little program that when run recreates what we've observed the Apperception engine may need to imagine another light it might need to imagine some kind of connection between the lights so the Apperception engine will in in formulating a causal Theory will do some abduction some imagining of

unobserved latent

objects

so we have a number of observations that
are remembered from previous time
frames we may uh we we don't accumulate
all observations forever we forget the
older

ones and then we feed these observations
into the a perception engine out of that
comes a little program a causal
theory that when applied to the latest
observation will produce
the next incoming expected

observations so that's a way of of of
making predictions of incoming uh
observations from current and past
observations from the current
observations given a causal Theory so
you feed the latest observation you feed
all the observations into the
Apperception engine it gives you a
causal Theory you feed the latest
observation into the the causal Theory
as input the output is incoming
predicted

observation so an app perception
Apperception means finding a causal
explains by all the rules of the theory
and predicts
observations

so a cognition actor accumulates
observations wants to make sense of them
in order to predict incoming
observations and for other reasons as
well which we will see and it will what
it does is okay it it gives the app
perception engine actor its observations
over time and says please explain to me

what I'm observing give me a causal Theory so I can predict future observations given this causal theory if if one is found successfully then the con actor can predict in coming observations and what does it observe it observes the beliefs of cognition actors in itself more more concrete contion actors and then if the prediction is correct if the causal theory is correct and accurate we don't get a prediction error if the causal theory is partially correct sometimes correct then we get a prediction error so getting prediction errors undermine confidence in the causal theory that the con actor has received from the aips and engine to explain what it sees in its umelt so it's quite possible that there's a moment in time when the observations when the causal theory is is quite effective and it's accurate but then something changes in in the state of the umel function actors such that the causal theory is is is missing is is very hit and miss and it may come a point where the contion actor says I don't like that causal Theory anymore it's it's producing too many prediction errors so it's going to ask the a perception engine actor give me a new causal Theory based on the latest observations that I've accumulated In My Memory okay so we looked at um how cognition actors predict income observations in order to uh accumulate those new observations

to update those new observations based on whether or not these uh predic the predictions uh are contested by prediction errors or not so now once we have a new set of current uh observations within this time frame that we are operating in it's time to update beliefs

beliefs are if you want um integrate are are the integration of observations into a a a more aggregated um analytical uh assertion so a cognition actor updates its belief from analyzing remembered observations and also uh remembered actions that it that that that it

took so and we'll see what kind of of integration of observations and Analysis of observations lead to what type of beliefs but whatever type of belief it is a belief will be desirable or undesirable depending on how it correlates with the agent's well-being if a belief is being held mostly while the a the agent as a whole is feeling bad it has low resources or low integrity or or or limited engagement then these these beliefs are going to be tainted by the uh wellbeing um levels of the agent at the time those beliefs are held now that's what makes a belief desirable or undesirable what makes a belief strongly held is if it is supported by observations those observations from which analysis the belief is derived if

it is supported by observations that are predicted by an accurate causal Theory we have an excellent causal Theory it's it's it rarely misses

so um the the cognition actor is feeling confident about its observations and it makes the the whatever belief is is is produced from analysis of these successfully predicted observations better supported more strongly held

now if a belief persists over many time frames thanks to or in spite of actions taken action taken to validate or invalidate it if the belief

persists then it's even more strongly held it's it's it's there's more there's increased confidence that the belief is uh represents uh what's really going on in the umelt of the cognition actor

now there are different kinds of beliefs beliefs are are are generated are produced from the analysis of uh current and past

observations and okay so there's

adduction one type of belief is is abduction and it's a direct consequence

of um having requested a causal Theory and using a causal Theory remember in in

a causal Theory May um imagine a a latent object or a latent relationship

between objects well that's that's a

form of belief I believe there's an

object that I don't see but must exist

in order for me for for a causal Theory to make sense

of actual observations I made so an

abduction belief would be like I assume

there's a hidden thing next to the one I
see another kind of belief is just
counting I'm I'm I'm next to two things
I have made two observations that
represent uh say being next to something
and so I have two of them so I'm I can
have a belief that I'm next to two
things account belief now a trend belief
is a it's a belief that represents the
the the um discovery of a trend in a c
certain observations over time for
example uh if um the contion actor
observes distances to some obstacle and
those distances are getting
smaller
um over time then that's a trend my
distance to a thing is getting
smaller and then there's another kind of
belief is is some
observation ending
stopping my getting closer to a thing
stopped so that's an ending kind of
belief and then if the conion actor has
uh taken some action well there's the
belief that it tried to do something
like I I tried to stop my distance to a
thing from getting smaller so if the
action was trying to end the belief that
the belief in a trend as you can see now
that beliefs are are are composed
they're getting more and more abstract
so I tried to stop my distance to a
thing from getting smaller is a kind of
another kind of belief it's an attempt
kind of belief so we have all these
different kinds of
beliefs and when a Cogan actor
synthesizes the belief from analyzing

observations it assigns of belief a
globally unique predicate
name parent cognition actors the
cognition actors in which umelt this
cognition actor
belongs um the parent conition actors
observe predict the beliefs of cogn
factors in in in their
owns and reference them by name so if
the there's a belief in umel actor that
some Trend
exists the um we're getting closer to an
obstacle well the this belief will be
exposed will be made available for for
observation to the parent as a named uh
predicate
why and how that belief uh came about
what the observations are by the umel K
actor that led to that
belief it's it's it's opaque there's a
name and that's the name that's
available for observation there's
an a statement a predicate a name
predicate that's available for
observation by the parent cognition
actor
only the cognition actor that holds the
belief it named knows which of its own
observations support the belief and how
so if I say to you I believe I I I
believe getting
closer that's something that can you can
observe as a parent contion actor ah my
child con actor believes in getting
closer but the parent cognition actor
doesn't know why I believe we're getting
closer doesn't have access to the set of
observations that led to me holding the

belief that we are getting closer all
these distance observations that led to
me having holding the belief that
getting closer all that parent knows is
that I hold the belief called getting
closer and it's able to observe it
observe it
you know ending or or starting Etc but
that's that's that's a key um a key
concept that beliefs are are essentially
named um there are predicates their
names if you wish that uh hide the C the
reason why the belief is held to parent
function actors that observe my beliefs
so there's information hiding going on
so a cognition actor observes the
beliefs of its children cognition actors
these these cognition these these
beliefs that are observed are the
consequences of themselves observations
about from of of actors one level below
so my belief is is uh
another if I'm a cognition actor my
belief is another cognition actor's
observation so it's really beliefs it's
it's beliefs about beliefs about beliefs
so it's beliefs all the way
down that's that's a maybe a little
tricky concept if I'm at the top as a
cognition actor I observe beliefs of
connection actors under me these beliefs
of connection actors under me are
supported by their observations of
beliefs of cognition cognition actors
under them so beliefs about beliefs
about
beliefs time to open a little
parenthesis here maybe take a

break the implementation that I'm
working on of
agency is a symbolic implementation so
it's its agency realized from symbolic
reasoning all the various agency actors
the the cognition actors the mid
cognition actors the well-being actors
the uh the perception engine actors all
run prologue programs prologue is the
programming language and those pro
programs do symbolic
inform a cognition actor as we've seen
invents symbols invents gives names to
beliefs it discovers from analyzing
observation so it invents symbols and it
Synthesizes these beliefs as symbolic
statements getting closer to
XYZ the app perception engine as we've
seen very kind of
briefly its job is to discover causal
models and these causal models are
symbolic programs they're little logic
programs so agency is built on symbolic
reasoning in my implementation closing
parenthesis now let's talk about
precision and belief
strength the more accurate a causal
Theory the more precise it are are are
considered its predictions so we saw
earlier now precise predictions if
they're not contradicted by by PR
errors do strengthen the beliefs the
predicted observations support so you
have a bunch of observations held by the
observations in the moment remembered
observations these observations are the
result of predictions and or prediction
errors and we analyze these these these

observations coming from predictions to
generate beliefs I'm getting closer to
an obstacle from observations of
distances now if my observations of
distances were predicted by a really
top-notch causal Theory one that almost
never misses then I have confidence in
the beliefs that have derived from my
predictions I feel confident that I am
indeed getting closer to an obstacle
given all the the the distance
observations that I've accumulated
because I've accumulated them by making
predictions using a very accurate
empirically accurate causal Theory so
precise
predictions strengthens the belief that
our der from these from observations
from predicted observations now the
stronger beliefs are because I have may
have strongly held beliefs and weakly
held beliefs the strongly the more
strongly held beliefs are the more
attention they will receive When
selecting when this the cognant actor
needs to wants to select one and say I
don't like this belief It's associated
with poor
well-being and I want to I want want to
change that belief I want I want to
impact it I want to I want to get rid of
that belief I want to make that belief
not true or if it's a very pleasant
belief oh I want to make sure that it is
I want to validate it I want to keep
that belief uh I want to keep holding
that belief because it's a good one but
I care whether this the belief is

strongly held or weakly held because
strongly held belief will get more
attention will be the ones that'll be
selected for validation or invalidation
above the the the weekly held
beliefs so how do we impact
beliefs that's the next step in the ud
next two steps in the UDA Loop of our
cognition actor within each time frame
it predicted based on its understanding
of past
observations it updated its belief from
uh from analyzing predict predictions
that were realized or or errors so those
predictions I.E those observations it
updated its beliefs and now it says it
needs to select a goal and or want may
need to select the goal that say I want
to get rid of this belief or I want to
validate this belief and how am I going
to accomplish this goal I'm going to
select the policy and then and and then
ask my unal to act to realize this
policy so that's what we're looking at
now so a cognition actor Acts via
policies it
formulates a cognition actor strives to
invalidate unpleasant beliefs and to
validate Pleasant beliefs but it will
prioritize strongly held beliefs whether
unpleasant or or
Pleasant a cognition actor will impact
the belief validate or invalidated it
holds by uh promoting or inhibiting the
observations that led to it now I have a
belief that we are getting closer to an
obstacle and somehow it's it's it's an
unpleasant belief maybe uh It's

associated with in the past with um
a a decrease in in in um
Integrity leads to bumping into things
so I want to I want to put an end to
getting closer to an obstacle I want to
I don't want this belief that I'm
getting closer an obstacle to go away
now I hold this belief because I had
observations of reducing distances over
time to some obstacle so I want to
change these observations basically I
want to I want to stop getting close
having um having uh distances that get
smaller I want to stop this
so my observations of distances happen
to be beliefs of function actors in my
umelt so what I'm going to do I'm going
to say okay I'm going to select a number
of of observations that support the
belief I want to get rid of and I'm
going to formulate a policy as a list of
goals to my intel CA I'm going to say I
want you to get rid of the fact that I
have small distances to an obstacle
that's that's that's your goal so I'm
going to I'm going to set a number of
goals me the cognition actors who
believes we're getting closer to
something and it's not does feel good
I'm going to tell I'm going to formulate
a policy such that my umelt con actors
will try to um impact their own beliefs
that we're we have a small distance to
an obstacle for
example so that's that's a little bit
subtle here so I'm as a cognition actor
I want to modify a belief that belief is
based on observations these observations

are beliefs of umelt cognition actors so
I'm going to say to my umelt cognition
actors I want you to to modify your own
beliefs that I've observed I want you
to change your beliefs beliefs that I'm
observing that lead me to believe that
I'm getting closer to an obstacle so I'm
I'm setting up a bunch of goals for my
umelt function actors so that if they
realize if they if they modify the these
beliefs then it will modify indirectly
my own belief that I'm trying to get rid
of okay so I'm basically telling my my
umelt these are the goals I want you to
realize so that I it I would indirectly
realize my own

goal okay so this Pol a policy that I
put together are a set of goals for my
umelt actors to
realize

now A nelt C

actor will then given the goal it
receives formulates its own
policy because we're I'm
asking a con actor in myelt to
modify some some belief or other okay
says the con actor then I will formulate
my own policies to do that
to to achieve the goal that you that my
parent C actor sent to
me

so a cognition actor will receive
goals from parent cognition actors and
these goals received from IR up will
have higher priority that whatever goals
the cognition actor might want to
formulate on its own because the umel
actor has its beliefs has belief that it

wants to validate has belief it wants to
to to
validate and left on its own to its own
devices will do just that but if it
receives goals from a parent cognition actor
these take priority and it will put
aside its own concerns to uh address the
concerns of the parent cognition
actor now a cognition actor can execute
only one policy at a time which is why
it would prioritize the ones that it
receives from its parent
and when all is said and done the
cognition actor that initiated the
policy will determine will will verify
if this was successful or failure over
time if the belief they wanted to impact
was indeed impacted and a policy policy
that's known to work will be reused
that's how cognition actors develop
habits okay

so where does the causal theory that the
con actor uh holds guide the formulation
of its policies because there was CA
causal Theory serves two roles one
is to predict incoming observations and
the other role is to formulate an
effective policy so let's say that
cognition actor is a belief that wants
to impact because
uh the well-being me levels are are are
low Associated to that belief so he
wants to modify that that belief get rid
of it well it needs to formulate a
candidate policy which of the
observations that I made that I've
analyzed into this belief which which
observations that support this belief I

want to
modify so which policy do I want to
formulate well what it will do is we'll
say okay what do I need to do given what
my what I'm observing right now so that
I will observe something differently
that will
either remove the
belief or or reduce it so it will use
the causal Theory so what would happen
if I did X if I I I I I if
I asked my umelt to do
XYZ so so the causal Theory will answer
that question we say
well plug that into the causal Theory
turn the crank and this is what you
would be uh be observing next if you
were doing this if you if if you
executed this action this is what you
would observe next and then given what
you would observe next you can
calculate ah this these will be the
anticipated beliefs out of these
anticipated observations
might the policy work with would the
belief that I want to impact be impacted
the way I want to impact it if so yeah
let's use that policy otherwise let's
look for another one so the cognition
actor searches for an effective policy
given its understanding of how things
work in its umelt its causal
Theory and and then applies the ca
theory predicts what would happen if it
took this action if it it it if it
requested this goal to be realized looks
at the the the the resulting anticipated
beliefs and decides was this a a a good

effective policy and then and then it
would Emmitt that policy and ask it umelt
please achieve these goals so that I
achieve my goal so and and that
percolates down because each COG actor
being given a goal from the parent con
actor says ha how should I best realize
that goal what policy should I
formulate for my
own child my own
umelt in order
to realize the goal that was given to me
by my parent so it's policies all the
way
down and it bottoms out when at the
level where um effects are produced
moving Motors turning moving forward
moving backward
whatever okay so that's a lot of um a
lot of content describing this cognitive
architecture this generative cognitive
architecture what I hope will happen is
through this growing shrinking evolving
um um defining of this Society of Mind
composed of of cognition actors
um with uh in the context of of of of
homeostatic measures provided by
actors I'm I'm hoping that with policies
going down beliefs percolating up
predictions going down prediction is
going up all this this this this changes
and evolution of the Society of Mind all
these all these activities from all of
this would
emerge agency and I I'm looking at that
as as a form of operational closure with
uh constraints flowing down constraints
flowing up and eventually hopefully the

um the Society of Mind will explore a space of possible organizations and and and and and and and space of of possible uh ways of defining cognition actors relating cognition actors together such that it leads to behaviors of the robot that preserves the life of the Society of Mind that Society of Mind never reaches a point where it runs out of fuel where it is damaged Beyond repair that's that's the overall goal I think of the Society of Mind as kind of autopoietic with all these um elements of the Society of Mind leading to other elements for example a causal Theory will lead to predictions predictions will lead to prediction errors prediction errors will lead to observations uh observations will be fed into apperceptions which will lead to an updated causal Theory and and observations lead to to beliefs um beliefs lead to policy um a causal Theory will determine which policy is more likely to work policy leads to effect effects lead to New Sensations Sensations lead to new observations and so forth and so on so this I look at this as a a kind of a um a a set of of of processes that support each other and in in something of a of a closed set of complex Loops leading to a Society of Mind that survives over time now there's a number of pH philosophical stenes that are taken by this project and by philosophical stance

I mean uh practic pragmatically
Justified perspectives a way of seeing
world first I look at active inference
as a normative framework I don't think
that's controversial at all I mean
that's that's what it's defined to be
and uh to persist an agent must behave
as if it acts to minimize surprisal now
clearly I'm not
implementing the uh Society of mine
using the mathematics of the free energy
principle but I'm implementing it
hopefully such that
doing an active inference analysis of
the robots observable behaviors in
states will demonstrate that the uh
Society of Mind the agent as a whole is
acting such as to reduce
surprisal so I'm
implementing an agent and also its
environment as a system of generative
processes the active inference analysis
will be done in terms of generative
models that are inferred uh from
observing the behaviors of the robot and
uh observing the states that are visible
to an outside
Observer other uh philosophical stenes
I'm just going to go through them very
quickly each one is worth an entire
presentation I I I take the stance that
cognition arises from an embodied agent
engaging its environment that sense
making is the Discovery
of unified causal
theories that meaning is grounded in the
agent's drive to survive no mortality no
meaning that intelligent emerges from

interactions between a collective of parts that's the Society of mind and that the parts constrain the whole and the whole constraints the part so constraint

closure I'm implementing I'm in the process of implementing this this this this system that will hopefully demonstrate artificial agency and here's the current progress there are three phases in this project phase one I developed in cognitive architecture that's what I presented today uh implemented a one version of the of the app perception engine and that's what I presented last year

I implemented uh the subsystem to uh realize a to connect to sensors and actuators that's the karma body and I implemented a simulated grid World in which a simulated um agent would would operate and and and interact that's to uh speed up uh implementation because it takes a long time to uh if I were to do this only in the real world the um the the edit run debug Loop would be extremely

long and so I'm in the process right now of implementing uh the agency uh capabilities in phase two um which would be about six months to a year from now I will have a a redesigned robot body I will um uh be testing in a real life and I will be collecting a lot of data through a a The Observer subsystem and in the third and final phase the uh data gathered from runs

both in real life and in virtual world
will be subjected to active inference
analysis to uh validate that my agent
does survive because it minimizes
surprisal that's it um if you are
interested you can contact me at uh this
email address JF do

Cloutier active. Institute you can also
uh reach me on uh the Sim Cog robotic
channel of the institute's

Discord and if you're interested in the
source code it is on GitHub so I'd like
to uh thank uh the active INF Institute
for uh support and encouragement